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MIL-HDBK-1010A
1 AUGUST 1992
SUPERSEDING
MIL-HDBK 1010
NOVEMBER 1989

MILITARY HANDBOOK

COST ENGINEERING: POLICY AND PROCEDURES

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ABSTRACT

Current criteria and procedures are presented for use in the preparation and review of the Activities' Project Data Sheet, Facility Study Cost Estimates, and the EFD's Cost Certification Estimates for MILCON construction and operational/maintenance projects at all naval shore facilities. General guidance for other types of cost estimates is also provided.

Projects have four parts of equal importance: requirement, justification, scope, and cost. The preparation of documents and the reviews are usually satisfactory with full attention given to these four parts.

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FOREWORD

This military handbook presents the Naval Facilities Engineering Command (NAVFACENGCOM) criteria and procedures related to the preparation and review of the activity's Project Data Sheet, Facility Study, and the Engineering Field Division's (EFD) Cost Certification estimates for the Military Construction Program, including Operations and Maintenance (O&M) projects. This handbook is intended to be a cost estimate preparation and review guide, and should not be used exclusive of full consideration of scope, additional functional features, and supporting facilities.

MIL-HDBK-1010A provides guidance, procedure, and examples. The guidance costs given are for two consecutive fiscal years based on an area cost factor of 1.00. The adjusted unit costs have been escalated to a common date for each fiscal year, and have been adjusted for size and area cost factors.

Recommendations for improvement of this handbook are encouraged and should be furnished to the appropriate EFD point of contact as noted in par. 1.5.

THIS HANDBOOK SHALL NOT BE USED AS A REFERENCE DOCUMENT FOR PROCUREMENT OF FACILITIES CONSTRUCTION. IT IS TO BE USED IN THE PURCHASE OF FACILITIES ENGINEERING STUDIES AND DESIGN (FINAL PLANS, SPECIFICATIONS, AND COST ESTIMATES). DO NOT REFERENCE IT IN MILITARY OR FEDERAL SPECIFICATIONS OR OTHER PROCUREMENT DOCUMENTS.

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COST ENGINEERING: POLICY AND PROCEDURES

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PART 1: PROJECT DATA SHEET (PDS) AND FACILITY STUDY COST GUIDANCE

Section 1: INTRODUCTION

1.1 Purpose. This handbook is a guide for the preparation of planning cost estimates in support of the Shore Facilities Planning System. It supplements instructions and direction provided in the instructions listed in par. 1.2.

1.2 Background. OPNAVINST 11010.16, Command Responsibility for Shore Activity Land and Facilities establishes the Shore Facilities Planning System. NAVFACINST 11010.44, Shore Facilities Planning Manual carries out the system and requires activity submission of a planning estimate with cost backup and other data, to propose a project for the Military Construction Requirements List.

1.3 Importance. The importance of an accurate planning cost estimate cannot be over emphasized, as it could be the determining factor as to whether or not the project is included in the budget year program. Although requirements and justifications are important parts of project submittals, this handbook deals only with scope and cost. Scope is included because an estimate is no better than the definition of scope that has been determined. An incomplete scope will result in an incomplete proposed project cost.

1.4 Cancellation. This handbook, MIL-HDBK-1010A, dated August 1992, cancels and supersedes MIL-HDBK-1010, dated October 1989.

1.5 Comments or Suggestions. You are encouraged to advise the Engineering Field Division (EFD) Planning Division (Code 20) or the Cost Engineering Branch (Code 407) of errors, conflicts, or omissions detected while using this handbook, as well as to suggest improvements which will help you produce higher quality planning cost estimates. Advice or comments should be forwarded to the EFD Code 407 to be considered for the MIL-HDBK-1010 update. Points of contact are as follows:

EFD	Code 407	Code 20
Atlantic Division	(804) 444-9907	(804) 445-2300
Chesapeake Division	(202) 433-3315	(202) 433-3387
Northern Division	(215) 897-6344	(215) 897-6270
Pacific Division	(808) 471-8436	(808) 471-3931
Southern Division	(803) 743-0463	(803) 743-0777
Western Division	(415) 742-7760	(415) 877-7544

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Section 2: SCOPE CONSIDERATIONS

2.1 Introductory Paragraph. The Primary Facility Unit Cost Development Sheet must include separate entries for specific features of additional cost. Certain features and their cost derivation are discussed below. Other specific features may not be discussed in this section, but should be a separate item to emphasize significant cost.

2.2 Operation and Maintenance Support Information (OMSI). The OMSI product is provided in the form of manuals prepared by the designer during construction to give the activity clear comprehensive data for use in operating and maintaining their new facilities and to ensure economical, safe, life cycle operation and maintenance. The principal OMSI elements requiring consideration are as follows:

a) Design Concept Data and Basis of Design

b) Basic Data needed for preparation of Facilities Support Contracts and Performance Work Statements (FSC/PWS)

c) Operating Procedures for Startup and Shutdown; Emergency and Safety Instructions

d) Operation and Maintenance Technical Data and As-built Product Documentation

e) Maintenance Requirements, Test Equipment, and Special Tools

f) Preventive Maintenance Requirements and Schedules

g) Troubleshooting Guide, Diagnostic Techniques, Repair Procedures

h) Supply Support (repair and spare parts, fuels, lubricants, etc.)

i) Extended Special Warranty Information

j) Environmental Considerations

k) Training Plan, Staffing Requirements, Personnel Qualifications

2.2.1 Types of OMSI. OMSI is valuable and should be included on most projects for which manufacturer's product data is to be submitted by the construction contractor. OMSI is usually needed on projects having complex built-in systems for which submittal of operating or operation and maintenance data is called for in the construction specifications. The degree of OMSI will vary with the kind of facility and complexity of systems in the facility. For identification, projects requiring OMSI can be classified as Type A, Type B, Type C, or Type D.

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a) Type A: Operation and Maintenance Manual plus User Manual

b) Type B: Operating Manual plus User Manual

c) Type C: User Manual only

d) Type D: No OMSI Manuals (As-built Drawings only)*
* As-built Drawing Listing is included in the User Manual for Types A, B, and C.

Examples of facilities which usually require a Type A are listed in Table 1, and those requiring a Type B, in Table 2. Rather simple facilities requiring only a User Manual are classified as Type C, samples of which are shown in Table 3. Projects listed in Table 4 such as paving and dredging would not require OMSI manuals and should be classified as Type D. For Type D projects, only As-built Drawings will be required in the same manner as presently handled.

The User Manual compiles and indexes as-built product and operation and maintenance data into a consolidated manual and also provides useful information such as special warranties and small scale floor plans showing utility shutoff locations. Facilities having complex systems which require special operating information should be classified as Type B and be provided an Operating Manual in addition to the User Manual. NOTE: An Operating Manual will not provide information on maintenance, troubleshooting, and repair of selected systems, therefore its use should be limited. Complex facilities having systems which require special operation, maintenance, and repair information should be classified as Type A and be provided an Operation and Maintenance Manual in addition to the User Manual. Depending on the complexity of a particular system, even a simple facility such as a Child Care Center could need a Type A OMSI to provide an OMSI Operation and Maintenance Manual for that system; e.g., variable air volume (VAV) or thermal storage system. A project such as a hospital may include 15 or more operational systems in the manual.

2.2.2 OMSI Estimate. In estimating the cost of OMSI for a specific project, use the best available information. If a good basis to make the estimate does not exist, Table 5 should be used for a preliminary estimate based upon a percentage of the total primary facility cost. If OMSI is required for the supporting facilities (such as utilities), add these costs to the primary facility costs before calculating the OMSI estimate. The percentage selected (within the ranges shown) should be based on the relative complexity of the facility and the number of systems to be covered by the Operation and Maintenance Manual or the Operating Manual. If OMSI is greater than $50,000, it shall be a separate line item on DD Form 1391 (Block 9) and Primary Facility Unit Cost Development Sheet. If OMSI is less than $50,000, it shall be part of the building cost and shall be shown on the Primary Facility Unit Cost Development Sheet. In either case, OMSI should be included in Block 10 of the DD Form 1391 under Description of Proposed Construction.

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Table 1
Examples of Facilities Usually Requiring Type A OMSI
(Operation and Maintenance Manual Plus User Manual)

FACILITY TYPE	BASIC CATEGORY CODE
PETROLEUM, OIL & LUBRICANT (POL) FACILITIES	120
COMPUTER OPERATIONS CENTER (LARGE/COMPLEX)	143
TRAINING FACILITIES WITH COMPLEX SYSTEMS	171
AIRCRAFT MAINTENANCE SHOPS	211
MISSILE ASSEMBLY & MAINTENANCE SHOPS	212
SHIP MAINTENANCE SHOPS	213
LABORATORIES	310
CENTRAL REFRIGERATION PLANTS	431 & 826
MEDICAL FACILITIES	500
POWER PLANTS (750 KW AND LARGER)	811
HEATING PLANTS (950 MBTUH AND LARGER)	821
INDUSTRIAL WASTE & SEWAGE TREATMENT PLANTS	831

Table 2
Examples of Facilities Usually Requiring Type B OMSI*
(Operating Manual Plus User Manual)

FACILITY TYPE	BASIC CATEGORY CODE
OPERATION & TRAINING FACILITIES	100
LIQUID FUEL STORAGE - BULK	411
MISCELLANEOUS UTILITIES	890

* May require Type A OMSI if maintenance and repair information is needed.

Table 3
Examples of Facilities Usually Requiring Type C OMSI
(User Manual Only)

FACILITY TYPE	BASIC CATEGORY CODE
WAREHOUSE - GENERAL	441
ADMINISTRATIVE FACILITIES	600
BEQ/BOQ	721 & 724
COMMUNITY, MORALE, RECREATION	740

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Table 4
Examples of Facilities Usually Requiring Type D OMSI
(As-built Drawings Only)

FACILITY TYPE	BASIC CATEGORY CODE
AIRFIELD PAVEMENTS/TAXIWAYS/APRONS	110-116
DREDGING	165
OPEN STORAGE	451
ROADS AND STREETS/PAVEMENTS	850/851

Table 5
Examples of OMSI Cost for Preliminary Estimates

	ESTIMATED OMSI COST BY TYPE (1)			
PRIMARY FACILITIES COST (6)	TYPE A (2)	TYPE B (3)	TYPE C (4)	TYPE D (5)
BELOW $5,000,000	1.20-2.35%	0.75-2.00%	0.55-1.60%	--
$5,000,000 to 20,000,000	0.95-1.85%	0.65-1.35%	0.35-0.75%	--
$20,000,000 to $50,000,000	0.70-1.35%	0.50-0.90%	0.30-0.60%	--
ABOVE $50,000,000	0.60-1.10%	0.45-0.75%	0.25-0.55%	--

NOTES:

(1) Table 5 shows estimated cost of OMSI as a percentage range of the total of primary facility plus applicable supporting facility costs. Use higher range for more complex projects. For example: Industrial Waste Treatment Plant with $18 million facility cost:
$18,000,000 x 0.0185 = $333,000 estimated cost for Type A OMSI (Operation and Maintenance Manual plus User Manual).
(2) The minimum estimate for a Type A OMSI should be approximately $22,000.
(3) The minimum estimate for a Type B OMSI should be approximately $18,000.
(4) The minimum estimate for a Type C OMSI should be approximately $8,000.
(5) No costs over those fees normally encountered for As-built Drawing preparation.
(6) Include applicable cost for supporting facilities which require OMSI consideration.

2.3 Intrusion Detection Systems (IDS). Interior "non-built-in" portions of IDS are no longer MCON funded per NAVCOMPT September 1987 decision. Effective with FY 90 Program, NAVCOMPT requires OPN funding for interior "non-built-in" IDS. The Naval Investigative Service Command (COMNISCOM) has the role of managing the design, procurement, and installation of IDS. NAVFACENGCOM message 181745Z June 1990 provides guidance on the IDS issue.

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Facility support, such as utilities, conduit runs, equipment space, building, and conventional fencing required to support IDS associated with a MCON project are properly chargeable to MCON funds and should be included in the MCON project scope. Refer to NAVFAC DM-13.02, Commercial Intrusion Detection Systems (IDS) for help in determining facility IDS requirements.

2.3.1 Non-MCON Fundable IDS. IDS installation can be divided into five general categories:

a) Category A: Navy MCON funded projects;

b) Category B: Nuclear ordnance storage sites;

c) Category C: Conventional arms, ammunition, and explosives storage sites (AA&E);

d) Category D: Marine Corps funded projects; and

e) Category E: All other special projects and non-MCON funded projects.

Planning, design, and installation of IDS Categories A, B, and C are the responsibility of COMNISCOM. The A&E shall coordinate IDS power requirements and conduit runs for IDS Categories A, B, and C with the appropriate contacts within COMNISCOM.

For Category D projects consult area project manager for guidance to determine if IDS will be A&E designed.

Category E projects are planned and designed by the A&E utilizing commercial systems for installation by the construction contractor. Identify areas requiring IDS and types of systems and sensors proposed.

Claimant, user, and COMNISCOM, depending on current policy, will budget for and fund non-MCON IDS requirements. These commands must be notified of a project's non-MCON IDS requirements and your need for data on facility support requirements. This should generally result in an IDS Engineering Plan (IDSEP).

2.3.2 Technical Support. Technical support for IDS design is available on a reimbursable basis from Naval Electronic Systems Engineering Center (NAVELEXCEN) Charleston, SC. This support includes guidance on scope and technical review of compiled IDS designs at appropriate steps in project development and design. NAVFACENGCOM will attempt to identify MCON projects requiring major IDS and request NAVELEXCEN Charleston, SC prepare an IDSEP for each.

2.4 Land Acquisition. Obtain instructions from major claimant and EFD Code 024 (Real Estate).

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2.5 Hyperbaric Requirements. CHESNAVFACENGCOM (FPO-1) is responsible for executing the design and procurement phases of shore based hyperbaric facilities. When the project scope includes provisions for hyperbaric facilities, you must contact CHESNAVFACENGCOM (FPO-1) to make appropriate arrangements for the design, procurement, and installation of the required hyperbaric facilities and the associated cost estimate.

2.6 Shielding Requirements

2.6.1 Radio Frequency Interference (RFI)/TEMPEST/HEMP (High Altitude Electromagnetic Pulse)/EMP (Electromagnetic Pulse) Shielding. NAVELEXSECCEN Washington, DC has been very effective in limiting requirements for shielding in MCON facilities. NAVFACENGCOM message 151455Z June 1984 provides the guidance on determining the extent of RF shielding required on projects involving installation of electronic equipment. For affected projects, ensure NAVELEXSECCEN Washington, DC has formally reviewed and commented on TEMPEST shielding requirements. If HEMP/EMP shielding is identified as a user requirement, consult with NAVFAC Code 04A1 prior to A&E scoping.

2.6.2 Coordination Required to Meet Objectives. Satisfying shielding objectives is often very difficult; it requires close coordination between the customer, the designer, and the construction contractor to establish this scope requirement. Good information transfer from designers to ROICC personnel is essential. The use of expert consultants during design and construction, and the development of both interim and final testing plans and criteria, should be evaluated and incorporated when appropriate. Design Policy Letter DPL-00-0002 dated 3 January 1989 established CHESNAVFACENGCOM (FPO-2) as the center of expertise for TEMPEST, HEMP, and RFI shielding. Consult MIL-HDBK-1195, Radio Frequency Shielded Enclosures for information on these subjects.

2.7 Uninterruptible Power System (UPS). If the UPS is required only to support personal property (electronic equipment) installations, it may not be procured and installed with MCON funds. Instead, its procurement and installation cost must be separately budgeted by the major claimant or user as an integral part of the associated electronic equipment package. Please ensure that the major claimant or user is aware of this requirement. Government furnished UPS are procured through CHESNAVFACENGCOM (FPO-2), and UPS requirements should be coordinated through them. However, facility items required to support the UPS such as power panels, flooring, and battery rooms are properly chargeable to MCON and should be included in the project scope.

2.8 Communication Requirements

2.8.1 Telephone and Communications Installations. It is the responsibility of the designer to determine the design requirements, and design the facility's telephone system. Based on NAVCOMPT guidance, the general policy is to include only the cost of the following as required to support future installation of telephone and other communication systems wiring internal to a MCON facility:

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a) Exterior: All supporting structures (e.g., telephone poles, underground conduits, underground ductbanks, manholes, and handholes).

b) Interior: Empty conduit with pull wire, outlet boxes, conduit bushings, backboards, and cable trays to support future installation of telephone wiring.

2.8.2 Funding Responsibilities. Normally, the wiring, instruments, and equipment are to be funded by other than MCON regardless of whether the system is leased or Government owned. The responsibility of Telephone Management Division East (TMD East) is to assist the user in identifying non-project funds to pay for the following items to be installed as a separate contract (not part of the MCON project):

a) Exterior: All cabling and telephone switching equipment.

b) Interior: All wiring, instruments, and equipment.

2.8.3 Funding Exceptions. In certain situations, such as medical facilities and Navy owned systems in overseas areas, entire systems may be MCON funded where practical and justifiable. These exceptions to the general policy should be clearly identified and justified.

2.9 Built-in Retrieval Equipment Racks (Warehouses). Policy on funding specialized equipment in automated warehouses is as follows:

2.9.1 MCON Funded. If the project includes a guide wire or guide rail built into the facility, include the cost of the guide wire or rail system, vehicles which ride on the system (system captive), and the storage racks serviced by the system in the MCON project.

2.9.2 OPN Funded. If the retrieval system does not include a built-in guide wire or rail system, the retrieval vehicles and racks are to be identified as collateral equipment.

2.9.3 Warehouse Design. The design of all warehouses must be coordinated with Naval Supply Systems Command (NAVSUPSYSCOM) as the Navy's expert in warehouses. NAVSUPSYSCOM must formally approve the design concept prior to your achieving a 35 percent design. To help with the design concept, NAVSUPSYSCOM has developed a computer program to determine the least cost alternative for various stacking heights and retrieval systems. As the foot print and stacking height affect the project cost, and some retrieval systems are MCON funded, these decisions must be made prior to starting design. Coordinate design with NAVSUPSYSCOM. The NAVFACENGCOM point of contact is Code 04B2, Autovon 221-0044, Commercial (703) 325-0044.

2.10 Asbestos Related Work. A&E firms are finding it difficult, if not impossible, to obtain professional liability insurance for work involving asbestos, hazardous waste, and pollution abatement. The Federal Acquisition Regulation (FAR) does not require A&E firms to carry professional liability

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insurance. NAVFACENGCOM does not have the authority to provide blanket indemnification for work involving hazardous waste. NAVFACENGCOM policy is that when asbestos or other hazardous waste will be encountered, or when there is a good probability of asbestos or other hazardous waste, note this fact in the Commerce Business Daily (CBD) synopsis for any affected contract. We should NOT include this warning in every CBD synopsis as a matter of routine; it should only be used where the hazards cited are known or probable. During interviews the willingness of each firm to undertake any required asbestos, hazardous waste, or pollution abatement work is to be clearly established and made a matter of record. Please continue to keep NAVFACENGCOM informed of problems in this area. The NAVFACENGCOM point of contact is Code 04BD, Autovon 221-0043, Commercial (703) 325-0043.

2.11 Computation of Gross Area. For all gross and net area calculations use Chapter 4 of MIL-HDBK-1190, Facility Planning and Design Guide. Gross area includes the total area of all floors, including mezzanines, basements, and penthouses as determined by the effective outside dimensions of the building. One-half the area shall be included in the gross area for balconies and porches, covered raised loading platforms, covered loading facilities (either depressed, ground level, or raised), covered but not enclosed passageways or walks, covered and uncovered but open stairs, and covered ramps. Excluded from the gross area are crawl spaces, exterior uncovered loading facilities (either depressed, ground level, or raised), open courtyards, roof overhangs and soffits, uncovered ramps and stoops, utility tunnels, and raceways. Separate central energy plants or utility buildings serving large complexes are not included in the gross module area shown in Table 5A. Separate support buildings should be programmed as a line item on DD Form 1391.

2.11.1 BEQ Area Requirements. For Bachelor Enlisted Quarters, refer to criteria shown in Table 5A.

2.11.2 BOQ Area Requirements. For Bachelor Officer Quarters, refer to MIL-HDBK-1190. The following requirements shall be used to program BOQ projects:

a) Grades 02 and below: Quarters shall provide a private suite of approximately 330 square feet net area, consisting of combination of living room and bedroom, bathroom, and pullman-type kitchen. The gross floor area shall approximate 475 square feet per officer.

b) Grades 03 and above: Quarters shall provide a private suite of approximately 460 square feet consisting of living room, bedroom, bathroom, and kitchen. The gross floor area shall approximate 650 square feet per officer.

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Table 5A
Bachelor Enlisted Quarters Area Table

MODULES (2 Rooms and Bath)	FLOORS (1)	AREA SF (2)	ELEVATORS (3)	SPRINKLER SYSTEM
60	3	800	---	---
72	3	785	---	---
96	4	800	F	(4)
180	6	800	F & P	YES

NOTES:

(1) Modules per floor must be divisible by 2, with no exceptions.
(2) Total gross module area including its share of the non-module areas.
(3) F (Freight) & P (Passenger)
(4) Provide a fire stand pipe system if sprinkler system is not required.
(5) The data in this table satisfies the net area requirements of MIL-HDBK-1190. Gross area for combinations not shown must be determined or verified by the EFD, Code 401.

GRADE	ACCOMMODATIONS
E-1 thru E-4	4 Men per Module
E-5 thru E-6	2 Men per Module
E-7 thru E-9	1 Man per Module

2.12 International Balance of Payments (IBOP). The IBOP Summary is only included when preparing the cost estimate for projects in specific foreign areas as shown in par. 4.2. Information on the exempt areas is available from the EFD Project Management Office. The IBOP Summary shall be developed on 8-1/2- by 11-inch plain bond paper as shown in Figure 1.

2.12.1 IBOP Implications. Current directives require action by cognizant addressees to maintain Foreign Exchange Costs (FEC) at a minimum. IBOP implications will be prepared to show the effect of construction under both normal and planned IBOP procedures. This effort is to permit analysis and response by NAVFACENGCOM to inquiries regarding items contributing to FEC.

2.12.2 Normal Procedures. Those construction procedures which would normally be used at a particular location if no effort were made to suppress FEC.

2.12.3 Planned Procedures. Those procedures utilized at a particular location to minimize FEC with specific support and emphasis on:

a) Adhering to pertinent executive orders, U.S. laws, status of forces agreements, and treaties which affect our contractual freedom abroad.

b) Maximum feasible use of U.S. contractors as allowed by par. 2.12.3.a).

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INTERNATIONAL BALANCE OF PAYMENTS (IBOP) IMPLICATIONS

1. Activity (Name and Location) (CAPITAL LETTERS)			Date	
2. Project Title (CAPITAL LETTERS)			3. P-No.	
	NORMAL PROCEDURES		PLANNED PROCEDURE	
	Construction	*Construction Total Material	Construction	*Construction Total Material
BUDGET COST	1,030,000	-	1,100,000	-
LESS RETURNS TO THE US				
a) US End Products	-	-	100,000	-
b) US Services	5,000	-	5,000	-
c) US Transportation	-	-	20,000	-
TOTAL RETURNS	5,000	0	125,000	0
FOREIGN EXCHANGE COST (FEC)	1,075,000		975,000	

This example makes the following assumptions:

1. Normal procedure; all foreign materials and contractors, only return to US is 6.5% SIOH.
2. Planned procedure; purchase of a pre-engineered metal building in the US at the same price as in the foreign country, shipping by a US line, 6.5% SIOH is returned to the US.

Returns to US include escalation and contingencies. Overhead and profit depends upon location of contractor.

SUMMARY OF IBOP DATA

*CWE is the same as total request rounded for selected procedures.

	NORMAL PROCEDURE	PLANNED PROCEDURE	PLANNED PROCEDURE EFFECT
CURRENT WORKING ESTIMATE	1,080	1,100	+ 20
FOREIGN EXCHANGE COST (FEC)	1,075	975	- 100

* Materials, supplies and equipment (including government furnished material procured abroad) acquired by the Navy from construction funds and provided to contractors for incorporation into foreign construction projects.

Figure 1
Sample IBOP Summary

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c) Maximum feasible use of U.S. materials, end products, and equipment as allowed by par. 2.12.3.a).

d) Maximum use of U.S. flag, ocean, and air carriers.

e) Utilization of competent military labor where available.

f) Designs incorporating portable van-type installations and structures prefabricated in the United States.

2.12.4 Basis for Choice of Construction Procedure. Generally, planned procedures will be recommended where for every dollar increase in Current Working Estimate (CWE) from NORMAL to PLANNED there is a resulting decrease in FEC of at least $2, and normal procedures will be recommended otherwise. An explanation for the deviation will be provided if the EFD recommends construction at the normal level and cost estimates justify programming at the planned level.

Exceptions to the above criteria are allowed where country-to-country agreements specifically establish construction procedures, or project urgency is an overwhelming factor.

2.12.5 Presentation of IBOP Data on DD Form 1391. The DD Form 1391 will be prepared to show costs at the recommended level. IBOP data will be shown on the DD Form 1391 with entries asterisked to show the corresponding CWE and FEC amounts for the recommended procedure. Such IBOP entries will be shown as:

IBOP DATA:	NORMAL	PLANNED	NORMAL	PLANNED
CWE	$550,000	$590,000*	FEC $330,000	$220,000*

*Recommended construction method

2.12.6 Requirements Regarding Returns to the United States. Other information requirements regarding returns to the United States are to be shown at the end of the cost estimates. This additional data is required in order that NAVFACENGCOM may provide estimates of IBOP expenditures as required by higher authority.

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Section 2A: PROJECT DATA SHEET

2.13 Project Data Sheet. The contents of the Project Data Sheet are described in Figure 2.

2.14 Project Submission Routes. Figures 3 and 4 show the procedures and documentation required for entering a project in the RL, for reviewing projects selected for the SYDP, and for submitting Facility Study and DD Form 1391 documentation.

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PROJECT DATA SHEET

COMPONENT	ESTIMATE YEAR	DATE PREPARED
NAVY	1995	15 August 1991

INSTALLATION AND LOCATION	PROJECT TITLE
U.S. NAVAL STATION, NORFOLK, VA*	ADMINISTRATIVE OFFICE**

PROGRAM ELEMENT	CATEGORY CODE	PROJECT NUMBER	PROJECT COST ($000)
*	610.10	P-000	3,550

ITEM	ESCALATED TO:	ACF: .92 FER: N.A.	U/M	QUANTITY	UNIT COST	COST ($000)
PRIMARY FACILITY			SF	30,000	91.30	2,739
SUPPORTING FACILITIES			LS	-	-	435 ***
SUBTOTAL						3,174
CONTINGENCY (5%)						159
TOTAL CONTRACT COST						3,333
SIOH (6.0%)						200
TOTAL REQUEST						3,533
TOTAL REQUEST ROUNDED						3,550
EQUIPMENT FROM OTHER APPROPRIATIONS				(NON-ADD)		(-)

DESCRIPTION OF PROPOSED CONSTRUCTION: Permanent two-story masonry and structural frame building. A passenger elevator and an exterior uncovered steel stairway are included. Fire protection, grounding, utilities, air conditioning, and OMSI are included. Buildings B15 and B25 are to be demolished.

REQUIREMENT:
The mission of the Naval Station, Norfolk, Virginia is to maintain and operate facilities and provide services and material to support operations of the Atlantic Fleet.

PROGRAMMING DATA:

ACTIVITY UIC: NXXXXX	ALTERNATE HOST: NYYYYY	SUP UNIT: NZZZZZ
INVESTMENT PROGRAM: 99	SIR: II	SPEC AREA:
MAJ/SUB CLAIMANT CODE: FA1/	INVESTMENT CATEGORY:14*	MOBILIZATION
ACTIVITY PRIORITY: 01	CLAIMANT PRIORITY: 12	INDICATOR: N
READINESS RATING: C4	RESOURCE SPONSOR: 04	FLEP PRIORITY: H

PROJECT DETAIL DATA

CAT CODE	DESCRIPTIONS	SCOPE	U/M	CC	VALIDATION INDICATOR
610-10	ADMINISTRATIVE	30,000	SF	3B	
610-10	ADMINISTRATIVE	22,000	SF	3B	
610-20	DATA PROCESSING	8,000	SF	3B	

REQUIREMENTS CERTIFICATION

ACTIVITY	(name) _____	DATE _____
MAJOR CLAIMANT	(name) _____	DATE _____
NAVFACENGCOMHQ	(name) _____	DATE _____

* COMPUTER GENERATED INFORMATION
** DEFAULTS TO CATEGORY CODE DIRECTORY TITLE IF NO ENTRY IS MADE
*** NO COST FOR SPECIAL FOUNDATION FEATURES IS INCLUDED SINCE FUNDING FOR TEST BORINGS IS NOT AVAILABLE AT THIS TIME, AND THEREFORE, THE REQUIREMENTS FOR FOUNDATION PILING WERE NOT KNOWN. COST FOR DEMOLITION OF BUILDINGS B15 AND B25 HAS BEEN INCLUDED.

Figure 2
Project Data Sheet

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LOCATION AND SITE PLAN

****
* PREPARATION NOTES:
* PROJECT DATA SHEET (PDS): The PDS is the initial activity submittal for a MCON project.[1]
*
* PREPARATION OF COST:
* PRIMARY FACILITY: Cost shall be prepared following the same instructions as for the preparation of the Facility Study primary estimate described in this instruction. Include the Primary Facility Unit Cost Development Sheets (see Figure 11), with the PDS submittal.
*
* SUPPORTING FACILITIES: Costs are to be a percentage of the primary facility cost. Use 15% when the project is to be constructed within a developed area. Use 20% for all other locations. Provide backup for other percentages. Special foundation features and demolition are to be separately priced and added to above cost.
****

SITING RATIONALE: Building sited outside the ammunition pier Explosive Safety Quantity Distance Arc and adjacent to another administrative building, Building 400, implementing the plan for consolidation of station administrative and data processing. All required utilities are immediately available at the site and are adequate to accommodate the additional connected load. Demolition will be required by the project.

EFD REVIEW ANALYSIS:

REQUIRES FURTHER ACTION ?	NO	YES	COMMENTS
EXPLOSIVE SAFETY	X		
AIRFIELD SAFETY	X		
ELECTROMAGNETIC RADIATION	X		
AICUZ VIOLATION	X		

NOTE: THESE ARE EXAMPLES. THE EFD WILL ENTER THE REQUIRED FACTORS DURING THE EFD REVIEW AND ANALYSIS

CULTURAL RESOURCES IMPACT	X		
OTHER SITE PROBLEMS	X		
UTILITIES SUPPORT	X		
ROADS, PARKING		X	Minor changes to existing
PRELIMINARY ENVIRONMENTAL ASSESSMENT		X	Submit when project enters SYDP
PRELIMINARY HAZARDS ANALYSIS		X	If RAC is a 1 or 2

EFD VALIDATION :

SITE APPROVED: YES_____ NO _____ DEFERRED _____

NAME_____ **DATE** _____

[1] A detailed discussion of the Project Data Sheet and the Capital Improvement Plan (CIP) is contained in NAVFACINST 11010.44 including Change 1 dated 1 October 1990

Figure 2 (Continued)
Project Data Sheet

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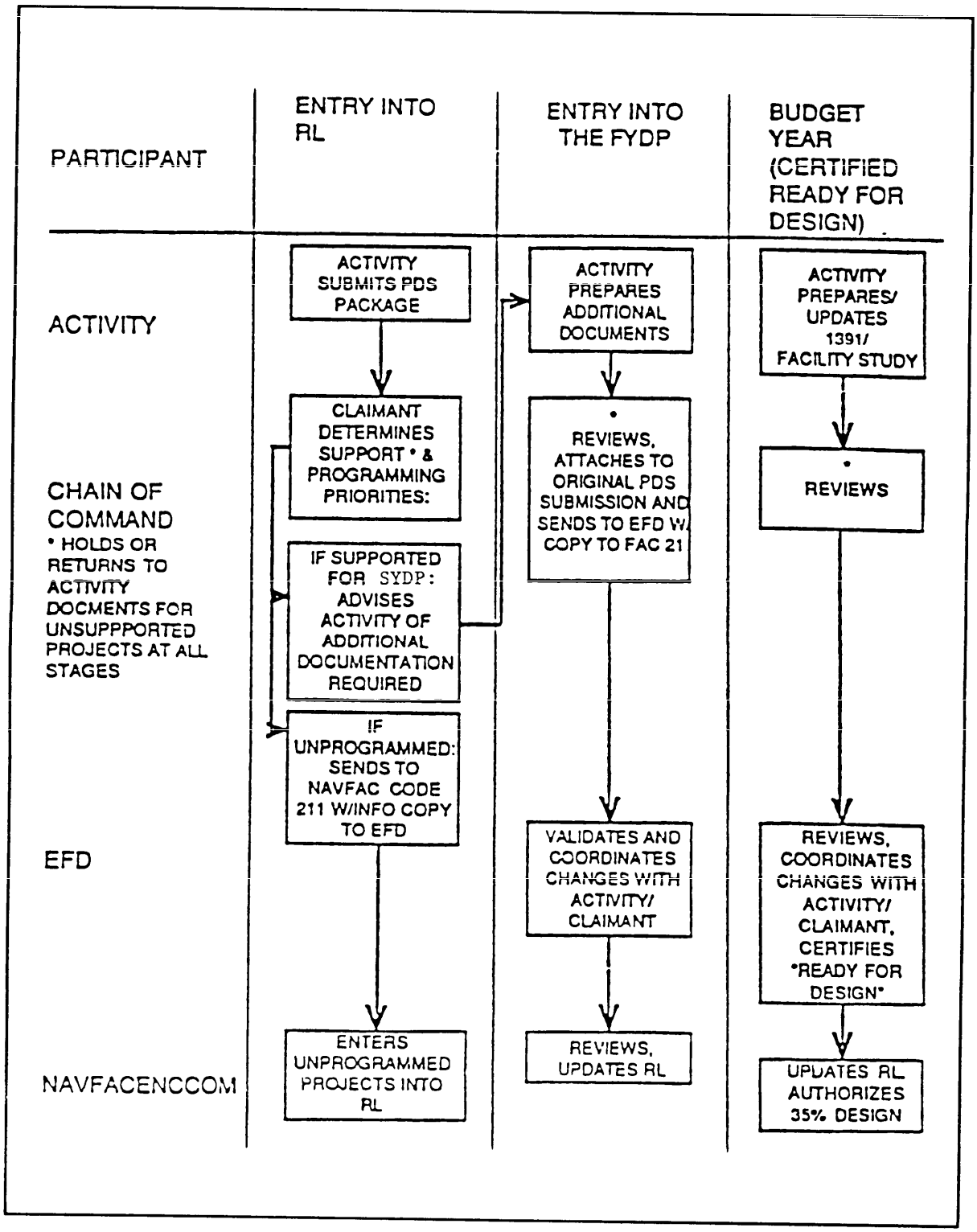


Figure 3
Project Submission Routes

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LEGEND

C: ALREADY COMPLETE: RESUBMIT ONLY IF DATA HAS CHANGED
CSMC: ALREADY COMPLETE: SUBMIT COPY TO MAJOR CLAIMANT W/PDS
E: EFD ACTION
S: SUBMIT TO EFD, COPY TO NAVFAC CODE 211
SMC: SUBMIT TO MAJOR CLAIMANT COPY TO NAVFAC CODE 211
X: COMPLETED AT PREVIOUS STAGE: NO SUBMISSION REQUIRED
* MAY NOT BE REQUIRED FOR ALL PROJECTS
** SUBMISSION REQUIRED AT THIS TIME FOR PROJECTS REQUIRING WASHINGTON-LEVEL SITE APPROVAL

DOCUMENTATION REQUIRED	ENTRY INTO RL	PDS REVIEW BY EFD FOR PROJECTS IN THE SYDP	BUDGET YEARS (CERTIFIED READY FOR DESIGN)
BFR IN SFPS	C	C	C
CURRENT EE	C	C	C
FPD	CSMC	C	C
PDS	SMC	C	C
PEA		S	C
PDS COST ESTIMATE		C	X
RISK ASSESSMENT CODE		S	X
QUICK SIR/PVA		S*	X*
SITE APPROVAL REQUEST		S**	S
LIFE CYCLE COST ANALYSIS SUMMARY		S*	C*
ECIP COST/ENGINEERING DATA		S*	C*
POLLUTION CONTROL REPORT		S*	C*
OSH CONTROL REPORT		S*	C*
EQUIVALENCY STATEMENT		S*	C*
MEDICAL ECONOMIC ANALYSIS		S*	C*
MEDICAL SPACE PROGRAM		S*	C*
NAF SUPPLEMENTAL INFORMATION	SMC*	X*	X*
FACILITY STUDY			S
FACILITY STUDY COST ESTIMATE			S
COLLATERAL EQUIPMENT LIST			S
BESEP/EQUIPMENT STUDY			S*
FACILITY STUDY 1391			S
1390s (RESERVES)			S
ENVIRONMENTAL ANALYSIS			S*
PRELIMINARY HAZARDS ANALYSIS			S*
TYPE I/II ECONOMIC ANALYSIS			S*
COST CERTIFICATION/ARCHITECTURAL PROGRAMMING/2 STEP PLANNING/DESIGN			E

Figure 4
Components of Project Documentation

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Section 3: FACILITIES STUDY COST DEVELOPMENT

3.1 Instructions for DD Form 1391. The following instructions correspond to the circled numbers shown in Figure 5.

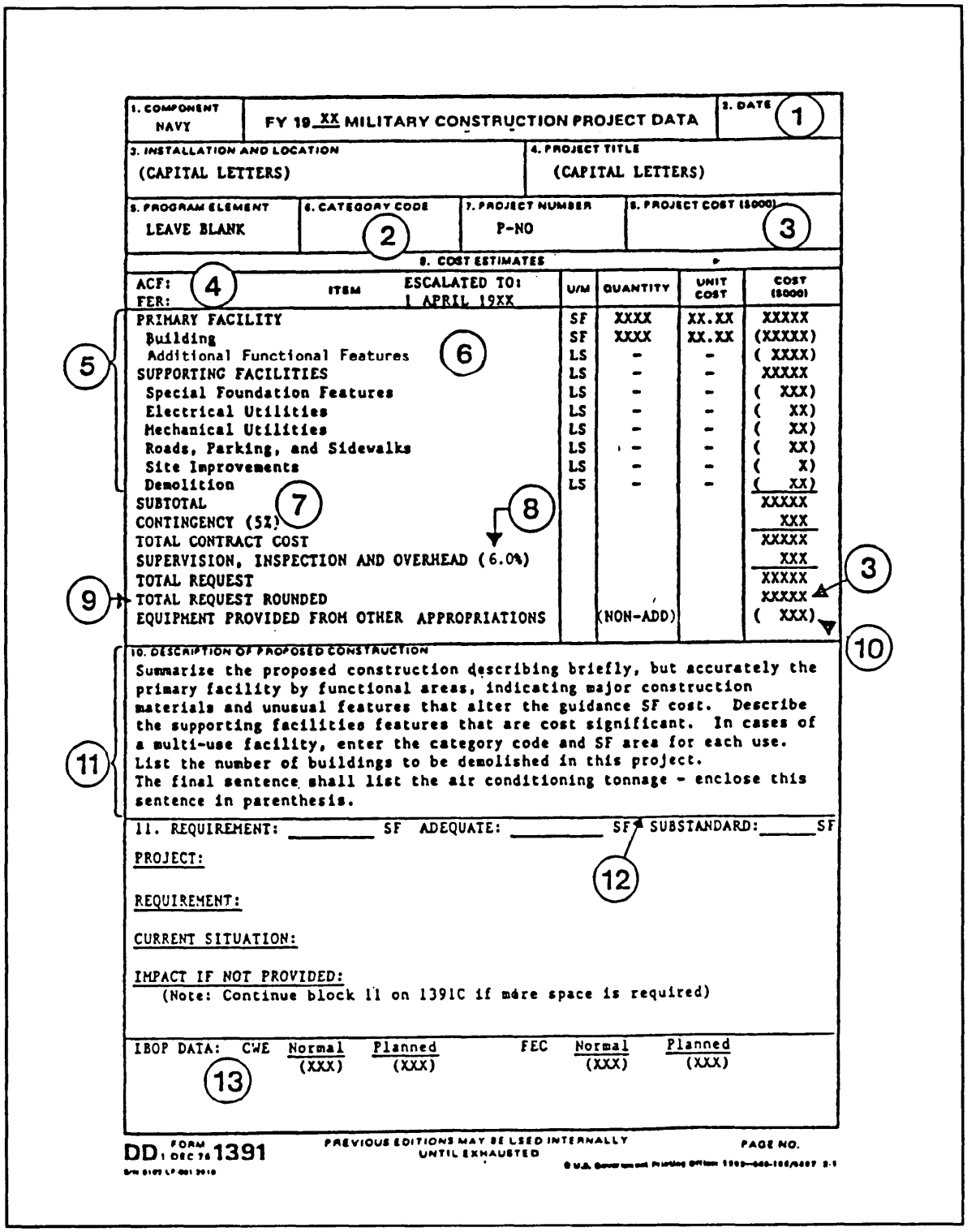
1. Date shall be the day submitted. Dates must be revised if a resubmittal is made.
2. Enter the five-digit category code number from NAVFAC P-72, Department of the Navy Facility Category Codes to identify the primary facility. In cases of multi-use facilities, enter the category code which represents the largest cost of the project.
3. Block 8 - "Project Cost" shall be the same as the "TOTAL REQUEST ROUNDED" in block 9. This cost shall be escalated to 1 April of the appropriation year, except assigned foreign areas, par. 4.2.
4. The Area Cost Factors for stateside areas are shown in par. 4.1. The Area Cost Factors and the corresponding Foreign Exchange Rates for various overseas locations are shown in par. 4.2.
5. The line items shown are generally suitable for most projects. They should be tailored to suit the individual project - discuss this with the EFD Cost Engineering Branch. The Cost figures in parentheses should add to equal "PRIMARY FACILITY" cost or "SUPPORTING FACILITIES" cost.
6. Include a list of the major use areas for multi-use facilities under the primary facility. Also, include a line item entitled, "Additional Functional Features." Show the unit cost as applicable to the total size of the primary facility as shown in Figure 6.
7. Contingency will be 5 percent for all projects.
8. Use 6.0 percent SIOH for all stateside projects and 6.5 percent SIOH for all overseas projects unless directed otherwise.
9. Rounded cost estimate:

less than $1,000,000	-	nearest 10,000
1,000,000 - 5,000,000	-	nearest 50,000
Over 5,000,000	-	nearest 100,000

10. Must agree with total for all other categories of equipment on collateral equipment list other than built-in equipment as defined in par. 7.1. If not available, type "(LIST NOT AVAILABLE)" in place of "(NON-ADD)." Built-in equipment is MCON funded and normal built-in equipment cost is included in guidance unit cost. Enclose cost figures in parentheses.
11. If the project includes more than one structure or location, address each in a separate paragraph in block 10 and separately in block 9.
12. Draw a line between blocks 10 and 11.
13. IBOP entry is required for assigned foreign areas.

NOTE: Complete the Supporting Facilities Cost Development, as shown in Figures 7 and 12.

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1. COMPONENT	FY 19_XX MILITARY CONSTRUCTION PROJECT DATA	2. DATE (1)
NAVY		

3. INSTALLATION AND LOCATION	4. PROJECT TITLE
(CAPITAL LETTERS)	(CAPITAL LETTERS)

5. PROGRAM ELEMENT	6. CATEGORY CODE (2)	7. PROJECT NUMBER	8. PROJECT COST ($000) (3)
LEAVE BLANK		P-NO	

9. COST ESTIMATES

ACF: (4) FER:	ITEM	ESCALATED TO: 1 APRIL 19XX	U/M	QUANTITY	UNIT COST	COST ($000)
(5)	PRIMARY FACILITY	(6)	SF	XXXX	XX.XX	XXXXX
	Building		SF	XXXX	XX.XX	(XXXXX)
	Additional Functional Features		LS	-	-	(XXXX)
	SUPPORTING FACILITIES		LS	-	-	XXXXX
	Special Foundation Features		LS	-	-	(XXX)
	Electrical Utilities		LS	-	-	(XX)
	Mechanical Utilities		LS	-	-	(XX)
	Roads, Parking, and Sidewalks		LS	-	-	(XX)
	Site Improvements		LS	-	-	(X)
	Demolition		LS	-	-	(XX)
	SUBTOTAL	(7)				XXXXX
	CONTINGENCY (5%)	(8)				XXX
	TOTAL CONTRACT COST					XXXXX
	SUPERVISION, INSPECTION AND OVERHEAD (6.0%)					XXX
	TOTAL REQUEST					XXXXX
(9)	TOTAL REQUEST ROUNDED					XXXXX (3)
	EQUIPMENT PROVIDED FROM OTHER APPROPRIATIONS			(NON-ADD)		(XXX) (10)

10. DESCRIPTION OF PROPOSED CONSTRUCTION (11)

Summarize the proposed construction describing briefly, but accurately the primary facility by functional areas, indicating major construction materials and unusual features that alter the guidance SF cost. Describe the supporting facilities features that are cost significant. In cases of a multi-use facility, enter the category code and SF area for each use. List the number of buildings to be demolished in this project. The final sentence shall list the air conditioning tonnage - enclose this sentence in parenthesis.

11. REQUIREMENT: _____ SF ADEQUATE: _____ SF SUBSTANDARD: _____SF (12)

PROJECT:

REQUIREMENT:

CURRENT SITUATION:

IMPACT IF NOT PROVIDED:

(Note: Continue block 11 on 1391C if more space is required)

IBOP DATA: (13)	CWE	Normal	Planned	FEC	Normal	Planned
		(XXX)	(XXX)		(XXX)	(XXX)

DD FORM 1 DEC 76 1391 PREVIOUS EDITIONS MAY BE USED INTERNALLY UNTIL EXHAUSTED PAGE NO.

Figure 5
DD Form 1391

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3.2 Instructions for Primary Facility Unit Cost Development. See Figure 6.

a) Guidance Cost Priorities:

(1) NAVFAC Primary Facility Guidance Costs, Table 6.

(2) Additional Primary Facility Guidance Costs, Table 6A.

(3) Guidance costs obtained from historical data of comparable projects (cost must be obtained from or verified by EFD Code 407).

b) The guidance cost must be adjusted to allow for differences in size (see Figure 8), area cost factors, and escalation (see Figure 9). The area cost factors are listed in pars 4.1 and 4.2.

c) The guidance cost represents a typical project, including all normal built-in equipment. Identification of built-in equipment may be obtained from the CES Standard Systems List shown in par. 7.1. The guidance cost must be adjusted for additional functional features which are considered to be additional cost. This adjustment is absolutely required to justify cost above guidance. Backup for the additional functional features must be attached. Examples of additional functional features are OMSI, hyperbaric requirements, TEMPEST/HEMP/EMP shielding, UPS support facilities, communications support facilities, retrieval equipment racks for warehouses, very high walls, emergency power, raised floor, and asbestos removal.

d) Gross area calculations (par. 2.11) require certain areas to be taken at zero or one-half the actual area. These excluded areas must be included in the project cost. This can be done by showing the actual cost as a functional feature on the Primary Facility Unit Cost Development Sheet.

e) The adjusted unit cost (u/c) for each functional feature is obtained by dividing the total cost of each feature by the total size of the proposed facility. (Include all multi-use areas.)

f) The percent change is obtained by dividing the total cost of each feature by the total adjusted historical cost of the facility. (Include all multi-use areas.)

g) The size adjustment factor is related to Figure 8.

h) The area cost factors are given in Section 4.

i) The escalation factors are given in Figure 9.

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PRIMARY FACILITY UNIT COST DEVELOPMENT

TITLE: _____ P#: _____

LOCATION: _____ DATE: _____

PREPARED BY: _____ PH#: _____

	CAT CODE	SIZE	ACF	DATE (mo/yr)
PROPOSED FACILITY				
GUIDANCE FACILITY				

HISTORICAL GUIDANCE SOURCE: _____ *

HISTORICAL GUIDANCE UNIT COST: _____ /SF

HISTORICAL UNIT COST	X	SIZE FACTOR	X	AREA COST FACTOR	X	ESCALATION FACTOR	=	ADJUSTED UNIT COST

* Show date of source as shown in Figure 11.

ADDITIONAL FUNCTIONAL FEATURES

FUNCTIONAL FEATURES	QTY	UNIT COST	TOTAL COST	ADJUSTED UC	%CHANGE

FUNCTIONAL FEATURES TOTAL: _____ _____

TOTAL PRIMARY FACILITY UNIT COST: _____
includes built-in equipment

Figure 6
Primary Facility Unit Cost Development

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SUPPORTING FACILITIES - COST DEVELOPMENT

TITLE: Administrative Office — DATE: Jan-24-1992

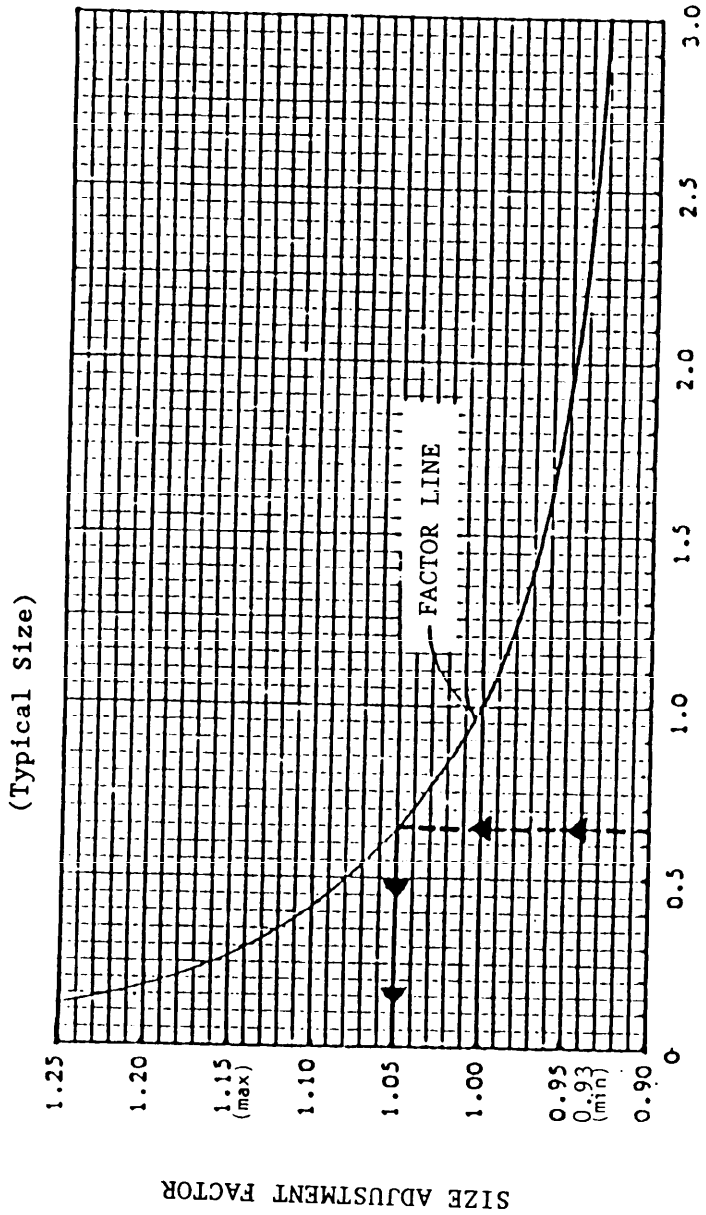
LOCATION: Naval Station, Norfolk, Virginia — P# 000 — FY: 95

CES SYS (1)	SYSTEM DESCRIPTION (2)	COMPOSITE QUANTITY	COMPOSITE UNIT COST (3)	TOTAL COST
SPECIAL FOUNDATION FEATURES (91X)				
911	Wood Piling, 40'-50', 12" butt, 7" Tip, 12 lb creosote	5000 Lf	$10.45	$52,250
ELECTRICAL UTILITIES (51X - 52X)				
511	3-1/C #2/0 AWG CU XLP EKV Primary w/o termin.	300 Lf	$25.48	$7,032
512	4-1/C #2/0 THW W/1-1/C#4 CU grnd secon w/termin	100 Lf	$7.44	$684
513	37.5 KVA 1 PH xfmr conc pad mounted	1 Ea	$6,392.00	$5,881
514	400w HPS on 35'-40' conc pole wire, excav, equip	4 Ea	$2,500.00	$11,720
				$25,318
MECHANICAL UTILITIES (54X - 59X)				
541	8" steam 4" condensate heat distri piping abovgrnd	400 Lf	$286.00	$105,248
551	8" PVC pipe AWWA C-900 class 150 w/excav	400 Lf	$19.30	$7,102
551	2" bit paving for pipe trench w/12" base & remo	100 Ea	$23.35	$2,148
554	6" PVC sewer pipe ASTM 3034, SDR-35 w/excav	600 Lf	$9.27	$5,117
554	2" bit paving for pipe trench w/12" base & remo	200 Sy	$23.35	$4,296
554	48" diam precast conc mnhole 8' deep w/frame/cover	1 Ea	$2,500.00	$2,925
				$126,837
ROADS, PARKING, AND SIDEWALKS (7XX)				
711	6" base course material	3000 Sy	$5.12	$14,131
711	8" concrete pavement, 5000 psi	3000 Sy	$57.65	$159,114
721	8" base course material	4000 Sy	$6.99	$25,723
721	2" asphalt wearing surface course	4000 Sy	$7.76	$28,557
721	Painting 4" line on pavement	1250 Lf	$0.14	$161
741	4" conc sidewalks no base	300 Sy	$29.54	$8,153
				$235,839
SITE IMPROVEMENTS (8XX)				
811	Site grading balance & cut, 75 hp dozer	12000 Sy	$2.30	$32,292
812	Borrow fill compact in place w/ 30mi rt haul	2100 Sy	$12.10	$29,730
813	4" topsoil/area prep/seeding/fertilize/lime	2000 Sy	$3.09	$5,686
814	Landscaping/miscel trees/shrubs/bed prep/edging	1 Ls		$20,000
818	Temporary silt fence, 36" high	1 Ac	$10,600.00	$12,402
851	18" rcp class III 5 ft deep w/excav	500 Lf	$15.10	$6,946
851	48" diam mnhole 5' deep w/frame & cover, excav	4 Ea	$2,398.00	$8,825
				$115,880
DEMOLITION (84X)				
843	Demo steel bldg frame complete w/disposal	90000 Cf	$0.22	$18,216
843	Removal of 6" conc slab non-reinf w/disposal	840 Sy	$10.23	$7,906
				$26,122

NOTES: Place additional information and excess composite items on back.
(1) Refer to Section 7.2 for CES System Identification
(2) Refer to Table 7 for Supporting Facilities Guidance Cost. Provide backup for cost of additional functional features.
(3) Composite Unit Cost Adjusted by Area Cost Factor, Escalation Factor & Size Factor

Figure 7
Supporting Facilities Cost Development

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Example: The Size Adjustment factor is desired for a 16,000 SF Academic Facility.

AREA RELATIONSHIP = $\frac{\text{PROPOSED FACILITY SIZE}}{\text{TYPICAL FACILITY SIZE}} = \frac{16,000}{25,000} = 0.64$

Find 0.64 on the horizontal axis. Trace a vertical line to the factor curve and then trace a horizontal line to the vertical axis' Size Adjustment Factor which is 1.05.

Figure 8
Size Adjustment Chart

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PROJECTED NAVFAC CONSTRUCTION COST ESCALATION INDEX
MONTHLY
APRIL 1992
Projected Indices are Followed by the Letter p

YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
1978	1609	1617	1620	1621	1652	1663	1696	1705	1720	1721	1732	1734
1979	1740	1740	1750	1749	1753	1809	1829	1849	1900	1899	1902	1909
1980	1895	1894	1915	1899	1888	1916	1950	1971	1976	1981	2000	2017
1981	2015	2016	2014	2064	2076	2083	2109	2118	2139	2156	2186	2184
1982	2184	2200	2195	2195	2220	2219	2233	2253	2249	2248	2260	2295
1983	2311	2348	2352	2347	2351	2388	2414	2428	2430	2416	2419	2406
1984	2402	2407	2412	2422	2419	2417	2418	2428	2430	2424	2421	2408
1985	2410	2414	2406	2405	2411	2429	2448	2442	2440	2441	2446	2439
1986	2440	2446	2447	2458	2479	2493	2499	2498	2504	2511	2511	2511
1987	2515	2510	2518	2523	2524	2525	2538	2557	2565	2569	2564	2589
1988	2574	2576	2586	2591	2592	2595	2598	2611	2612	2612	2616	2617
1989	2619	2613	2616	2620	2621	2626	2631	2640	2668	2672	2675	2679
1990	2673	2674	2684	2684	2697	2725	2725	2721	2729	2728	2730	2719
1991	2719	2716	2726	2736	2723	2733	2757	2792	2785	2786	2791	2784
1992	2780	2779	7991	2809	2818p	2826p	2835p	2844p	2853p	2861p	2870p	2887p
1993	2887p	2895p	2904p	2912p	2921p	2939p	2938p	2947p	2955p	2964p	2973p	2981p
1994	2990p	2999p	3007p	3016p	3025p	3033p	3042p	3051p	3059p	3068p	3077p	3085p
1995	3094p	3103p	3111p	3120p	3129p	3137p	3146p	3155p	3163p	3172p	3181p	3190p
1996	3199p	3208p	3217p	3226p	3235p	3244p	3253p	3262p	3271p	3280p	3289p	3299p

Figure 9
Projected NAVFAC Cost Index

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PROJECTED NAVFAC CONSTRUCTION COST ESCALATION INDEX
MONTHLY
APRIL 1992
Projected Indices are Followed by the Letter p

YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
1997	3308p	3317p	3327p	3336p	3345p	3355p	3364p	3373p	3382p	3392p	3402p	3411p
1998	3421p	3430p	3440p	3449p	3459p	3469p	3478p	3488p	3497p	3507p	3517p	3527p
1999	3537p	3547p	3557p	3567p	3577p	3586p	3596p	3606p	3616p	3626p	3636p	3646p

HISTORICAL INDICES ARE BASED UPON ENGINEERING NEWS RECORD BUILDING COST INDEX.

DIVIDE THE INDEX NUMBER OF THE PROJECTED DATE BY THE INDEX NUMBER OF THE BASE DATE TO DETERMINE THE COST ESCALATION MULTIPLIER. PROJECTED INDICES ARE BASED ON OFFICIAL OASD PROJECTED RATES AS FOLLOWS:

FY	% PER YEAR	% PER MONTH
FY92	3.70	0.31
FY93	3.60	0.30
FY94	3.50	0.29
FY95	3.40	0.28
FY96	3.40	0.28
FY97	3.40	0.28
FY98	3.40	0.28
FY99	3.40	0.28

FOR MORE INFORMATION CONTACT: MR. WILLIAM C. O' CONNOR, P.E., NAVFAC COST ENGINEER, DSN 221-0069 COMMERCIAL (703) 325-0069

Figure 9 (Continued)
Projected NAVFAC Cost Index

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3.3 Supporting Facilities. (See Figure 7.) Supporting facilities include all construction items outside the building 5-foot line and all special foundation features of significant cost unique to the building. All supporting facility items should be included under one of the following major categories. Each sub-item included under these major categories shall have a separate line item entry on the Supporting Facility Cost Development Sheet. Refer to par. 7.2 for identification by system number of supporting facilities.

3.3.1 Special Foundation Features. Include special foundation requirements (e.g., piling, drilled piers, vibroflotation, dewatering, etc.). (Systems 91X)

3.3.2 Electrical Utilities. Include electrical distribution, substations, emergency power generators, fire alarms, exterior lighting, telephone distribution, etc., beyond the 5-foot line. (Systems 51X, 52X)

3.3.3 Mechanical Utilities. Include those utilities outside the 5-foot lin covered in Divisions 2 and 15 of the specifications (e.g., gas, steam, water, sanitary, etc.). (Systems 54X through 59X)

3.3.4 Roads, Parking, and Sidewalks. Include subgrade preparation, subbase and surface courses, curb and gutters, and valley gutters. (Systems 7XX)

3.3.5 Site Improvements. Include seeding, sodding, riprap, landscaping, irrigation, fencing, storm drainage, and other site enhancements. Do not include normal excavation which is within the 5-foot line. (Systems 8XX)

3.3.6 Demolition. Include real property inventory (building or structure), major utility, or major paving demolition only. Utility, pavement, or other miscellaneous demolition associated with installation of new construction should be included with the applicable trade category (e.g., electrical demolition with Electrical Utilities, steam line demolition with Mechanical Utilities, paving demolition with Roads, Parking, and Sidewalks, etc.). Provide the total number of buildings to be demolished in par. 10 of DD Form 1391. (Systems 84X)

3.3.7 Contaminated Soil. Cost to remove or treat contaminated soil in areas such as old industrial sites.

3.3.8 Asbestos Removal. Discussions with a contractor specialist or Code 407 is encouraged.

3.3.9 Supporting Facilities - Cost. Each supporting facility system must be priced separately.

3.3.10 Cost Model. A completed cost model (Systems 5XX through 9XX) must be included. Refer to Appendix A.

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3.4 Proposed Site Plan. In order to reasonably estimate the supporting facility requirements, the proposed site should be investigated and station utility maps reviewed. It is suggested that a scaled, single line site plan be developed for the project. For most projects, this site plan can be made on an 8-1/2 by 11-inch sheet and should be attached to the estimate. The site plan should show most, if not all, of the following:

a) Existing elevations

b) Proposed building(s) and structure(s)

c) Finish floor elevation

d) Existing mechanical and electrical utility runs

e) New mechanical and electrical services to the facility

f) Paving

g) Demolition

3.5 Pricing Sources. Supporting facility costs should be estimated using the following pricing sources:

a) Supporting facility guidance cost reference provides composite in-place cost for typical systems. See Table 7.

b) Systems cost obtained from recently bid local projects. This is considered a reliable source.

c) NAVFACENGCOM CES unit price file including assembly, and the composite prices.

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Section 4: AREA COST FACTORS (6/90)

4.1 Stateside Area Cost Factors. The following list outlines area cost factors for various areas within the United States.

ALABAMA	.82
ALABAMA AAP	.79
ANNISTON AD	.79
BIRMINGHAM	.83
FORT MCCLELLAN	.79
FORT RUCKER	.85
MAXWELL AFB	.77
MOBILE AREA	.87
MONTGOMERY	.77
REDSTONE ARS	.89
ALASKA	2.18
ADAK	2.93
ADAK NAVAL STA	2.93
ANCHORAGE	1.69
EIELSON AFB	1.96
ELMENDORF AFB	1.69
FAIRBANKS	1.93
FORT GREELY	2.23
FT RICHARDSON	1.69
FORT WAINWRIGHT	1.95
SHEMYA AFB	3.21
ARIZONA	.95
FLAGSTAFF	1.00
FORT HUACHUCA	1.12
NAVAJO AD	1.00
PHOENIX	.96
TUCSON	.90
YUMA MAR.A.S.	1.11
YUMA PG	1.11
ARKANSAS	.83
EAKER AFB	.80
FORT CHAFFEE	.87
FORT SMITH	.87
LITTLE ROCK	.79
PINEBLUFF	.84
PINEBLUFF ARS	.84
CALIFORNIA	1.24
BEALE AFB	1.24
CAMP PENDLETON	1.18
CHINA LAKE NWC	1.40
EDWARDS AFB	1.38
EL CENTRO NAF	1.21
EL TORO MCAS	1.23
FT HUNT LIGGETT	1.44
FORT IRWIN	1.30
FORT ORD	1.21
LOS ANGELES	1.18
LOS ANGELES AREA	1.18
MARCH AFB	1.26
MATHER AFB	1.14
MCCLELLAN AFB	1.14
MONTEREY AREA	1.20
NORTON AFB	1.26
OAKLAND AB	1.37
PORT HUENEME AREA	1.18
PRESIDIO MONTEREY	1.23
PRESIDIO SAN FRAN	1.39
RIVERBANK AAP	1.19
SACRAMENTO AD	1.12
SAN DIEGO	1.16
SAN DIEGO AREA	1.16
SAN FRANCISCO	1.37
*SAN FRAN AREA	1.34
SHARPE AD	1.16
TWENTY-NINE PALMS	1.38
VANDENBURG AFB	1.36
COLORADO	.97
AIR FORCE ACADEMY	1.05
CHEYENNE MOUNTAIN	1.03
COLORADO SPRINGS	.96
DENVER	.96
FALCON AFS	.96
FITZSIMMONS AMC	.98
FORT CARSON	1.03
PETERSON AFB	.96
PUEBLO	.98
PUEBLO AD	.98
ROCKY MOUNTAIN ARS	.97
CONNECTICUT	1.19
BRIDGEPORT	1.21
HARTFORD	1.16
NEW LONDON	1.21
NEW LONDON AREA	1.21
STRATFORD ENGR P	1.21
DELAWARE	1.00
DOVER	1.02
DOVER AFB	1.02
LEWES	.99
FLORIDA	.89
CAPE CANAVERAL	.91

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CAPE KENNEDY	.91
EGLIN AFB	.83
HOMESTEAD	.90
HOMESTEAD AFB	.90
JACKSONVILLE AREA	.90
KEY WEST NAS	1.10
MCDILL AFB	.94
MIAMI	.88
ORLANDO	.82
PANAMA CITY AREA	.86
PENSACOLA AREA	.84
TAMPA	.94
TYNDALL AFB	.86
GEORGIA	.80
ALBANY	.80
ALBANY AREA	.80
ATLANTA	.83
ATLANTA, MARIETTA	.84
FORT BENNING	.79
FORT GILLEM	.83
FORT GORDON	.88
FORT MCPHERSON	.83
FORT STEWART	.81
KINGS BAY	.92
MACON	.77
WARNER ROBINS AFB	.77
HAWAII	1.46
ALIAMANU	1.37
BARBERS PT NAS	1.40
BARKING SANDS	1.80
EWA BEACH	1.40
FORD ISLAND	1.53
FORT DERUSSY	1.36
FORT SHAFTER	1.37
HELEMANO	1.42
HICKAM AFB	1.34
HILO	1.55
HONOLULU	1.36
KANEOHE	1.46
KANEOHE MCAS	1.39
LUALUALEI	1.43
MAUNA LOA	1.36
PEARL CITY	1.36
PEARL HARBOR	1.36
POHAKULOA	1.65
SCHOFIELD BRKS	1.42
TRIPLER AMC	1.36
WHEELER AFB	1.42
IDAHO	1.06
BOISE	1.04
IDAHO FALLS	1.03
MTN HOME	1.10
MTN HOME AFB	1.10
ILLINOIS	1.08
BELLEVILLE	.98
CHICAGO	1.27
CHICAGO AREA	1.27
FORT SHERIDAN	1.28
GREAT LAKES NTC	1.28
ROCK ISLAND	.99
ROCK ISLAND ARS	1.02
SAVANNAH AD	1.02
SCOTT AFB	1.05
INDIANA	1.00
CRANE AAP	1.06
CRANE NWSC	1.06
FT BEN HARRISON	1.07
GRISSOM AFB	1.06
INDIANA AAP	.93
INDIANAPOLIS	1.04
INDIANAPOLIS AREA	1.05
JEFFERSON PG	.95
LOGANSPORT	1.00
MADISON	.96
NEWPORT AAP	.96
IOWA	1.02
BURLINGTON	1.07
CEDAR RAPIDS	.99
DES MOINES	.99
IOWA AAP	1.08
KANSAS	.92
FORT LEAVENWORTH	1.06
FORT RILEY	.90
KANSAS AAP	.92
MANHATTAN	.90
MCCONNELL AFB	.92
TOPEKA	.93
WICHITA	.92
KENTUCKY	.95
BOWLING GREEN	.91
FORT CAMPBELL	1.02
FORT KNOX	.98
LEXINGTON	1.01
LEX/BLUEGRASS AD	1.08
LOUISVILLE	.92
LOUISVILLE NAS	.92
LOUISIANA	.89
ALEXANDRIA	.88
BARKSDALE AFB	.86
ENGLAND AFB	.88

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FORT POLK	.95
LOUISIANA AAP	.86
NEW ORLEANS	.93
NEW ORLEANS AB	.93
NEW ORLEANS AREA	.93
SHREVEPORT	.86
MAINE	1.11
BANGOR	1.05
BRUNSWICK	1.07
CARIBOU	1.21
LORING AFB	1.19
PORTLAND	1.07
PROSPECT HARBOR	1.11
WINTER HARBOR	1.11
MARYLAND	.95
ABERDEEN PG	.96
ANDREWS AFB	1.05
BALTIMORE	.96
FORT DETRICK	.94
FORT HOLABIRD	.96
FORT MEADE	1.05
FORT RITCHIE	.94
FREDERICK	.95
HARRY DIAMOND LAB	1.05
INDIAN HEAD	.99
LEXINGTON PARK	.95
PATUXENT RIV AR	.95
MASSACHUSETTS	1.20
ARMY MATL & MECH	1.22
BOSTON	1.23
BOSTON AREA	1.23
FITCHBURG	1.21
FORT DEVENS	1.28
HANSCOMB AFB	1.23
NATICK	1.22
MICHIGAN	1.12
BAY CITY	1.06
DETROIT	1.18
DETROIT AREA	1.18
DETROIT ARS	1.18
MARQUETTE	1.13
REPUBLIC ELFCOM	1.21
SAWYER AFB	1.21
WURTSMITH AFB	1.05
MINNESOTA	1.20
DULUTH	1.18
MINNEAPOLIS	1.22
ROSEMONT	1.22
ST. CLOUD	1.19
TWIN CITY AP	1.22
MISSISSIPPI	.82
BILOXI	.84
COLUMBUS	.82
COLUMBUS AFB	.82
GULFPORT	.84
JACKSON	.79
KEESLER AFB	.84
MERIDIAN NAS	.92
MISSISSIPPI AAP	.84
MISSOURI	.98
FT LEONARD WOOD	.96
KANSAS CITY	1.01
KANSAS CITY AREA	1.01
LAKE CITY AAP	1.00
ROLLA	.91
ST. LOUIS	1.03
ST. LOUIS AAP	1.08
WHITEMAN AFB	1.11
MONTANA	1.22
BILLINGS	1.22
BUTTE	1.23
GREAT FALLS	1.20
MALMSTROM AFB	1.20
NEBRASKA	.93
CORNHUSKER AAP	.88
GRAND ISLAND	.88
LINCOLN	.91
OFFUTT AFB	.99
OMAHA	.99
NEVADA	1.12
FALLON	1.19
HAWTHORNE	1.17
HAWTHORNE AAP	1.17
LAS VEGAS	1.10
NELLIS AFB	1.10
RENO	1.09
NEW HAMPSHIRE	1.02
CONCORD	1.01
NASHUA	1.04
PEASE AFB	1.02
PORTSMOUTH	1.02
PORTSMOUTH AREA	1.02
NEW JERSEY	1.14
BAYONNE MOT	1.17
EARLE	1.17
FORT DIX	1.15
FORT MONMOUTH	1.13
LAKEHURST	1.16
MCGUIRE AFB	1.15
NEWARK	1.17

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PICATINNY ARS	1.25
RED BANK	1.12
TRENTON	1.14
TRENTON AREA	1.14
NEW MEXICO	.95
ALAMAGORDO	.92
ALBUQUERQUE	.92
CANNON AFB	1.10
FORT WINGATE	1.00
GALLUP	1.00
HOLLOMAN AFB	1.01
KIRTLAND AFB	.92
WHITE SAND MIS RN	1.10
NEW YORK	1.11
ALBANY	1.05
BROOKLYN	1.25
FORT DRUM	1.18
GARDEN CITY	1.25
NEW YORK CITY	1.25
NEW YORK CITY AREA	1.25
SENECA AD	1.15
STATEN ISLAND	1.32
SYRACUSE	1.04
US MIL ACADEMY	1.13
WATERVLIET ARS	1.05
NORTH CAROLINA	.81
CAMP LEJEUNE AREA	.82
CHERRY POINT	.83
FAYETTEVILLE	.79
FORT BRAGG	.80
GREENSBORO	.86
NEW RIVER	.85
POPE AFB	.80
SEYMOUR JOHN. AFB	.80
SUNNY POINT	.85
WILMINGTON	.79
NORTH DAKOTA	1.00
BISMARCK	.97
GRAND FORKS	.96
GRAND FORKS AFB	.96
MINOT	1.07
MINOT AFB	1.07
OHIO	.99
COLUMBUS	.98
DAYTON	1.00
RAVENNA AAP	.99
WRGT-PATTERSON AFB	1.00
YOUNGSTOWN	.99
OKLAHOMA	.83
ALTUS AFB	.86
FORT SILL	.81
LAWTON	.81
MCALESTER	.81
MCALESTER AAP	.81
OKLAHOMA CITY	.87
TINKER AFB	.87
OREGON	1.04
PENDLETON	1.12
PORTLAND	1.00
SALEM	.99
UMATILLA AD	1.19
PENNSYLVANIA	1.01
CARLISLE BRKS	.94
HARRISBURG	.94
INDIANTOWN GAP MIS	1.01
LETTERKENNY AD	1.00
MECHANICSBURG AREA	.94
NEW CUMBERLAND AD	.94
PHILADELPHIA	1.13
PHILADELPHIA AREA	1.13
PITTSBURGH	.97
SCRANTON	1.05
TOBYHANNA AD	1.14
WARMINSTER AREA	1.11
RHODE ISLAND	1.15
BRISTOL	1.16
DAVISVILLE	1.22
NEWPORT	1.12
NEWPORT AREA	1.12
PROVIDENCE	1.16
SOUTH CAROLINA	.89
BEAUFORT AREA	.94
CHARLESTON	.91
CHARLESTON AFB	.91
CHARLESTON AREA	.91
COLUMBIA	.82
FORT JACKSON	.82
MYRTLE BEACH	.93
SHAW AFB	.83
SOUTH DAKOTA	1.02
ABERDEEN	1.04
ELLSWORTH AFB	1.02
RAPID CITY	1.01
SIOUX FALLS	1.00
TENNESSEE	.80
ARNOLD AFB	.90
CHATTANOOGA	.74
HOLSTON ORDNANCE	.79
KINGSPORT	.79
MEMPHIS	.86

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MEMPHIS AREA	.86
MILAN ARS	.79
VOLUNTEER ORDNANCE	.79
TEXAS	.89
BROOKS AFB	.88
CAMP BULLIS	.88
CARSWELL AFB	.95
CHASE FIELD	.93
CORPUS CHRISTI	.84
CORPUS CHRISTI AD	.84
DYESS AFB	.92
FORT BLISS	.89
FORT HOOD	.90
FORT SAM HOUSTON	.88
FORT WORTH	.95
GOODFELLOW AFB	.89
INGLESIDE	.86
KELLY AFB	.88
KINGSVILLE	.98
LACKLAND AFB	.88
LONE STAR AAP	.88
LONGHORN AAP	.81
RANDOLPH AFB	.88
RED RIVER AD	.88
REESE AFB	.86
SAN ANGELO	.83
SAN ANTONIO	.88
SHEPPARD AFB	.94
UTAH	.96
DUGWAY PG	1.00
FORT DOUGLAS	.93
HILL AFB	.98
OGDEN	.92
SALT LAKE CITY	.93
TOOELE	1.02
TOOELE AD	1.02
VERMONT	1.06
BURLINGTON	1.09
MONTPELIER	1.08
RUTLAND	1.01
VIRGINIA	.92
ARLINGTON	1.05
ARLINGTON HS	1.05
CAMERON STA	1.05
DAHLGREN	.94
FORT BELVOIR	1.05
FORT EUSTIS	.92
FORT AP HILL	.94
FORT LEE	.87
FORT MONROE	.92
FORT MYER	1.05
FORT PICKETT	.94
FORT STORY	.92
NORFOLK	.92
NORFOLK AREA	.92
QUANTICO	.93
RADFORD	.89
RADFORD AAP	.95
RICHMOND	.94
VINT HILL FARMS	.94
WASHINGTON	1.00
BREMERTON	.98
FAIRCHILD AFB	1.00
FORT LEWIS	1.00
INDIAN ISLAND	.99
MCCHORD AFB	1.00
SILVERDALE	.94
SPOKANE	.99
TACOMA	1.00
WHIDBEY ISLAND	1.09
YAKIMA	1.00
YAKIMA FR	1.07
WEST VIRGINIA	.99
BLUEFIELD	.96
CHARLESTON	.98
CLARKSBURG	1.02
SUGAR GROVE	
(Buildings)	1.29
(Heavy Constr)	1.50
WISCONSIN	1.04
BADGER AAP	.99
FORT MCCOY	1.08
LA CROSSE	1.02
MADISON	.99
MILWAUKEE	1.10
WYOMING	1.08
CASPER	1.08
CHEYENNE	1.08
F.E. WARREN AFB	1.08
LARAMIE	1.07
WASHINGTON, DC	1.05
FORT MCNAIR	1.05
WALTER REED AMC	1.05
WASHINGTON DC AR	1.05

4.2 Overseas Area Cost Information (6/90). The following list outlines area cost information for various areas outside of the United States.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:54Z
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LOCATION	ACF/FER	NOTES
ANTIGUA	1.73/2.78	
AUSTRALIA	1.55/1.30	
DARWIN	1.90	
PERTH	1.42	
SYDNEY	1.34	
AZORES	1.04/168.10	2
BAHAMAS	1.79/1.00	2
ANDROS IS	1.79	2
BAHRAIN	2.07/.38	
BELGIUM	1.25/42.55	
BRUSSELS	1.25	
BERMUDA	1.61/1.00	2
CANADA	1.55/1.16	
ARGENTIA, NEWFOUNDLAND	1.55	
CUBA	1.60	2
GUANTANAMO	1.60	2
GTMO PAVING	3.00	2
DIEGO GARCIA	3.00	
EGYPT	1.32/2.65	
CAIRO	1.32	
GREECE	1.13/155.30	
ATHENS	1.13	
GREENLAND	2.48/7.81	
GODTHAAB(NUUK)	2.48	
GUAM	2.24	
HONDURAS	.64/4.50	
TEGUCIGALPA	.64	
ICELAND	3.38/62.50	1
KEFLAVIK	3.38	1
REYKJAVIK	3.38	1
ISRAEL	1.18/2.03	
TEL AVIV	1.18	
ITALY	1.44/1132.50	1
ISOLA DI CAPO RIZZUTO	1.43	1
LA MADDALENA	1.50	1
NAPLES	1.44	1
SIGONELLA	1.44	1
VENICE	1.39	1
JAPAN	1.60/149.05	
ATSUGI	1.66	
MISAWA	1.57	
OKINAWA	1.49	
TOKYO	1.66	
JOHNSTON ATOLL	2.32	
KOREA	1.12/664.01	
CHINHAE	1.11	
DMZ ZONE AREA	1.18	
KUNSAN	1.12	
OSAN	1.10	
SEOUL	1.08	
KWAJALEIN	2.54	
MIDWAY ISLAND	2.07	
MOROCCO	1.55/8.67	
CASABLANCA	1.55	
NETHERLANDS	1.25/2.26	
OSS	1.25	
NEW ZEALAND	2.07/1.74	
WELLINGTON	2.07	
OMAN	1.58/.39	
RUWI	1.58	
PANAMA	1.24	
PANAMA CIT	1.24	
PHILLIPINES	1.10/22.25	
MANILA	1.08	
SUBIC BAY	1.11	
PUERTO RICO	1.05	2
SAN JUAN	1.05	2
SEYCHELLES IS	2.50	
SPAIN	1.18/95.15	
ROTA	1.18	
TURKEY	.71/2185.52	
ANKARA	.71	
INCIRLICK	.70	
UNITED KINGDOM	1.71/.52	1
LONDON	1.75	1
MANCHESTER	1.64	1
ST. MAWGAN	1.71	1
EDZELL, SCOTLAND	1.75	1
WEST GERMANY	1.38/2.03	
FRANKFURT	1.54	
KAISERSLAUTERN	1.40	
TRIER	1.20	

NOTES:
(1) Escalate project costs to the one-third time point of construction.
(2) Allow for Defense Base Act Insurance Costs.
(3) International Balance of Payments (IBOP) Summary Sheet is required.
ACF: Area Cost Factor FER: Foreign Exchange Rate

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(8) Minimally corrosive

Section 5: GUIDANCE COSTS

5.1 Primary Facility Guidance Costs. Tables 6 and 6A give the current guidance costs for various facilities identified by category code. The costs have been adjusted for an ACF of 1.00, and the time has been adjusted for FY 94 and FY 95.

5.2 Supporting Facilities Guidance Cost. Table 7 gives the current total composite contract cost for typical supporting facilities systems. Refer to par. 7.2 for CES systems identification. The cost has been adjusted for an ACF of 1.00, and the time has been adjusted for FY 94 and FY 95.

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Table 6
Primary Facility Guidance Costs

NAVFACENGCOM FY 94 & FY 95 GUIDANCE COST (ACF = 1.00) AS OF JULY 1991.

ALL UNIT COSTS INCLUDE BUILT-IN EQUIPMENT TYPICAL FOR THE FACILITY.
UNIT COSTS INCLUDE QAP & PCAS EXCEPT HIGH TECH AND HIGH RISK FACILITIES.

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
133.72	RADAR AIR TRAFFIC CONTROL CENTER	6500 SF	147	151
137.10	OCEANOGRAPHIC BUILDING	37000 SF	97	100
141.25	ACFT FIRE/RESCUE/STRUC/STATION	8000 SF	106	109
141.40	AIRCRAFT OPERATIONS BUILDING	20000 SF	101	105
141.70	CONTROL TOWER (ATTACHED/FREESTANDING)	3000 SF	173	178
143.15	RANGE OPERATION CENTER	13000 SF	198	204
143.20	ORDNANCE OPERATIONS BUILDING	13000 SF	86	89
143.41	AMPHIBIOUS OPERATIONS BUILDING	25000 SF	100	104
143.45	ARMORY	10000 SF	89	92
143.65	OPCON CENTER FOR FLEET OPERATIONS	36000 SF	146	150
143.77	OPERATIONAL STORAGE (READY ISSUE)	4000 SF	53	55
151.20	GENERAL PURPOSE/BERTHING PIER	10000 SY	1017	1048
151.80	DEPERMING PIER	3240 SY	1360	1403
156.10	WATERFRONT TRANSIT SHED	5000 SF	77	79
159.64	WATERFRONT OPERATIONS BUILDING	20000 SF	83	86
171.10	ACADEMIC INSTRUCTION BUILDING	30000 SF	82	85
171.20	APPLIED INSTRUCTION BUILDING	25000 SF	88	91
171.35	OPERATIONAL TRAINER FACILITY	22000 SF	108	111
171.45	MOCK-UP & TRNG. AID PREPAR CENTER	25000 SF	98	101
171.60	RECRUIT PROCESSING BUILDING	80000 SF	79	81
179.45	FIREFIGHTING TRAINING FACILITY	8100 SF	244	251
179.55	COMBAT TRAINING POOL/TANK	22000 SF	112	115
211.03	CORROSION CONTROL HANGAR	35000 SY	236	244
211.05	MAINTENANCE HANGAR (1 MODULE)	38000 SF	84	87
211.06	MAINTENANCE HANGAR (1 MODULE)	38000 SF	83	86
211.14	AIRCRAFT REWORK SHOP (NADEP)	160800 SF	105	108
211.21	ENGINE MAINTENANCE SHOP	34000 SF	78	80
211.24	DEDICATED A/C ENGINE O/H (NADEP)	80000 SF	87	90
211.34	DYNAMIC COMPONENTS SHOP (NADEP)	8000 SF	95	98
211.45	AVIONICS SHOP (NON-NADEP)	23000 SF	84	87
211.54	AVIATION ARMAMENT SHOP (NON-NADEP)	16000 SF	79	81
211.81	ENGINE TEST CELL (NON-NADEP)	7065 SF	749	772
212.10	GUIDED MISSILES INTEGRATION FACILITY	5000 SF	84	87
212.30	MISSILE ASSEMBLY & TEST BUILDING	60000 SF	127	131
214.10	COMBAT VEHICLE MAINTENANCE SHOP	20000 SF	89	92
214.20	AUTOMOTIVE VEHICLE MAINTENANCE SHOP	20000 SF	79	81

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Table 6 (Continued)
Primary Facility Guidance Costs

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
214.30	REFUELING VEHICLE SHOP	2600 SF	91	94
214.51	AUTOMOTIVE ORGANIZAT'L SHOP (MARC)	30000 SF	76	78
216.10	AMMO REWORK AND OVERHAUL SHOP	14170 SF	147	151
216.40	TORPEDO SHOP	26000 SF	138	142
217.10	ELEC/COMM MAINTENANCE SHOP	17000 SF	93	96
219.10	PUBLIC WORKS SHOP	20000 SF	72	74
310.23	COMBINED RESEARCH LAB	60000 SF	106	109
310.37	OCEAN SCIENCES LABORATORY	56000 SF	105	108
315.20	UNDERWATER WEAPON SYSTEMS LABORATORY	20000 SF	137	141
315.30	WEAPONS SYSTEMS INTEGRATION LABORATORY	60000 SF	138	142
317.15	DETECTION SYSTEMS LAB	17000 SF	112	115
317.20	ELECTRICAL/ELECTRONICS SYS LAB	60000 SF	123	127
317.25	ELECTRICAL/COMM. SYS. INTEGRATION LAB	66000 SF	119	123
320.10	UNDERWATER EQUIPMENT LABORATORY	6000 SF	132	136
411.50	JET ENGINE FUEL STORAGE	600000 BL	11	11
421.22	HIGH EXPLOSIVE MAGAZINE	7000 SF	148	153
421.32	INERT STOREHOUSE	16000 SF	69	71
421.72	MISSILE MAGAZINE	9300 SF	140	144
441.10	GENERAL WAREHOUSE	40000 SF	46	47
441.11	GENERAL PURPOSE WAREHOUSE (MARINE CORPS DSSC)	40000 SF	58	60
441.30	HAZARDOUS/FLAMMABLE STOREHOUSE	10000 SF	107	110
441.35	GENERAL STORAGE SHED	20000 SF	32	33
441.72	SERVMART	39000 SF	53	55
550.10	MEDICAL CLINIC	30000 SF	124	128
610.10	ADMINISTRATIVE OFFICE (INCL HQ)	25000 SF	84	87
610.1A	PASS OFFICE	6000 SF	103	106
610.20	DATA PROCESSING CENTER	21000 SF	113	116
610.72	BATTALION/SQUADRON HEADQUARTERS	30000 SF	72	74
721.11	BEQ (E-1 THRU E-4)	100000 SF	71	73
721.12	BEQ (E-5 AND E-6)	100000 SF	79	81
722.10	ENLISTED DINING FACILITY	16000 SF	152	157
724.11	BOQ (W-1 THRU O-2)	44000 SF	76	78
724.12	BOQ (O-3 AND ABOVE)	44000 SF	71	73
730.10	FIRE STATION (COMMUNITY)	3500 SF	107	110
730.15	CONFINEMENT FACILITY	30000 SF	104	107
730.20	POLICE STATION	6500 SF	99	103
730.83	CHAPEL	10000 SF	122	126
740.23	COMMISSARY (INCLUDES BACK-UP STORAGE)	60000 SF	61	63
740.43	GYMNASIUM	20000 SF	100	104
740.74	CHILD CARE CENTER	8000 SF	90	93
831.41	HAZARDOUS WASTE FACILITY	3600 SF	90	93

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Table 6A
Additional Primary Facility Guidance Costs *

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
113-20	AIRCRAFT PARKING APRON	65000 SY	45	47
116-15	AIRCRAFT RINSE FACILITY	1310 SY	142	146
116-56	COMBAT AIRCRAFT ORDNANCE LOADING AREA	55281 SY	62	64
125-10	POL PIPELINE	2689 LF	161	166
131-15	COMMUNICATIONS CENTER	6000 SF	243	250
131-25	TELEMETRY BUILDING	9000 SF	201	207
131-35	RECEIVER BUILDING	20550 SF	308	318
131-40	TELEPHONE EXCHANGE BUILDING	6000 SF	170	176
131-45	TERMINAL EQUIPMENT BUILDING	70000 SF	114	118
131-55	CIRCULARLY DISPOSED ANTENNA ARRAY BUILDING	15000 SF	132	136
132-10	ANTENNA - COMMUNICATIONS	1 EA	2635	2718
141-30	AIRCRAFT LINE OPERATIONS BUILDING	15620 SF	65	67
141-42	AIR INTELLIGENCE SUPPORT CENTER	20000 SF	100	103
141-87	LIQUID OXYGEN/NITROGEN FACILITY (NON-INDUSTRIAL)	5582 SF	104	107
141-88	HARDENED AIRCRAFT SHELTER	13991 SF	78	80
143-10	EMERGENCY VEHICLE GARAGE	3066 SF	59	61
143-25	SEAL TEAM BUILDING	57569 SF	87	90
143-40	COMPUTER PROGRAMMING OPERATIONS CENTER	21000 SF	102	105
143-47	ALERT FORCE BUILDING	22500 SF	69	71
143-80	NAVAL SECURITY GROUP REGIONAL REPORTING CENTER	26000 SF	158	163
159-20	DEGAUSSING BUILDING	1600 SF	185	190
171-15	RESERVE TRAINING BUILDING	33834 SF	69	71
179-40	SMALL ARMS RANGE - OUTDOOR	4011 SF	84	87
211-01	AIRCRAFT ACOUSTICAL ENCLOSURE (NON-NADEP)	7950 SF	685	706
211-08	AIRFRAMES SHOP (NON-NADEP)	23000 SF	147	152
211-12	PAINT AND FINISHING HANGAR (NADEP)	69708 SF	221	228
211-32	METAL COMPONENTS SHOP (NADEP)	104000 SF	97	100
211-36	ELECTRICAL COMPONENTS SHOP (NADEP)	15500 SF	94	97
211-52	AIRCRAFT WEAPON OVERHAUL AND TEST SHOP (NADEP)	19480 SF	131	135
211-60	SUPPORT EQUIPMENT REWORK SHOP CENTER (NADEP)	57680 SF	96	99
211-70	MANUFACTURING AND REPAIR SHOP (NADEP)	81000 SF	69	71
211-72	METAL FABRICATION/MANUFACTURING SHOP (NADEP)	81000 SF	117	120

* Unit costs derived from Award Cost Data or NAVFAC FY 92 & FY 93 Guidance Costs.

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Table 6A (Continued)
Additional Primary Facility Guidance Costs

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
211-73	METAL TREATMENT SHOP (NADEP)	81000 SF	166	171
211-75	PARACHUTE SURVIVAL EQUIPMENT SHOP (NON-NADEP)	8000 SF	88	90
211-89	POWER CHECK PAD WITHOUT SOUND SUPPRESSION (NON-NADEP)	1 EA	**364	**376
211-89I	POWER CHECK PAD (VSTOL)	1 EA	**610	**629
211-93	ENGINEERING LABORATORY (NADEP)	16000 SF	153	157
211-94	AIRCRAFT POWER CHECK FACILITIES (NADEP)	9000 SF	480	495
211-96	MAINTENANCE, AIRCRAFT SPARES STORAGE (READY ISSUE/SHOP STORAGE MISCELLANEOUS)	13150	36	37
211-98	AIRCRAFT ACOUSTICAL ENCLOSURE (NADEP)	12653 SF	557	574
212-20	MISSILE EQUIPMENT MAINTENANCE SHOP	10462 SF	97	100
213-55	PIPEFITTING SHOP (56) (J)	78000 SF	105	109
213-59	ABRASIVE BLAST FACILITY	20000 SF	169	174
213-61	RIGGING SHOP - (72) (T)	155500 SF	64	66
214-53	FIELD MAINTENANCE SHOP (COMBAT/AUTOMOTIVE/TRACK) (MARINE CORPS)	20000 SF	92	95
215-30	HEAVY GUN (6IN TO 16IN) SHOP	25000 SF	122	126
216-60	QUALITY EVALUATION LAB	27000 SF	90	93
217-77	ELECTRONICS - SPARES AND STORAGE (READY ISSUE/SHOP STORES/MISCELLANEOUS)	12928 SF	106	109
218-20	CONSTRUCTION/WEIGHT HANDLING EQUIPMENT SHOP	9600 SF	68	70
218-25	MARINE AIR-BASE SQUADRON (MABS) FACILITY	48330 SF	95	98
218-45	INSTRUMENT CALIBRATION SHOP	12000 SF	74	77
218-50	BATTERY SHOP	27000 SF	90	93
218-60	AIRCRAFT GROUND SUPPORT EQUIPMENT SHOP	32000 SF	85	87
218-61	GROUND SUPPORT EQUIPMENT HOLDING SHED	5051 SF	30	31
218-62	SHIPBOARD - AIRCRAFT SUPPORT - EQUIPMENT FACILITY	57784 SF	95	98

** Figures represent cost in thousands.

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Table 6A (Continued)
Additional Primary Facility Guidance Costs

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
218-80	FIELD MAINTENANCE SHOP (GENERAL SUPPLY)	40993 SF	61	63
219-77	PUBLIC WORKS MAINTENANCE STORAGE (READY ISSUE/SHOP STORES/MISC.)	42830 SF	68	70
222-10	MISSILE ASSEMBLY PLANT	3430 SF	116	119
226-65	PROPELLANT AND RELATED CHEMICAL PLANT	6000 SF	369	381
227-20	SONAR EQUIPMENT PLANT	93200 SF	96	99
310-17	OPTICS LABORATORY	37912 SF	248	256
310-33	COMPUTATION AND ANALYSIS LAB	28000 SF	116	119
311-25	AIRCRAFT SYSTEMS INTEGRATION LABORATORY	26500 SF	95	98
312-25	SPACECRAFT/SATELLITE LABORATORY	80000 SF	158	163
313-10	SHIP AND MARINE LABORATORY	20000 SF	100	103
313-15	SHIPS AND MARINE NAVIGATIONAL EQUIPMENT LABORATORY	30200 SF	158	163
316-10	AMMUNITION, EXPLOSIVES, AND TOXICS LABORATORY	6000 SF	490	505
318-10	PROPULSION SYSTEMS LABORATORY	13000 SF	305	315
319-10	MISCELLANEOUS EQUIPMENT AND ITEMS LABORATORY	38000 SF	67	69
319-40	RANGE OPERATIONS AND INSTRUMENTATION LABORATORY	31000 SF	85	87
321-10	TECHNICAL SERVICES LABORATORY	56500 SF	69	71
371-17	ELECTRONIC, COMMUNICATION AND ELECTRICAL SYSTEMS RANGE FACILITY	15238 SF	53	54
412-25	LUBRICANT STORAGE	16750 SF	50	51
421-48	SMALL ARMS/PYROTECHNICS MAGAZINE	3000 SF	158	163
431-10	COLD STORAGE WAREHOUSE	6000 SF	93	96
441-71	INTEGRATED LOGISTICS OVERHAUL AND OUTFITTING BUILDING	10000 SF	105	109
510-77	HOSPITAL/MEDICAL STORAGE (READY ISSUED/SHOP STORES/ MISCELLANEOUS)	36217 SF	36	37
540-10	DENTAL CLINIC	23597 SF	200	206
610-40	COURT ROOM FACILITY	6000 SF	100	103
610-70	DIVISION/WING HEADQUARTERS (MARINE CORPS)	30000 SF	95	98
610-71	REGIMENTAL/GROUP HEADQUARTERS (MARINE CORPS)	30000 SF	95	98

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Table 6A (Continued)
Additional Primary Facility Guidance Costs

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
610-73	COMPANY/BATTERY HEADQUARTERS (MARINE CORPS)	81145 SF	81	84
711-70	FUND HOUSING AFTER 1969, WO, O-1 THRU O-3	353136 SF	39	40
721-14	CLASS A STUDENT BARRACKS	108800 SF	100	103
723-30	LAUNDRY, DETACHED	5265 SF	52	54
724-22	CIVILIAN QUARTERS, GS-7 AND ABOVE	36420 SF	55	57
730-60	DEPENDENT SCHOOL	100000 SF	75	77
730-77	PERSONNEL SUPPORT STORAGE (READY ISSUE/SHOP STORES/MISCELLANEOUS	15795 SF	79	81
730-84	RELIGIOUS EDUCATION BUILDING	15000 SF	100	103
740-01	EXCHANGE RETAIL STORE	38080 SF	59	61
740-02	LOCATION EXCHANGE	5829 SF	56	58
740-03	EXCHANGE CENTRAL ADMINISTRATION	7200 SF	75	77
740-20	TEMPORARY LODGING	25397 SF	63	65
740-25	FAMILY SERVICES CENTER	5000 SF	110	114
740-36	HOBBY SHOP, ARTS/CRAFTS	16500 SF	112	116
740-38	HOBBY SHOP - AUTOMOTIVE	5068 SF	67	69
740-40	BOWLING ALLEY	16789 SF	72	74
740-50	FIELD HOUSE	38330 SF	105	109
740-53	INDOOR SWIMMING POOL	25260 SF	111	114
740-55	YOUTH CENTER	6974 SF	101	104
740-63	ENLISTED PERSONNEL CLUB	15836 SF	112	116
740-71	CLASS VI PACKAGE STORE	9640 SF	55	57
740-77	COMMUNITY STORAGE (READY ISSUE/ SHOP STORES/MISCELLANEOUS	4538 SF	45	46
740-80	GOLF CLUB HOUSE	10012 SF	103	106
740-84	INDOOR PLAYING COURTS	6438 SF	107	110
740-89	BATHHOUSE	2580 SF	69	71
821-50	NON-NUCLEAR STEAM PLANT	720000 PPH	167	172
831-14	INDUSTRIAL WASTE TREATMENT BUILDING	10000 SF	80	83
841-40	GROUND LEVEL POTABLE WATER STORAGE TANK	750000 GAL	1.84	1.90
843-10	FIRE PROTECTION PIPELINE	7200 LF	160	165
843-30	FIRE PROTECTION WATER TANK	750000 GAL	1.66	1.71
872-20	GUARD AND WATCH TOWERS	100 SF	231	238

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Table 7
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($) FY 94	FY 95
511, 512, & 516	ELECTRICAL DIST.			
	Note: Normal duct size is 4". For 35KV circuit use 5" duct			
	Note: For quantities less than 200 LF add 20%; less than 100 LF add 50%			
WK-G	4-1/C #2 THW str al. w/1-1/C#8 CU ground secondary w/termination	LF	4.08	4.20
WK-I	4-1/C #1/0 THW str al. w/1-1/C#6 CU ground secondary w/termination	LF	5.92	6.10
WK-J	4-1/C #2/0 THW str al. w/1-1/C#4 CU ground secondary w/termination	LF	7.22	7.44
WK-K	4-1/C #4/0 THW str al. w/1-1/C#2 CU ground secondary w/termination	LF	9.53	9.82
WK-250	4-1/C 250 MCM THW str al. w/1-1/C#2 CU ground secondary w/termination	LF	10.87	11.20
WK-500	4-1/C 500 MCM THW str al. w/1-1/C#1/0 CU ground secondary w/termination	LF	18.23	18.78
WK-750	4-1/C 750 MCM THW str al. w/1-1/C#1/0 CU ground secondary w/termination	LF	24.54	25.28
VF-G	4-1/C #2 CU USE wire 600V w/1-1/C#8 CU ground secondary w/termination	LF	10.33	10.64
VF-I	4-1/C #1/0 CU USE wire 600V w/1-1/C#6 CU ground secondary w/termination	LF	14.51	14.95
VF-J	4-1/C #2/0 CU USE wire 600V w/1-1/C#4 CU ground secondary w/termination	LF	17.70	18.23
VF-L	4-1/C #4/0 CU USE wire 600V w/1-1/C#2 CU ground secondary w/termination	LF	25.30	26.06
VF-250	4-1/C 250 MCM CU USE wire 600V w/1-1/C#2 CU ground secondary w/termination	LF	22.25	22.92
VF-500	4-1/C 500 MCM CU USE wire 600V w/1-1/C#1/0 CU grd. secondary w/termination	LF	38.87	40.04
VG-E	3-1/C #4 AWG CU XLP 5KV shielded cable primary w/o termination	LF	17.45	17.97
VG-G	3-1/C #2 AWG CU XLP 5KV shielded cable primary w/o termination	LF	21.02	21.65
VG-I	3-1/C #1/0 AWG CU XLP 5KV shielded cable primary w/o termination	LF	23.88	24.60
VG-J	3-1/C #2/0 AWG CU XLP 5KV shielded cable primary w/o termination	LF	24.74	25.48
VG-L	3-1/C #4/0 AWG CU XLP 5KV shielded cable primary w/o termination	LF	34.62	35.66

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
VG-250	3-1/C 250 MCM AWG CU XLP 5KV shielded cable primary w/o termination	LF	44.99	46.34
VG-500	3-1/C 500 MCM AWG CU XLP 5KV shielded cable primary w/o termination	LF	95.14	97.99
VH-G	3-1/C #2 AWG CU XLP 15KV shielded cable primary w/o termination	LF	22.74	23.42
VH-H	3-1/C #1 AWG CU XLP 15KV shielded cable primary w/o termination	LF	25.23	25.99
VH-I	3-1/C #1/0 AWG CU XLP 15KV shielded cable primary w/o termination	LF	29.44	30.32
VH-J	3-1/C #2/0 AWG CU XLP 15KV shielded cable primary w/o termination	LF	32.46	33.43
VH-L	3-1/C #4/0 AWG CU XLP 15KV shielded cable primary w/o termination	LF	42.35	43.62
VH-250	3-1/C 250 MCM AWG CU XLP 15KV shielded cable primary w/o termination	LF	47.26	48.68
VH-500	3-1/C 500 MCM AWG CU XLP 15KV shielded cable primary w/o termination	LF	70.06	72.16
VI-B47	Cable termination size no. 2 thru 4/0 0-15KV porcelain	EA	311.00	320.00
VI-47A	Cable termination size 250 MCM thru 500 MCM 0-15KV porcelain	EA	418.00	431.00
VI-47B	Cable termination size 500 MCM thru 750 MCM 0-15KV porcelain	EA	606.00	624.00
VL-012	Duct bank 1 way 2" grs duct w/exca & backfill does not incl pave remvl	LF	9.22	9.50
VL-013	Duct bank 1 way 3" grs duct w/exca & backfill does not incl pave remvl	LF	15.60	16.07
VL-014	Duct bank 1 way 4" grs duct w/exca & backfill does not incl pave remvl	LF	21.73	22.38
VL-015	Duct bank 1 way 5" grs duct w/exca & backfill does not incl pave remvl	LF	38.17	39.32
VL-016	Duct bank 1 way 6" grs duct w/exca & backfill does not incl pave remvl	LF	52.71	54.29
VL-022	Duct bank 2 way 2" grs duct w/exca & backfill does not incl pave remvl	LF	17.47	17.99
VL-023	Duct bank 2 way 3" grs duct w/exca & backfill does not incl pave remvl	LF	30.03	30.93
VL-024	Duct bank 2 way 4" grs duct w/exca & backfill does not incl pave remvl	LF	42.15	43.41

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
VL-025	Duct bank 2 way 5" grs duct w/exca & backfill does not incl pave remvl	LF	74.45	76.68
VL-026	Duct bank 2 way 6" grs duct w/exca & backfill does not incl pave remvl	LF	104.00	107.12
VL-032	Duct bank 3 way 2" grs duct w/exca & backfill does not incl pave remvl	LF	25.63	26.40
VL-033	Duct bank 3 way 3" grs duct w/exca & backfill does not incl pave remvl	LF	44.33	45.66
VL-034	Duct bank 3 way 4" grs duct w/exca & backfill does not incl pave remvl	LF	54.86	55.89
VL-035	Duct bank 3 way 5" grs duct w/exca & backfill does not incl pave remvl	LF	111.75	115.10
VL-036	Duct bank 3 way 6" grs duct w/exca & backfill does not incl pave remvl	LF	155.30	159.96
VL-042	Duct bank 4 way 2" grs duct w/exca & backfill does not incl pave remvl	LF	32.76	33.74
VL-043	Duct bank 4 way 3" grs duct w/exca & backfill does not incl pave remvl	LF	53.13	54.72
VL-044	Duct bank 4 way 4" grs duct w/exca & backfill does not incl pave remvl	LF	81.43	83.87
VL-045	Duct bank 4 way 5" grs duct w/exca & backfill does not incl pave remvl	LF	146.94	151.35
VL-046	Duct bank 4 way 6" grs duct w/exca & backfill does not incl pave remvl	LF	204.97	211.12
VN-012	Duct bank 1 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	4.52	4.66
VN-013	Duct bank 1 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	5.50	5.67
VN-014	Duct bank 1 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	6.78	6.98
VN-015	Duct bank 1 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	8.27	8.52
VN-016	Duct bank 1 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	9.96	10.26
VN-022	Duct bank 2 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	8.09	8.33
VN-023	Duct bank 2 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	9.83	10.12
VN-024	Duct bank 2 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	12.24	12.61

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
VN-025	Duct bank 2 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	14.99	15.44
VN-026	Duct bank 2 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	18.47	19.02
VN-032	Duct bank 3 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	8.29	8.54
VN-033	Duct bank 3 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	14.15	14.57
VN-034	Duct bank 3 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	17.73	18.26
VN-035	Duct bank 3 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	21.68	22.33
VN-036	Duct bank 3 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	27.00	27.81
VN-042	Duct bank 4 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	13.99	14.41
VN-043	Duct bank 4 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	17.18	17.70
VN-044	Duct bank 4 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	21.60	22.25
VN-045	Duct bank 4 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	26.90	27.71
VN-046	Duct bank 4 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	33.97	34.99
VN-062	Duct bank 6 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	19.99	20.59
VN-063	Duct bank 6 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	24.54	25.28
VN-064	Duct bank 6 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	31.05	31.98
VN-065	Duct bank 6 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	38.80	39.96
VN-066	Duct bank 6 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	49.53	51.02
VN-082	Duct bank 8 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	26.01	26.79
VN-083	Duct bank 8 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	31.13	32.06
VN-084	Duct bank 8 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	40.41	41.62

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF - 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($) FY 94	FY 95
VN-085	Duct bank 8 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	50.70	52.22
VN-086	Duct bank 8 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	64.99	66.94
VN-122	Duct bank 12 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	37.71	38.84
VN-123	Duct bank 12 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	46.48	47.87
VN-124	Duct bank 12 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	59.04	60.81
VN-125	Duct bank 12 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	74.27	76.50
VN-126	Duct bank 12 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	95.60	98.47
VN-162	Duct bank 16 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	49.50	50.99
VN-163	Duct bank 16 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	60.93	62.76
VN-164	Duct bank 16 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	77.66	79.99
VN-165	Duct bank 16 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	97.00	100.73
VN-166	Duct bank 16 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	126.08	129.86
VN-15	15KV feeder w/4 way 4" DB, 4/0 CP grd, 3-6'x8'x6'D MHS & 3-250 MCM cab w/spl	LF	75.31	77.57
VN-35	35KV feeder w/4 way 6" DB, 4/0 CP grd, 3-9'x12'x6'D MHS & 3-500 MCM cab w/spl	LF	493.92	508.74
513	ELECTRICAL SUBSTATION			
YC@C1	15KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	5,137.00	5,291.00
YC@C10	25KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	6,038.00	6,219.00
YC@C11	37.5KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	6,206.00	6,392.00
YC@C12	50KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	6,416.00	6,608.00

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF - 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
YC@C13	75KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	9,476.00	9,760.00
YC@C14	100KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	7,899.00	8,136.00
YC@C15	167KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	8,208.00	8,454.00
YC@C16	75KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	13,027.00	13,418.00
YC@C17	112KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	13,739.00	14,151.00
YC@C18	150KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	14,563.00	15,000.00
YC@C19	225KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	16,546.00	17,042.00
YC@C20	300KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	17,091.00	17,604.00
YC@C21	500KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	20,627.00	21,246.00
YC@C22	750KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	24,035.00	24,756.00
YC@C23	1000KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	27,414.00	28,236.00
YC@C24	1500KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	34,559.00	35,596.00
VU-1	1, 1PH 10KVA xfmr w/term cutouts arresters 2" rscriser 40' pole w/guy	EA	5,366.00	5,527.00
VU-15	1, 1PH 15KVA xfmr w/term cutouts arresters 2" rscriser 40' pole w/guy	EA	5,581.00	5,748.00
VU-25	1, 1PH 25KVA xfmr w/term cutouts arresters 2" rscriser 40' pole w/guy	EA	13,169.00	13,564.00
541 & 542	HEAT DISTRIBUTION			
DK-1AA	10" steam & 6" condensate heat distribution piping aboveground	LF	330.00	336.00
DK-3AA	8" steam & 4" condensate heat distribution piping aboveground	LF	260.00	265.00
DK-4AA	6" steam & 4" condensate heat distribution piping aboveground	LF	218.00	222.00

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
DK-5AA	4" steam & 2.5" condensate heat distribution piping aboveground	LF	192.00	195.00
DK-7AA	3" steam & 2" condensate heat distribution piping aboveground	LF	162.00	165.00
DK-9AA	2.5" steam & 2" condensate heat distribution piping aboveground	LF	159.00	162.00
DK-92A	2" steam & 2" condensate heat distribution piping aboveground	LF	154.00	157.00
DK-94A	2" steam & 1" condensate heat distribution piping aboveground	LF	146.00	149.00
DK-97A	1" steam & 1" condensate heat distribution piping aboveground	LF	138.00	140.00
	For areas with unstable soil conditions increase unit prices by 20%.			
	551, 552, & 553 WATER DISTRIBUTION			
DF-DJ	2" bituminous paving for pipe trench w/12" base (incl removal)	SY	22.67	23.35
DC-EZ	6" conc pavement for pipe trench with 6" base (incl removal)	SY	89.31	91.99
DJ-LA	Fire hydrant standard 3 way post type w/6" valve and thrust block	EA	2,696.00	2,777.00
DJ-AE	4" 90 deg bend CI MJ	EA	179.00	184.00
DJ-AI	6" 90 deg bend CI MJ	EA	236.00	243.00
DJ-AM	8" 90 deg bend CI MJ	EA	282.00	290.00
DJ-AQ	10" 90 deg bend CI MJ	EA	380.00	391.00
DJ-AU	12" 90 deg bend CI MJ	EA	493.00	508.00
DJ-BA	4" PVC water pipe AWWA C-900 Class 150 w/excav backfill & compaction	LF	9.85	10.15
DJ-BB	6" PVC water pipe AWWA C-900 Class 150 w/excav backfill & compaction	LF	12.72	13.10
DJ-BC	8" PVC water pipe AWWA C-900 Class 150 w/excav backfill & compaction	LF	18.74	19.30
DJ-BD	10" PVC water pipe AWWA C-900 Class 150 w/excav backfill & compaction	LF	25.91	26.69
DJ-BE	12" PVC water pipe AWWA C-900 Class 150 w/excav backfill & compaction	LF	34.45	35.48
QL-JE	1" Type K copper pipe w/3' deep trenching, backfill & compaction	LF	8.42	8.67

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
QL-JM	1-1/2" Type K copper pipe w/3' deep trenching, backfill & compaction	LF	14.10	14.52
QL-KA	2" Type K copper pipe w/3' deep trenching, backfill & compaction	LF	19.67	20.26
DJ-LN	4" tapping saddle tap size to 2"	EA	113.00	116.00
DJ-LO	6" tapping saddle tap size to 2"	EA	133.00	137.00
DJ-LP	8" tapping saddle tap size to 2"	EA	148.00	152.00
DJ-LQ	10" tapping saddle tap size to 2"	EA	171.00	176.00
DJ-LR	12" tapping saddle tap size to 2"	EA	201.00	207.00
DJ-LT	4" x 4" tapping sleeve, 4" tapping valve & 4" tapped hole in pipe	EA	1,490.00	1,535.00
DJ-LV	6" x 6" tapping sleeve, 6" tapping valve & 6" tapped hole in pipe	EA	2,089.00	2,152.00
DJ-LY	8" x 8" tapping sleeve, 8" tapping valve & 8" tapped hole in pipe	EA	2,631.00	2,710.00
DJ-LYD	10" x 10" tapping sleeve, 10" tapping valve & 10" tapped hole in pipe	EA	4,202.00	4,328.00
DJ-LYI	12" x 12" tapping sleeve, 12" tapping valve & 12" tapped hole in pipe	EA	5,460.00	5,624.00
	Underground tanks for hazardous waste oil storage tanks increase cost by 120%			
	554 SANITARY SEWERS			
DP-KA	Precast 48" dia manhole 5' deep w/frame & cover (incl poured invert)	EA	2,213.00	2,279.00
DP-KB	Precast 48" dia manhole 8' deep w/frame & cover (incl poured invert)	EA	2,878.00	2,964.00
DF-DJ	2" bituminous paving for pipe trench w/12" base (incl removal)	SY	22.67	23.35
DC-EZ	6" conc paving for pipe trench with 6" base (incl removal)	SY	89.31	91.99
DP-EC	4" exterior sewer cleanout	EA	800.00	824.00
DP-ED	6" exterior sewer cleanout	EA	904.00	931.00
DP-EE	8" exterior sewer cleanout	EA	1,140.00	1,174.00
DP-DA	4" PVC sewer pipe ASTM 3034, SDR-35 w/excav, backfill & compaction	LF	7.06	7.27
DP-DB	6" PVC sewer pipe ASTM 3034, SDR-35 w/excav, backfill & compaction	LF	9.00	9.27

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
DP-DC	8" PVC sewer pipe ASTM 3034, SDR-35 w/excav, backfill & compaction	LF	11.59	11.94
DP-DD	10" PVC sewer pipe ASTM 3034, SDR-35 w/excav, backfill & compaction	LF	18.25	18.80
DP-DE	12" PVC sewer pipe ASTM 3034, SDR-35 w/excav, backfill & compaction	LF	22.67	23.35
561 & 562	FUEL STORAGE			
QV-BF	Underground oil storage tank 550 gal fiberglass double wall (incl excav)	GAL	21.73	22.38
QV-BG	Underground oil storage tank 1000 gal fiberglass double wall (incl excav)	GAL	16.01	16.49
QV-BJ	2000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	1.12	1.15
QV-BI	4000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	8.00	8.24
QV-BK	6000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	6.29	6.48
QV-BL	8000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	5.31	5.47
QV-BLA	10000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	5.52	5.69
QV-BM	12000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	5.57	5.74
QV-BN	15000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	5.58	5.75
QV-BO	20000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	5.22	5.38
71X	ROADS			
BW-AH	Stone aggregate base 6" thick	SY	4.97	5.12
BW-AJ	Stone aggregate base 8" thick	SY	6.79	6.99
DF-DD	Asphalt paving 1-1/2" overlay	SY	5.77	5.94
DF-DE	Asphalt paving 2" overlay	SY	7.53	7.76
DB-BD	Single bituminous surface treatment	SY	1.49	1.53
DB-BE	Double bituminous surface treatment	SY	1.97	2.03
DF-FF	Slurry seal	SY	3.82	3.93

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
DC-EC	Conc paving non-reinf 5000 psi 6 inch thick	SY	41.51	42.76
DC-EE	Conc paving non-reinf 5000 psi 8 inch thick	SY	55.97	57.65
DC-EG	Conc paving non-reinf 5000 psi 10 inch thick	SY	63.30	65.20
DC-EI	Conc paving non-reinf 5000 psi 12 inch thick	SY	70.61	72.73
DF-AA	Painting 4" lines on pavement	LF	0.14	0.14
DF-EA	Inlaid pavement lane line markings 4" wide	LF	1.41	1.45
EA-C	Concrete curb 6x18 cast-in-place	LF	9.11	9.38
EA-E	Concrete curb & gutter cast-in-place	LF	12.77	13.15
EA-F	Concrete valley gutter cast-in-place	SY	36.94	38.05
72X	PARKING			
BW-AH	Stone aggregate base 6" thick	SY	4.97	5.12
BW-AJ	Stone aggregate base 8" thick	SY	6.79	6.99
DF-DD	Asphalt paving 1-1/2" overlay	SY	5.77	5.94
DF-DE	Asphalt paving 2" overlay	SY	7.53	7.76
DB-BD	Single bituminous surface treatment	SY	1.49	1.53
DB-BE	Double bituminous surface treatment	SY	1.97	2.03
DF-FF	Slurry seal	SY	3.82	3.93
DF-AA	Painting 4" lines on pavement	LF	0.14	0.14
DF-EA	Inlaid pavement lane line markings 4" wide	LF	1.41	1.45
EA-C	Concrete curb 6x18 cast-in-place	LF	9.11	9.38
EA-E	Concrete curb & gutter cast-in-place	LF	12.77	13.15
EA-F	Conc valley gutter cast-in-place	SY	36.94	38.05
EA-G	Precast concrete parking bumpers 6"x10"x6' incl dowels	EA	36.11	37.19
74X	SIDEWALKS			
EA-D	Conc sidewalks 4 inch cast-in-place no base	SY	28.68	29.54

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
75X & 76X AIRCRAFT PAVING				
BW-AH	Stone aggregate base 6" thick	SY	4.97	5.12
BW-AJ	Stone aggregate base 8" thick	SY	6.79	6.99
DF-DD	Asphalt paving 1-1/2" overlay	SY	5.77	5.94
DF-DE	Asphalt paving 2" overlay	SY	7.53	7.76
DB-BD	Single bituminous surface treatment	SY	1.49	1.53
DB-BE	Double bituminous surface treatment	SY	1.97	2.03
DF-FF	Slurry seal	SY	3.82	3.93
DC-EC	Conc paving non-reinf 5000 psi 6 inch thick	SY	41.51	42.76
DC-EE	Conc paving non-reinf 5000 psi 8 inch thick	SY	55.97	57.65
DC-EG	Conc paving non-reinf 5000 psi 10 inch thick	SY	63.30	65.20
DC-EI	Conc paving non-reinf 5000 psi 12 inch thick	SY	70.61	72.73
DF-AA	Painting 4" lines on pavement	LF	0.14	0.14
DF-EA	Inlaid pavement lane line markings 4" wide	LF	1.41	1.45
812 BORROW				
BG-BC	Borrow fill in place at Sewells Point Area from Willoughby borrow pit	CY	10.17	10.48
BG-BL	Borrow fill in place at Cherry Pt/Camp Lejeunefrom station borrow pit	CY	10.17	10.48
BG-BJ	Borrow fill in place from commercial Tidewater pit 30 mile round trip	CY	14.82	15.26
BG-BK	Borrow fill in place from commercial Tidewater pit 20 miles round trip	CY	12.49	17.30
813 TOPSOIL SEED				
DV-AA	4" topsoil from piles on site w/seed, fertilize lime, & fine grade	SY	3.00	3.09

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
84X	DEMOLITION			
BE-BB	Clear & grub medium dense areas w/disposal	AC	3,659.00	3,769.00
BB-EA	Demolition of masonry building complete w/disposal	CF	0.15	0.15
BB-EB	Demolition of wood frame building complete w/disposal	CF	0.15	0.15
BB-EE	Demolition of steel frame building complete w/disposal	CF	0.21	0.22
BC-AB	Removal of bituminous paving up to 3" thick w/disposal	SY	4.93	5.08
BC-AD	Removal of 6" thick non-reinforced concrete pavement w/disposal	SY	9.93	10.23
BC-AJ	Removal of 8" thick non-reinforced concrete pavement w/disposal	SY	22.61	23.29
BC-JA	Removal of 11" thick non-reinforced concrete pavement w/disposal	SY	33.95	34.97
BC-AL	Removal of conc sidewalk 4" thick w/disposal	SY	8.88	9.15
BC-BF	Removal of concrete curb & gutter	LF	3.14	3.23
85X	STORM DRAINAGE			
CG-BA	12" RCP Class III storm drainage 4 ft deep w/excav, backfill & compact	LF	14.66	15.10
CG-BB	15" RCP Class III storm drainage 4 ft deep w/excav, backfill & compaction	LF	19.05	19.62
CG-BC	18" RCP Class III storm drainage 5 ft deep w/excav, backfill & compaction	LF	25.48	26.24
CG-BE	24" RCP Class III storm drainage 5 ft deep w/excav, backfill & compaction	LF	35.62	36.69
CG-BG	30" RCP Class III storm drainage 5 ft deep w/excav, backfill & compaction	LF	49.10	50.57
CG-BH	36" RCP Class III storm drainage 5 ft deep w/excav, backfill & compaction	LF	64.82	66.76
CG-MW	Precast 48" dia manhole 5' deep w/frame & cover	EA	2,328.00	2,398.00
CG-OA	Precast 48" dia manhole 8' deep w/frame & cover	EA	2,847.00	2,932.00

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Table 7 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
CG-MN	Standard catch basin 5 ft deep	EA	1,656.00	1,706.00
CG-OH	Curb inlet max depth 8' 4' throat width, 12" to 30 " pipe w/excav & backfill	EA	2,134.00	2,198.00
DF-DJ	2" bituminous paving for pipe trench w/12" base (incl removal)	SY	22.67	23.35
DC-EZ	6" concrete paving for pipe trench w/6" base (incl removal)	SY	89.31	91.99
86X	FENCING			
CI-AC	Chain link fence 9 ga galv steel 4' high w/3 strand barbwire	LF	17.30	17.82
CI-AE	Chain link fence 9 ga galv steel 6' high w/3 strand barbwire	LF	19.81	20.40
CI-AF	Chain link fence 9 ga galv steel 7' high w/3 strand barbwire	LF	21.48	22.12
CI-AG	Chain link fence 9 ga galv steel 8' high w/3 strand barbwire	LF	22.80	23.48
CI-CC	Gate chain link 9 ga galv steel 4' high w/gate posts & barbwire	LF	57.62	59.35
CI-CE	Gate chain link 9 ga galv steel 6' high w/gate posts & barbwire	LF	70.77	72.89
CI-CF	Gate chain link 9 ga galv steel 7' high w/gate posts & barbwire	LF	76.61	78.91
CI-CG	Gate chain link 9 ga galv steel 8' high w/gate posts & barbwire	LF	108.40	110.44
911, 912, & 913	LAND PILING			
CB-BC	Treated wood piling 40' to 50' long 12" butt 7" tip 12 lb creosote	LF	10.15	10.45
CD-AB	Piling prestressed conc 10" sq, more than 60 LF long, truck delivery	LF	12.42	12.79
CD-CB	Piling prestressed conc 12" sq, more than 60 LF long, truck delivery	LF	15.41	15.87

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Section 6: FACILITY STUDY SAMPLE ESTIMATE

6.1 DD Form 1391. Figure 10 is a sample of a facility study estimate.

6.2 Primary Facility Unit Cost Development. Figure 11 is a sample of primary facility unit cost development.

6.3 Supporting Facilities Cost Development. Figure 12 is a facilities study for supporting facilities cost development.

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1. COMPONENT	FY 19 95 MILITARY CONSTRUCTION PROJECT DATA	2. DATE
NAVY		15 AUG 1991

3. INSTALLATION AND LOCATION	4. PROJECT TITLE
U.S. NAVAL STATION NORFOLK, VIRGINIA	ADMINISTRATIVE OFFICE

5. PROGRAM ELEMENT	6. CATEGORY CODE	7. PROJECT NUMBER	8. PROJECT COST ($000)
	610-10	P-000	3,500

9. COST ESTIMATES

ACF: 0.92 ITEM ESCALATED TO: 1 APRIL 1995
FER: NA

	U/M	QUANTITY	UNIT COST	COST ($000)
PRIMARY FACILITY				
Administrative Office	SF	30,000	91.30	2,739
Data Processing Center	SF	22,000	80.84	(1,778)
Additional Functional Features	SF	8,000	117.39	(939)
	SF	30,000	.74	(22)
SUPPORTING FACILITIES				
Special Foundation Features	LS	-	-	427
Electrical Utilities	LS	-	-	(48)
Mechanical Utilities	LS	-	-	(15)
Roads, Parking, and Sidewalks	LS	-	-	(116)
Site Improvements	LS	-	-	(157)
Demolition	LS	-	-	(73)
SUB-TOTAL	LS	-	-	(18)
				3,166
CONTINGENCY (5%)				158
TOTAL CONTRACT COST				3,324
SUPERVISION, INSPECTION & OVERHEAD (6.0%)				199
TOTAL REQUEST				3,523
TOTAL REQUEST (ROUNDED)				3,500
EQUIPMENT PROVIDED FROM OTHER APPROPRIATIONS (LIST NOT AVAILABLE)				(-)

10. DESCRIPTION OF PROPOSED CONSTRUCTION

Construct a permanent three-story masonry and structural frame building on a pile foundation. A passenger elevator and an exterior uncovered steel stairway are included. The facility will provide 22,000 SF of administration space (610-10) and 8,000 SF data processing space (610-20). Supporting facilities consist of the foundation wood piling, utility connections, parking and road construction, and site improvements including storm drainage. Two buildings are to be demolished. OMSI is included. (Air Conditioning: 140 tons)

11. REQUIREMENT:

```
**********************************************************
*  NOTE 1:  The sample estimate (Figures 10 and 11)      *
*           was based on the escalation rate, area       *
*           cost factor, and the historical unit         *
*           costs which were in effect as of             *
*           August 1991.  Current estimates must be      *
*           based on current criteria.                   *
*  NOTE 2:  Refer to Section 3 for DD Form 1391          *
*           instructions.                                *
**********************************************************
```

DD FORM 1 DEC 76 1391

Figure 10
DD Form 1391

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PRIMARY FACILITY UNIT COST DEVELOPMENT

TITLE: ADMINISTRATIVE OFFICE P#: P-000

LOCATION: NAVAL STATION, NORFOLK, VA DATE: 8/15/91

PREPARED BY: U. R. ABLE PH#: 100-0000

	CAT CODE	SIZE	ACF	DATE (mo/yr)
PROPOSED FACILITY	610-10	22,000 SF	0.92	4/95
GUIDANCE FACILITY	610-10	25,000 SF	1.00	4/95

HISTORICAL GUIDANCE SOURCE: MIL-HDBK-1010 (10/91) *

HISTORICAL GUIDANCE UNIT COST: $87.00 /SF

HISTORICAL UNIT COST	X SIZE FACTOR	X AREA COST FACTOR	X ESCALATION FACTOR	= ADJUSTED UNIT COST
87.00	1.01	0.92	1.00	= $80.84/SF

* Show date of source as shown in Figure 11.

ADDITIONAL FUNCTIONAL FEATURES

FUNCTIONAL FEATURES	QTY	UNIT COST	TOTAL COST	ADJUSTED U/C	%CHANGE
OMSI (C - .6%)	$2,717,600**	0.006	16,306	0.54	0.6%
UNCOVERED EXTERIOR STAIRS	1 EA	6,000	6,000	0.20	0.2%

ADDITIONAL FUNCTIONAL FEATURES TOTAL: 0.74 0.8%

** NOTE: This figure is the total primary cost of both elements of the multi-use building (i.e., $80.84/SF X 22,000 SF plus $117.39/SF X 8,000 SF).

Figure 11
Primary Facility Unit Cost Development

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:54Z
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PRIMARY FACILITY UNIT COST DEVELOPMENT

TITLE: ADMINISTRATIVE OFFICE P#: P-000

LOCATION: NAVAL STATION, NORFOLK, VA DATE: 8/15/91

PREPARED BY: U. R. ABLE PH#: 100-0000

	CAT CODE	SIZE	ACF	DATE (mo/yr)
PROPOSED FACILITY	610-20	8,000 SF	0.92	4/95
GUIDANCE FACILITY	610-20	21,000 SF	1.00	4/95

HISTORICAL GUIDANCE SOURCE: MIL-HDBK-1010 (10/91) *

HISTORICAL GUIDANCE UNIT COST: $116.00 /SF

HISTORICAL UNIT COST	X SIZE FACTOR	X AREA COST FACTOR	X ESCALATION FACTOR	= ADJUSTED UNIT COST
116.00	1.10	0.92	1.00	= $117.39/SF

* Show date of source as shown in Figure 11.

ADDITIONAL FUNCTIONAL FEATURES

FUNCTIONAL FEATURES	QTY	UNIT COST	TOTAL COST	ADJUSTED U/C	%CHANGE
OMSI (C - .6%)	$2,717,600**	0.006	16,306	0.54	0.6%
UNCOVERED EXTERIOR STAIRS	1 EA	6,000	6,000	0.20	0.2%

ADDITIONAL FUNCTIONAL FEATURES TOTAL: _____ _____

Figure 11 (Continued)
Primary Facility Unit Cost Development

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SUPPORTING FACILITIES - COST DEVELOPMENT

TITLE: Administrative Office

DATE: Jan-24-1992

LOCATION: Naval Station, Norfolk, Virginia

P# 000 FY: 95

CESSYS (1)	SYSTEM DESCRIPTION (2)	COMPOSITE QUANTITY	COMPOSITE UNIT COST (3)	TOTAL COST
SPECIAL FOUNDATION FEATURES (91X)				
911	Wood Piling, 40'-50', 12" butt, 7" Tip, 12 lb creosote	5000 Lf	$10.45	$52,250
ELECTRICAL UTILITIES (51X - 52X)				
511	3-1/C #2/0 AWG CU XLP EKV Primary w/o termin.	300 Lf	$25.48	$7,032
512	4-1/C #2/0 THW W/1-1/C#4 CU grnd secon w/termin	100 Lf	$7.44	$684
513	37.5 KVA 1 PH xfmr conc pad mounted	1 Ea	$6,392.00	$5,881
514	400w HPS on 35'-40' conc pole wire, excav, equip	4 Ea	$2,500.00	$11,720
				$25,318
MECHANICAL UTILITIES (54X - 59X)				
541	8" steam 4" condensate heat distri piping abovgrnd	400 Lf	$286.00	$105,248
551	8" PVC pipe AWWA C-900 class 150 w/excav	400 Lf	$19.30	$7,102
551	2" bit paving for pipe trench w/12" base & remo	100 Ea	$23.35	$2,148
554	6" PVC sewer pipe ASTM 3034, SDR-35 w/excav	600 Lf	$9.27	$5,117
554	2" bit paving for pipe trench w/12" base & remo	200 Sy	$23.35	$4,296
554	48" diam precast conc mnhole 8' deep w/frame/cover	1 Ea	$2,500.00	$2,925
				$126,837
ROADS, PARKING, AND SIDEWALKS (7XX)				
711	6" base course material	3000 Sy	$5.12	$14,131
711	8" concrete pavement, 5000 psi	3000 Sy	$57.65	$159,114
721	8" base course material	4000 Sy	$6.99	$25,723
721	2" asphalt wearing surface course	4000 Sy	$7.76	$28,557
721	Painting 4" line on pavement	1250 Lf	$0.14	$161
741	4" conc sidewalks no base	300 Sy	$29.54	$8,153
				$235,839
SITE IMPROVEMENTS (8XX)				
811	Site grading balance & cut, 75 hp dozer	12000 Sy	$2.30	$32,292
812	Borrow fill compact in place w/ 30mi rt haul	2100 Sy	$12.10	$29,730
813	4" topsoil/area prep/seeding/fertilize/lime	2000 Sy	$3.09	$5,686
814	Landscaping/miscel trees/shrubs/bed prep/edging	1 Ls		$20,000
818	Temporary silt fence, 36" high	1 Ac	$10,600.00	$12,402
851	18" rcp class III 5 ft deep w/excav	500 Lf	$15.10	$6,946
851	48" diam mnhole 5' deep w/frame & cover, excav	4 Ea	$2,398.00	$8,825
				$115,880
DEMOLITION (84X)				
843	Demo steel bldg frame complete w/disposal	90000 Cf	$0.22	$18,216
843	Removal of 6" conc slab non-reinf w/disposal	840 Sy	$10.23	$7,906
				$26,122

NOTES: Place additional information and excess composite items on back.
(1) Refer to Section 7.2 for CES System Identification
(2) Refer to Table 7 for Supporting Facilities Guidance Cost.
(3) Composite Unit Cost Adjusted by Area Cost Factor, Escalation Factor & Size Factor

Figure 12
Supporting Facilities Cost Development

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Section 7: CES STANDARD SYSTEMS LIST (10/29/87)

7.1 Interior. The interior systems are shown in the following list.

0XX General
000 Building Construction
00X Overhead
001 Field Overhead
002 Home Office OH
01X Additive Bid Items
02X Deductive Bid Items
03X Base Bids
04X Government Cost
041 OMSI
042 PCAS
043 Contingencies
044 SIOH

1XX General Construction
11X Structural
111 Foundation (SF)
112 Slab on Grade (SF)
113 Structural (SF)
114 Supported Floor (SF)
115 Stairs (EA)
116 Roof Structure (SF)
117 Pre-Engd Building (SF)
118 Vault (SF)
119 Write-in
13X Structural Write-in
14X Architectural
141 Roofing (SF)
142 Exterior Walls (SF)
143 Interior Walls (SF)
144 Interior Finishes (SF)
145 Doors (SF)
146 Windows (SF)
147 Specialties (SF)
148 BEQ Wardrobes (EA)
149 Write-in
16X Architectural Write-in
17X Demolition Interior
171 Demo Interior (SF)
172 Asbestos Removal Interior (LS)
173 - 175 Write-in

2XX Interior Mechanical
21X Plumbing Domestic
211 Plumbing, domestic (EA)
212 Domestic Hot Water Solar (MBTU)
213 Roof Drainage (EA)
214 - 216 Write-in
22X Space Conditioning
221 HVAC (MBTU/TNS)
222 Air Conditioning (TNS)
223 Heating, Ventilation (MBTU/CFM)
224 Mechanical Ventilation (CFM)
225 Dehumidification (PPH)
226 Heating Solar (SF)
227 Write-in
23X Weight Handling Equipment
231 Bridge Cranes (TNS)(1)
232 Monorails (TNS)(1)
233 Fixed Hoist (TNS)(1)
234 Vehicle Lifts (TNS)(1)
235 Elevators (TNS)(1)
236 Escalators (TNS)(1)
237 Conveyors (TNS)(1)
238 Write-in
24X Special Mechanical Piping System
241 Vacuum (CFM)(1)
242 Oxygen (EA)(1)
243 Nitrogen (EA)(1)
244 Compressed Air (CFM)(1)
245 Interior Steam (PPH)(1)
246 - 247 Write-in
25X Special Mechanical, Other
251 Dust Collection (CFM)(1)
252 Engine Exhaust (CFM)(1)
253 - 255 Write-in
27X Interior Fire Protection
271 Sprinklers (SF)
272 AFFF (SF)
273 Carbon Dioxide (LBS)
274 Halon (LBS)
275 Fire Alarm (SF)
276 - 277 Write-in

(1) Built-in Equipment

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3XX Interior Electrical
- 31X Power & Lighting
  - 311 Power (KW)
  - 312 Lighting (SF)
  - 313 400HZ (KW)(1)
  - 314 Direct Current (KW)(1)
  - 315 UPS (KVA)(1)
  - 316 Electrical Generators (KW)(1)
  - 317 - 318 Write-in
- 32X Special Equipment Protection
  - 321 Grounding (SF)(1)
  - 322 Lightning Protection (SF)(1)
- 33X Special Electrical Electronics
  - 331 Security Detection
  - 332 EMCS (EA)(1)
  - 333 Computer Communication
  - 334 - 336 Write-in
- 34X Communication
  - 341 Telephone (SF)
  - 342 Intercom (SF)
  - 343 Television (SF)
  - 344 Public Announcement (SF)
  - 345 Master Clock (EA)
  - 346 - 347 Write-in
- 35X Write-in

4XX Special Equipment
- 41X Medical Equipment
  - 411 Hospital Equip (SF) (1)
  - 412 Dental Equip (EA)(1)
  - 413 - 414 Write-in
  - 415 Install Gov't Furnished Medical Equip
  - 416 Gov't Furn Med Equip
- 42X Food Service Equipment
  - 421 Food Service Equip (SF)
  - 422 - 424 Write-in
  - 425 Install Gov't Furnished Food Service Equip
  - 426 Gov't Furnished Food Service Equip
- 43X Miscellaneous Equipment
  - 431 Chapel Equip (SF)(1)
  - 432 Movie Theater Equip (SF)(1)
  - 433 Rifle Range Equip (EA)(1)
  - 434 Lab Equip (EA)(1)
  - 435 Install Gov't Furnished Lab Equip
  - 436 Gov't Furnished Lab Equip
  - 437 Waste Disposal (PPH)(1)
  - 438 Write-in
- 44X Special Warehouse Equipment
  - 441 Special Warehouse Equip (SF)(1)
  - 442 - 444 Write-in
  - 445 Install Gov't Furnished Warehouse Equip
  - 446 Gov't Furnished Warehouse Equip
- 45X Special Construction
  - 451 Hangar Doors (SF)
  - 452 Vault Doors (EA)
  - 453 Blast Doors (SF)
  - 454 Raised Floor (SF)*
  - 455 Radio Frequency Shielding (SF)(1)
  - 456 Bowling Lanes (EA)
  - 457 Paint Spray Booth (SF)(1)
  - 458 Write-in
- 49X Other Equipment
  - 491 Other Miscellaneous Equip
  - 492 Snow Melting (SF)
  - 495 Install Other Gov't Furnished Equip
  - 496 Other Gov't Furnished Equip (SF)(1)
  - 499 Building Work Not Included Elsewhere (LS)

(1) Built-in Equipment

* Considered built-in equipment for category code 610-20 (Data Processing Center) only.

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7.2 Exterior. The exterior systems are shown in the following lists.

SUPPORTING FACILITIES

UTILITIES

EXTERIOR ELECTRICAL

511	Electrical Distribution, Primary	Length of Run	Lin. Ft.

Includes overhead power distribution, i.e., poles, crossarms, insulators, guying, terminations, lightning protection, wire and cable, and underground distribution, i.e., excavation and backfill, concrete encased duct bank, direct burial duct, manholes, handholes, cable, terminations, stress cones, and grounding. All for primary distribution to transformer or substation.

512	Electrical Distribution, Secondary	Length of Run	Lin. Ft.

Includes overhead power distribution, i.e. poles, crossarms, insulators, guying, terminations, lighting protection, wire and cable and underground distribution, i.e., excavation, backfill concrete encased and direct burial duct bank, manholes, handholes, cable terminations, stress cones, and grounding.

513	Substation/Transformer	Capacity	KVA

Includes overhead transformer, i.e. fuse cutouts, transformers, substation, including concrete fence, steel framing, fencing and enclosure, and pad mounted transformer; i.e. disconnect services, transformers, substation including concrete steel framing, fencing, and enclosure.

514	Area Lighting	Number of Fixtures	Each

Includes poles, fixtures, excavation and backfill, concrete work, wire, duct, and conduit.

515	Airfield Lighting	Length of Run	Lin. Ft.

Includes excavation and backfill, duct, conduit, devices, transformer, relays, and wiring necessary to install the airfield lighting.

516	Lightning Protection	Points of Protection	Each

Includes base plate, lighting mast, concrete foundations, ground rods, cables, and connections for a complete exterior lighting protection system.

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EXTERIOR COMMUNICATION

521 Fire Alarm — Length of Run — Lin. Ft.

Includes excavation, backfill, duct, conduit, devices, and wire necessary to install an exterior fire alarm system.

522 Security Alarm — Length of Run — Lin. Ft.

Includes excavation and backfill, duct, conduit, and junction boxes. Wiring cable, devices, and their installation are not MCON funded.

523 Communication, Telephone — Length of Run — Lin. Ft.

Includes excavation and backfill, duct, conduit, devices and wire necessary to install an exterior communication system.

524 Exterior EMCS — Length of Run — Lin. Ft.

Includes cable, duct, conduit necessary for EMCS systems outside the building 5-foot line.

EXTERIOR MECHANICAL

541 Heat Distribution, Overhead — Length of Run — Lin. Ft.

Includes supports, piping, insulation, and valves to install overhead steam and hot water distribution systems outside the 5-foot line.

542 Heat Distribution, Underground — Length of Run — Lin. Ft.

Includes excavation, backfill, manholes, supports, piping insulation, and valves to install underground steam and hot water distribution systems outside the 5-foot line.

543 Chilled Water Distribution — Length of Run — Lin. Ft.

Includes excavation, backfill, manholes, supports, piping, insulation, and valves to install chilled water distribution systems outside the 5-foot line.

544 Condensate Collection (only) — Length of Run — Lin. Ft.

Includes excavation, backfill, manholes, supports, piping, and valves to install a condensate collection system.

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545 Gas Distribution — Length of Run — Lin. Ft.

Includes excavation, backfill, piping, valves, and other miscellaneous work to install a complete gas system outside the 5-foot line.

546 Compressed Air Distribution — Length of Run — Lin. Ft.

Includes excavation, backfill, piping, valves, and devices to install a complete compressed air distribution system outside the 5-foot line.

547 Fuel Distribution — Length of Run — Lin. Ft.

Includes excavation, backfill, piping, pipe coatings, valves, and devices to install a complete fuel distribution system outside the 5-foot line.

548 Exterior Foam System — Length of Run — Lin. Ft.

Includes distribution piping, foam storage and proportioning equipment, booster pump, and emergency power.

551 Water Distribution — Length of Run — Lin. Ft.

Includes complete potable water distribution system, i.e., utility service connections, fire hydrants, excavation and backfill, pipe, valves, and fittings. Use when domestic and fire water System 552 are combined in one line.

552 Fire Protection Water Distribution — Length of Run — Lin. Ft.

Includes complete fire water distribution system, i.e., utility service connection fire hydrants, excavation and backfill, pipe, valves, and fittings.

553 Salt Water Distribution — Length of Run — Lin. Ft.

Includes complete salt water distribution system, i.e., utility service connections, fire hydrants, excavation and backfill, pipe, valves, and fittings.

554 Sanitary Sewers — Length of Run — Lin. Ft.

Includes complete sanitary sewer system, i.e., utility service connections, excavation and backfill, sheeting and shoring, dewatering, pipe and fittings, manholes, cleanouts, septic disposal and process, and acid waste system outside the 5-foot line.

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BULK STORAGE

561	Fuel Storage, Bulk	Capacity	BBL

Includes tanks, foundation, coatings, fire protection, and level controls. Does not include piping or other special foundation work.

562	Fuel Storage, Ready Issue	Capacity	Gal

Includes excavation, backfill, gravel fill, concrete, tie-downs, tanks, accessories, piping at tank, coatings, and monitoring wells. (Does not include long runs of supply piping.)

563	Water Storage	Capacity	Gal

Includes ground level and elevated storage tanks, foundation, coatings, lighting, valves, and piping required at tanks. Does not include piling or other special foundation work.

FUELING

571	Vehicle Fueling	Number of Outlets	Each

Includes concrete island, pumps, piping to 5-foot of tank and electrical work to panel.

572	Aircraft Fueling	Number of Hydrants	Each

Includes hydrants, filter, supports, loading arms, piping, and canopy. System includes truck loading stand for loading of fueling trucks.

573	Marine Fueling	Number of Outlets	Each

Includes hydrants, filters, supports, loading arms, piping, and canopy.

PUMPING

581	Fuel Pumping Station	Pump Capacity	GPM

Includes all work to install a complete fuel pumping station to the 5-foot line. Includes excavation and backfill, pit, pump, piping, valves, structure and electrical work at pump, and other miscellaneous work.

582	Water Pumping Station	Pump Capacity	GPM

Includes all work to install a complete water pumping station to the 5-foot line. Includes building, pit excavation and backfill, piping, pumps, electrical, and other miscellaneous work.

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583 Fire Booster Pump — Pump Capacity — GPM

Includes all work to install a complete water booster pumping station to the 5-foot line, if within a separate building. If pump is contained within the primary building, include in System 271. Includes pump, piping valves, electrical work at pump, and other miscellaneous work to install a complete fire booster pump.

584 Sewage Pump Station — Pump Capacity — GPM

Includes all work to install a complete sewage pumping station to the 5-foot line. Includes building, wet well, dry well, excavation and backfill, piping, pumps, electrical, and other miscellaneous work.

585 Sewage Lift Station — Pump Capacity — GPM

Includes all work to install a complete sewage lift station to the 5-foot line. Includes structure, wet well, excavation, backfill, piping, pumps, electrical, and other miscellaneous work.

TREATMENT FACILITY

591 Water Treatment — Capacity — MGD

(Million gal/day) Includes excavation, backfill, piping, pumps, equipment, controls, and other necessary accessories to install a complete water treatment system.

592 Domestic Sewage Treatment — Capacity — MGD

(Million gal/day) Includes excavation, backfill, piping, pumps, equipment, controls, and other necessary accessories to install a complete domestic sewage treatment system.

593 Industrial Waste Treatment — Capacity — MGD

(Million gal/day) Includes excavation, backfill, piping, pumps, equipment, controls, and other necessary accessories to install a complete industrial waste treatment system.

EQUIPMENT BUILDING

611 Mechanical Equipment Building — Gross Bldg. Area (MEB) — Sq. Ft.

Includes all work to construct a mechanical equipment building as part of a larger building project. Includes building construction, mechanical, and electrical work.

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612 Electrical Equipment Building — Gross Bldg. Area (EEB) — Sq. Ft.

Includes all work to construct an electrical equipment building as part of a larger building project. Includes building construction, mechanical, and electrical work.

SUPPORTING STRUCTURES MISCELLANEOUS

621 Security Gate — Opening Width — Lin. Ft.

Includes mechanically operated gate, foundation, supports, operating mechanism controls, and electrical work at gate.

622 Guard House — Gross Bldg. Area — Sq. Ft.

Includes structure, canopy, apron, curb, lighting, and mechanical and electrical work. Use for entrance gate structure as well as other guard house. This system is not intended for large structures that are a primary facility.

POLLUTION ABATEMENT STRUCTURES

641 Oil Water Separators — Capacity — Gal.

Includes excavation, backfill, concrete structure, miscellaneous metals, thru wall piping, and other work necessary to install a complete oil water separator.

642 Electrostatic Precipitator — Airflow — SCFM

Includes all work to install precipitator. Includes breeching (inlet and outlet), structural supports, control system, and ash hopper.

SHOP SUPPORT STRUCTURES

651 Vehicle Grease Rack — Number of Vehicle Racks — Each

Includes all work necessary to install an exterior grease rack. Includes excavation, backfill, concrete, miscellaneous metal, and other required items.

652 Vehicle Wash Platform — Number of Vehicle Bays — Each

Includes all work to install a complete vehicle wash platform. Includes excavation, backfill, paving, piping, accessories, and electrical work to edge of platform. Does not include distribution piping.

AIRCRAFT CLEANING FACILITY

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661 Aircraft Washing Facility — Paved Area — Sq. Yd.

Includes paving, drainage, water supply, washing equipment controls, and other necessary items to install a complete aircraft washing facility. Do not include access paving or long utility supply runs.

662 Aircraft Rinsing Facility — Paved Area — Sq. Yd.

Includes paving, drainage, water supply rinsing equipment, controls, and other necessary items to install a complete aircraft rinsing facility. Do not include access paving or long utility supply runs.

SITE PREPARATION & IMPROVEMENTS ROADS

711 Concrete Roads — Paved Area — Sq. Yd.

Includes concrete paving, curbs, curbs and gutters, subgrade preparation, fine grading, compaction, sub-base course, base course, edge forms, concrete reinforcement, finishing, expansion and control joints, rails and barriers, wheels stops, and pavement markings.

712 Flexible Roads — Paved Area — Sq. Yd.

Includes bituminous concrete paving, tack and seal coats, curbs, curbs and gutters, subgrade preparation, fine grading, compaction, sub-base course, base course, wearing course, finish course, rails and barriers, wheels stops, and pavement markings.

713 Overlay Road — Paved Area — Sq. Yd.

Includes flexible paving overlay of existing roads. Includes cleaning, sweeping, leveling courses, minor patching, tack coat, and pavement marking.

714 Surface Treatment - Road — Paved Area — Sq. Yd.

Includes bituminous surface treatment over existing pavement or base course. Includes cleaning, sweeping, minor patching, tack or prime coat, and pavement marking.

715 Slurry Seal Road — Paved Area — Sq. Yd.

Includes slurry seal over existing road. Includes cleaning, sweeping, minor patching, and pavement marking.

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PARKING

721 Concrete Parking — Paved Area — Sq. Yd.

Includes concrete paving, curbs, curbs and gutters, fine grading, compaction, sub-base course, base course, edge forms, concrete reinforcement, curbing, finishing, expansion and control joints, rails and barriers, wheel stops, and pavement markings.

722 Flexible Parking — Paved Area — Sq. Yd.

Includes bituminous concrete paving, tack and seal coats, curbs, curbs and gutters, fine grading, compaction, sub-base course, base course, wearing course, finish course, rails and barriers, wheel stops, and pavement markings.

723 Overlay Parking — Paved Area — Sq. Yd.

Includes flexible paving overlay of existing parking areas. Includes cleaning, sweeping, leveling course, minor patching, tack coat, and pavement marking.

724 Surface Treatment - Parking — Paved Area — Sq. Yd.

Includes bituminous surface treatment over existing pavement or base course. Includes cleaning, sweeping, leveling courses, minor patching, tack or prime coat, and pavement marking.

725 Slurry Seal Parking — Paved Area — Sq. Yd.

Includes slurry seal over existing parking area. Includes cleaning, sweeping, minor patching, and pavement marking.

WALKS

741 Concrete Walks — Paved Area — Sq. Yd.

Includes concrete paving, base course, edge forms, concrete reinforcement, finishing, edging, and exterior steps.

742 Bituminous Walks — Paved Area — Sq. Yd.

Includes bituminous paving, base course, edging, and exterior steps.

743 Special Walks — Paved Area — Sq. Yd.

Includes brick paving, limestone paving, flagstone paving, other paving materials, base course, and exterior steps.

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AIRCRAFT PAVING CONCRETE

751 Concrete Runways — Paved Area — Sq. Yd.

Includes concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course, edge forms, reinforcing, finishing, expansion and control joints, rails and barriers, wheel stops, and pavement markings when constructed for aircraft runways.

752 Concrete Taxiways — Paved Area — Sq. Yd.

Includes concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course, edge forms, reinforcing, finishing, expansion and control joints, rails and barriers, wheel stops, and pavement markings when constructed for aircraft taxiing.

753 Concrete Aprons — Paved Area — Sq. Yd.

Includes concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course, edge forms, reinforcing, finishing, expansion and control joints, rails and barriers, wheel stops, and pavement markings when constructed for aircraft parking.

754 Repairs Concrete Paving — Paved Area — Sq. Ft.

Includes removing and replacing individual slabs, partial slab removal and replacement, spall repairs, full depth repairs, and other miscellaneous patching.

755 Crack and Joint Repair — Length of Repair — Lin. Ft.

Includes cleaning and resealing joints and cracks repairs.

AIRCRAFT PAVING FLEXIBLE

761 Flexible Runways — Paved Area — Sq. Yd.

Includes bituminous concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course, wearing course, finish course, rails and barriers, wheel stops, and pavement markings when constructed for aircraft runways.

762 Flexible Taxiways — Paved Area — Sq. Yd.

Includes bituminous concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course, wearing course, finish course, rails and barriers, wheel stops, and pavement markings when constructed for aircraft taxiing.

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763 Flexible Aprons — Paved Area — Sq. Yd.

Includes bituminous concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course, wearing course, finish course, rails and barriers, wheel stops, and pavement markings when constructed for aircraft parking.

764 Repairs Aircraft Flexible Paving — Paved Area Patched — Sq. Yd.

Includes removing and replacing small area of flexible paving including sub-base course, base course, tack or prime coat, and surface material.

765 Aircraft Paving Overlay — Paved Area — Sq. Yd.

Includes flexible overlays of aircraft paving. Includes cleaning, sweeping, tack coat, milling, minor patching, leveling courses, and pavement marking.

766 Aircraft Paving Slurry Seal — Paved Area — Sq. Yd.

Includes slurry seal applied over existing aircraft paving. Includes cleaning, sweeping, minor patching, tack coat, and pavement markings.

INCIDENTAL PAVING WORK

771 Pavement Milling — Milled Area — Sq. Yd.

Includes bituminous and concrete milling with removal and/or disposal cost. Savings from recycled materials should be reflected in other systems.

EARTHWORK

811 Earthwork — Volume — Cu. Yd.

Includes site grading, site excavation, soil stabilization, soil treatment, and site clearing.

812 Borrow — Volume in Place — Cu. Yd.

Includes all work to provide off site borrow material compacted in place.

813 Topsoil, Seed, Sod — Area Graded — Sq. Yd.

Includes fine grading and leveling, fertilizer and limestone application, spreading and leveling topsoil, seeding, mulching, and sodding.

814 Landscaping — Area Planted — Sq. Yd.

Includes trees, shrubs, ground covers, and planters.

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815	Site Irrigation	Number Sprinkler Heads	Ea.

Includes underground piping, sprinkler heads, and controls to install a site irrigation system.

816	Surcharge	Fill in Surcharge	Cu. Yd.

Includes cost for obtaining, placing, monitoring, and removing to disposal area all required surcharge. If used for site fill does not include cost of compaction.

817	Earth Replacement	Earth Replaced (In place Measurement)	Cu. Yd.

Includes cost of removing and disposal of existing material, obtaining, placing, compacting, and proof rolling new material for improved bearing capacity. (See 846 for contaminated earth removal.)

818	Environmental Protection	Area Protected	Acres

Includes all cost for temporary environmental protection work required due to clearing, grubbing, earthwork, or other related work. (Measure to nearest tenth of acre.)

SITE IMPROVEMENTS

821	Site Improvements	Area Developed	Sq. Yd.

Includes retaining walls, terrace and perimeter walls, signs, site furnishings, fountains, pools and watercourse, flagpoles, and other miscellaneous related items.

SITE PREPARATION

831	Site Dewatering (Major)	Header Pipe	Lin. Ft.

Includes all costs for site dewatering where the dewatering is a major item and is required for several systems. Includes furnishing, installing, operating, and removal of well pointing system.

DEMOLITION

841	Remove Utilities	Length of Run	Lin. Ft.

Includes excavation, utility removal, capping, and backfill as required to remove existing utility lines.

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842 Remove Paving and Slabs — Area Removed — Sq. Yd.

Includes removal and disposal of material from both bituminous and portland cement concrete paved areas.

843 Remove Structures — Structure Volume — Cu. Ft.

Includes complete cost for removal of structure, foundation, and interior equipment. Removal of hazardous materials is not included. See Systems 844 and 845.

844 Remove/Dispose of Asbestos (Exterior) — Total Cost — Lump Sum

Includes costs for removal of asbestos from any exterior areas including buildings to be demolished. Included are costs for sealing off areas, asbestos removal, monitoring testing disposal, change areas, disposable suits, respirators, and other related costs. (See System 172 for interior asbestos removal.)

845 Remove/Dispose of P.C.B. — Total Cost — Lump Sum

Includes costs for removing and disposing of P.C.B. contaminated areas. Included are costs for sealing off area, removal of contaminated materials, monitoring, testing change area, disposal suits, respirators, disposal, and other related cost.

846 Remove/Dispose of Contaminated Earth — Earth Removed — Cu. Yd.

Includes cost of removing and disposing of existing industrial and hazardous materials.

STORM DRAINAGE

851 Storm Drainage Piping — Length of Run — Lin. Ft.

Includes utility service connections, excavation and backfill, sheeting and shoring, dewatering, pipe and fittings, manholes, catch basins, curb inlets, dry wells, ditches and culverts, retention ponds, and headwalls.

852 Box and Arch Culvert — Length of Culvert — Lin. Ft.

Includes excavation, backfill, forming, concrete, reinforcing steel accessories, steel arch, and bedding materials.

853 Drainage Facing Materials — Surface Area — Sq. Yds.

Includes stone riprap, fabrics, and other materials used as facing materials for outfalls and slope protection from storm water runoff.

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FENCING

861 Fencing (Perimeter) — Length of Fence — Lin. Ft.

Includes footings, posts, fencing materials, gates, and turnstiles for perimeter fencing. Includes station perimeter and individual facility. Does not include fencing for fields in Systems 872, 874, 875. See 621 for mechanically operated security gate.

862 Fencing (Alarm) — Length of Fence — Lin. Ft.

Includes footing, post fencing materials, gate and turnstiles when fence is part of a security alarm system.

RECREATION EQUIPMENT/FIELDS

871 Playground Equipment — Pieces of Equipment — Each

Includes swings, slides, and other items of recreation-type equipment for children's playgrounds.

872 Playing Field Equipment — Pieces of Equipment — Each

Includes costs for playing field equipment not shown elsewhere.

873 Tennis Courts — Number of Courts — Each

Includes paving, fencing, lighting, and nets for a complete tennis court.

874 Softball/Baseball Fields — Number of Fields — Each

Includes grading, topsoil and seeding drainage, infield, dugouts, fencing, and lighting as required for a complete softball or baseball field.

875 Football/Soccer Fields — Number of Fields — Each

Includes grading, topsoil and seeding, drainage, fencing, goals, benches, and lighting as required for a complete football or soccer field.

HEAVY CONSTRUCTION

SPECIAL BUILDING FOUNDATIONS

911 Piling - Land — Length of Piling — Lin. Ft.

Piling driven from land-based equipment used for building foundations or bulkhead tieback systems. Includes mobilization and demobilization, pile cutoff, and piling. Includes pile tests if not a separate bid item.

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(8) Minimally corrosive

912	Caissons	Number of Caissons	Each

Includes excavation, concrete, and reinforcing for foundation caissons.

913	Pressure Injected Footings	Number of Footings	Each

Includes drilling, concrete, and reinforcing a complete pressure injected footing system.

914	Vibroflotation	Number of Locations	Each

Includes specialized vibratory probe for penetration and extraction, and furnishing and placing sand backfill.

915	Chemical Soil Stabilization	Number of Locations	Each

Includes grouting equipment and specialized chemicals for injection into unstable sub-surface materials.

916	Dynamic Consolidation	Number of Locations	Each

Includes equipment, consolidation effort, sand, backfill, and testing.

917	Piling - Cast-in-Place	Length of Piling	Lin. Ft.

Includes drilling, casing concrete, and reinforcing for a complete cast-in-place piling system.

RAILROAD

921	Track Work - Railroad	Length of Track	Lin. Ft.

Includes subgrade preparation, ballast, ties, rail, switches, and other accessories for a complete track system.

MARINE STRUCTURES PIERS

931	Piling - Marine	Length of Piling	Lin. Ft.

Piling driven from floating equipment or land equipment used to support piers or similar waterfront structures. Includes mobilization and demobilization, pile cutoff and piling. Includes pile tests if not a separate bid item.

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932	Dolphins	Number of Dolphins	Each

Includes piling, cap and accessories for a complete dolphin.

933	Med Mooring	Total Cost	Lump Sum

Includes chain, anchors, buoys and other accessories for a complete Med mooring.

934	Pier Deck	Area of Deck	Sq. Ft.

Includes mobilization and demobilization, cast-in-place or precast concrete pile caps, trench sections, deck panels, manholes or transformer vaults, embedded conduit, drains, and expansion joints.

935	Fenders	Length of Fender	Lin. Ft.

Includes mobilization and demobilization, wales, chocks, fender piles, rubber fenders, pneumatic fenders, and associated hardware.

936	Pile Clusters	Number of Clusters	Each

Includes piles and accessories required for a complete pile cluster.

937	Marine Specialties	Total Cost	Lump Sum

Includes mobilization and demobilization, bollards, cleats, brows, camels, oil booms, and other marine specialities.

MARINE STRUCTURES SPECIAL

941	Bulkheads/Seawalls	Total Face Area	Sq. Ft.

Includes mobilization and demobilization, sheet piles, concrete cap, relieving platform, and tieback system.

942	Erosion Protection Marine	Area Covered	Sq. Yd.

Includes mobilization and demobilization, filled bags, concrete and asphalt surfaces, ground mats, sand fences, precast structural items (dolos, tetrapads, etc.), and filter cloth.

943	Riprap (Marine)	Area Covered	Sq. Yd.

Includes bed preparation, all layers of stone, and other required work to place riprap for slope protection from wave action.

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MARINE REPAIRS

951	Underdeck Repair	Total Cost	Lump Sum

Includes mobilization and demobilization, repair of cracks, spalls, beam, and cap repairs.

952	Pile Repairs Marine	Length Repaired	Lin. Ft.

Includes costs for removal and replacement of existing concrete including forms and reinforcing and repair for timber piling. All for marine structures.

DREDGING

961	Mobilization/Demobilization Dredging	Total Cost	Lump Sum

Includes outfitting and moving equipment to and from site, dike work, spillways, pipeline laying, and other work not directly associated with actual moving of dredged materials.

962	Hydraulic Dredging	Volume	Cu. Yd.

Includes costs for actual moving of material by hydraulic means from in-place location to disposal area.

963	Bucket Dredging	Volume	Cu. Yd.

Includes pumping of material from scow to disposal area.

TOWERS/ANTENNAS

971	Towers	Length	Lin. Ft.

Includes foundations, ground screens, tower lighting, and accessories.

972	Antennas	Total Cost	Lump Sum

Includes cost for installing or repairing antennas.

TUNNEL/PIPE JACKING

981	Tunneling	Total Cost	Lump Sum

Includes drilling equipment, excavation, disposal, liners, conduits, and other required work.

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982	Pipe Jacking	Length Jacked	Lin. Ft.

Includes jacking pits, jacking equipment, conduit, and other required items.

HEAVY CONSTRUCTION MISCELLANEOUS

991	Bridges	Deck Area	Sq. Ft.

Includes foundation, supports, caps, beams, deck, rails, lights, and accessories required for a complete bridge. Foundation piling not included.

992	Overpasses	Deck Area	Sq. Ft.

Includes foundation, supports, caps, beams, deck, rails, lights, and accessories required for a complete bridge. Foundation piling not included.

999	Supporting Facilities	Total Cost	Lump Sum

Includes all work outside the building 5-foot line. This system is not used for estimates.

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Section 8: SUPPLEMENTAL INFORMATION

8.1 Blank Forms and References. Refer to Appendix A.

8.1.1 Project Data Sheet. Refer to Appendix A.

8.1.2 DD Form 1391. Refer to Appendix A.

8.1.3 Primary Facility Unit Cost Development. Refer to Appendix A.

8.1.4 Supporting Facilities Cost Development. Refer to Appendix A.

8.1.5 Cost Model (Facility Study). Refer to Appendix A.

8.2 SI Conversion Factors. Refer to Appendix B.

8.3 Units of Measure and Their Symbols. Refer to Appendix C.

8.4 Acronyms and Abbreviations. Refer to Appendix D.

8.5 Air Conditioning Tonnage Guide. Refer to Appendix E.

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PART 2: COST CERTIFICATION GUIDANCE

Section 9: INTRODUCTION

9.1 Purpose and Background. This handbook is a guide to assist in the preparation of cost certification estimates for the appropriate fiscal year Military Construction, Navy (MCON) NAVFACENGCOM projects. NAVFACINST 11010.44, Shore Facilities Planning Manual requires a facility study to be prepared by the activity. The PDS and the facility study provide the initial conceptual cost. Many items described herein and required as part of the cost certification were also required as part of the facility. It should be recognized that this does not represent inadvertent duplication; rather, it is an effort to evaluate and refine the facility study as the project scope is defined.

9.2 Importance. The importance of an accurate planning cost estimate can not be over emphasized, as it could be the determining factor as to whether or not the project is included in the budget year program. Although requirements and justifications are important parts of project submittals, this publication deals only with scope and cost. Scope is included because an estimate is no better than the definition of scope that has been determined. An incomplete scope will result in an incomplete proposed project cost.

9.3 Contents. The cost certification components which are described in this part of MIL-HDBK-1010A are as follows:

a) DD Form 1391 (Figure 14)

b) Budget Estimate Summary Sheet and Backup Estimate (Figure 15)

c) Primary Facility Unit Cost Development Sheet (Figure 16)

d) Cost Model Questionnaire (Section 15)

9.4 Comments or Suggestions. You are encouraged to advise the Engineering Field Division (EFD) Cost Engineering Branch (Code 407) of errors, conflicts, or omissions detected while using this handbook, as well as to suggest improvements which will help produce higher quality cost certification estimates. Advice or comments should be forwarded to the EFD Code 407 to be considered for the MIL-HDBK-1010 update. Points of contact are as follows:

EFD	Code 407
Atlantic Division	(804) 444-9907
Chesapeake Division	(202) 433-3315
Northern Division	(215) 897-6344
Pacific Division	(808) 471-8436
Southern Division	(803) 743-0463
Western Division	(415) 742-7760

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Section 10: SCOPE CONSIDERATIONS

10.1 Introductory Paragraph. The Primary Facility Unit Cost Development Sheet must include separate entries for specific features of additional cost. Certain features and their cost derivation are discussed below. Other specific features may not be discussed in this section, but should be a separate item to emphasize significant cost.

10.2 Operation and Maintenance Support Information (OMSI). The OMSI product is provided in the form of manuals prepared by the designer during construction to give the activity clear comprehensive data for use in operating and maintaining their new facilities and to ensure economical, safe, life cycle operation and maintenance. The principal OMSI elements requiring consideration are as follows:

a) Design Concept Data and Basis of Design

b) Basic Data needed for preparation of Facilities Support Contracts and Performance Work Statements (FSC/PWS)

c) Operating Procedures for Startup and Shutdown; Emergency and Safety Instructions

d) Operation and Maintenance Technical Data and As-built Product Documentation

e) Maintenance Requirements, Test Equipment, and Special Tools

f) Preventive Maintenance Requirements and Schedules

g) Troubleshooting Guide, Diagnostic Techniques, Repair Procedures

h) Supply Support (repair and spare parts, fuels, lubricants, etc.)

i) Extended Special Warranty Information

j) Environmental Considerations

k) Training Plan, Staffing Requirements, Personnel Qualifications

10.2.1 Types of OMSI. OMSI is valuable and should be included on most projects for which manufacturer's product data is to be submitted by the construction contractor. OMSI is usually needed on projects having complex built-in systems for which submittal of operating or operation and maintenance data is called for in the construction specifications. The degree of OMSI will vary with the kind of facility and complexity of systems in the facility. For identification, projects requiring OMSI can be classified as Type A, Type B, Type C, or Type D.

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a) Type A: Operation and Maintenance Manual plus User Manual

b) Type B: Operating Manual plus User Manual

c) Type C: User Manual only

d) Type D: No OMSI Manuals (As-built Drawings only)*
* As-built Drawing Listing is included in the User Manual for Types A, B, and C.

Examples of facilities which usually require a Type A are listed in Table 8, and those requiring a Type B, in Table 9. Rather simple facilities requiring only a User Manual are classified as Type C, samples of which are shown in Table 10. Projects listed in Table 11 such as paving and dredging would not require OMSI manuals and should be classified as Type D. For Type D projects, only As-built Drawings will be required in the same manner as presently handled.

The User Manual compiles and indexes as-built product and operation and maintenance data into a consolidated manual and also provides useful information such as special warranties and small scale floor plans showing utility shutoff locations. Facilities having complex systems which require special operating information should be classified as Type B and be provided an Operating Manual in addition to the User Manual. NOTE: An Operating Manual will not provide information on maintenance, troubleshooting, and repair of selected systems, therefore its use should be limited. Complex facilities having systems which require special operation, maintenance, and repair information should be classified as Type A and be provided an Operation and Maintenance Manual in addition to the User Manual. Depending on the complexity of a particular system, even a simple facility such as a Child Care Center could need a Type A OMSI to provide an OMSI Operation and Maintenance Manual for that system; e.g., variable air volume (VAV) or thermal storage system. A project such as a hospital may include 15 or more operational systems in the manual.

10.2.2 OMSI Estimate. In estimating the cost of OMSI for a specific project, use the best available information. If a good basis to make the estimate does not exist, Table 12 should be used for a preliminary estimate based upon a percentage of the total primary facility cost. If OMSI is required for the supporting facilities (such as utilities), add these costs to the primary facility costs before calculating the OMSI estimate. The percentage selected (within the ranges shown) should be based on the relative complexity of the facility and the number of systems to be covered by the Operation and Maintenance Manual or the Operating Manual. If OMSI is greater than $50,000, it shall be a separate line item on DD Form 1391 (Block 9), the Estimate Summary Sheet, and Primary Facility Unit Cost Development Sheet. If OMSI is less than $50,000, it shall be part of the building cost and shall be shown on the Primary Facility Unit Cost Development Sheet. In either case, OMSI should be included in Block 10 of the DD Form 1391 under Description of Proposed Construction.

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Table 8
Examples of Facilities Usually Requiring Type A OMSI
(Operation and Maintenance Manual Plus User Manual)

FACILITY TYPE	BASIC CATEGORY CODE
PETROLEUM, OIL & LUBRICANT (POL) FACILITIES	120
COMPUTER OPERATIONS CENTER (LARGE/COMPLEX)	143
TRAINING FACILITIES WITH COMPLEX SYSTEMS	171
AIRCRAFT MAINTENANCE SHOPS	211
MISSILE ASSEMBLY & MAINTENANCE SHOPS	212
SHIP MAINTENANCE SHOPS	213
LABORATORIES	310
CENTRAL REFRIGERATION PLANTS	431 & 826
MEDICAL FACILITIES	500
POWER PLANTS (750 KW AND LARGER)	811
HEATING PLANTS (950 MBTUH AND LARGER)	821
INDUSTRIAL WASTE & SEWAGE TREATMENT PLANTS	831

Table 9
Examples of Facilities Usually Requiring Type B OMSI*
(Operating Manual Plus User Manual)

FACILITY TYPE	BASIC CATEGORY CODE
OPERATION & TRAINING FACILITIES	100
LIQUID FUEL STORAGE - BULK	411
MISCELLANEOUS UTILITIES	890

* May require Type A OMSI if maintenance and repair information is needed.

Table 10
Examples of Facilities Usually Requiring Type C OMSI
(User Manual Only)

FACILITY TYPE	BASIC CATEGORY CODE
WAREHOUSE - GENERAL	441
ADMINISTRATIVE FACILITIES	600
BEQ/BOQ	721 & 724
COMMUNITY, MORALE, RECREATION	740

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Table 11
Examples of Facilities Usually Requiring Type D OMSI
(As-built Drawings Only)

FACILITY TYPE	BASIC CATEGORY CODE
AIRFIELD PAVEMENTS/TAXIWAYS/APRONS	110-116
DREDGING	165
OPEN STORAGE	451
ROADS AND STREETS/PAVEMENTS	850/851

Table 12
Examples of OMSI Cost for Preliminary Estimates

	ESTIMATED OMSI COST BY TYPE (1)			
PRIMARY FACILITIES COST (6)	TYPE A (2)	TYPE B (3)	TYPE C (4)	TYPE D (5)
BELOW $5,000,000	1.20-2.35%	0.75-2.00%	0.55-1.60%	--
$5,000,000 to 20,000,000	0.95-1.85%	0.65-1.35%	0.35-0.75%	--
$20,000,000 to $50,000,000	0.70-1.35%	0.50-0.90%	0.30-0.60%	--
ABOVE $50,000,000	0.60-1.10%	0.45-0.75%	0.25-0.55%	--

NOTES:

(1) Table 12 shows estimated cost of OMSI as a percentage range of the total of primary facility plus applicable supporting facility costs. Use higher range for more complex projects, for example:
Industrial Waste Treatment Plant with $18 million facility cost:
$18,000,000 x 0.0185 = $333,000 estimated cost for Type A OMSI (Operation and Maintenance Manual plus User Manual).
(2) The minimum estimate for a Type A OMSI should be approximately $22,000.
(3) The minimum estimate for a Type B OMSI should be approximately $18,000.
(4) The minimum estimate for a Type C OMSI should be approximately $8,000.
(5) No costs over those fees normally encountered for As-built Drawing preparation.
(6) Include applicable cost for supporting facilities which require OMSI consideration.

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10.3 Intrusion Detection Systems (IDS). Interior "non-built-in" portions of IDS are no longer MCON funded per NAVCOMPT September 1987 decision. Effective with FY 90 Program, NAVCOMPT requires OPN funding for interior "non-built-in" IDS. The Naval Investigative Service Command (COMNISCOM) has the role of managing the design, procurement, and installation of IDS. NAVFACENGCOM message 181745Z June 1990 provides guidance on the IDS issue.

Facility support, such as utilities, conduit runs, equipment space, building, and conventional fencing required to support IDS associated with a MCON project are properly chargeable to MCON funds and should be included in the MCON project scope. Refer to NAVFAC DM-13.02, Commercial Intrusion Detection Systems (IDS) for help in determining facility IDS requirements.

10.3.1 Non-MCON Fundable IDS. IDS installation can be divided into five general categories:

a) Category A: Navy MCON funded projects;

b) Category B: Nuclear ordnance storage sites;

c) Category C: Conventional arms, ammunition, and explosives storage sites (AA&E);

d) Category D: Marine Corps funded projects; and

e) Category E: All other special projects and non-MCON funded projects.

Planning, design, and installation of IDS Categories A, B, and C are the responsibility of COMNISCOM. The A&E shall coordinate IDS power requirements and conduit runs for IDS Categories A, B, and C with the appropriate contacts within COMNISCOM.

For Category D projects consult area project manager for guidance to determine if IDS will be A&E designed.

Category E projects are planned and designed by the A&E utilizing commercial systems for installation by the construction contractor. Identify areas requiring IDS and types of systems and sensors proposed.

Claimant, user, and COMNISCOM, depending on current policy, will budget for and fund non-MCON IDS requirements. These commands must be notified of a project's non-MCON IDS requirements and your need for data on facility support requirements. This should generally result in an IDS Engineering Plan (IDSEP).

10.3.2 Technical Support. Technical support for IDS design is available on a reimbursable basis from Naval Electronic Systems Engineering Center (NAVELEXCEN) Charleston, SC. This support includes guidance on scope and

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technical review of compiled IDS designs at appropriate steps in project development and design. NAVFACENGCOM will attempt to identify MCON projects requiring major IDS and request NAVELEXCEN Charleston, SC prepare an IDSEP for each.

10.4 Land Acquisition. Obtain instructions from major claimant and EFD Code 024 (Real Estate).

10.5 Hyperbaric Requirements. CHESNAVFACENGCOM (FPO-1) is responsible for executing the design and procurement phases of shore based hyperbaric facilities. When the project scope includes provisions for hyperbaric facilities, you must contact CHESNAVFACENGCOM (FPO-1) to make appropriate arrangements for the design, procurement, and installation of the required hyperbaric facilities and the associated cost estimate.

10.6 Shielding Requirements

10.6.1 Radio Frequency Interference (RFI)/TEMPEST/HEMP (High Altitude Electromagnetic Pulse)/EMP (Electromagnetic Pulse) Shielding. Naval Electronic Systems Security Engineering Center (NAVELEXSECCEN), Washington, DC has been very effective in limiting requirements for shielding in MCON facilities. NAVFACENGCOM message 151455Z June 1984 provides the guidance on determining the extent of RF shielding required on projects involving installation of electronic equipment. For affected projects, ensure NAVELEXSECCEN Washington, DC has formally reviewed and commented on TEMPEST shielding requirements. If HEMP/EMP shielding is identified as a user requirement, consult with NAVFAC Code 04A1 prior to A&E scoping.

10.6.2 Coordination Required to Meet Objectives. Satisfying shielding objectives is often very difficult; it requires close coordination between the customer, the designer, and the construction contractor to establish this scope requirement. Good information transfer from designers to ROICC personnel is essential. The use of expert consultants during design and construction, and the development of both interim and final testing plans and criteria, should be evaluated and incorporated when appropriate. Design Policy Letter DPL-00-0002 dated 3 January 1989 established CHESNAVFACENGCOM (FPO-2) as the center of expertise for TEMPEST, HEMP, and RFI shielding. Consult MIL-HDBK-1195, Radio Frequency Shielded Enclosures for information on these subjects.

10.7 Uninterruptible Power System (UPS). If the UPS is required only to support personal property (electronic equipment) installations, it may not be procured and installed with MCON funds. Instead, its procurement and installation cost must be separately budgeted by the major claimant or user as an integral part of the associated electronic equipment package. Please ensure that the major claimant or user is aware of this requirement. Government furnished UPS are procured through CHESNAVFACENGCOM (FPO-2), and UPS requirements should be coordinated through them. However, facility items required to support the UPS such as power panels, flooring, and battery rooms are properly chargeable to MCON and should be included in the project scope.

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10.8 Communication Requirements

10.8.1 Telephone and Communications Installations. It is the responsibility of the designer to determine the design requirements, and design the facility's telephone system. Based on NAVCOMPT guidance, the general policy is to include only the cost of the following as required to support future installation of telephone and other communication systems wiring internal to a MCON facility:

a) Exterior: All supporting structures (e.g., telephone poles, underground conduits, underground ductbanks, manholes, and handholes).

b) Interior: Empty conduit with pull wire, outlet boxes, conduit bushings, backboards, and cable trays to support future installation of telephone wiring.

10.8.2 Funding Responsibilities. Normally, the wiring, instruments, and equipment are to be funded by other than MCON regardless of whether the system is leased or Government owned. The responsibility of Telephone Management Division East (TMD East) is to assist the user in identifying non-project funds to pay for the following items to be installed as a separate contract (not part of the MCON project):

a) Exterior: All cabling and telephone switching equipment.

b) Interior: All wiring, instruments, and equipment.

10.8.3 Funding Exceptions. In certain situations, such as medical facilities and Navy owned systems in overseas areas, entire systems may be MCON funded where practical and justifiable. These exceptions to the general policy should be clearly identified and justified.

10.9 Built-in Retrieval Equipment Racks (Warehouses). Policy on funding specialized equipment in automated warehouses is as follows:

10.9.1 MCON Funded. If the project includes a guide wire or guide rail built into the facility, include the cost of the guide wire or rail system, vehicles which ride on the system (system captive), and the storage racks serviced by the system in the MCON project.

10.9.2 OPN Funded. If the retrieval system does not include a built-in guide wire or rail system, the retrieval vehicles and racks are to be identified as collateral equipment.

10.9.3 Warehouse Design. The design of all warehouses must be coordinated with Naval Supply Systems Command (NAVSUPSYSCOM) as the Navy's expert in warehouses. NAVSUPSYSCOM must formally approve the design concept prior to your achieving a 35 percent design. To help with the design concept, NAVSUPSYSCOM has developed a computer program to determine the least cost alternative for various stacking heights and retrieval systems. As the foot

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print and stacking height affect the project cost, and some retrieval systems are MCON funded, these decisions must be made prior to starting design. Coordinate design with NAVSUPSYSCOM. The NAVFACENGCOM point of contact is Code 04B2, Autovon 221-0044, Commercial (703) 325-0044.

10.10 Asbestos Related Work. A&E firms are finding it difficult, if not impossible, to obtain professional liability insurance for work involving asbestos, hazardous waste, and pollution abatement. The Federal Acquisition Regulation (FAR) does not require A&E firms to carry professional liability insurance. NAVFACENGCOM does not have the authority to provide blanket indemnification for work involving hazardous waste. NAVFACENGCOM policy is that when asbestos or other hazardous waste will be encountered, or when there is a good probability of asbestos or other hazardous waste, note this fact in the Commerce Business Daily (CBD) synopsis for any affected contract. We should NOT include this warning in every CBD synopsis as a matter of routine; it should only be used where the hazards cited are known or probable. During interviews the willingness of each firm to undertake any required asbestos, hazardous waste, or pollution abatement work is to be clearly established and made a matter of record. Please continue to keep NAVFACENGCOM informed of problems in this area. The NAVFACENGCOM point of contact is Code 04BD, Autovon 221-0043, Commercial (703) 325-0043.

10.11 Computation of Gross Area. For all gross and net area calculations use Chapter 4 of MIL-HDBK-1190, Facility Planning and Design Guide. Gross area includes the total area of all floors, including mezzanines, basements, and penthouses as determined by the effective outside dimensions of the building. One-half the area shall be included in the gross area for balconies and porches, covered raised loading platforms, covered loading facilities (either depressed, ground level, or raised), covered but not enclosed passageways or walks, covered and uncovered but open stairs, and covered ramps. Excluded from the gross area are crawl spaces, exterior uncovered loading facilities (either depressed, ground level, or raised), open courtyards, roof overhangs and soffits, uncovered ramps and stoops, utility tunnels, and raceways. Separate central energy plants or utility buildings serving large complexes are not included in the gross module area shown in Table 12A. Separate support buildings should be programmed as a line item on DD Form 1391.

10.11.1 BEQ Area Requirements. For Bachelor Enlisted Quarters, refer to criteria shown in Table 12A.

10.11.2 BOQ Area Requirements. For Bachelor Officer Quarters, refer to MIL-HDBK-1190. The following requirements shall be used to program BOQ projects:

a) Grades 02 and below: Quarters shall provide a private suite of approximately 330 square feet net area, consisting of combination of living room and bedroom, bathroom, and pullman-type kitchen. The gross floor area shall approximate 475 square feet per officer.

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b) Grades 03 and above: Quarters shall provide a private suite of approximately 460 square feet consisting of living room, bedroom, bathroom, and kitchen. The gross floor area shall approximate 650 square feet per officer.

10.12 International Balance of Payments (IBOP). The IBOP Summary is only included when preparing the cost estimate for projects in specific foreign areas as shown in par. 4.2. Information on the exempt areas is available from the EFD Project Management Office. The IBOP Summary shall be developed on 8-1/2- by 11-inch plain bond paper as shown in Figure 13.

10.12.1 IBOP Implications. Current directives require action by cognizant addressees to maintain Foreign Exchange Costs (FEC) at a minimum. IBOP implications will be prepared to show the effect of construction under both normal and planned IBOP procedures. This effort is to permit analysis and response by NAVFACENGCOM to inquiries regarding items contributing to FEC.

10.12.2 Normal Procedures. Those construction procedures which would normally be used at a particular location if no effort were made to suppress FEC.

Table 12A
Bachelor Enlisted Quarters Area Table

MODULES (2 Rooms and Bath)	FLOORS (1)	AREA SF (2)	ELEVATORS (3)	SPRINKLER SYSTEM
60	3	800	---	---
72	3	785	---	---
96	4	800	F	(4)
180	6	800	F & P	YES

NOTES:

(1) Modules per floor must be divisible by 2, with no exceptions.
(2) Total gross module area including its share of the non-module areas.
(3) F (Freight) & P (Passenger).
(4) Provide a fire stand pipe system if sprinkler system is not required.
(5) The data in this table satisfies the net area requirements of MIL-HDBK-1190. Gross area for combinations not shown must be determined or verified by the EFD, Code 401.

GRADE	ACCOMMODATIONS
E-1 thru E-4	4 Men per Module
E-5 thru E-6	2 Men per Module
E-7 thru E-9	1 Man per Module

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INTERNATIONAL BALANCE OF PAYMENTS (IBOP) IMPLICATIONS

1. Activity (Name and Location) (CAPITAL LETTERS)			Date
2. Project Title (CAPITAL LETTERS)			3. P-No.

	NORMAL PROCEDURES		PLANNED PROCEDURE	
	Construction	*Construction Total Material	Construction	*Construction Total Material
BUDGET COST	1,030,000	-	1,100,000	-
LESS RETURNS TO THE US				
a) US End Products	-	-	100,000	-
b) US Services	5,000	-	5,000	-
c) US Transportation	-	-	20,000	-
TOTAL RETURNS	5,000	0	125,000	0
FOREIGN EXCHANGE COST (FEC)	1,075,000		975,000	

This example makes the following assumptions:

1. Normal procedure; all foreign materials and contractors, only return to US is 6.5% SIOH.
2. Planned procedure; purchase of a pre-engineered metal building in the US at the same price as in the foreign country, shipping by a US line, 6.5% SIOH is returned to the US.

Returns to US include escalation and contingencies. Overhead and profit depends upon location of contractor.

SUMMARY OF IBOP DATA

*CWE is the same as total request rounded for selected procedures.

	NORMAL PROCEDURE	PLANNED PROCEDURE	PLANNED PROCEDURE EFFECT
CURRENT WORKING ESTIMATE	1,080	1,100	+ 20
FOREIGN EXCHANGE COST (FEC)	1,075	975	- 100

* Materials, supplies and equipment (including government furnished material procured abroad) acquired by the Navy from construction funds and provided to contractors for incorporation into foreign construction projects.

Figure 13
Sample IBOP Summary

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10.12.3 Planned Procedures. Those procedures utilized at a particular location to minimize FEC with specific support and emphasis on:

a) Adhering to pertinent executive orders, U.S. laws, status of forces agreements, and treaties which affect our contractual freedom abroad.

b) Maximum feasible use of U.S. contractors as allowed by par. 10.12.3.a).

c) Maximum feasible use of U.S. materials, end products, and equipment as allowed by par. 10.12.3.a).

d) Maximum use of U.S. flag, ocean, and air carriers.

e) Utilization of competent military labor where available.

f) Designs incorporating portable van-type installations and structures prefabricated in the United States.

10.12.4 Basis for Choice of Construction Procedure. Generally, planned procedures will be recommended where for every dollar increase in Current Working Estimate (CWE) from NORMAL to PLANNED there is a resulting decrease in FEC of at least $2, and normal procedures will be recommended otherwise. An explanation for the deviation will be provided if the EFD recommends construction at the normal level and cost estimates justify programming at the planned level.

Exceptions to the above criteria are allowed where country-to-country agreements specifically establish construction procedures, or project urgency is an overwhelming factor.

10.12.5 Presentation of IBOP Data on DD Form 1391. The DD Form 1391 will be prepared to show costs at the recommended level. IBOP data will be shown on the DD Form 1391 with entries asterisked to show the corresponding CWE and FEC amounts for the recommended procedure. Such IBOP entries will be shown as:

IBOP DATA:	NORMAL	PLANNED	NORMAL	PLANNED
CWE	$550,000	$590,000*	FEC $330,000	$220,000*

*Recommended construction method

10.12.6 Requirements Regarding Returns to the United States. Other information requirements regarding returns to the United States are to be shown at the end of the cost estimates. This additional data is required in order that NAVFACENGCOM may provide estimates of IBOP expenditures as required by higher authority.

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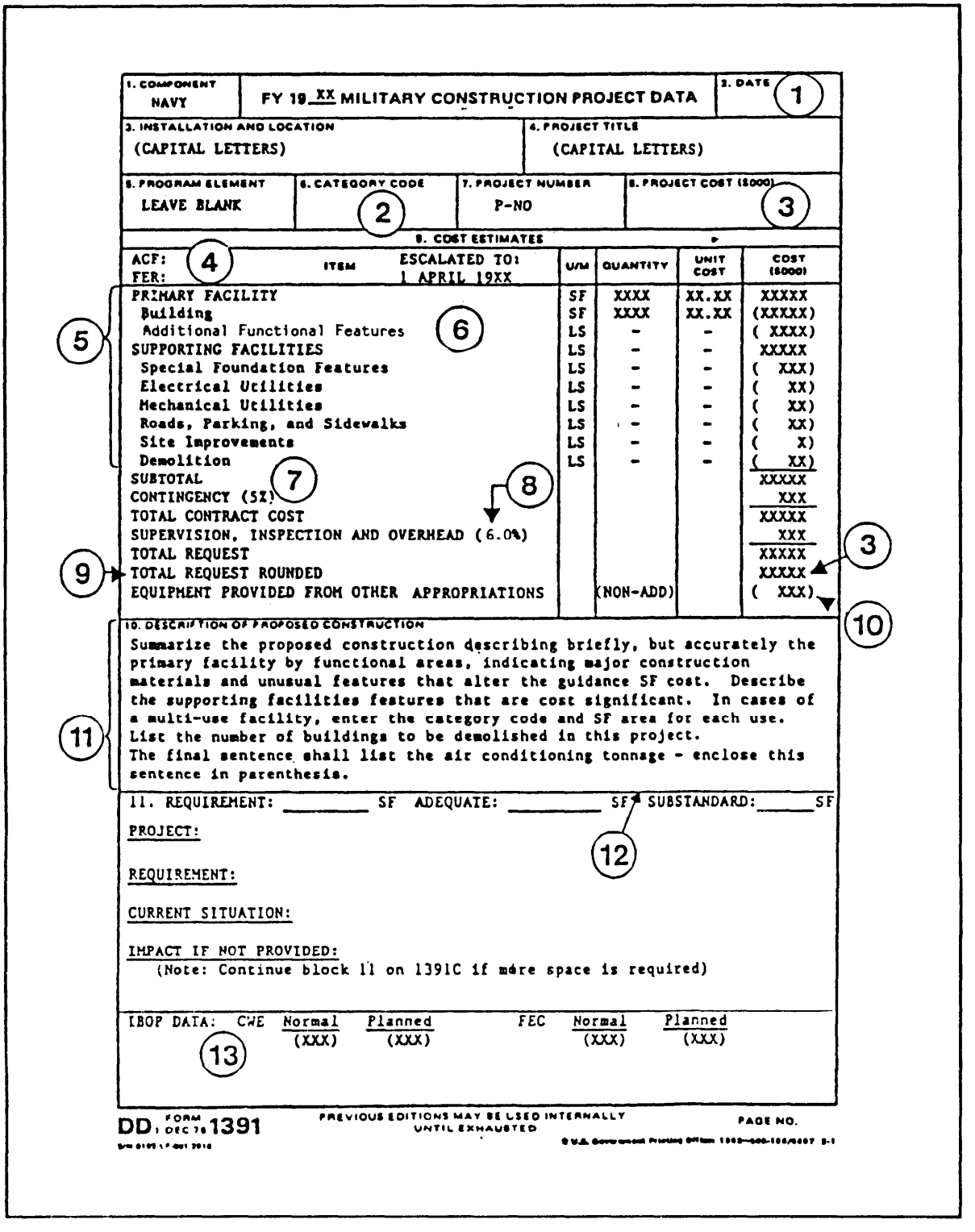
10.13 Instructions for DD Form 1391. The following instructions correspond to the circled numbers shown in Figure 14.

1. Date shall be the day submitted. Dates must be revised if a resubmittal is made.
2. Enter the five-digit category code number from NAVFAC P-72, Department of the Navy Facility Category Codes to identify the primary facility. In cases of multi-use facilities, enter the category code which represents the largest cost of the project.
3. Block 8 - "Project Cost" shall be the same as the "TOTAL REQUEST ROUNDED" in block 9. This cost shall be escalated to 1 April of the appropriation year, except assigned foreign areas, par. 12.2.
4. The Area Cost Factors for stateside areas are shown in par. 12.1. The Area Cost Factors and the corresponding Foreign Exchange Rates for various overseas locations are shown in par. 12.2.
5. The line items shown are generally suitable for most projects. They should be tailored to suit the individual project - discuss this with the EFD Cost Engineering Branch. The Cost figures in parentheses should add to equal "PRIMARY FACILITY" cost or "SUPPORTING FACILITIES" cost.
6. Include a list of the major use areas for multi-use facilities under the primary facility. Also, include a line item entitled, "Additional Functional Features." Show the unit cost as applicable to the total size of the primary facility as shown in Figure 14.
7. Contingency will be 5 percent for all projects.
8. Use 6.0 percent SIOH for all stateside projects and 6.5 percent SIOH for all overseas projects unless directed otherwise.
9. Rounded cost estimate:

less than	$1,000,000	- nearest 10,000
1,000,000 -	5,000,000	- nearest 50,000
Over	5,000,000	- nearest 100,000

10. Must agree with total for all other categories of equipment on collateral equipment list other than built-in equipment as defined in par. 15.2.1. If not available, type "(LIST NOT AVAILABLE)" in place of "(NON-ADD)." Built-in equipment is MCON funded and normal built-in equipment cost is included in guidance unit cost. Enclose cost figures in parentheses.
11. If the project includes more than one structure or location, address each in a separate paragraph in block 10 and separately in block 9.
12. Draw a line between blocks 10 and 11.
13. IBOP entry is required for assigned foreign areas (par. 12.2).

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1. COMPONENT: NAVY | FY 19XX MILITARY CONSTRUCTION PROJECT DATA | 2. DATE (1)

3. INSTALLATION AND LOCATION: (CAPITAL LETTERS) | 4. PROJECT TITLE: (CAPITAL LETTERS)

5. PROGRAM ELEMENT: LEAVE BLANK | 6. CATEGORY CODE (2) | 7. PROJECT NUMBER: P-NO | 8. PROJECT COST ($000) (3)

9. COST ESTIMATES

ACF: / FER: (4) ITEM	ESCALATED TO: 1 APRIL 19XX	U/M	QUANTITY	UNIT COST	COST ($000)
(5)	PRIMARY FACILITY	SF	XXXX	XX.XX	XXXXX
	Building	SF	XXXX	XX.XX	(XXXXX)
	Additional Functional Features (6)	LS	-	-	(XXXX)
	SUPPORTING FACILITIES	LS	-	-	XXXXX
	Special Foundation Features	LS	-	-	(XXX)
	Electrical Utilities	LS	-	-	(XX)
	Mechanical Utilities	LS	-	-	(XX)
	Roads, Parking, and Sidewalks	LS	-	-	(XX)
	Site Improvements	LS	-	-	(X)
	Demolition	LS	-	-	(XX)
	SUBTOTAL (7)				XXXXX
	CONTINGENCY (5%) (8)				XXX
	TOTAL CONTRACT COST				XXXXX
	SUPERVISION, INSPECTION AND OVERHEAD (6.0%)				XXX
	TOTAL REQUEST				XXXXX
(9)	TOTAL REQUEST ROUNDED				XXXXX (3)
	EQUIPMENT PROVIDED FROM OTHER APPROPRIATIONS		(NON-ADD)		(XXX)

(10)

10. DESCRIPTION OF PROPOSED CONSTRUCTION (11)

Summarize the proposed construction describing briefly, but accurately the primary facility by functional areas, indicating major construction materials and unusual features that alter the guidance SF cost. Describe the supporting facilities features that are cost significant. In cases of a multi-use facility, enter the category code and SF area for each use. List the number of buildings to be demolished in this project. The final sentence shall list the air conditioning tonnage - enclose this sentence in parenthesis.

11. REQUIREMENT: _____ SF ADEQUATE: _____ SF SUBSTANDARD: _____ SF (12)

PROJECT:

REQUIREMENT:

CURRENT SITUATION:

IMPACT IF NOT PROVIDED:

(Note: Continue block 11 on 1391C if more space is required)

IBOP DATA: (13)	CWE	Normal	Planned	FEC	Normal	Planned
		(XXX)	(XXX)		(XXX)	(XXX)

DD FORM 1 DEC 76 1391 — PREVIOUS EDITIONS MAY BE USED INTERNALLY UNTIL EXHAUSTED — PAGE NO.

Figure 14
Sample DD Form 1391 for Cost Certification Estimate

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10.14 Instructions for Estimate Summary Sheet. The following instructions correspond to the circled numbers shown in Figure 15.

1. Show project title in conformance with NAVFAC P-72, Department of the Navy Facility Category Codes. If summary sheet is for one building (or any portion of a project) so indicate. In that case, provide one summary sheet for totals transferred from sub-summary sheets.
2. Enter the area cost factor and the official foreign exchange rate.
3. Escalate to 1 April of the fiscal year of planned appropriation. See overseas area cost factors for requirement of escalation to one-third point of construction. NAVFAC escalation chart will be used for all projects.
4. Enter unit of measure.
5. Quantity - Quantities applicable to the primary and supporting facilities. Use the CES Standard System List. For primary facilities, use gross building area.
6. Unit cost for all line items shown based upon the building gross area and systems quantity.
7. Round cost to nearest thousand.
8. Contingency will be 5 percent for all projects.
9. Enter 6.0 percent Supervision, Inspection and Overhead (SIOH) for stateside projects and 6.5 percent for foreign projects.
10. Rounding - See DD Form 1391 Instructions. This figure must agree with block 8 on DD Form 1391.

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be attempted without thoroughly b) another substitute should not

BUDGET ESTIMATE SUMMARY SHEET

P.NO.: P-____ FY 19__

TITLE: (1) _____ DATE OF ESTIMATE: _____

LOCATION: _____ DESIGN STATUS (%): _____

PREPARED BY: _____ DATE ESCALATED TO: (3) _____

A.C.F. /FER: (2) _____

DESCRIPTION	UM	QUANTITY	UNIT COST	ESCALATED COST ROUNDED	COST TRANSFRD TO 1391
PRIMARY FACILITIES					
Building	(4)	(5)	(6)	(7)	
Additional Functional Features					
SUPPORTING FACILITIES					
Special Foundation Features					
Electrical Utilities					
Mechanical Utilities					
Roads, Parking, Sidewalks					
Site Improvements					
Demolition					

Total Contract Cost W/O Contingency _____ (8)

Contingency (5%) _____

TOTAL CONTRACT COST _____ (9)

SIOH (6.0%) _____

TOTAL BUDGET COST _____ (10)

ROUNDED _____

Figure 15
Building Budget Estimate Summary Sheet

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10.15 Instructions for Primary Facility Unit Cost Development Sheet. See Figure 16.

a) Guidance Cost Priorities:

(1) NAVFAC Primary Facility Guidance Costs, Table 13.

(2) Additional Primary Facility Guidance Costs, Table 13A.

(3) Guidance cost obtained from historical data of comparable projects.

(4) Guidance cost obtained from historical data of comparable projects (cost must be obtained from or verified by EFD Code 407).

b) The guidance cost must be adjusted to allow for differences in size (see Figure 17), area cost factors, and escalation (see Figure 18). The area cost factors are listed in pars. 12.1 and 12.2.

c) The guidance cost represents a typical project, including all normal built-in equipment. Identification of built-in equipment may be obtained from the CES Standard Systems List shown in par. 15.2.1. The guidance cost must be adjusted for additional functional features which are considered to be additional cost. This adjustment is absolutely required to justify cost above guidance. Backup for the additional functional features must be attached. Examples of additional functional features are OMSI, hyperbaric requirements, TEMPEST/HEMP/EMP shielding, UPS support facilities, communications support facilities, retrieval equipment racks for warehouses, very high walls, emergency power, raised floor, and asbestos removal.

d) Gross area calculations (par. 10.11) require certain areas to be taken at zero or one-half the actual area. These excluded areas must be included in the project cost. This can be done by showing the actual cost as a functional feature on the Primary Facility Unit Cost Development Sheet.

e) The adjusted unit cost (u/c) for each functional feature is obtained by dividing the total cost of each feature by the total size of the proposed facility. (Include all multi-use areas.)

f) The percent change is obtained by dividing the total cost of each feature by the total adjusted historical cost of the facility. (Include all multi-use areas.)

g) The size adjustment factor is related to Figure 17.

h) The area cost factors are given in Section 12.

i) The escalation factors are given in Figure 18.

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PRIMARY FACILITY UNIT COST DEVELOPMENT

TITLE: _____ P#: _____

LOCATION: _____ DATE: _____

PREPARED BY: _____ PH#: _____

	CAT CODE	SIZE	ACF	DATE (mo/yr)
PROPOSED FACILITY				
GUIDANCE FACILITY				

HISTORICAL GUIDANCE SOURCE: _____ *

HISTORICAL GUIDANCE UNIT COST: _____ /SF

HISTORICAL UNIT COST X SIZE FACTOR X AREA COST FACTOR X ESCALATION FACTOR = ADJUSTED UNIT COST

* Show date of source as shown in Figure 11.

ADDITIONAL FUNCTIONAL FEATURES

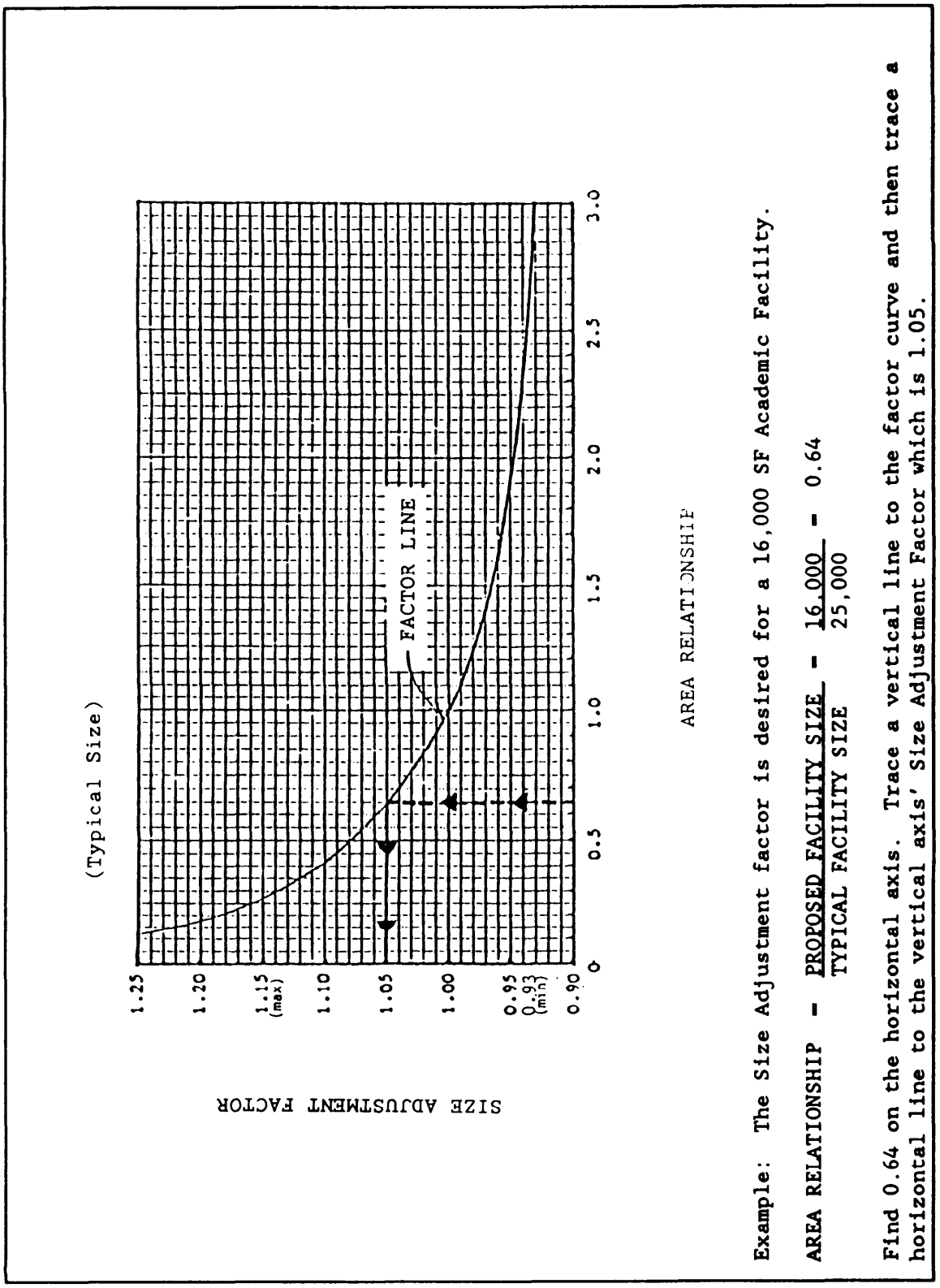
FUNCTIONAL FEATURES	QTY	UNIT COST	TOTAL COST	ADJUSTED UC	%CHANGE

FUNCTIONAL FEATURES TOTAL: _____ _____

TOTAL PRIMARY FACILITY UNIT COST: _____
includes built-in equipment

Figure 16
Primary Facility Unit Cost Development

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Example: The Size Adjustment factor is desired for a 16,000 SF Academic Facility.

$$\text{AREA RELATIONSHIP} = \frac{\text{PROPOSED FACILITY SIZE}}{\text{TYPICAL FACILITY SIZE}} = \frac{16,000}{25,000} = 0.64$$

Find 0.64 on the horizontal axis. Trace a vertical line to the factor curve and then trace a horizontal line to the vertical axis' Size Adjustment Factor which is 1.05.

Figure 17
Size Adjustment Chart

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PROJECTED NAVFAC CONSTRUCTION COST ESCALATION INDEX
MONTHLY
APRIL 1992
Projected Indices are Followed by the Letter p

YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
1978	1609	1617	1620	1621	1652	1663	1696	1705	1720	1721	1732	1734
1979	1740	1740	1750	1749	1753	1809	1829	1849	1900	1899	1902	1909
1980	1895	1894	1915	1899	1888	1916	1950	1971	1976	1981	2000	2017
1981	2015	2016	2014	2064	2076	2083	2109	2118	2139	2156	2186	2184
1982	2184	2200	2195	2195	2220	2219	2233	2253	2249	2248	2260	2295
1983	2311	2348	2352	2347	2351	2388	2414	2428	2430	2416	2419	2406
1984	2402	2407	2412	2422	2419	2417	2418	2428	2430	2424	2421	2408
1985	2410	2414	2406	2405	2411	2429	2448	2442	2440	2441	2446	2439
1986	2440	2446	2447	2458	2479	2493	2499	2498	2504	2511	2511	2511
1987	2515	2510	2518	2523	2524	2525	2538	2557	2565	2569	2564	2589
1988	2574	2576	2586	2591	2592	2595	2598	2611	2612	2612	2616	2617
1989	2619	2613	2616	2620	2621	2626	2631	2640	2668	2672	2675	2679
1990	2673	2674	2684	2684	2697	2725	2725	2721	2729	2728	2730	2719
1991	2719	2716	2726	2736	2723	2733	2757	2792	2785	2786	2791	2784
1992	2780	2779	7991	2809	2818p	2826p	2835p	2844p	2853p	2861p	2870p	2887p
1993	2887p	2895p	2904p	2912p	2921p	2939p	2938p	2947p	2955p	2964p	2973p	2981p
1994	2990p	2999p	3007p	3016p	3025p	3033p	3042p	3051p	3059p	3068p	3077p	3085p
1995	3094p	3103p	3111p	3120p	3129p	3137p	3146p	3155p	3163p	3172p	3181p	3190p
1996	3199p	3208p	3217p	3226p	3235p	3244p	3253p	3262p	3271p	3280p	3289p	3299p

Figure 18
Projected NAVFAC Cost Index

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PROJECTED NAVFAC CONSTRUCTION COST ESCALATION INDEX
MONTHLY
APRIL 1992
Projected Indices are Followed by the Letter p

YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
1997	3308p	3317p	3327p	3336p	3345p	3355p	3364p	3373p	3382p	3392p	3402p	3411p
1998	3421p	3430p	3440p	3449p	3459p	3469p	3478p	3488p	3497p	3507p	3517p	3527p
1999	3537p	3547p	3557p	3567p	3577p	3586p	3596p	3606p	3616p	3626p	3636p	3646p

HISTORICAL INDICES ARE BASED UPON ENGINEERING NEWS RECORD BUILDING COST INDEX.

DIVIDE THE INDEX NUMBER OF THE PROJECTED DATE BY THE INDEX NUMBER OF THE BASE DATE TO DETERMINE THE COST ESCALATION MULTIPLIER. PROJECTED INDICES ARE BASED ON OFFICIAL OASD PROJECTED RATES AS FOLLOWS:

FY	% PER YEAR	% PER MONTH
FY92	3.70	0.31
FY93	3.60	0.30
FY94	3.50	0.29
FY95	3.40	0.28
FY96	3.40	0.28
FY97	3.40	0.28
FY98	3.40	0.28
FY99	3.40	0.28

FOR MORE INFORMATION CONTACT: MR. WILLIAM C. O'CONNOR, P.E., NAVFAC COST ENGINEER, DSN 221-0069 COMMERCIAL (703) 325-0069

Figure 18 (Continued)
Projected NAVFAC Cost Index

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Section 11: PRIMARY AND SUPPORTING FACILITIES DEFINITIONS

11.1 Primary Facility. The unit cost of the primary facility shall be derived using a guidance cost that is adjusted for size, location (area cost factor), escalation, and additional functional features, see Figure 21 (Primary Facility Unit Cost Development). Built-in equipment is included in the guidance primary facility unit cost. All non-identical buildings shall be listed separately. However, completion of the cost model is required.

11.2 Supporting Facilities. Supporting facilities include all construction items outside the building 5-foot line and all special foundation features of significant cost unique to the building. All supporting facility items should be included under one of the following major categories. Each sub-item included under these major categories shall have a separate line item entry on the Budget Estimate Summary Sheet. Refer to the CES Standard Systems List and the systems narratives for identification by system number of supporting facilities. Completion of the cost model is required.

11.2.1 Special Foundation Features. Include special foundation requirements (e.g., piling, drilled piers, vibroflotation, dewatering, etc.). (Systems 91X)

11.2.2 Electrical Utilities. Include electrical distribution, substations, emergency power generators, fire alarms, exterior lighting, telephone distribution, etc., beyond the 5-foot line. (Systems 51X, 52X)

11.2.3 Mechanical Utilities. Include those utilities outside the 5-foot line covered in Divisions 2 and 15 of the specifications (e.g., gas, steam, water, sanitary, etc.). (Systems 54X through 59X)

11.2.4 Roads, Parking, and Sidewalks. Include subgrade preparation, subbase and surface courses, curb and gutters, and valley gutters (Systems 7XX)

11.2.5 Site Improvements. Include seeding, sodding, riprap, landscaping, irrigation, fencing, storm drainage, and other site enhancements. Do not include normal excavation which is within the 5-foot line. (Systems 8XX)

11.2.6 Demolition. Include real property inventory (building or structure), major utility, or major paving demolition only. Utility, pavement, or other miscellaneous demolition associated with installation of new construction should be included with the applicable trade category (e.g., electrical demolition with Electrical Utilities, steam line demolition with Mechanical Utilities, paving demolition with Roads, Parking, and Sidewalks, etc.). Provide the total number of buildings to be demolished in par. 10 of DD Form 1391.

11.2.7 Contaminated Soil. Cost to remove or treat contaminated soil in areas such as old industrial sites.

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11.2.8 Asbestos Removal. Discussions with a contract specialist or the EFD Code 407 is encouraged.

11.3 Supporting Facilities - Cost. Each supporting facility system must be priced separately and backup work sheets included.

11.4 Proposed Site Plan. In order to reasonably estimate the supporting facility requirements, the proposed site should be investigated and station utility maps reviewed. It is suggested that a scaled, single line site plan be developed for the project. For most projects, this site plan can be made on an 8-1/2 by 11-inch sheet and should be attached to the estimate. The site plan should show most, if not all, of the following:

a) Existing elevations

b) Proposed building(s) and structure(s)

c) Finish floor elevation

d) Existing mechanical and electrical utility runs

e) New mechanical and electrical services to the facility

f) Paving

g) Demolition

11.5 Pricing Sources. Supporting facility costs should be estimated using the following pricing sources:

a) Supporting facility guidance cost reference provides composite in-place cost for typical systems. See Table 14.

b) Systems cost obtained from recently bid local projects. This is considered a reliable source.

c) NAVFACENGCOM CES unit price file including assembly, and the composite prices.

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Section 12: AREA COST FACTORS (6/90)

12.1 Stateside Area Cost Factors. The following list outlines area cost factors for various areas within the United States.

ALABAMA	.82
ALABAMA AAP	.79
ANNISTON AD	.79
BIRMINGHAM	.83
FORT MCCLELLAN	.79
FORT RUCKER	.85
MAXWELL AFB	.77
MOBILE AREA	.87
MONTGOMERY	.77
REDSTONE ARS	.89
ALASKA	2.18
ADAK	2.93
ADAK NAVAL STA	2.93
ANCHORAGE	1.69
EIELSON AFB	1.96
ELMENDORF AFB	1.69
FAIRBANKS	1.93
FORT GREELY	2.23
FT RICHARDSON	1.69
FORT WAINWRIGHT	1.95
SHEMYA AFB	3.21
ARIZONA	.95
FLAGSTAFF	1.00
FORT HUACHUCA	1.12
NAVAJO AD	1.00
PHOENIX	.96
TUCSON	.90
YUMA MAR.A.S.	1.11
YUMA PG	1.11
ARKANSAS	.83
EAKER AFB	.80
FORT CHAFFEE	.87
FORT SMITH	.87
LITTLE ROCK	.79
PINEBLUFF	.84
PINEBLUFF ARS	.84
CALIFORNIA	1.24
BEALE AFB	1.24
CAMP PENDLETON	1.18
CHINA LAKE NWC	1.40
EDWARDS AFB	1.38
EL CENTRO NAF	1.21
EL TORO MCAS	1.23
FT HUNT LIGGETT	1.44
FORT IRWIN	1.30
FORT ORD	1.21
LOS ANGELES	1.18
LOS ANGELES AREA	1.18
MARCH AFB	1.26
MATHER AFB	1.14
MCCLELLAN AFB	1.14
MONTEREY AREA	1.20
NORTON AFB	1.26
OAKLAND AB	1.37
PORT HUENEME AREA	1.18
PRESIDIO MONTEREY	1.23
PRESIDIO SAN FRAN	1.39
RIVERBANK AAP	1.19
SACRAMENTO AD	1.12
SAN DIEGO	1.16
SAN DIEGO AREA	1.16
SAN FRANCISCO	1.37
*SAN FRAN AREA	1.34
SHARPE AD	1.16
TWENTY-NINE PALMS	1.38
VANDENBURG AFB	1.36
COLORADO	.97
AIR FORCE ACADEMY	1.05
CHEYENNE MOUNTAIN	1.03
COLORADO SPRINGS	.96
DENVER	.96
FALCON AFS	.96
FITZSIMMONS AMC	.98
FORT CARSON	1.03
PETERSON AFB	.96
PUEBLO	.98
PUEBLO AD	.98
ROCKY MOUNTAIN ARS	.97
CONNECTICUT	1.19
BRIDGEPORT	1.21
HARTFORD	1.16
NEW LONDON	1.21
NEW LONDON AREA	1.21
STRATFORD ENGR P	1.21
DELAWARE	1.00
DOVER	1.02
DOVER AFB	1.02
LEWES	.99
FLORIDA	.89
CAPE CANAVERAL	.91

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CAPE KENNEDY	.91
EGLIN AFB	.83
HOMESTEAD	.90
HOMESTEAD AFB	.90
JACKSONVILLE AREA	.90
KEY WEST NAS	1.10
MCDILL AFB	.94
MIAMI	.88
ORLANDO	.82
PANAMA CITY AREA	.86
PENSACOLA AREA	.84
TAMPA	.94
TYNDALL AFB	.86
GEORGIA	.80
ALBANY	.80
ALBANY AREA	.80
ATLANTA	.83
ATLANTA, MARIETTA	.84
FORT BENNING	.79
FORT GILLEM	.83
FORT GORDON	.88
FORT MCPHERSON	.83
FORT STEWART	.81
KINGS BAY	.92
MACON	.77
WARNER ROBINS AFB	.77
HAWAII	1.46
ALIAMANU	1.37
BARBERS PT NAS	1.40
BARKING SANDS	1.80
EWA BEACH	1.40
FORD ISLAND	1.53
FORT DERUSSY	1.36
FORT SHAFTER	1.37
HELEMANO	1.42
HICKAM AFB	1.34
HILO	1.55
HONOLULU	1.36
KANEOHE	1.46
KANEOHE MCAS	1.39
LUALUALEI	1.43
MAUNA LOA	1.36
PEARL CITY	1.36
PEARL HARBOR	1.36
POHAKULOA	1.65
SCHOFIELD BRKS	1.42
TRIPLER AMC	1.36
WHEELER AFB	1.42
IDAHO	1.06
BOISE	1.04
IDAHO FALLS	1.03
MTN HOME	1.10
MTN HOME AFB	1.10
ILLINOIS	1.08
BELLEVILLE	.98
CHICAGO	1.27
CHICAGO AREA	1.27
FORT SHERIDAN	1.28
GREAT LAKES NTC	1.28
ROCK ISLAND	.99
ROCK ISLAND ARS	1.02
SAVANNAH AD	1.02
SCOTT AFB	1.05
INDIANA	1.00
CRANE AAP	1.06
CRANE NWSC	1.06
FT BEN HARRISON	1.07
GRISSOM AFB	1.06
INDIANA AAP	.93
INDIANAPOLIS	1.04
INDIANAPOLIS AREA	1.05
JEFFERSON PG	.95
LOGANSPORT	1.00
MADISON	.96
NEWPORT AAP	.96
IOWA	1.02
BURLINGTON	1.07
CEDAR RAPIDS	.99
DES MOINES	.99
IOWA AAP	1.08
KANSAS	.92
FORT LEAVENWORTH	1.06
FORT RILEY	.90
KANSAS AAP	.92
MANHATTAN	.90
MCCONNELL AFB	.92
TOPEKA	.93
WICHITA	.92
KENTUCKY	.95
BOWLING GREEN	.91
FORT CAMPBELL	1.02
FORT KNOX	.98
LEXINGTON	1.01
LEX/BLUEGRASS AD	1.08
LOUISVILLE	.92
LOUISVILLE NAS	.92
LOUISIANA	.89
ALEXANDRIA	.88
BARKSDALE AFB	.86
ENGLAND AFB	.88

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FORT POLK	.95
LOUISIANA AAP	.86
NEW ORLEANS	.93
NEW ORLEANS AB	.93
NEW ORLEANS AREA	.93
SHREVEPORT	.86
MAINE	1.11
BANGOR	1.05
BRUNSWICK	1.07
CARIBOU	1.21
LORING AFB	1.19
PORTLAND	1.07
PROSPECT HARBOR	1.11
WINTER HARBOR	1.11
MARYLAND	.95
ABERDEEN PG	.96
ANDREWS AFB	1.05
BALTIMORE	.96
FORT DETRICK	.94
FORT HOLABIRD	.96
FORT MEADE	1.05
FORT RITCHIE	.94
FREDERICK	.95
HARRY DIAMOND LAB	1.05
INDIAN HEAD	.99
LEXINGTON PARK	.95
PATUXENT RIV AR	.95
MASSACHUSETTS	1.20
ARMY MATL & MECH	1.22
BOSTON	1.23
BOSTON AREA	1.23
FITCHBURG	1.21
FORT DEVENS	1.28
HANSCOMB AFB	1.23
NATICK	1.22
MICHIGAN	1.12
BAY CITY	1.06
DETROIT	1.18
DETROIT AREA	1.18
DETROIT ARS	1.18
MARQUETTE	1.13
REPUBLIC ELFCOM	1.21
SAWYER AFB	1.21
WURTSMITH AFB	1.05
MINNESOTA	1.20
DULUTH	1.18
MINNEAPOLIS	1.22
ROSEMONT	1.22
ST. CLOUD	1.19
TWIN CITY AP	1.22
MISSISSIPPI	.82
BILOXI	.84
COLUMBUS	.82
COLUMBUS AFB	.82
GULFPORT	.84
JACKSON	.79
KEESLER AFB	.84
MERIDIAN NAS	.92
MISSISSIPPI AAP	.84
MISSOURI	.98
FT LEONARD WOOD	.96
KANSAS CITY	1.01
KANSAS CITY AREA	1.01
LAKE CITY AAP	1.00
ROLLA	.91
ST. LOUIS	1.03
ST. LOUIS AAP	1.08
WHITEMAN AFB	1.11
MONTANA	1.22
BILLINGS	1.22
BUTTE	1.23
GREAT FALLS	1.20
MALMSTROM AFB	1.20
NEBRASKA	.93
CORNHUSKER AAP	.88
GRAND ISLAND	.88
LINCOLN	.91
OFFUTT AFB	.99
OMAHA	.99
NEVADA	1.12
FALLON	1.19
HAWTHORNE	1.17
HAWTHORNE AAP	1.17
LAS VEGAS	1.10
NELLIS AFB	1.10
RENO	1.09
NEW HAMPSHIRE	1.02
CONCORD	1.01
NASHUA	1.04
PEASE AFB	1.02
PORTSMOUTH	1.02
PORTSMOUTH AREA	1.02
NEW JERSEY	1.14
BAYONNE MOT	1.17
EARLE	1.17
FORT DIX	1.15
FORT MONMOUTH	1.13
LAKEHURST	1.16
MCGUIRE AFB	1.15

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NEWARK	1.17
PICATINNY ARS	1.25
RED BANK	1.12
TRENTON	1.14
TRENTON AREA	1.14
NEW MEXICO	.95
ALAMAGORDO	.92
ALBUQUERQUE	.92
CANNON AFB	1.10
FORT WINGATE	1.00
GALLUP	1.00
HOLLOMAN AFB	1.01
KIRTLAND AFB	.92
WHITE SAND MIS RN	1.10
NEW YORK	1.11
ALBANY	1.05
BROOKLYN	1.25
FORT DRUM	1.18
GARDEN CITY	1.25
NEW YORK CITY	1.25
NEW YORK CITY AREA	1.25
SENECA AD	1.15
STATEN ISLAND	1.32
SYRACUSE	1.04
US MIL ACADEMY	1.13
WATERVLIET ARS	1.05
NORTH CAROLINA	.81
CAMP LEJEUNE AREA	.82
CHERRY POINT	.83
FAYETTEVILLE	.79
FORT BRAGG	.80
GREENSBORO	.86
NEW RIVER	.85
POPE AFB	.80
SEYMOUR JOHN. AFB	.80
SUNNY POINT	.85
WILMINGTON	.79
NORTH DAKOTA	1.00
BISMARCK	.97
GRAND FORKS	.96
GRAND FORKS AFB	.96
MINOT	1.07
MINOT AFB	1.07
OHIO	.99
COLUMBUS	.98
DAYTON	1.00
RAVENNA AAP	.99
WRGT-PATTERSON AFB	1.00
YOUNGSTOWN	.99
OKLAHOMA	.83
ALTUS AFB	.86
FORT SILL	.81
LAWTON	.81
MCALESTER	.81
MCALESTER AAP	.81
OKLAHOMA CITY	.87
TINKER AFB	.87
OREGON	1.04
PENDLETON	1.12
PORTLAND	1.00
SALEM	.99
UMATILLA AD	1.19
PENNSYLVANIA	1.01
CARLISLE BRKS	.94
HARRISBURG	.94
INDIANTOWN GAP MIS	1.01
LETTERKENNY AD	1.00
MECHANICSBURG AREA	.94
NEW CUMBERLAND AD	.94
PHILADELPHIA	1.13
PHILADELPHIA AREA	1.13
PITTSBURGH	.97
SCRANTON	1.05
TOBYHANNA AD	1.14
WARMINSTER AREA	1.11
RHODE ISLAND	1.15
BRISTOL	1.16
DAVISVILLE	1.22
NEWPORT	1.12
NEWPORT AREA	1.12
PROVIDENCE	1.16
SOUTH CAROLINA	.89
BEAUFORT AREA	.94
CHARLESTON	.91
CHARLESTON AFB	.91
CHARLESTON AREA	.91
COLUMBIA	.82
FORT JACKSON	.82
MYRTLE BEACH	.93
SHAW AFB	.83
SOUTH DAKOTA	1.02
ABERDEEN	1.04
ELLSWORTH AFB	1.02
RAPID CITY	1.01
SIOUX FALLS	1.00
TENNESSEE	.80
ARNOLD AFB	.90
CHATTANOOGA	.74
HOLSTON ORDNANCE	.79
KINGSPORT	.79

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MEMPHIS	.86
MEMPHIS AREA	.86
MILAN ARS	.79
VOLUNTEER ORDNANCE	.79
TEXAS	.89
BROOKS AFB	.88
CAMP BULLIS	.88
CARSWELL AFB	.95
CHASE FIELD	.93
CORPUS CHRISTI	.84
CORPUS CHRISTI AD	.84
DYESS AFB	.92
FORT BLISS	.89
FORT HOOD	.90
FORT SAM HOUSTON	.88
FORT WORTH	.95
GOODFELLOW AFB	.89
INGLESIDE	.86
KELLY AFB	.88
KINGSVILLE	.98
LACKLAND AFB	.88
LONE STAR AAP	.88
LONGHORN AAP	.81
RANDOLPH AFB	.88
RED RIVER AD	.88
REESE AFB	.86
SAN ANGELO	.83
SAN ANTONIO	.88
SHEPPARD AFB	.94
UTAH	.96
DUGWAY PG	1.00
FORT DOUGLAS	.93
HILL AFB	.98
OGDEN	.92
SALT LAKE CITY	.93
TOOELE	1.02
TOOELE AD	1.02
VERMONT	1.06
BURLINGTON	1.09
MONTPELIER	1.08
RUTLAND	1.01
VIRGINIA	.92
ARLINGTON	1.05
ARLINGTON HS	1.05
CAMERON STA	1.05
DAHLGREN	.94
FORT BELVOIR	1.05
FORT EUSTIS	.92
FORT AP HILL	.94
FORT LEE	.87
FORT MONROE	.92
FORT MYER	1.05
FORT PICKETT	.94
FORT STORY	.92
NORFOLK	.92
NORFOLK AREA	.92
QUANTICO	.93
RADFORD	.89
RADFORD AAP	.95
RICHMOND	.94
VINT HILL FARMS	.94
WASHINGTON	1.00
BREMERTON	.98
FAIRCHILD AFB	1.00
FORT LEWIS	1.00
INDIAN ISLAND	.99
MCCHORD AFB	1.00
SILVERDALE	.94
SPOKANE	.99
TACOMA	1.00
WHIDBEY ISLAND	1.09
YAKIMA	1.00
YAKIMA FR	1.07
WEST VIRGINIA	.99
BLUEFIELD	.96
CHARLESTON	.98
CLARKSBURG	1.02
SUGAR GROVE	1.29
(Buildings)	
(Heavy Constr)	1.50
WISCONSIN	1.04
BADGER AAP	.99
FORT MCCOY	1.08
LA CROSSE	1.02
MADISON	.99
MILWAUKEE	1.10
WYOMING	1.08
CASPER	1.08
CHEYENNE	1.08
F.E. WARREN AFB	1.08
LARAMIE	1.07
WASHINGTON, DC	1.05
FORT MCNAIR	1.05
WALTER REED AMC	1.05
WASHINGTON DC AR	1.05

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12.2 Overseas Area Cost Information (6/90). The following list outlines area cost information for various areas outside of the United States.

LOCATION	ACF/FER	NOTES
ANTIGUA	1.73/2.78	
AUSTRALIA	1.55/1.30	
DARWIN	1.90	
PERTH	1.42	
SYDNEY	1.34	
AZORES	1.04/168.10	2
BAHAMAS	1.79/1.00	2
ANDROS IS	1.79	2
BAHRAIN	2.07/.38	
BELGIUM	1.25/42.55	
BRUSSELS	1.25	
BERMUDA	1.61/1.00	2
CANADA	1.55/1.16	
ARGENTIA, NEWFOUNDLAND	1.55	
CUBA	1.60	2
GUANTANAMO	1.60	2
GTMO PAVING	3.00	2
DIEGO GARCIA	3.00	
EGYPT	1.32/2.65	
CAIRO	1.32	
GREECE	1.13/155.30	
ATHENS	1.13	
GREENLAND	2.48/7.81	
GODTHAAB(NUUK)	2.48	
GUAM	2.24	
HONDURAS	.64/4.50	
TEGUCIGALPA	.64	
ICELAND	3.38/62.50	1
KEFLAVIK	3.38	1
REYKJAVIK	3.38	1
ISRAEL	1.18/2.03	
TEL AVIV	1.18	
ITALY	1.44/1132.50	1
ISOLA DI CAPO RIZZUTO	1.43	1
LA MADDALENA	1.50	1
NAPLES	1.44	1
SIGONELLA	1.44	1
VENICE	1.39	1
JAPAN	1.60/149.05	
ATSUGI	1.66	
MISAWA	1.57	
OKINAWA	1.49	
TOKYO	1.66	
JOHNSTON ATOLL	2.32	
KOREA	1.12/664.01	
CHINHAE	1.11	
DMZ ZONE AREA	1.18	
KUNSAN	1.12	
OSAN	1.10	
SEOUL	1.08	
KWAJALEIN	2.54	
MIDWAY ISLAND	2.07	
MOROCCO	1.55/8.67	
CASABLANCA	1.55	
NETHERLANDS	1.25/2.26	
OSS	1.25	
NEW ZEALAND	2.07/1.74	
WELLINGTON	2.07	
OMAN	1.58/.39	
RUWI	1.58	
PANAMA	1.24	
PANAMA CITY	1.24	
PHILLIPINES	1.10/22.25	
MANILA	1.08	
SUBIC BAY	1.11	
PUERTO RICO	1.05	2
SAN JUAN	1.05	2
SEYCHELLES IS	2.50	
SPAIN	1.18/95.15	
ROTA	1.18	
TURKEY	.71/2185.52	
ANKARA	.71	
INCIRLICK	.70	
UNITED KINGDOM	1.71/.52	1
LONDON	1.75	1
MANCHESTER	1.64	1
ST. MAWGAN	1.71	1
EDZELL, SCOTLAND	1.75	1
WEST GERMANY	1.38/2.03	
FRANKFURT	1.54	
KAISERSLAUTERN	1.40	
TRIER	1.20	

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NOTES:

1. Escalate project costs to the one-third time point of construction.

2. Allow for Defense Base Act Insurance Costs.

3. International Balance of Payments (IBOP) Summary Sheet is required.

ACF: Area Cost Factor

FER: Foreign Exchange Rate

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Section 13: GUIDANCE COSTS

13.1 Primary Facility Guidance Costs. Tables 13 and 13A give the current guidance costs for various facilities identified by category code. The costs have been adjusted for an ACF of 1.00, and the time has been adjusted for FY 94 and FY 95.

13.2 Supporting Facilities Guidance Cost. Table 14 gives the current total composite contract cost for typical supporting facilities systems. Refer to par. 15.2 for CES systems identification. The cost has been adjusted for an ACF of 1.00, and the time has been adjusted for FY 94 and FY 95.

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Table 13
Primary Facility Guidance Costs

NAVFACENGCOM FY 94 & FY 95 GUIDANCE COST (ACF = 1.00) AS OF JULY 1991.

ALL UNIT COSTS INCLUDE BUILT-IN EQUIPMENT TYPICAL FOR THE FACILITY.
UNIT COSTS INCLUDE QAP & PCAS EXCEPT HIGH TECH AND HIGH RISK FACILITIES.

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
133.72	RADAR AIR TRAFFIC CONTROL CENTER	6500 SF	147	151
137.10	OCEANOGRAPHIC BUILDING	37000 SF	97	100
141.25	ACFT FIRE/RESCUE/STRUC/STATION	8000 SF	106	109
141.40	AIRCRAFT OPERATIONS BUILDING	20000 SF	101	105
141.70	CONTROL TOWER (ATTACHED/FREESTANDING)	3000 SF	173	178
143.15	RANGE OPERATION CENTER	13000 SF	198	204
143.20	ORDNANCE OPERATIONS BUILDING	13000 SF	86	89
143.41	AMPHIBIOUS OPERATIONS BUILDING	25000 SF	100	104
143.45	ARMORY	10000 SF	89	92
143.65	OPCON CENTER FOR FLEET OPERATIONS	36000 SF	146	150
143.77	OPERATIONAL STORAGE (READY ISSUE)	4000 SF	53	55
151.20	GENERAL PURPOSE/BERTHING PIER	10000 SY	1017	1048
151.80	DEPERMING PIER	3240 SY	1360	1403
156.10	WATERFRONT TRANSIT SHED	5000 SF	77	79
159.64	WATERFRONT OPERATIONS BUILDING	20000 SF	83	86
171.10	ACADEMIC INSTRUCTION BUILDING	30000 SF	82	85
171.20	APPLIED INSTRUCTION BUILDING	25000 SF	88	91
171.35	OPERATIONAL TRAINER FACILITY	22000 SF	108	111
171.45	MOCK-UP & TRNG. AID PREPAR CENTER	25000 SF	98	101
171.60	RECRUIT PROCESSING BUILDING	80000 SF	79	81
179.45	FIREFIGHTING TRAINING FACILITY	8100 SF	244	251
179.55	COMBAT TRAINING POOL/TANK	22000 SF	112	115
211.03	CORROSION CONTROL HANGAR	35000 SY	236	244
211.05	MAINTENANCE HANGAR (1 MODULE)	38000 SF	84	87
211.06	MAINTENANCE HANGAR (1 MODULE)	38000 SF	83	86
211.14	AIRCRAFT REWORK SHOP (NADEP)	160800 SF	105	108
211.21	ENGINE MAINTENANCE SHOP	34000 SF	78	80
211.24	DEDICATED A/C ENGINE O/H (NADEP)	80000 SF	87	90
211.34	DYNAMIC COMPONENTS SHOP (NADEP)	8000 SF	95	98
211.45	AVIONICS SHOP (NON-NADEP)	23000 SF	84	87
211.54	AVIATION ARMAMENT SHOP (NON-NADEP)	16000 SF	79	81
211.81	ENGINE TEST CELL (NON-NADEP)	7065 SF	749	772
212.10	GUIDED MISSILES INTEGRATION FACILITY	5000 SF	84	87
212.30	MISSILE ASSEMBLY & TEST BUILDING	60000 SF	127	131
214.10	COMBAT VEHICLE MAINTENANCE SHOP	20000 SF	89	92
214.20	AUTOMOTIVE VEHICLE MAINTENANCE SHOP	20000 SF	79	81

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Table 13 (Continued)
Primary Facility Guidance Costs

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
214.30	REFUELING VEHICLE SHOP	2600 SF	91	94
214.51	AUTOMOTIVE ORGANIZAT'L SHOP (MARC)	30000 SF	76	78
216.10	AMMO REWORK AND OVERHAUL SHOP	14170 SF	147	151
216.40	TORPEDO SHOP	26000 SF	138	142
217.10	ELEC/COMM MAINTENANCE SHOP	17000 SF	93	96
219.10	PUBLIC WORKS SHOP	20000 SF	72	74
310.23	COMBINED RESEARCH LAB	60000 SF	106	109
310.37	OCEAN SCIENCES LABORATORY	56000 SF	105	108
315.20	UNDERWATER WEAPON SYSTEMS LABORATORY	20000 SF	137	141
315.30	WEAPONS SYSTEMS INTEGRATION LABORATORY	60000 SF	138	142
317.15	DETECTION SYSTEMS LAB	17000 SF	112	115
317.20	ELECTRICAL/ELECTRONICS SYS LAB	60000 SF	123	127
317.25	ELECTRICAL/COMM. SYS. INTEGRATION LAB	66000 SF	119	123
320.10	UNDERWATER EQUIPMENT LABORATORY	6000 SF	132	136
411.50	JET ENGINE FUEL STORAGE	600000 BL	11	11
421.22	HIGH EXPLOSIVE MAGAZINE	7000 SF	148	153
421.32	INERT STOREHOUSE	16000 SF	69	71
421.72	MISSILE MAGAZINE	9300 SF	140	144
441.10	GENERAL WAREHOUSE	40000 SF	46	47
441.11	GENERAL PURPOSE WAREHOUSE (MARINE CORPS DSSC)	40000 SF	58	60
441.30	HAZARDOUS/FLAMMABLE STOREHOUSE	10000 SF	107	110
441.35	GENERAL STORAGE SHED	20000 SF	32	33
441.72	SERVMART	39000 SF	53	55
550.10	MEDICAL CLINIC	30000 SF	124	128
610.10	ADMINISTRATIVE OFFICE (INCL HQ)	25000 SF	84	87
610.1A	PASS OFFICE	6000 SF	103	106
610.20	DATA PROCESSING CENTER	21000 SF	113	116
610.72	BATTALION/SQUADRON HEADQUARTERS	30000 SF	72	74
721.11	BEQ (E-1 THRU E-4)	100000 SF	71	73
721.12	BEQ (E-5 AND E-6)	100000 SF	79	81
722.10	ENLISTED DINING FACILITY	16000 SF	152	157
724.11	BOQ (W-1 THRU O-2)	44000 SF	76	78
724.12	BOQ (O-3 AND ABOVE)	44000 SF	71	73
730.10	FIRE STATION (COMMUNITY)	3500 SF	107	110
730.15	CONFINEMENT FACILITY	30000 SF	104	107
730.20	POLICE STATION	6500 SF	99	103
730.83	CHAPEL	10000 SF	122	126
740.23	COMMISSARY (INCLUDES BACK-UP STORAGE)	60000 SF	61	63
740.43	GYMNASIUM	20000 SF	100	104
740.74	CHILD CARE CENTER	8000 SF	90	93
831.41	HAZARDOUS WASTE FACILITY	3600 SF	90	93

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Table 13A
Additional Primary Facility Guidance Costs *

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
113-20	AIRCRAFT PARKING APRON	65000 SY	45	47
116-15	AIRCRAFT RINSE FACILITY	1310 SY	142	146
116-56	COMBAT AIRCRAFT ORDNANCE LOADING AREA	55281 SY	62	64
125-10	POL PIPELINE	2689 LF	161	166
131-15	COMMUNICATIONS CENTER	6000 SF	243	250
131-25	TELEMETRY BUILDING	9000 SF	201	207
131-35	RECEIVER BUILDING	20550 SF	308	318
131-40	TELEPHONE EXCHANGE BUILDING	6000 SF	170	176
131-45	TERMINAL EQUIPMENT BUILDING	70000 SF	114	118
131-55	CIRCULARLY DISPOSED ANTENNA ARRAY BUILDING	15000 SF	132	136
132-10	ANTENNA - COMMUNICATIONS	1 EA	2635	2718
141-30	AIRCRAFT LINE OPERATIONS BUILDING	15620 SF	65	67
141-42	AIR INTELLIGENCE SUPPORT CENTER	20000 SF	100	103
141-87	LIQUID OXYGEN/NITROGEN FACILITY (NON-INDUSTRIAL)	5582 SF	104	107
141-88	HARDENED AIRCRAFT SHELTER	13991 SF	78	80
143-10	EMERGENCY VEHICLE GARAGE	3066 SF	59	61
143-25	SEAL TEAM BUILDING	57569 SF	87	90
143-40	COMPUTER PROGRAMMING OPERATIONS CENTER	21000 SF	102	105
143-47	ALERT FORCE BUILDING	22500 SF	69	71
143-80	NAVAL SECURITY GROUP REGIONAL REPORTING CENTER	26000 SF	158	163
159-20	DEGAUSSING BUILDING	1600 SF	185	190
171-15	RESERVE TRAINING BUILDING	33834 SF	69	71
179-40	SMALL ARMS RANGE - OUTDOOR	4011 SF	84	87
211-01	AIRCRAFT ACOUSTICAL ENCLOSURE (NON-NADEP)	7950 SF	685	706
211-08	AIRFRAMES SHOP (NON-NADEP)	23000 SF	147	152
211-12	PAINT AND FINISHING HANGAR (NADEP)	69708 SF	221	228
211-32	METAL COMPONENTS SHOP (NADEP)	104000 SF	97	100
211-36	ELECTRICAL COMPONENTS SHOP (NADEP)	15500 SF	94	97
211-52	AIRCRAFT WEAPON OVERHAUL AND TEST SHOP (NADEP)	19480 SF	131	135
211-60	SUPPORT EQUIPMENT REWORK SHOP CENTER (NADEP)	57680 SF	96	99
211-70	MANUFACTURING AND REPAIR SHOP (NADEP)	81000 SF	69	71
211-72	METAL FABRICATION/MANUFACTURING SHOP (NADEP)	81000 SF	117	120

* Unit costs derived from Award Cost Data or NAVFAC FY 92 & FY 93 Guidance Costs.

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Table 13A (Continued)
Additional Primary Facility Guidance Costs

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
211-73	METAL TREATMENT SHOP (NADEP)	81000 SF	166	171
211-75	PARACHUTE SURVIVAL EQUIPMENT SHOP (NON-NADEP)	8000 SF	88	90
211-89	POWER CHECK PAD WITHOUT SOUND SUPPRESSION (NON-NADEP)	1 EA	**364	**376
211-89I	POWER CHECK PAD (VSTOL)	1 EA	**610	**629
211-93	ENGINEERING LABORATORY (NADEP)	16000 SF	153	157
211-94	AIRCRAFT POWER CHECK FACILITIES (NADEP)	9000 SF	480	495
211-96	MAINTENANCE, AIRCRAFT SPARES STORAGE (READY ISSUE/SHOP STORAGE MISCELLANEOUS)	13150	36	37
211-98	AIRCRAFT ACOUSTICAL ENCLOSURE (NADEP)	12653 SF	557	574
212-20	MISSILE EQUIPMENT MAINTENANCE SHOP	10462 SF	97	100
213-55	PIPEFITTING SHOP (56) (J)	78000 SF	105	109
213-59	ABRASIVE BLAST FACILITY	20000 SF	169	174
213-61	RIGGING SHOP - (72) (T)	155500 SF	64	66
214-53	FIELD MAINTENANCE SHOP (COMBAT/AUTOMOTIVE/TRACK) (MARINE CORPS)	20000 SF	92	95
215-30	HEAVY GUN (6IN TO 16IN) SHOP	25000 SF	122	126
216-60	QUALITY EVALUATION LAB	27000 SF	90	93
217-77	ELECTRONICS - SPARES AND STORAGE (READY ISSUE/SHOP STORES/MISCELLANEOUS)	12928 SF	106	109
218-20	CONSTRUCTION/WEIGHT HANDLING EQUIPMENT SHOP	9600 SF	68	70
218-25	MARINE AIR-BASE SQUADRON (MABS) FACILITY	48330 SF	95	98
218-45	INSTRUMENT CALIBRATION SHOP	12000 SF	74	77
218-50	BATTERY SHOP	27000 SF	90	93
218-60	AIRCRAFT GROUND SUPPORT EQUIPMENT SHOP	32000 SF	85	87
218-61	GROUND SUPPORT EQUIPMENT HOLDING SHED	5051 SF	30	31
218-62	SHIPBOARD - AIRCRAFT SUPPORT - EQUIPMENT FACILITY	57784 SF	95	98

** Figures represent cost in thousands.

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Table 13A (Continued)
Additional Primary Facility Guidance Costs

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
218-80	FIELD MAINTENANCE SHOP (GENERAL SUPPLY)	40993 SF	61	63
219-77	PUBLIC WORKS MAINTENANCE STORAGE (READY ISSUE/SHOP STORES/MISC.)	42830 SF	68	70
222-10	MISSILE ASSEMBLY PLANT	3430 SF	116	119
226-65	PROPELLANT AND RELATED CHEMICAL PLANT	6000 SF	369	381
227-20	SONAR EQUIPMENT PLANT	93200 SF	96	99
310-17	OPTICS LABORATORY	37912 SF	248	256
310-33	COMPUTATION AND ANALYSIS LAB	28000 SF	116	119
311-25	AIRCRAFT SYSTEMS INTEGRATION LABORATORY	26500 SF	95	98
312-25	SPACECRAFT/SATELLITE LABORATORY	80000 SF	158	163
313-10	SHIP AND MARINE LABORATORY	20000 SF	100	103
313-15	SHIPS AND MARINE NAVIGATIONAL EQUIPMENT LABORATORY	30200 SF	158	163
316-10	AMMUNITION, EXPLOSIVES, AND TOXICS LABORATORY	6000 SF	490	505
318-10	PROPULSION SYSTEMS LABORATORY	13000 SF	305	315
319-10	MISCELLANEOUS EQUIPMENT AND ITEMS LABORATORY	38000 SF	67	69
319-40	RANGE OPERATIONS AND INSTRUMENTATION LABORATORY	31000 SF	85	87
321-10	TECHNICAL SERVICES LABORATORY	56500 SF	69	71
371-17	ELECTRONIC, COMMUNICATION AND ELECTRICAL SYSTEMS RANGE FACILITY	15238 SF	53	54
412-25	LUBRICANT STORAGE	16750 SF	50	51
421-48	SMALL ARMS/PYROTECHNICS MAGAZINE	3000 SF	158	163
431-10	COLD STORAGE WAREHOUSE	6000 SF	93	96
441-71	INTEGRATED LOGISTICS OVERHAUL AND OUTFITTING BUILDING	10000 SF	105	109
510-77	HOSPITAL/MEDICAL STORAGE (READY ISSUED/SHOP STORES/ MISCELLANEOUS)	36217 SF	36	37
540-10	DENTAL CLINIC	23597 SF	200	206
610-40	COURT ROOM FACILITY	6000 SF	100	103
610-70	DIVISION/WING HEADQUARTERS (MARINE CORPS)	30000 SF	95	98
610-71	REGIMENTAL/GROUP HEADQUARTERS (MARINE CORPS)	30000 SF	95	98

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Table 13A (Continued)
Additional Primary Facility Guidance Costs

CAT CODE	FACILITY DESCRIPTION	GUIDANCE SIZE	FY 94 GUID U/C	FY 95 GUID U/C
610-73	COMPANY/BATTERY HEADQUARTERS (MARINE CORPS)	81145 SF	81	84
711-70	FUND HOUSING AFTER 1969, WO, O-1 THRU O-3	353136 SF	39	40
721-14	CLASS A STUDENT BARRACKS	108800 SF	100	103
723-30	LAUNDRY, DETACHED	5265 SF	52	54
724-22	CIVILIAN QUARTERS, GS-7 AND ABOVE	36420 SF	55	57
730-60	DEPENDENT SCHOOL	100000 SF	75	77
730-77	PERSONNEL SUPPORT STORAGE (READY ISSUE/SHOP STORES/MISCELLANEOUS	15795 SF	79	81
730-84	RELIGIOUS EDUCATION BUILDING	15000 SF	100	103
740-01	EXCHANGE RETAIL STORE	38080 SF	59	61
740-02	LOCATION EXCHANGE	5829 SF	56	58
740-03	EXCHANGE CENTRAL ADMINISTRATION	7200 SF	75	77
740-20	TEMPORARY LODGING	25397 SF	63	65
740-25	FAMILY SERVICES CENTER	5000 SF	110	114
740-36	HOBBY SHOP, ARTS/CRAFTS	16500 SF	112	116
740-38	HOBBY SHOP - AUTOMOTIVE	5068 SF	67	69
740-40	BOWLING ALLEY	16789 SF	72	74
740-50	FIELD HOUSE	38330 SF	105	109
740-53	INDOOR SWIMMING POOL	25260 SF	111	114
740-55	YOUTH CENTER	6974 SF	101	104
740-63	ENLISTED PERSONNEL CLUB	15836 SF	112	116
740-71	CLASS VI PACKAGE STORE	9640 SF	55	57
740-77	COMMUNITY STORAGE (READY ISSUE/ SHOP STORES/MISCELLANEOUS	4538 SF	45	46
740-80	GOLF CLUB HOUSE	10012 SF	103	106
740-84	INDOOR PLAYING COURTS	6438 SF	107	110
740-89	BATHHOUSE	2580 SF	69	71
821-50	NON-NUCLEAR STEAM PLANT	720000 PPH	167	172
831-14	INDUSTRIAL WASTE TREATMENT BUILDING	10000 SF	80	83
841-40	GROUND LEVEL POTABLE WATER STORAGE TANK	750000 GAL	1.84	1.90
843-10	FIRE PROTECTION PIPELINE	7200 LF	160	165
843-30	FIRE PROTECTION WATER TANK	750000 GAL	1.66	1.71
872-20	GUARD AND WATCH TOWERS	100 SF	231	238

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Table 14
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($) FY 94	FY 95
511, 512, & 516	ELECTRICAL DIST.			
	Note: Normal duct size is 4". For 35KV circuit use 5" duct			
	Note: For quantities less than 200 LF add 20%; less than 100 LF add 50%			
WK-G	4-1/C #2 THW str al. w/1-1/C#8 CU ground secondary w/termination	LF	4.08	4.20
WK-I	4-1/C #1/0 THW str al. w/1-1/C#6 CU ground secondary w/termination	LF	5.92	6.10
WK-J	4-1/C #2/0 THW str al. w/1-1/C#4 CU ground secondary w/termination	LF	7.22	7.44
WK-K	4-1/C #4/0 THW str al. w/1-1/C#2 CU ground secondary w/termination	LF	9.53	9.82
WK-250	4-1/C 250 MCM THW str al. w/1-1/C#2 CU ground secondary w/termination	LF	10.87	11.20
WK-500	4-1/C 500 MCM THW str al. w/1-1/C#1/0 CU ground secondary w/termination	LF	18.23	18.78
WK-750	4-1/C 750 MCM THW str al. w/1-1/C#1/0 CU ground secondary w/termination	LF	24.54	25.28
VF-G	4-1/C #2 CU USE wire 600V w/1-1/C#8 CU ground secondary w/termination	LF	10.33	10.64
VF-I	4-1/C #1/0 CU USE wire 600V w/1-1/C#6 CU ground secondary w/termination	LF	14.51	14.95
VF-J	4-1/C #2/0 CU USE wire 600V w/1-1/C#4 CU ground secondary w/termination	LF	17.70	18.23
VF-L	4-1/C #4/0 CU USE wire 600V w/1-1/C#2 CU ground secondary w/termination	LF	25.30	26.06
VF-250	4-1/C 250 MCM CU USE wire 600V w/1-1/C#2 CU ground secondary w/termination	LF	22.25	22.92
VF-500	4-1/C 500 MCM CU USE wire 600V w/1-1/C#1/0 CU grd. secondary w/termination	LF	38.87	40.04
VG-E	3-1/C #4 AWG CU XLP 5KV shielded cable primary w/o termination	LF	17.45	17.97
VG-G	3-1/C #2 AWG CU XLP 5KV shielded cable primary w/o termination	LF	21.02	21.65
VG-I	3-1/C #1/0 AWG CU XLP 5KV shielded cable primary w/o termination	LF	23.88	24.60
VG-J	3-1/C #2/0 AWG CU XLP 5KV shielded cable primary w/o termination	LF	24.74	25.48
VG-L	3-1/C #4/0 AWG CU XLP 5KV shielded cable primary w/o termination	LF	34.62	35.66

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
VG-250	3-1/C 250 MCM AWG CU XLP 5KV shielded cable primary w/o termination	LF	44.99	46.34
VG-500	3-1/C 500 MCM AWG CU XLP 5KV shielded cable primary w/o termination	LF	95.14	97.99
VH-G	3-1/C #2 AWG CU XLP 15KV shielded cable primary w/o termination	LF	22.74	23.42
VH-H	3-1/C #1 AWG CU XLP 15KV shielded cable primary w/o termination	LF	25.23	25.99
VH-I	3-1/C #1/0 AWG CU XLP 15KV shielded cable primary w/o termination	LF	29.44	30.32
VH-J	3-1/C #2/0 AWG CU XLP 15KV shielded cable primary w/o termination	LF	32.46	33.43
VH-L	3-1/C #4/0 AWG CU XLP 15KV shielded cable primary w/o termination	LF	42.35	43.62
VH-250	3-1/C 250 MCM AWG CU XLP 15KV shielded cable primary w/o termination	LF	47.26	48.68
VH-500	3-1/C 500 MCM AWG CU XLP 15KV shielded cable primary w/o termination	LF	70.06	72.16
VI-B47	Cable termination size no. 2 thru 4/0 0-15KV porcelain	EA	311.00	320.00
VI-47A	Cable termination size 250 MCM thru 500 MCM 0-15KV porcelain	EA	418.00	431.00
VI-47B	Cable termination size 500 MCM thru 750 MCM 0-15KV porcelain	EA	606.00	624.00
VL-012	Duct bank 1 way 2" grs duct w/exca & backfill does not incl pave remvl	LF	9.22	9.50
VL-013	Duct bank 1 way 3" grs duct w/exca & backfill does not incl pave remvl	LF	15.60	16.07
VL-014	Duct bank 1 way 4" grs duct w/exca & backfill does not incl pave remvl	LF	21.73	22.38
VL-015	Duct bank 1 way 5" grs duct w/exca & backfill does not incl pave remvl	LF	38.17	39.32
VL-016	Duct bank 1 way 6" grs duct w/exca & backfill does not incl pave remvl	LF	52.71	54.29
VL-022	Duct bank 2 way 2" grs duct w/exca & backfill does not incl pave remvl	LF	17.47	17.99
VL-023	Duct bank 2 way 3" grs duct w/exca & backfill does not incl pave remvl	LF	30.03	30.93
VL-024	Duct bank 2 way 4" grs duct w/exca & backfill does not incl pave remvl	LF	42.15	43.41

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
VL-025	Duct bank 2 way 5" grs duct w/exca & backfill does not incl pave remvl	LF	74.45	76.68
VL-026	Duct bank 2 way 6" grs duct w/exca & backfill does not incl pave remvl	LF	104.00	107.12
VL-032	Duct bank 3 way 2" grs duct w/exca & backfill does not incl pave remvl	LF	25.63	26.40
VL-033	Duct bank 3 way 3" grs duct w/exca & backfill does not incl pave remvl	LF	44.33	45.66
VL-034	Duct bank 3 way 4" grs duct w/exca & backfill does not incl pave remvl	LF	54.86	55.89
VL-035	Duct bank 3 way 5" grs duct w/exca & backfill does not incl pave remvl	LF	111.75	115.10
VL-036	Duct bank 3 way 6" grs duct w/exca & backfill does not incl pave remvl	LF	155.30	159.96
VL-042	Duct bank 4 way 2" grs duct w/exca & backfill does not incl pave remvl	LF	32.76	33.74
VL-043	Duct bank 4 way 3" grs duct w/exca & backfill does not incl pave remvl	LF	53.13	54.72
VL-044	Duct bank 4 way 4" grs duct w/exca & backfill does not incl pave remvl	LF	81.43	83.87
VL-045	Duct bank 4 way 5" grs duct w/exca & backfill does not incl pave remvl	LF	146.94	151.35
VL-046	Duct bank 4 way 6" grs duct w/exca & backfill does not incl pave remvl	LF	204.97	211.12
VN-012	Duct bank 1 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	4.52	4.66
VN-013	Duct bank 1 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	5.50	5.67
VN-014	Duct bank 1 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	6.78	6.98
VN-015	Duct bank 1 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	8.27	8.52
VN-016	Duct bank 1 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	9.96	10.26
VN-022	Duct bank 2 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	8.09	8.33
VN-023	Duct bank 2 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	9.83	10.12
VN-024	Duct bank 2 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	12.24	12.61

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
VN-025	Duct bank 2 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	14.99	15.44
VN-026	Duct bank 2 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	18.47	19.02
VN-032	Duct bank 3 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	8.29	8.54
VN-033	Duct bank 3 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	14.15	14.57
VN-034	Duct bank 3 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	17.73	18.26
VN-035	Duct bank 3 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	21.68	22.33
VN-036	Duct bank 3 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	27.00	27.81
VN-042	Duct bank 4 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	13.99	14.41
VN-043	Duct bank 4 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	17.18	17.70
VN-044	Duct bank 4 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	21.60	22.25
VN-045	Duct bank 4 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	26.90	27.71
VN-046	Duct bank 4 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	33.97	34.99
VN-062	Duct bank 6 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	19.99	20.59
VN-063	Duct bank 6 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	24.54	25.28
VN-064	Duct bank 6 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	31.05	31.98
VN-065	Duct bank 6 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	38.80	39.96
VN-066	Duct bank 6 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	49.53	51.02
VN-082	Duct bank 8 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	26.01	26.79
VN-083	Duct bank 8 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	31.13	32.06
VN-084	Duct bank 8 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	40.41	41.62

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($) FY 94	FY 95
VN-085	Duct bank 8 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	50.70	52.22
VN-086	Duct bank 8 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	64.99	66.94
VN-122	Duct bank 12 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	37.71	38.84
VN-123	Duct bank 12 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	46.48	47.87
VN-124	Duct bank 12 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	59.04	60.81
VN-125	Duct bank 12 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	74.27	76.50
VN-126	Duct bank 12 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	95.60	98.47
VN-162	Duct bank 16 way 2" PVC duct w/exca & backfill does not incl pave remvl	LF	49.50	50.99
VN-163	Duct bank 16 way 3" PVC duct w/exca & backfill does not incl pave remvl	LF	60.93	62.76
VN-164	Duct bank 16 way 4" PVC duct w/exca & backfill does not incl pave remvl	LF	77.66	79.99
VN-165	Duct bank 16 way 5" PVC duct w/exca & backfill does not incl pave remvl	LF	97.00	100.73
VN-166	Duct bank 16 way 6" PVC duct w/exca & backfill does not incl pave remvl	LF	126.08	129.86
VN-15	15KV feeder w/4 way 4" DB, 4/0 CP grd, 3-6'x8'x6'D MHS & 3-250 MCM cab w/spl	LF	75.31	77.57
VN-35	35KV feeder w/4 way 6" DB, 4/0 CP grd, 3-9'x12'x6'D MHS & 3-500 MCM cab w/spl	LF	493.92	508.74
513	ELECTRICAL SUBSTATION			
YC@C1	15KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	5,137.00	5,291.00
YC@C10	25KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	6,038.00	6,219.00
YC@C11	37.5KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	6,206.00	6,392.00
YC@C12	50KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	6,416.00	6,608.00

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
YC@C13	75KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	9,476.00	9,760.00
YC@C14	100KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	7,899.00	8,136.00
YC@C15	167KVA 1PH pad mtd xfmr w/conc pad & grd'g	EA	8,208.00	8,454.00
YC@C16	75KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	13,027.00	13,418.00
YC@C17	112KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	13,739.00	14,151.00
YC@C18	150KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	14,563.00	15,000.00
YC@C19	225KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	16,546.00	17,042.00
YC@C20	300KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	17,091.00	17,604.00
YC@C21	500KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	20,627.00	21,246.00
YC@C22	750KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	24,035.00	24,756.00
YC@C23	1000KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	27,414.00	28,236.00
YC@C24	1500KVA 3PH pad mtd xfmr w/conc pad & grd'g	EA	34,559.00	35,596.00
VU-1	1, 1PH 10KVA xfmr w/term cutouts arresters 2" rscriser 40' pole w/guy	EA	5,366.00	5,527.00
VU-15	1, 1PH 15KVA xfmr w/term cutouts arresters 2" rscriser 40' pole w/guy	EA	5,581.00	5,748.00
VU-25	1, 1PH 25KVA xfmr w/term cutouts arresters 2" rscriser 40' pole w/guy	EA	13,169.00	13,564.00
541 & 542 HEAT DISTRIBUTION				
DK-1AA	10" steam & 6" condensate heat distribution piping aboveground	LF	330.00	336.00
DK-3AA	8" steam & 4" condensate heat distribution piping aboveground	LF	260.00	265.00
DK-4AA	6" steam & 4" condensate heat distribution piping aboveground	LF	218.00	222.00

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($) FY 94	FY 95
DK-5AA	4" steam & 2.5" condensate heat distribution piping aboveground	LF	192.00	195.00
DK-7AA	3" steam & 2" condensate heat distribution piping aboveground	LF	162.00	165.00
DK-9AA	2.5" steam & 2" condensate heat distribution piping aboveground	LF	159.00	162.00
DK-92A	2" steam & 2" condensate heat distribution piping aboveground	LF	154.00	157.00
DK-94A	2" steam & 1" condensate heat distribution piping aboveground	LF	146.00	149.00
DK-97A	1" steam & 1" condensate heat distribution piping aboveground	LF	138.00	140.00
	For areas with unstable soil conditions increase unit prices by 20%.			
	551, 552, & 553 WATER DISTRIBUTION			
DF-DJ	2" bituminous paving for pipe trench w/12" base (incl removal)	SY	22.67	23.35
DC-EZ	6" conc pavement for pipe trench with 6" base (incl removal)	SY	89.31	91.99
DJ-LA	Fire hydrant standard 3 way post type w/6" valve and thrust block	EA	2,696.00	2,777.00
DJ-AE	4" 90 deg bend CI MJ	EA	179.00	184.00
DJ-AI	6" 90 deg bend CI MJ	EA	236.00	243.00
DJ-AM	8" 90 deg bend CI MJ	EA	282.00	290.00
DJ-AQ	10" 90 deg bend CI MJ	EA	380.00	391.00
DJ-AU	12" 90 deg bend CI MJ	EA	493.00	508.00
DJ-BA	4" PVC water pipe AWWA C-900 Class 150 w/excav backfill & compaction	LF	9.85	10.15
DJ-BB	6" PVC water pipe AWWA C-900 Class 150 w/excav backfill & compaction	LF	12.72	13.10
DJ-BC	8" PVC water pipe AWWA C-900 Class 150 w/excav backfill & compaction	LF	18.74	19.30
DJ-BD	10" PVC water pipe AWWA C-900 Class 150 w/excav backfill & compaction	LF	25.91	26.69
DJ-BE	12" PVC water pipe AWWA C-900 Class 150 w/excav backfill & compaction	LF	34.45	35.48
QL-JE	1" Type K copper pipe w/3' deep trenching, backfill & compaction	LF	8.42	8.67

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
QL-JM	1-1/2" Type K copper pipe w/3' deep trenching, backfill & compaction	LF	14.10	14.52
QL-KA	2" Type K copper pipe w/3' deep trenching, backfill & compaction	LF	19.67	20.26
DJ-LN	4" tapping saddle tap size to 2"	EA	113.00	116.00
DJ-LO	6" tapping saddle tap size to 2"	EA	133.00	137.00
DJ-LP	8" tapping saddle tap size to 2"	EA	148.00	152.00
DJ-LQ	10" tapping saddle tap size to 2"	EA	171.00	176.00
DJ-LR	12" tapping saddle tap size to 2"	EA	201.00	207.00
DJ-LT	4" x 4" tapping sleeve, 4" tapping valve & 4" tapped hole in pipe	EA	1,490.00	1,535.00
DJ-LV	6" x 6" tapping sleeve, 6" tapping valve & 6" tapped hole in pipe	EA	2,089.00	2,152.00
DJ-LY	8" x 8" tapping sleeve, 8" tapping valve & 8" tapped hole in pipe	EA	2,631.00	2,710.00
DJ-LYD	10" x 10" tapping sleeve, 10" tapping valve & 10" tapped hole in pipe	EA	4,202.00	4,328.00
DJ-LYI	12" x 12" tapping sleeve, 12" tapping valve & 12" tapped hole in pipe	EA	5,460.00	5,624.00
	Underground tanks for hazardous waste oil storage tanks increase cost by 120%			
	554 SANITARY SEWERS			
DP-KA	Precast 48" dia manhole 5' deep w/frame & cover (incl poured invert)	EA	2,213.00	2,279.00
DP-KB	Precast 48" dia manhole 8' deep w/frame & cover (incl poured invert)	EA	2,878.00	2,964.00
DF-DJ	2" bituminous paving for pipe trench w/12" base (incl removal)	SY	22.67	23.35
DC-EZ	6" conc paving for pipe trench with 6" base (incl removal)	SY	89.31	91.99
DP-EC	4" exterior sewer cleanout	EA	800.00	824.00
DP-ED	6" exterior sewer cleanout	EA	904.00	931.00
DP-EE	8" exterior sewer cleanout	EA	1,140.00	1,174.00
DP-DA	4" PVC sewer pipe ASTM 3034, SDR-35 w/excav, backfill & compaction	LF	7.06	7.27
DP-DB	6" PVC sewer pipe ASTM 3034, SDR-35 w/excav, backfill & compaction	LF	9.00	9.27

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
DP-DC	8" PVC sewer pipe ASTM 3034, SDR-35 w/excav, backfill & compaction	LF	11.59	11.94
DP-DD	10" PVC sewer pipe ASTM 3034, SDR-35 w/excav, backfill & compaction	LF	18.25	18.80
DP-DE	12" PVC sewer pipe ASTM 3034, SDR-35 w/excav, backfill & compaction	LF	22.67	23.35
561 & 562	FUEL STORAGE			
QV-BF	Underground oil storage tank 550 gal fiberglass double wall (incl excav)	GAL	21.73	22.38
QV-BG	Underground oil storage tank 1000 gal fiberglass double wall (incl excav)	GAL	16.01	16.49
QV-BJ	2000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	1.12	1.15
QV-BI	4000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	8.00	8.24
QV-BK	6000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	6.29	6.48
QV-BL	8000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	5.31	5.47
QV-BLA	10000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	5.52	5.69
QV-BM	12000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	5.57	5.74
QV-BN	15000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	5.58	5.75
QV-BO	20000 gal underground fiberglass fuel oil tank double wall (incl excav)	GAL	5.22	5.38
71X	ROADS			
BW-AH	Stone aggregate base 6" thick	SY	4.97	5.12
BW-AJ	Stone aggregate base 8" thick	SY	6.79	6.99
DF-DD	Asphalt paving 1-1/2" overlay	SY	5.77	5.94
DF-DE	Asphalt paving 2" overlay	SY	7.53	7.76
DB-BD	Single bituminous surface treatment	SY	1.49	1.53
DB-BE	Double bituminous surface treatment	SY	1.97	2.03
DF-FF	Slurry seal	SY	3.82	3.93

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
DC-EC	Conc paving non-reinf 5000 psi 6 inch thick	SY	41.51	42.76
DC-EE	Conc paving non-reinf 5000 psi 8 inch thick	SY	55.97	57.65
DC-EG	Conc paving non-reinf 5000 psi 10 inch thick	SY	63.30	65.20
DC-EI	Conc paving non-reinf 5000 psi 12 inch thick	SY	70.61	72.73
DF-AA	Painting 4" lines on pavement	LF	0.14	0.14
DF-EA	Inlaid pavement lane line markings 4" wide	LF	1.41	1.45
EA-C	Concrete curb 6x18 cast-in-place	LF	9.11	9.38
EA-E	Concrete curb & gutter cast-in-place	LF	12.77	13.15
EA-F	Concrete valley gutter cast-in-place	SY	36.94	38.05
72X	PARKING			
BW-AH	Stone aggregate base 6" thick	SY	4.97	5.12
BW-AJ	Stone aggregate base 8" thick	SY	6.79	6.99
DF-DD	Asphalt paving 1-1/2" overlay	SY	5.77	5.94
DF-DE	Asphalt paving 2" overlay	SY	7.53	7.76
DB-BD	Single bituminous surface treatment	SY	1.49	1.53
DB-BE	Double bituminous surface treatment	SY	1.97	2.03
DF-FF	Slurry seal	SY	3.82	3.93
DF-AA	Painting 4" lines on pavement	LF	0.14	0.14
DF-EA	Inlaid pavement lane line markings 4" wide	LF	1.41	1.45
EA-C	Concrete curb 6x18 cast-in-place	LF	9.11	9.38
EA-E	Concrete curb & gutter cast-in-place	LF	12.77	13.15
EA-F	Conc valley gutter cast-in-place	SY	36.94	38.05
EA-G	Precast concrete parking bumpers 6"x10"x6' incl dowels	EA	36.11	37.19
74X	SIDEWALKS			
EA-D	Conc sidewalks 4 inch cast-in-place no base	SY	28.68	29.54

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
75X & 76X AIRCRAFT PAVING				
BW-AH	Stone aggregate base 6" thick	SY	4.97	5.12
BW-AJ	Stone aggregate base 8" thick	SY	6.79	6.99
DF-DD	Asphalt paving 1-1/2" overlay	SY	5.77	5.94
DF-DE	Asphalt paving 2" overlay	SY	7.53	7.76
DB-BD	Single bituminous surface treatment	SY	1.49	1.53
DB-BE	Double bituminous surface treatment	SY	1.97	2.03
DF-FF	Slurry seal	SY	3.82	3.93
DC-EC	Conc paving non-reinf 5000 psi 6 inch thick	SY	41.51	42.76
DC-EE	Conc paving non-reinf 5000 psi 8 inch thick	SY	55.97	57.65
DC-EG	Conc paving non-reinf 5000 psi 10 inch thick	SY	63.30	65.20
DC-EI	Conc paving non-reinf 5000 psi 12 inch thick	SY	70.61	72.73
DF-AA	Painting 4" lines on pavement	LF	0.14	0.14
DF-EA	Inlaid pavement lane line markings 4" wide	LF	1.41	1.45
812 BORROW				
BG-BC	Borrow fill in place at Sewells Point Area from Willoughby borrow pit	CY	10.17	10.48
BG-BL	Borrow fill in place at Cherry Pt/Camp Lejeunefrom station borrow pit	CY	10.17	10.48
BG-BJ	Borrow fill in place from commercial Tidewater pit 30 mile round trip	CY	14.82	15.26
BG-BK	Borrow fill in place from commercial Tidewater pit 20 miles round trip	CY	12.49	17.30
813 TOPSOIL SEED				
DV-AA	4" topsoil from piles on site w/seed, fertilize lime, & fine grade	SY	3.00	3.09

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
84X	DEMOLITION			
BE-BB	Clear & grub medium dense areas w/disposal	AC	3,659.00	3,769.00
BB-EA	Demolition of masonry building complete w/disposal	CF	0.15	0.15
BB-EB	Demolition of wood frame building complete w/disposal	CF	0.15	0.15
BB-EE	Demolition of steel frame building complete w/disposal	CF	0.21	0.22
BC-AB	Removal of bituminous paving up to 3" thick w/disposal	SY	4.93	5.08
BC-AD	Removal of 6" thick non-reinforced concrete pavement w/disposal	SY	9.93	10.23
BC-AJ	Removal of 8" thick non-reinforced concrete pavement w/disposal	SY	22.61	23.29
BC-JA	Removal of 11" thick non-reinforced concrete pavement w/disposal	SY	33.95	34.97
BC-AL	Removal of conc sidewalk 4" thick w/disposal	SY	8.88	9.15
BC-BF	Removal of concrete curb & gutter	LF	3.14	3.23
85X	STORM DRAINAGE			
CG-BA	12" RCP Class III storm drainage 4 ft deep w/excav, backfill & compact	LF	14.66	15.10
CG-BB	15" RCP Class III storm drainage 4 ft deep w/excav, backfill & compaction	LF	19.05	19.62
CG-BC	18" RCP Class III storm drainage 5 ft deep w/excav, backfill & compaction	LF	25.48	26.24
CG-BE	24" RCP Class III storm drainage 5 ft deep w/excav, backfill & compaction	LF	35.62	36.69
CG-BG	30" RCP Class III storm drainage 5 ft deep w/excav, backfill & compaction	LF	49.10	50.57
CG-BH	36" RCP Class III storm drainage 5 ft deep w/excav, backfill & compaction	LF	64.82	66.76
CG-MW	Precast 48" dia manhole 5' deep w/frame & cover	EA	2,328.00	2,398.00
CG-OA	Precast 48" dia manhole 8' deep w/frame & cover	EA	2,847.00	2,932.00
CG-MN	Standard catch basin 5 ft deep	EA	1,656.00	1,706.00

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Table 14 (Continued)
Supporting Facilities Guidance Cost (1 July 1991)

PRICE INCLUDES CONTRACTOR OVERHEAD AND PROFIT ESCALATED TO APRIL 1 / ACF = 1.0

CODE	DESCRIPTION	U/M	TOTAL COST ($)	
			FY 94	FY 95
CG-OH	Curb inlet max depth 8' 4' throat width, 12" to 30 " pipe w/excav & backfill	EA	2,134.00	2,198.00
DF-DJ	2" bituminous paving for pipe trench w/12" base (incl removal)	SY	22.67	23.35
DC-EZ	6" concrete paving for pipe trench w/6" base (incl removal)	SY	89.31	91.99
86X	FENCING			
CI-AC	Chain link fence 9 ga galv steel 4' high w/3 strand barbwire	LF	17.30	17.82
CI-AE	Chain link fence 9 ga galv steel 6' high w/3 strand barbwire	LF	19.81	20.40
CI-AF	Chain link fence 9 ga galv steel 7' high w/3 strand barbwire	LF	21.48	22.12
CI-AG	Chain link fence 9 ga galv steel 8' high w/3 strand barbwire	LF	22.80	23.48
CI-CC	Gate chain link 9 ga galv steel 4' high w/gate posts & barbwire	LF	57.62	59.35
CI-CE	Gate chain link 9 ga galv steel 6' high w/gate posts & barbwire	LF	70.77	72.89
CI-CF	Gate chain link 9 ga galv steel 7' high w/gate posts & barbwire	LF	76.61	78.91
CI-CG	Gate chain link 9 ga galv steel 8' high w/gate posts & barbwire	LF	108.40	110.44
911, 912, & 913	LAND PILING			
CB-BC	Treated wood piling 40' to 50' long 12" butt 7" tip 12 lb creosote	LF	10.15	10.45
CD-AB	Piling prestressed conc 10" sq, more than 60 LF long, truck delivery	LF	12.42	12.79
CD-CB	Piling prestressed conc 12" sq, more than 60 LF long, truck delivery	LF	15.41	15.87

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Section 14: COST CERTIFICATION SAMPLE ESTIMATE

14.1 DD Form 1391. Figure 19 shows the DD Form 1391 segment of the cost certification estimate.

14.2 Estimate Summary Sheet. Figure 20 shows the budget estimate summaries for the cost certification estimate.

14.3 Primary Facility Unit Cost Development. See Figure 21.

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1. COMPONENT	FY 1994 MILITARY CONSTRUCTION PROJECT DATA	2. DATE:
NAVY		01 OCT 91

3. INSTALLATION AND LOCATION	4. PROJECT TITLE
NAS, OCEANA, VIRGINIA BEACH, VIRGINIA	WEAPONS SYSTEM TRAINER BUILDING

5. PROGRAM ELEMENT	6. CAT. CODE	7. PROJECT NUMBER	8. PROJECT COST ($000)
	171-35	P- 000	3,500

9. COST ESTIMATES

A.C.F. .92
F.E.R. N/A

ITEM (DATE ESCALATED TO: 01 APR 94)	U/M	QUANTITY	UNIT COST	COST ($000)
PRIMARY FACILITY	SF	25,480	105.57	2,690
WEAPONS SYSTEM TRAINER	SF	25,480	98.37	(2,506)
ADDITIONAL FUNCTIONAL FEATURES	LS	1	184000	(184)
SUPPORTING FACILITIES	LS			438)
SPECIAL FOUNDATION FEATURES	LS			(105)
ELECTRICAL UTILITIES	LS			(181)
MECHANICAL UTILITIES	LS			(57)
ROADS,PARKING,SIDEWALKS	LS			(22)
SITE IMPROVEMENTS	LS			(73)
SUBTOTAL				3,128
CONTINGENCIES (5%)				156
TOTAL CONTRACT COST				3,284
SUPERVISION, INSPECTION, OVERHEAD (6%)				197
TOTAL REQUEST				3,481
TOTAL REQUEST (ROUNDED)				3,500
EQUIPMENT PROVIDED FROM OTHER APPROPRIATIONS (NON ADD)				

10. DESCRIPTION OF PROPOSED CONSTRUCTION

THE TWO STORY PRIMARY FACILITY IS A 25,480 SF STEEL FRAME AND MASONRY BUILDING. THE PILE FOUNDATION WILL SUPPORT CONCRETE PILE CAPS AND GRADE BEAMS. THE ROOF WILL BE METAL DECK WITH INSULATION ON METAL BAR JOIST FOR THE MAJORITY OF THE ROOF; HOWEVER, THE TRAINER, MECHANICAL, ELECTRICAL, AND HYDRAULICS AREAS WILL HAVE CONCRETE OVER THE METAL DECK. THE ROOFING SYSTEM WILL BE A SINGLE-PLY MEMBRANE. THE EXTERIOR WALLS WILL BE EXTERIOR FINISHING SYSTEM OVER MASONRY, INTERIOR WALLS TO BE MASONRY. RAISED FLOOR SYSTEMS ON FIRST AND SECOND FLOOR FOR COMPUTER/EQUIPMENT AREAS. THIS BUILDING WILL PROVIDE FOR FUTURE FIRST AND SECOND FLOOR EXPANSIONS. THERE ARE REQUIREMENTS OF OMSI, IDS, FIRE SPRINKLER, FIRE ALARM, AND SECURITY ALARM SYSTEMS. THE SUPPORTING SYSTEMS INCLUDES; PILE FOUNDATIONS, MECHANICAL, AND ELECTRICAL DISTRIBUTION SYSTEMS, AREA LIGHTING, PARKING AREAS, SIDEWALKS, AND LANDSCAPING.
(AIR CONDITIONING: 125 TONS)

11. REQUIREMENT: 141,400 SF ADEQUATE: 38381 SF SUBSTANDARD: 17,878 SF

PROJECT:
PROVIDES AN APPLIED INSTRUCTION BUILDING TO HOUSE ONE WEAPONS SYSTEM TRAINER USED TO TRAIN PILOTS AND FLIGHT OFFICERS. THIS PROJECT SUPPORTS THE IMPROVED F-14 AIRCRAFT. (NEW MISSION)

REQUIREMENT:
A BUILDING TO HOUSE WEAPONS SYSTEMS TRAINER SCHEDULED FOR DELIVERY BEGINNING IN 1994. OCEANA IS THE HOMEPORT FOR ALL ATLANTIC FLEET F-14A FIGHTER AIRCRAFT. THE F-14 IS THE LINCH-PIN OF THE CARRIER BATTLE GROUP'S AIR DEFENSE. THE F-14A WAS INTRODUCED IN 1972 AND HAS PROVEN TO BE A VERY EFFECTIVE AND POTENT WEAPON.

DD FORM 1391 FILE: ESTRELLA

Figure 19
DD Form 1391 for Cost Certification Estimate

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BUDGET ESTIMATE SUMMARY SHEET

P. NO. : P-000 FY 94

TITLE: WEAPONS SYSTEM TRAINER BUILDING DATE OF ESTIMATE: 01 OCT 91

LOCATION: NAS, OCEANA, VIRGINIA BEACH, VIRGINIA DESIGN STATUS (%): COST CERT

PREPARED BY: R. O. ESTRELLA DATE ESCALATED TO: 01 APR 94

A.C.F.: .92

DESCRIPTION	UM	QUANTITY	UNIT COST	ESCALATED COST ROUNDED	COST TRANSFRD TO 1391
PRIMARY FACILITY	SF	25,480	105.57	2,690,000	2,690,000
WEAPONS SYSTEM TRAINER	SF	25,480	98.37	2,506,000	
ADDITIONAL FUNCTIONAL FEATURES	LS	1	184,000.00	184,000	
SUPPORTING FACILITIES					438,000
SPECIAL FOUNDATION FEATURES	LS	------	------	------	(105,000)
911 LAND PILING	LF	7,360	14.28	105,000	
ELECTRICAL UTILITIES	LS	------	------	------	(181,000)
511 ELECTRICAL DISTRIBUTION, PRIMARY	LF	748	111.76	84,000	
512 ELECTRICAL DISTRIBUTION, SECONDARY	LF	75	60.10	5,000	
513 ELECTRICAL SUBSTATION	KVA	750	121.26	91,000	
523 COMMUNICATION, TELEPHONE	LF	110	10.58	1,000	
MECHANICAL UTILITIES	LS	------	------	------	(57,000)
541 STEAM DISTRIBUTION	LF	150	247.74	37,000	
551 WATER DISTRIBUTION	LF	20	220.94	4,000	
554 SANITARY SEWER	LF	440	35.72	16,000	
ROADS,PARKING,SIDEWALKS, TAXIWAY, AND APRON	LS	------	------	------	(22,000)
555 CONCRETE TRENCHWAY	LF	135	98.84	13,000	
721 CONCRETE PARKING	SY	80	77.17	6,000	
741 CONCRETE SIDEWALKS	SY	180	16.10	3,000	
SITE IMPROVEMENTS	LS	------	------	------	(73,000)
811 EARTHWORK	CY	1,800	7.41	13,000	
812 BORROW	CY	2,500	10.42	26,000	
813 TOPSOIL/SEED/SOD	SY	600	8.55	5,000	
814 LANDSCAPING	SY	270.00	24.66	7,000	
818 ENVIRONMENTAL PROTECTION	ACRE	2.00	2,000.00	4,000	
851 STORM DRAINAGE	LF	800	22.18	18,000	
			TOTAL CONTRACT COST W/O CONTINGENCY		3128000.00
			CONTINGENCY (5%)		156000.00
			TOTAL CONTRACT COST		3284000.00
			SIOH (6.0%)		197000.00
			TOTAL BUDGET COST		3481000.00
			TOTAL BUDGET COST ROUNDED		3500000.00

Figure 20
Building Budget Estimate Summary Sheet

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PRIMARY FACILITY COST DEVELOPMENT

TITLE: WEAPONS SYSTEM TRAINER BUILDING P#: 000

LOCATION: NAVAL AIR STATION, OCEANA, VIRGINIA BEACH, VIRGINIA DATE: 01 OCT 91

PREPARED BY: R. O. ESTRELLA PH#: (804)444-9776

COST COMPARISON	CAT CODE	SIZE (SF)	ACF	TIME -mon-	TIME ---yr---
PROPOSED FACILITY	171-35	25,480	0.92	4	94
GUIDANCE FACILITY	171-35	22,000	1.00	4	94

HISTORICAL GUIDANCE SOURCE: NAVFACENGCOM GUIDANCE DATED JULY 1991

HISTORICAL GUIDANCE UNIT COST: $108.00 / S.F.

HISTORICAL UNIT COST		SIZE FACTOR		AREA COST FACTOR		ESCALATION FACTOR		ADJUSTED UNIT COST
$108.00	X	0.99	X	0.92	X	1.00	=	$98.37

ADDITIONAL FUNCTION FEATURES
(Additions/Deletions to adjusted unit costs - attach backup for each feature)

UNUSUAL FEATURES	QUANTITY	UNIT COST	TOTAL COST	ADJUSTED U/C	% CHANGE
OMSI (A - 1.2%)	$2,506,467.00	0.012	$30,078	1.18	1.20
FUTURE 2ND FLOOR-STRUCTURAL	6800.00	5.45	$37,060	1.45	1.47
EXTRA P-3 PILE CAPS FOR FUTURE ADDIT.	9.00	850.00	$7,650	0.30	0.30
M.G. POWER CONDITIONER	1.00	87000.00	$87,000	3.41	3.47
TEMPORARY PARTITION WALL	667.00	34.00	$22,678	0.89	0.90
UNUSUAL FEATURES TOTAL:			184,466	7.23	7.35

PRIMARY FACILTIY UNIT COST: $105.60 / S.F.

Figure 21
Primary Facility Unit Cost Development

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Section 15: COST MODEL QUESTIONNAIRE

15.1 Cost Model Questionnaire. The Cost Model Questionnaire (see Figure 22) serves as a checklist for the identification of project components (CES systems). The project scope is a summation of the applicable systems. Project unique features may require non-typical systems or affect the cost of typical systems. This data should be noted by the preparer. Pricing of the stated scope, by systems, requires pricing criteria for each system. The questionnaire requests that minimum pricing criteria for each system be entered. The data entered should be considered, at that time, to be the most probable pricing criteria. Emphasize scope features that are uncommon or high cost. This questionnaire, coordinated by the A&E and completed by the activity, must be reviewed by the design team prior to initial scoping meetings. This effort will generate early collection of information by the activity and will be a basis for the A&E prepared questionnaires. Questionnaires for A&E submittals to LANTDIV should define scope and should support the estimate.

15.2 CES Systems List

15.2.1 Interior. The interior systems are shown in the following list.

0XX General
000 Building Construction
00X Overhead
001 Field Overhead
002 Home Office OH
01X Additive Bid Items
02X Deductive Bid Items
03X Base Bids
04X Government Cost
041 OMSI
042 PCAS
043 Contingencies
044 SIOH

1XX General Construction
11X Structural
111 Foundation (SF)
112 Slab on Grade (SF)
113 Structural (SF)
114 Supported Floor (SF)
115 Stairs (EA)
116 Roof Structure (SF)
117 Pre-Engd Building (SF)
118 Vault (SF)
119 Write-in
13X Structural Write-in
14X Architectural
141 Roofing (SF)
142 Exterior Walls (SF)
143 Interior Walls (SF)
144 Interior Finishes (SF)
145 Doors (SF)
146 Windows (SF)
147 Specialties (SF)
148 BEQ Wardrobes (EA)
149 Write-in
16X Architectural Write-in
17X Demolition Interior
171 Demo Interior (SF)
172 Asbestos Removal Interior (LS)
173 - 175 Write-in

2XX Interior Mechanical
21X Plumbing Domestic
211 Plumbing, domestic (EA)
212 Domestic Hot Water Solar (MBTU)
213 Roof Drainage (EA)
214 - 216 Write-in
22X Space Conditioning
221 HVAC (MBTU/TNS)

(1) Built-in Equipment

Source: https://assist.dla.mil -- Downloaded: 202610-26T13:54Z
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222 Air Conditioning (TNS)
223 Heating, Ventilation (MBTU/CFM)
224 Mechanical Ventilation (CFM)
225 Dehumidification (PPH)
226 Heating Solar (SF)
227 Write-in

23X Weight Handling Equipment
231 Bridge Cranes (TNS)(1)
232 Monorails (TNS)(1)
233 Fixed Hoist (TNS)(1)
234 Vehicle Lifts (TNS)(1)
235 Elevators (TNS)(1)
236 Escalators (TNS)(1)
237 Conveyors (TNS)(1)
238 Write-in

24X Special Mechanical Piping System
241 Vacuum (CFM)(1)
242 Oxygen (EA)(1)
243 Nitrogen (EA)(1)
244 Compressed Air (CFM)(1)
245 Interior Steam (PPH)(1)
246 - 247 Write-in

25X Special Mechanical, Other
251 Dust Collection (CFM)(1)
252 Engine Exhaust (CFM)(1)
253 - 255 Write-in

27X Interior Fire Protection
271 Sprinklers (SF)
272 AFFF (SF)
273 Carbon Dioxide (LBS)
274 Halon (LBS)
275 Fire Alarm (SF)
276 - 277 Write-in

3XX Interior Electrical

31X Power & Lighting
311 Power (KW)
312 Lighting (SF)
313 400HZ (KW)(1)
314 Direct Current (KW)(1)
315 UPS (KVA)(1)
316 Electrical Generators (KW)(1)
317 - 318 Write-in

32X Special Equipment Protection
321 Grounding (SF)(1)
322 Lightning Protection (SF)(1)

33X Special Electrical Electronics
331 Security Detection
332 EMCS (EA)(1)
333 Computer Communication
334 - 336 Write-in

34X Communication
341 Telephone (SF)
342 Intercom (SF)
343 Television (SF)
344 Public Announcement (SF)
345 Master Clock (EA)
346 - 347 Write-in

35X Write-in

4XX Special Equipment

41X Medical Equipment
411 Hospital Equip (SF) (1)
412 Dental Equip (EA)(1)
413 - 414 Write-in
415 Install Gov't Furnished Medical Equip
416 Gov't Furn Med Equip

42X Food Service Equipment
421 Food Service Equip (SF)
422 - 424 Write-in
425 Install Gov't Furnished Food Service Equip
426 Gov't Furnished Food Service Equip

43X Miscellaneous Equipment
431 Chapel Equip (SF)(1)
432 Movie Theater Equip (SF)(1)
433 Rifle Range Equip (EA)(1)
434 Lab Equip (EA)(1)
435 Install Gov't Furnished Lab Equip
436 Gov't Furnished Lab Equip
437 Waste Disposal (PPH)(1)
438 Write-in

44X Special Warehouse Equipment
441 Special Warehouse Equip (SF)(1)
442 - 444 Write-in
445 Install Gov't Furnished Warehouse Equip
446 Gov't Furnished Warehouse Equip

45X Special Construction

(1) Built-in Equipment

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451 Hangar Doors (SF)
452 Vault Doors (EA)
453 Blast Doors (SF)
454 Raised Floor (SF)*
455 Radio Frequency Shielding (SF)(1)
456 Bowling Lanes (EA)
457 Paint Spray Booth (SF)(1)
458 Write-in

49X Other Equipment
491 Other Miscellaneous Equip
492 Snow Melting (SF)
495 Install Other Gov't Furnished Equip
496 Other Gov't Furnished Equip (SF)(1)
499 Building Work Not Included Elsewhere (LS)

(1) Built-in Equipment

15.2.2 Exterior. The exterior systems are shown in the following lists.

5XX Exterior Utilities

51X Exterior Electrical
511 Electrical Distribution Primary (LF)
512 Electrical Distribution Secondary (LF)
513 Substation/ Transformer (KVA)
514 Area Lighting (EA)
515 Airfield Lighting (LF)
516 Lightning Protection (EA)
517 - 518 Write-in

52X Exterior Communication
521 Fire Alarm (LF)
522 Security Alarm (LF)
523 Communication, Phone (LF)
524 Exterior EMCS (LF)
525 - 526 Write-in

54X Exterior Mechanical
541 Heat Distribution Overhead (LF)
542 Heat Distribution Underground (LF)
543 Chilled Water Dist (LF)
544 Condensate Collect (LF)
545 Gas Distribution (LF)
546 Compressed Air Dist (LF)
547 Fuel Distribution (LF)
548 Exterior Foam (LF)
549 Write-in

55X Exterior Mechanical
551 Water Distribution (LF)
552 Fire Protection Water Distribution (LF)
553 Salt Water Distribution (LF)
554 Sanitary Sewers (LF)
555 - 556 Write-in

56X Bulk Storage
561 Fuel Storage Bulk (BBL)
562 Fuel Storage Ready Issue (GAL)
563 Water Storage (GAL)
564 - 566 Write-in

57X Fueling
571 Vehicle Fueling (EA)
572 Aircraft Fueling (EA)
573 Marine Fueling (EA)
574 - 576 Write-in

58X Pumping
581 Fuel Pumping Station (GPM)
582 Water Pumping Station (GPM)
583 Fire Booster Pump (GPM)
584 Sewage Pump Station (GPM)

(1) Built-in Equipment

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585 Sewage Lift Station (GPM)
586 Write-in

59X Treatment Facility
591 Water Treatment (MGD)
592 Domestic Sewage (MGD)
593 Industrial Waste (MGD)
594 - 596 Write-in

6XX Exterior Structures
61X Equipment Building
611 Mechanical Equip Building (SF)
612 Electrical Equip Building (SF)
613 - 615 Write-in

62X Supporting Structures Miscellaneous
621 Security Gate (LF)
622 Guard House (SF)
623 - 625 Write-in

64X Pollution Abatement Structures
641 Oil Water Separators (GAL)
642 Electrostatic Precipitator (SCFM)
643 - 645 Write-in

65X Shop Support Structures
651 Vehicle Grease Racks (EA)
652 Vehicle Wash Platforms (EA)
653 - 656 Write-in

66X AC Cleaning Facility
661 AC Washing Facility (SY)
662 AC Rinsing Facility (SY)
663 - 666 Write-in

7XX Paving
71X Roads
711 Concrete Roads (SY)
712 Flexible Roads (SY)
713 Overlay Roads (SY)
714 Surface Treatment (SY)
715 Slurry Seal (SY)
716 - 717 Write-in

72X Parking
721 Concrete (SY)
722 Flexible (SY)
723 Overlay (SY)
724 Surface Treatment (SF)
725 Slurry Seal (SY)
726 - 727 Write-in

73X Write-in

74X Walks
741 Concrete Walks (SY)
742 Bituminous Walks (SY)
743 Special Walks (SY)
744 - 746 Write-in

75X Aircraft Paving Concrete
751 Runways (SY)
752 Taxiways (SY)
753 Aprons (SY)
753 Aprons (SY)
754 Repairs Paving (SY)
755 Crack and Joint Repairs (LF)
756 - 757 Write-in

76X Aircraft Paving Flexible
761 Flexible Runways (SY)
762 Flexible Taxiways (SY)
763 Flexible Aprons (SY)
764 Repair A/C Flexible Paving (SY)
765 AC Paving Overlay (SY)
766 AC Paving Slurry Seal (SY)
767 - 768 Write-in

77X Incidental Paving Work
771 Pavement Milling (SY)
772 - 776 Write-in

8XX Site Preparation
81X Earthwork
811 Earthwork (CY)
812 Borrow (CY)
813 Topsoil/Seed/Sod (SY)
814 Landscaping (SY)

(1) Built-in Equipment

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Check the source to verify that this is the current version before use.

815 Site Irrigation (EA)
816 Surcharge (CY)
817 Earth Replacement (CY)
818 Environmental Protection (AC)
819 Write-in

82X Site Improvements
821 Site Improvements (SY)
822 - 825 Write-in

83X Site Preparation Miscellaneous
831 Site Dewatering (Major) (LF)
832 - 835 Write-in

84X Demolition
841 Remove Utilities (LF)
842 Remove Paving and Slabs (SY)
843 Remove Structures (CF)
844 Remove/Dispose of Asbestos (Exterior) (LS)
845 Remove/Dispose of P.C.B. (LS)
846 Remove/Dispose of Contaminated Earth (CY)
847 Write-in

85X Storm Drainage
851 Storm Drainage Piping (LF)
852 Box Arch Culvert (LF)
853 Drainage Facing Materials (SY)
854 - 856 Write-in

86X Fencing
861 Fencing (Perimeter) (LF)
862 Fencing Alarm (LF)
863 - 865 Write-in

87X Recreation Equipment/Fields
871 Playground Equip (EA)
872 Playing Field Equip (EA)
873 Tennis Courts (EA)
874 Softball/Baseball (EA)
875 Football/Soccer (EA)
876 - 877 Write-in

9XX Heavy Construction

91X Special Building Foundations
911 Piling - Land (LF)
912 Caissons (EA)
913 Pressure Injected Footings (EA)
914 Vibrofloatation (EA)
915 Chemical Soil Stabilization (EA)
916 Dynamic Consolidation (EA)
917 Piling Cast-in-place (LF)

92X Railroad
921 Track Work Railroad (LF)
922 - 925 Write-in

93X Marine Structures Piers
931 Piling-Marine (LF)
932 Dolphins (EA)
933 Med Mooring (LS)
934 Pier Deck (SF)

(1) Built-in Equipment

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935 Fenders (LF)
936 Pile Clusters (EA)
937 Marine Specialties (LS)
938 Write-in

94X Marine Structures Special
941 Bulkhead/Seawall (SF)
942 Erosion Protection-Marine (SY)
943 Riprap (Marine) (SY)
944 - 946 Write-in

95X Marine Repairs
951 Underdeck Repairs (LS)
952 Pile Repairs Marine (LF)
953 - 957 Write-in

96X Dredging
961 Mob/Demob Dredging (LS)
962 Hydraulic Dredging (CY)
963 Bucket Dredging (CY)
964 - 966 Write-in

97X Towers/Antennas
971 Towers (LF)
972 Antennas (LS)
973 - 976 Write-in

98X Tunnel/Pipe Jacking
981 Tunneling (LS)
982 Pipe Jacking (LF)
983 - 986 Write-in

99X Heavy Construction Miscellaneous
991 Bridges (SF)
992 Overpasses (SF)
993 - 996 Write-in
999 Supporting Facilities

(1) Built-in Equipment

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Project Title _____

Location _____

Contract Number _____

P# _____ FY _____ Date _____

Completed by _____

Phone Number _____

COST ENGINEERING SYSTEM (CES)
COST MODEL QUESTIONNAIRE

APRIL 1991

000 Building Construction Gross Building Area (SF) _____

Function or Category Code	% or Area
_____	_____
_____	_____
_____	_____

Where a single building includes areas of significant cost variation as identified by category code or functional area, show a breakdown by function or category code. Show area in sq. ft. or by percentage to nearest 5%.

20	Number of Floors	_____ EA
21	Floor to Floor Height	_____ LF
22	Length - Nearest Foot	_____ LF
23	Width - Nearest Foot	_____ LF
25	Irregular - Yes/No	_____
26	Building Perimeter	_____ LF

40 Describe any known special considerations; scheduling, security, asbestos removal, hazardous materials, etc.

50 Finish floor elevation above existing grade - nearest foot _____ LF

Figure 22
Cost Model Questionnaire

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000 Building Construction Summary

Includes the total cost of all systems 001 thru 499. Total building cost to the 5-foot line. This system is not to be used for estimate preparation.

OVERHEAD

001 Field Overhead Total Cost Lump Sum

Includes Project Manager, Superintendent, Timekeeper, Field Engineer, Clerical Staff, Field Offices, Utilities, Field Storage, and any other field incurred cost necessary to administer the project in the field. For projects outside the United States includes shipping, transportation of personnel per diem and temporary housing.

002 Home Office Overhead Total Cost Lump Sum

Includes the portion of Home office costs allocated to the project being priced.

003 Contractor Quality Control Total Cost Lump Sum

Includes costs for the Field Contractor Quality Control Staff. Testing requirements are included with other systems as applicable.

*011 Additive 1 Total Cost Lump Sum

Includes all costs associated with Additive Item 1 as described in the specifications. When an additive is a separate building or Involves substantial work it may be advisable to run a separate estimate to develop the cost and transfer only the total to the master estimate.

*Typical for all Additive and Deductive items.

041 OMSI Total Cost Lump Sum

Includes costs associated with preparation of an Operation and Maintenance Service Instruction.

042 PCAS Percent Lump Sum

Includes costs associated with Post Contract Award Services.

043 Contingencies Percent Lump Sum

Includes contingency allowance. The contingency entered is not anestimating contingency but a contingency for change orders.

044 SIOH Percent Lump Sum

Includes an allowance for government administration of the construction contract. Supervision, Inspection, and Overhead.

Figure 22 (Continued)
Cost Model Questionnaire

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BUILDING SYSTEMS INTERIOR

GENERAL CONSTRUCTION

111 Foundation — Ground Floor — Sq. Ft.

Includes excavation and backfill for foundation and basement construction, pile caps, footings, grade beams, piers, foundation walls, basement walls, fill under floor slabs, and all required construction to the first floor elevation, excluding all structural floor slabs, ground slabs, basement structural framing, piling, special foundations and soil treatment. See 84X for removing unsuitable materials and 816 for surcharging.

	Footing Depth from Existing Grade	
_01 Spread Footings	_11	_21 Borrowfill 1'
_02 Thickened Slab	_12	_22 Borrowfill 2'
_03 Pile Foundation	_13	_23 Borrowfill 3'
_04 Caissons	_14	_24 Borrowfill 4'
_05 Continuous wall footing	_15	_25 Borrowfill 5'
_06 Grade Beams		_26 Borrowfill over 5'

***** For convenience pilings are described with building foundation.
***** Price under the supporting facilities.

911 Land Piling-Driven — Length of Piling — Lin. Ft.

	Length of Piling
_02 Timber	_20 Under 25'
_03	_21 26' - 35'
_04 Concrete 10" Sq	_22 36' - 45'
_05 Concrete 12" Sq	_23 46' - 55'
_06 Concrete 14" Sq	_24 56' - 65'
_07	_25 66' - 75'
_08 Steel	_26 76' - 85'
	_27 86' - 95'
	_28 Over 95'

112 Slab on Grade — Slab on Grade — Sq. Ft.

Includes all ground slabs and vapor barrier, waterproofing, wire mesh, capillary fill, and soil treatment. Includes ground slab, reinforcing steel, waterproofing and soil treatment for structural slab placed on fill where fill is used as form. Borrow fill under slab is included in System 111.

		Thickness Floorload 30 Under 100 PSF
_01 Floating	_11 Under 6"	
_02 Grade Beam Supported	_12 6"	_30 Under 100 PSF
_03 Pile Supported	_13 8"	_31 101-200 PSF
	_14 Over 8"	_32 201-300 PSF
		_33 301-400 PSF
		_34 401-500 PSF
		_35 Over 501 PSF

Figure 22 (Continued)
Cost Model Questionnaire

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113 Structural — Gross Bldg. Area — Sq. Ft.

Includes structural frame consisting of; skeleton frame of building, i.e., columns, girders, cantilevered members extending beyond exterior walls, and fireproofing. Excludes framing in direct support of floor or roof construction.

_01 Bearing Wall
_02 Steel Framing
_03 Concrete Cast in Place
_04 Wood
_05
_06 Concrete Precast
_07 Air Inflatable

_10 Seismic Zone 0
_11 Seismic Zone 1
_12 Seismic Zone 2
_13 Seismic Zone 3
_14 Seismic Zone 4

_21 Windload 80 mph
_22 Windload 92 mph
_23 Windload 103 mph
_24 Windload 115 mph
_25 Windload 127 mph

_32 Roofload 20 PSF
_33 Roofload 30 PSF
_34 Roofload 40 PSF
_35 Roofload 50 PSF
_36 Roofload 60 PSF
_37 Roofload 70 PSF
_38 Roofload 80 PSF

114 Supported Floor — Supported Floor — Sq. Ft.

Includes construction of structurally integrated or independently supported floors, i.e., steel decking, joists, beams, slabs, precast concrete decking with topping steel reinforcing and other related items to provide a complete structural floor. Excludes applied finishes.

_01 Concrete Cast in Place
_02 Concrete on Steel Joists
_03 Concrete on Steel Framing
_04 Precast Concrete
_05 Wood

Floorload
_30 Under 40 PSF
_31 41-60 PSF
_32 61-80 PSF
_33 81-100 PSF
_34 101-150 PSF
_35 151-200 PSF
_36 201-250 PSF
_37 251-300 PSF
_38 Over 300 PSF

Span
_40 Under 26'
_41 26'-35'
_42 36'-45'
_43 46'-55'
_44 56'-65'
_45 66'-75'
_46 Over 76'

If more than one entry give percentage of each - nearest 10%.

115 Stairs — Number of Risers — Each

Includes interior and exterior stairs, landings, platforms, and railings.

_01 Exposed
_02 Enclosed
_03
_04
_05 Exterior
_06 Interior

_10 Concrete
_11 Steel
_12 Galvanized
_13 Painted

_20 Closed Riser
_21 Open Riser
_22 Steel Pan
_23 Checkered Plate
_24 Grate

Figure 22 (Continued)
Cost Model Questionnaire

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:54Z
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116 Roof Structure — Roof Area Sq. Ft.

Includes construction of structurally integrated or independently supported roofs, i.e., precast concrete roof slabs, concrete topping, steel decking, joists, beams. Roofing is included in 141.

Framing	Deck	Span
_01 Conc Cast in Place	_20 Other	_40 Under 26'
_02	_21 Steel	_41 26'-35'
_03 Precast Hollow Core	_22 Conc Slab	_42 36'-45'
_04 Precast Concrete	_23 Wood	_43 46'-55'
_05 Wood	_24 Gypsum	_44 56'-65'
_06 Steel Joist		_45 66'-75'
_07 Steel Framing		_46 Over 76'
_08		

117 Pre-Engineered Building — Gross Bldg. Area Sq. Ft.

Includes steel frame, siding, roofing, insulation, doors, windows, and all other work normally furnished by the pre-engineered building sub contractor. Does not include sliding or roll-up doors.

_01 Eave height under 12'	_10 Seismic Zone 0	_32 Roofload 20 PSF
_02 Eave height 12-20'	_11 Seimic Zone 1	_33 Roofload 30 PSF
_03 Eave height over 20	_12 Seimic Zone 2	_33 Roofload 40 PSF
_04	_13 Seimic Zone 3	_34 Roofload 50 PSF
	_14 Seimic Zone 4	_35 Roofload 60 PSF
		_36 Roofload 70 PSF
		_37 Roofload 80 PSF
	_21 Windload 80mph	
	_22 Windload 92mph	
	_23 Windload 103mph	
	_24 Windload 115mph	
	_25 Windload 127mph	

118 Vault — Gross Floor Area Sq. Ft.

Includes walls, ceiling construction and floor slab. Only includes work incurred to construct vault. See system 452 for vault door.

_ 01 Concrete
_ 02 Reinforced CMU

119 Write-in System

ARCHITECTURAL

141 Roofing — Roof Area Sq. Ft.

Includes roof curbing, roof insulation, roofing, gravel stops, gutters, and downspouts, flashing, skylights, roof-access hatches, and other related roofing items. Roof structure is included in 116.

Surface	Insulation	
_10		_20 U = 0.03
_11 Built-up	_21 U = 0.04	
_12 Shingles	_22 U = 0.05	
_13 Elast.	_23	
_14 Sprayed	_24	
_15 Metal Roofing		
Surface		

Figure 22 (Continued)
Cost Model Questionnaire

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142 Exterior Walls — Exterior Wall Area Sq. Ft.

Includes bearing or non bearing walls from inside rough wall to outside finish wall, parapet walls, damp proofing, flashing, insulation waterproofing, balcony walls, and handrails. Includes exterior finishes, caulking, and painting.

Ext. Surface	Backup	Height
_01 Brick	_21 CMU	_30 Under 12'
_02 CMU	_22 Wood Studs	_31 12' - 20'
_03 Drivit	_23 Steel Studs	_32 over 20'
_04 Metal Panels	_24 Conc Cast in Place	
_05 Stucco	_25 Precast Concrete	
_06 Wood	_26 Furring	
_07 Conc Cast In Place		
_08 Precast Concrete		

If more than one height is shown give percentage of each - nearest 10%.

143 Interior Walls — Interior Wall Sq. Ft.

Includes partitions, bearing or non bearing walls, extending from floor-to-floor or floor-to-ceiling excluding finishes. Includes masonry walls, steel or wood stud walls, studs only and excludes painting, gypsum board, or other applied finish.

	Height
_01 Concrete	_21 8'
_02 Steel Studs	_22 9'
_03	_23 10'
_04 Concrete Cast in Place	
_05 Wood Studs	

If more than one entry under height, give percentage of each - nearest 10%. Give narrative description of type and size of rooms.

144 Interior Finishes — Gross Building Area Sq. Ft.

Includes finishes applied to floors, walls, ceilings, stairs and ramps such as; wall covering, resilient flooring tile, terrazzo, wood, carpeting, acoustical tile, plaster, paint, gypsum board, suspended ceiling systems, caulking, and all related trim work.

	Floor	Ceiling
_01 Gypsum Board	_11	_21 Acoustical
_02 CMU	_12 Ceramic Tile	_22 Gypsum Board
_03 Ceramic Tile	_13 Quarry Tile	_23 Plaster
_04 Wood Panels	_14 Carpet	_24 Concrete
_05 Plaster	_15 Concrete	_25 Spray on
_06 Vinyl Wall Covering	_16 Terrazzo	
	_17 Special Toppings	
	_18 VCT	

Figure 22 (Continued)
Cost Model Questionnaire

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:54Z
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145 Doors — Surface Area one Side — Sq. Ft.

Includes all exterior and interior doors, frames, hardware, caulking, and painting, see 45X for 451 Hangar Doors, 452, Vault Doors, and 453 Blast Doors.

_01 Hollow Metal Exterior (Steel or Aluminum)

_02 Alum Entrance Store Front
_03 Folding
_04 Roll Up
_05 Overhead
_06
_07
_08 Wood Interior
_09 Hollow Metal Interior

146 Windows — Surface Area One-Side — Sq. Ft.

Includes windows, glazed wall systems, glazing, caulking, and painting.

Windows	Special
_11 Double Hung	_20 Reglazing
_12 Projected	_21 Bullet Proof
_13 Casement	_22 Hardware
_14 Sliding	_23 Thermal Break
_15 Storm	
_16 Awning	
_17 Jalousie	

147 Specialities — Gross Bldg. Area — Sq. Ft.

Includes chalk and tack boards, signs and plaques, flag poles, telephone enclosures, ladders, lockers, storage shelving, toilet and bath accessories, fireplaces, compartments and cubicles, movable partitions, identifying devices, protective covers, postal specialities, scales, exterior sun control devices, and wardrobe specialties, excluding special mechanical or electrical equipment.

_01 Wardrobes	_08
_02 Jail	_09 Wire Partitions
_03 Clean Room	_10 Metal Walkways
_04 Case Work	_11 X-Ray
_05 Dark Rooms	
_06	
_07 Loading Dock	

Give estimating quantities for required ITEMS OF SIGNIFICANT COST. List shown is not complete. Detailing of toilet accessories and partitions is not required. Very large items may be more suited to using a write-in system.

148 BEQ Wardrobes — Number — Each

Includes furnishing and installing prefabricated wardrobes and chests for BEQs.

_01 With Chest
_02 Without Chest

149 Write-in System

Figure 22 (Continued)
Cost Model Questionnaire

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:54Z
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DEMOLITION INTERIOR

171 Demolition Interior — Gross Building Area — Sq. Ft.

Includes all interior building demolition connected with new construction or alterations. Also include any work on or in, the exterior wall. Do not include complete building demolition, see System 84X.

_01 Complete Interior Partition
_02 Complete Interior Finishes
_03 Complete Interior Mechanical
_04 Complete Interior Electrical
_05 Complete Interior of Bldg. 01 thru 04

172 Asbestos Removal Interior — Total Cost — Lump Sum

Includes costs for removal of asbestos from any interior areas of buildings to be rehabilitated. Includes all costs for sealing off areas, asbestos removal, monitoring, testing, disposal, change areas, disposal suits, respirators, and other related costs. (See System 844 for asbestos removal, exterior.)

_01 Pipe Insulation __LF
_02 Equipment Insulation __SF
_03

173 - 175 Write-in Systems

PLUMBING

211 Plumbing-Domestic — Number of Fixtures — Each

Includes water supply and treatment, waste water disposal and treatment, plumbing equipment, fixtures and trim, and insulation, i.e., hot and cold water pipes, waste, soil and vent pipes, hot water heaters, water coolers, and floor drains.

_01 Flush Tank WC floor mntd.
_02 Flush Tank WC Piping wall mntd.
_03 Flush Valve WC floor mntd.
_04 Flush Valve WC wall mntd.
_05 Water Heater Electric
_06 Water Heater Steam
_07 Instantaneous W.H., Elec.
_08 Instantaneous W.H., Steam

__09 Mop Sink
__10 Urinal
__11 Lavatory

__21 Tub
__22 Shower Fiberglass
__23 Shower/Receptor
__24 Shower Multi-head
__25 Emergency Shower
__26 Emergency Eyewash
__27 Emergency Shower/ Eyewash

_31 Saltwater Flush
_32 Gray-water Flush
_33 Acid Resistant

_34 Fixture Rough-ins
_35 Solar DHW

_50 Cast Iron
_51 PVC

Note: Refer to MIL-HDBK-1010 to determine number of fixtures for systems 211 and 213

212 Domestic Hot Water Solar — Capacity — MBTH

Includes piping, solar panels, supports and controls necessary for solar heated domestic hot water system. Does not include piping normal to System 211 Plumbing-Domestic. When the solar system heats both the building and domestic hot water use System 226.

Figure 22 (Continued)
Cost Model Questionnaire

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213 Roof Drainage — Number of Drains — Each

Includes roof drains, pipes and fittings.

214 - 216 Write-in Systems

SPACE CONDITIONING

221 Heating, Ventilation, and Air Conditioning — Capacity — MBTU or Tons

Includes heating, ventilating and air conditioning systems, i.e., heat generating equipment, refrigeration, air distribution, piping, controls and instrumentation, and insulation.

221

_01 Chilled Wtr Recip. Air Cooled
_02 Chilled Wtr Recip. Water Cooled
_03 Chilled Wtr Centrif. Air Cooled
_04 Chilled Wtr Centrif. Water Cooled
_05 Direct Expansion Air Cooled
_06 Heat Pump Air Cooled
_07 Heat Pump Water Cooled
_08 Chilled Water Steam Absorption
_09 Chilled Water Rotary (Screw)
_10 Fan Coil Units

HVAC

_11 VAV Single Duct
_12 TAB
_13 AHU w/duct
_14 Computer Rm DX
_15 Computer Rm CW
_16 Roof Top
_17 H&V Units
_18 Pkgd. thru the wall
_19 Dual duct
_20 Bldg. Htng Plant HW, oil
_21 Bldg. Htng Plant Steam, oil
_22 Bldg. Htng Plant Hot air, oil
_23 Bldg. Htng Plant Hot air, gas
_24 Bldg. Htng Plant HW, gas
_25 Bldg. Htng Plant steam, gas
_26 Bldg. Htng Plant Elec.
_27 Base Plant Steam
_28 Base Plant HW
_29 Base Plant HTHW
_30 Heat Reclaim
_31 Fixture Radiation & Unit Htrs.

Fans

_40 Power Roof Exh Fans
_41 Inline Exh Fans
_42 Inline Supply Fans
_43 Power Roof Supply
_44 Wall Exhaust Fans

Give tonnage, refer to MIL-HDBK-1010 if better information is not available. If building contains partial A/C, use systems 222 and 223 for each respective area.

222 Air Conditioning — Capacity — Tons

Includes air conditioning systems, i.e., refrigeration, air distribution, piping, controls and instrumentation, and insulation.

_01 Chilled Wtr Recip. Air Cooled
_02 Chilled Wtr Recip. Water Cooled
_03 Chilled Wtr Centrif. Air Cooled
_04 Chilled Wtr Centrif. Water Cooled
_05 Direct Expansion Air Cooled
_06 Heat Pump Air Cooled
_07 Heat Pump Water Cooled
_08 Chilled Water Steam Absorption
_09 Chilled Water Rotary (Screw)
_10 Fan Coil Units
_11 VAV Single Duct
_12 TAB
_13 AHU w/duct
_14 Computer Rm DX
_15 Computer Rm CW
_16 Roof Top
_17 H&V Units
_18 Pkgd. through the wall
_19 Dual duct
_20 Power Roof Exh Fans
_21 Inline Exh Fans
_22 Inline Supply Fans
_23 Power Roof Sup Fans
_24 Wall Exhaust Fans

Figure 22 (Continued)
Cost Model Questionnaire

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223 Heating and Ventilation — Capacity MBTH

Includes heating, ventilating systems, i.e., heating generating equipment, air distribution, piping, controls and instrumentation, and insulation. See System 224 for ventilation only.

_01 H&V Units HW, oil
_02 Unit Heaters
_03 Fin Tube Radiation
_04 Cabinet Unit Heaters
_05 Unit Htrs & Fin Tube Radiation
_06 Roof Top H&V
_07 Duct Mntd. Coils
_10 Power Roof Exh Fans
_11 Inline Exh Fans
_12 Inline Supply Fans
_13 Power Roof Supply Fans
_14 Wall Exhaust Fans
_20 Bldg. Htng Plant
_21 Bldg. Htng Plant Steam, oil
_22 Bldg. Htng Plant Hot Air, oil
_23 Bldg. Htng Plant Hot air, gas
_24 Bldg. Htng Plant HW, gas
_25 Bldg. Htng Plant Steam, gas
_26 Bldg. Htng Plant Elec.
_27 Base Plant Steam
_28 Base Plant HW
_29 Base Plant HTHW

224 Mechanical Ventilation — Fan Capacity CFM

Includes fans, supports, duct work and controls for a mechanical ventilation system. See System 223 for combined heating and ventilation.

_10 Power Roof Exhaust Fans
_11 In Line Exhaust Fans
_12 In Line Supply Fans
_13 Power Roof Supply Fans
_14 Wall Exhaust & Fans

225 Dehumidification — Capacity PPH

Includes equipment, duct work and controls for a dehumidification system. Use system 221 or 222 where dehumidification is an integral part of an air conditioning system.

_01 Desiccant
_02 Refrigeration
_11 Regenerative
_12 Non-Regenerative

226 Heating, Solar — Collector Area Sq. Ft.

Includes piping, solar panels, supports, and controls necessary for a solar-heated building heating system. Does not include work normal to the building and included under system 221 or 223. (See note under system 212.)

_02 Heating
_03 Heating & Domestic HW
_11 Glycol
_12 Water

227 Write-in System

Figure 22 (Continued)
Cost Model Questionnaire

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WEIGHT HANDLING EQUIPMENT

231 Bridge Cranes — Capacity — Tons

Includes--Crane rails, bridge, hoists, trolleys, and electricification. Power supply at hoist area only.

_01 Span under 50'
_02 Span 51-75'
_03 Span over 75'

_10 Capacity under 10 T
_11 Capacity 10-20 T
_12 Capacity 20-30 T
_13 Capacity 30-40 T
_14 Capacity over 40 T

_21 Run under 50'
_22 Run 50-100'
_23 Run over 100'

232 Monorails — Capacity — Tons

Includes rail, hoist, trolley and electrification, and air supply. Power supply at hoist area only.

_01 Manual
_02 Electric
_03 Air Operated

_10 Capacity under 5 T
_11 Capacity 5-10 T
_12 Capacity over 10 T

_21 Run under 50'
_22 Run 50 to 100'
_23 Run over 100'

233 Fixed Hoist — Capacity — Tons

Includes hoist, support electrification and air supply. Power supply at hoist area only.

_01 Manual
_02 Electric
_03 Air Operated

234 Vehicle Lifts — Capacity — Tons

Includes lift, pit, electrification and air supply. Power supply at hoist area only.

_10 Capacity under 5 T
_11 Capacity 5-10 T
_12 Capacity over 10 T

235 Elevators — Number of Stops — Each

Passenger or freight elevators including conveyor cab, doors, controls, and rails.

_01 Electric
_02 Hydraulic
_03

_11 Passenger
_12 Freight

236 Escalators — Flights — Each

Includes equipment and power supply at escalator area.

237 Conveyors — Capacity — Tons/hr

Includes equipment, foundations, controls and power supply at conveyor area only.

238 Write-in System

Figure 22 (Continued)
Cost Model Questionnaire

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SPECIAL MECHANICAL PIPING

241 Vacuum — Capacity CFM

Includes vacuum pump, piping, connectors, and controls. (Industrial or medical.)

_01 Medical
_02 Industrial
_03

242 Oxygen — Outlets Each

Includes piping, tanks, and accessories.

243 Nitrogen — Outlets Each

Includes piping, tanks, and accessories.

244 Compressed Air — Capacity CFM

Includes air compressors, dryers, after coolers, piping, controls, valves, gauges, and filters.

_01 Low Pressure 150 below
_02
_03 High Pressure 151 up

245 Interior Steam — Capacity PPH

Includes steam generator, piping, and outlets. Does not include steam source that serves other uses. Do not use this system for steam heating.

_01 High Pressure
_02 Medium Pressure
_03 Low Pressure
_04
_05 Chemical Treatments
_06 Feed Water Equipment
_07 With Condensate Return
_08 Without Condensate Return
_11 Oil Fired
_12 Electric Fired
_13 Prefabricated Stack
_14 Fire Tube
_15 Water Tube
_16
_17 Controls
_20 Fuel Oil Storage

246, 247 Write-in Systems

SPECIAL MECHANICAL OTHER

251 Dust Collection — Capacity CFM

Includes equipment supports and duct work associated with an interior dust collection system.

252 Engine Exhaust — Fan Capacity CFM

Includes door exhaust ports, fans, and exhaust duct (flexible and rigid).

_01 Overhead
_02 Underfloor
_03 Through Door

253 - 255 Write-in Systems

Figure 22 (Continued)
Cost Model Questionnaire

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:54Z
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INTERIOR FIRE PROTECTION

271 Sprinklers — Gross Area Sprinkled — Sq. Ft.

Includes sprinkler pipe, fittings, valves, pumping equipment, tanks, sprinkler heads, and controls.

_01 Dry
_02 Wet
_03 Preaction
_04 Deluge
_10 Light Hazard
_11 Ordinary Hazard
_12 Extra Hazard
_13
_14 Includes Booster Pump

272 AFFF (Aqueous Film Forming Foam) — Area Protected — Sq. Ft.

Includes equipment, piping, and controls necessary for an AFFF System. Includes storage tanks and foam when only the facility being priced is served.

_01 Deluge
_02 Pre-action
_03
_04 Oscillating Monitors
_05 Heat Detection
_06 Ultravoilet/Infrared Detection
_11 with 100% Back-up Storage
_12 without 100% Back-up Storage
_13 Bladder Tanks
_14 Standard Tank, Pump, Proportioner
_15
_16 6% Concentration
_17 3% Concentration

273 Carbon Dioxide — Storage Capacity — Pounds

Includes storage cylinders, hose reels, piping, nozzles, and controls.

_01 Hose Reel
_02 Flooding Area
_03 Flooding Total
_04

274 Halon — Storage Capacity — Pounds

Includes storage cylinder, controls, piping, and nozzles.

_01
_02

275 Fire Alarm — Gross Bldg. Area — Sq. Ft.

Includes control panels, power supplies, detectors, manual alarm station, alarm devices, conduit, and wire.

_01 Manual
_02 Automatic Detectors
_03 Mechanical & Electrical
_04 Extend Existing

276, 277 Write-in Systems

Figure 22 (Continued)
Cost Model Questionnaire

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POWER & LIGHTING

311 Power — Connected Load — KW

Includes all interior distribution for power and special electrical systems, i.e., switchboards, transformers, motor controls, distribution switches, motor starters, feeders, branch-circuit wiring and devices, panels, and lightning protection. Exclude all interior distribution for lighting fixtures and emergency lighting, i.e., light fixtures, panels, branch circuit wiring, and devices for lighting.

_01 120/208
_02 277/480/120/208
_03 277/480
_04 120/240
_05 Alteration to Existing
_09 Explosion Proof

312 Lighting — Gross Bldg. Area — Sq. Ft.

Includes all interior lighting fixtures, exit and emergency lighting, branch circuit wiring, conduit, and devices for light fixtures only. Panels are included in System 311.

_01 Incand, Fluor., HID (Standard Spec) w/Bat-Op Emerg.
_02 Special System
_03 High Ind Discharge (HID) High Bay
_04 High Ind discharge (HID) low Bay
_05 Explosion Proof

313 400 HZ — Converter Capacity — KW

Includes frequency changer, control system, wire, conduit, and devices to install a 400 HZ power system.

_01 Solid State
_02 M G Set

314 Direct Current — Converter Capacity — KW

Includes rectifier battery charger, battery bank, wire, conduit, and devices to install a direct current power system.

_01 Rectifier

315 Uninterruptable Power Supply (UPS) — Converter Capacity — KVA

Includes battery bank, rectifier, and static inverter for solid state equipment and flywheel, motor, generator, and battery bank for rotary equipment plus all associated wiring, conduit, switches, panels, and switchgear.

_01 Static

Figure 22 (Continued)
Cost Model Questionnaire

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316 Electrical Generators — Equipment Capacity — KW

Includes diesel, gas and gasoline engines, day tanks, valves, piping and fittings, generators, operating and synchronizing controls, transfer switches, and wiring.

_01 Intermittent	_30	_40
_02 Continuous	_31 120/240V, 1PH, 60HZ	_41 600 RPM
_03 Electrical	_32 120/208V, 3PH, 60HZ	_42 720 RPM
_04 Gas	_33 277/480V, 3PH, 60HZ	_43 900 RPM
_05 Diesel	_34 347/600V, 3PH, 60HZ	_44 1200 RPM
_06 Turbine	_35 4160V/2400V, 3PH, 60HZ	_45 1800 RPM
_07	_36 11.5/6.5KV, 3PH, 60HZ	

_10 Overhaul existing units
_11 Generators GFM

_21 Integral Radiators
_22 Remote Radiators

317, 318 Write-in Systems

SPECIAL ELECTRICAL PROTECTION

321 Grounding — Gross Building Area — Sq. Ft.

Includes cable, ground rods, clamps, and connections for grounding system.

_01 Lightning Grounding
_02 Electronic Grounding
_03 Protective Grounding

322 Lightning Protection — Gross Building Area — Sq. Ft.

Includes base plate, lightning mast, concrete foundations, ground rods, cables, and connections for a complete building lightning protection system.

323 - 326 Write-in Systems

SPECIAL ELECTRICAL/ELECTRONICS

331 Security Detection — Gross Bldg. Area — Sq. Ft.

Includes control panels, power supplies, detectors, alarm stations, alarm devices, wire, and conduit.

_01 Intrusion Alarm for Access Control
_02 Access Control
_03 TV Camera & Monitor
_04 Conduit
_05 Conduit & wire

If layout is not available describe area covered, where does alarm sound etc. Also describe intended function of system.

332 Energy Monitoring and Control System — Points of Controls — Each

Includes control panels, devices, wiring, and conduit.

_01 Local Control
_02 Remote Control

Figure 22 (Continued)
Cost Model Questionnaire

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333 Computer Communication — Cable Run — Lin. Ft.

Includes wiring, cable tray, and devices necessary for a complete computer communication system.

334 - 336 Write-in Systems

COMMUNICATION

341 Telephone — Gross Bldg. Area — Sq. Ft.

Includes control panels, devices, wire, and conduit.

_01 Government Owned System
_02 Conduit Only
_03 Conduit & Wire

342 Intercom — Gross Bldg. Area — Sq. Ft.

Includes master stations, substations, control panels, speakers, wire, and conduit. Includes System 344 public announcement when it is combined with the intercommunication system.

_01 Theater Sound
_03 Two-way communication listening
_04 Special System (Describe)
_05 Includes PA Systems
_06 Conduit
_07 Conduit & Wire
_08 Leased System
_09 Government Owned System

343 Television — Gross Bldg. Area — Sq. Ft.

Includes antenna, booster amplifiers, circuit wiring, conduit, cameras, control consoles, and monitors.

_01 Government Owned System
_02 Leased Cable System
_03 Conduit Only
_04 Conduit & Wire

344 Public Announcement — Gross Bldg. Area — Sq. Ft.

Includes master stations, substations, control panels, speakers, wire, and conduit. See System 342 for combined intercommunication and Public Announcement systems.

_01 Government Owned System
_02 Conduit only
_03 Conduit & Wire

345 Master Clock — Number of Outlets — Each

_01 Government Owned System
_02 Conduit & Wire

Includes wire, conduit, and devices for a master clock system.

346, 347 Write-in Systems

Figure 22 (Continued)
Cost Model Questionnaire

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4XX Special Equipment

Note: Equipment may be obtained for various funding sources and by different procurement methods. See Standard System Descriptions 4x5 - 4x6 Use: Install Government Furnished Equipment to allow for contractor cost of installation for government furnished equipment; use Government furnished Equipment to allow for the cost of procuring equipment when financed from construction funds.

MEDICAL EQUIPMENT

411 Hospital Equipment — Gross Bldg. Area — Sq. Ft.

Includes contractor furnished and installed built-in hospital equipment. Mechanical and electrical work are not included.

412 Dental Equipment — Number of Chairs — Each

Includes contractor furnished and installed built-in dental equipment. Mechanical and electrical work are not included.

413, 414 Write-in Systems

415 Install Government Furnished Medical Equipment

416 Government Furnished Medical Equipment

FOOD SERVICE EQUIPMENT

421 Food Service Equipment — Pieces of Food Service Equip — Each

Includes contractor furnished and installed built-in food service equipment.

422 - 425 Write-in Systems

425 Install Government Furnished Food Service Equipment

426 Government Furnished Food Service Equipment

MISCELLANEOUS EQUIPMENT

431 Chapel Equipment — Gross Bldg. Area — Sq. Ft.

Includes contractor furnished and installed built-in chapel equipment.

432 Movie Theater Equipment — Gross Bldg. Area — Sq. Ft.

Includes contractor furnished and installed built-in equipment for showing films, such as projection booth windows, curtains, and projection screens.

Small rear projection screens should be included under 491 Other Equipment.

433 Rifle Range Equipment — Firing Lanes — Each

Includes targets, guides, baffles, and bullet stops.

Figure 22 (Continued)
Cost Model Questionnaire

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434 Laboratory Equipment — Number Pieces — Each

Includes case work, benches, testing equipment apparatus, and other equipment designated as laboratory equipment. Mechanical and electrical work not included.

435 Install Government Furnished Laboratory Equipment

436 Government Furnished Laboratory Equipment

437 Waste Disposal Equipment — Equipment Capacity — PPH

Includes built-in compactors, incinerators, and shredders with installation cost. Enclosures, mechanical, and electrical work not included.

438 Write-in System

SPECIAL WAREHOUSE EQUIPMENT

441 Special Warehouse Equipment — Bldg. Area — Sq. Ft.

Includes built-in storage racks and conveying systems, in rack special sprinkler system, dock leveler, dock door seals, and rubber bumpers with installation cost. Building mechanical and electrical work not included.

442 - 444 Write-in Systems

445 Install Government Furnished Warehouse Equipment

446 Government Furnished Warehouse Equipment

SPECIAL CONSTRUCTION

451 Hanger Doors — Surface Area One Side — Sq. Ft.

Includes tracks, guides, operators doors, and controls. Does not include structural steel supports, concrete. Electrical work at door only.

01 Sliding	10 Manual
02 Vertical Lift	11 Power Operated

452 Vault Doors — Number — Each
Size

Includes vault door and frame.

453 Blast Doors — Surface Area One Side — Sq. Ft.

Includes doors, frames track, operators, and controls.

Span (TNT Equivalent)	Charge Weight	Distance
_01	_11	_21
_02 3 feet	_12 0 - 10 LBS	_22 0 - 20 FT
_03 5 feet	_13	_23
_04 10 feet	_14 10 - 500 LBS	_24 20 - 100 FT
_05 15 feet	_15	_25
_06 20 feet	_16 Over 500 LBS	_26 Over 100 FT

Figure 22 (Continued)
Cost Model Questionnaire

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454 Raised Floor — Area of Raised Floor — Sq. Ft.

Includes floor system, panel covering such as carpet, ramps, steps, closure panels, cable cutouts, and air flow panels, with installation cost. Railing to be a speciality item.

455 Radio Frequency Shielding — Surface Area Shielded — Sq. Ft.

Includes wall floors and ceiling panels, doors, filters, filter cabinets, wave guides, and enclosures for telephone cypher locks, and fire alarm pull box with installation cost. Mechanical and electrical services not included.

_01	_10 Spray on Metal	_20
_02 Interferance	_11 Foil Backed Gyp Board	_21
_03	_12 Wire Mesh	_22 20 Db
_04 Tempest	_13 Copper Sheet	_23
_05	_14 Aluminum Sheet	_24
_06 Hemp	_15 Single Sheet Steel	_25 50 Db
	_16 Steel Prefab Room	_26
	_17 Stainless Steel	_27
	_18	_28 80 Db
		_29
		_30 100 Db

456 Bowling Lanes — Number of Lanes — Each

Includes lanes and installation with pinsetter, returns, and other associated equipment.

_01 New	_10 W/O Auto Pin Setter
_02 Refinished	_11 W/ Auto Pin Setters

457 Paint Spray Booth — Floor Area — Sq. Ft.

Includes booth and equipment procured with booth. Mechanical and electrical hookup at booth area area included.

458 Write-in System

OTHER EQUIPMENT

491 Other Miscellaneous Equipment — Total Cost — Lump Sum
Includes special equipment not included above. Also refer to System

147 Specialities.

492 Snow Melting Area — Protected — Sq. Ft.

_01 Electrical
_02 Glycol
_03 Geothermal

495 Install Government Furnished Miscellaneous Equipment

496 Government Furnished Miscellaneous Equipment

Describe any equipment not shown elsewhere that is purchased from construction funds. Usually this is either from equipment purchased by the government and furnished to the construction contractor, or it is furnished and installed under a seperate contract. Also show, if not shown elsewhere equipment not from construction funds for which installation costs will be required.

Figure 22 (Continued)
Cost Model Questionnaire

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499 Building work not included elsewhere — Total Cost — Lump Sum

Used when partial data is loaded and it is desirable to combine several systems of no historical value. (Rehabilitation and Alteration projects are the ones for which this system is normally used; very seldom, if ever, is this used for new work.)

SUPPORTING FACILITIES

(Systems 500 through 999)

Note: 1. Show the following criteria suitable for the application of historical cost; under each applicable system heading;

a. Size (assume if unknown) and length of utility runs. Include storm drainage requirement. Indicate valves, fire hydrants, manholes, catch basins and other significant extras. Depth of cut (when over 4 feet) and dewatering requirements must be shown.

b. Area and assumed section for paved areas. Identify each type and section variation.

c. Description of work required for grading, cut and borrow. Describe assumed nature of materials to be moved. If rock excavation is required, state assumptions on removal requirements (e.g., rip, blast, etc.).

d. Indicate approximate quantities of clearing and grubbing in acres, and whether heavy, medium, or light.

e. Describe any functional features required outside the building 5-foot line such as: sewage lift stations, wash racks and water tanks, and unsuitable soil conditions

f. If hazardous materials are present, state type, extent and method of disposal.

g. See System 911 for Land Piling–Driven.

EXTERIOR ELECTRICAL
SUPPORTING FACILITIES

511 Electrical Distribution, Primary — Length of Run — Lin. Ft. Size

Includes overhead power distribution, i.e., poles, crossarms, insulators, guying, terminations, lightning protection, wire and cable, and underground distribution, i.e., excavation and backfill, concrete encased duct bank, direct burial duct, manholes, handholes, cable, terminations, stress cones, and grounding. All for primary distribution to transformer or substation.

_01 Underground
_02 Overhead

512 Electrical Distribution, Secondary — Length of Run — Lin. Ft. Size

Includes overhead power distribution, i.e. poles, crossarms, insulators, guying, terminations, lightning protection, wire and cable and underground distribution, i.e., excavation, backfill concrete encased and direct burial duct bank, manholes, handholes, cable terminations, stress cones, and grounding.

_01 Overhead
_02 Pad Mounted

Figure 22 (Continued)
Cost Model Questionnaire

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513 Substation/Transformer — Capacity — KVA

Includes overhead transformer, i.e. fuse cutouts, transformers, substation, including concrete fence, steel framing, fencing and enclosure, and pad mounted transformer; i.e. disconnect services, transformers, substation including concrete steel framing, fencing, and enclosure.

_01 Overhead
_02 Pad Mounted

514 Area Lighting — Number of Fixtures — Each

Includes poles, fixtures, excavation and backfill, concrete work, wire, duct, and conduit.

_01 Incandescent
_02 High pressure sodium
_03 Low pressure sodium
_04 Mercury vapor
_05 Other
_11 Overhead Distribution
_12 Underground Distribution

515 Airfield Lighting — Length of Run — Lin. Ft.

Includes excavation and backfill, duct, conduit, devices, transformer, relays, and wiring necessary to install the airfield lighting.

516 Lightning Protection — Points of Protection — Each

Includes base plate, lighting mast, concrete foundations, ground rods, cables, and connections for a complete exterior lightning protection system.

517, 518 Write-in Systems

EXTERIOR COMMUNICATION

521 Fire Alarm — Length of Run — Lin. Ft.

Includes excavation, backfill, duct, conduit, devices, and wire necessary to install an exterior fire alarm system.

522 Security Alarm — Length of Run — Lin. Ft.

Includes excavation and backfill, duct, conduit, and junction boxes. Wiring cable, devices, and their installation are not MCON funded.

523 Communication, Telephone — Length of Run — Lin. Ft.

Includes excavation and backfill, duct, conduit, devices and wire necessary to install an exterior communication system.

524 Exterior EMCS — Length of Run — Lin. Ft.

Includes cable, duct, conduit necessary for EMCS systems outside the building 5-foot line.

525, 526 Write-in Systems

Figure 22 (Continued)
Cost Model Questionnaire

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EXTERIOR MECHANICAL

541 Heat Distribution, Overhead — Length of Run — Lin. Ft. / Size

Includes supports, piping, insulation, and valves to install overhead steam/condensate and hot water distribution systems outside the 5-foot line.

_01 Concrete supports
_02 Timber supports

542 Heat Distribution, Underground — Length of Run — Lin. Ft. / Size

Includes excavation, backfill, manholes, supports, piping insulation, and valves to install underground steam/condensate and hot water distribution systems outside the 5-foot line.

_01 # of Manholes
_02 Size of manholes

543 Chilled Water Distribution — Length of Run — Lin. Ft. / Size

Includes excavation, backfill, manholes, supports, piping, insulation, and valves to install chilled water distribution systems outside the 5-foot line.

544 Condensate Collection (only) — Length of Run — Lin. Ft. / Size

Includes excavation, backfill, manholes, supports, piping, and valves to install a condensate collection system.

545 Gas Distribution — Length of Run — Lin. Ft. / Size

Includes excavation, backfill, piping, valves, and other miscellaneous work to install a complete gas system outside the 5-foot line.

546 Compressed Air Distribution — Length of Run — Lin. Ft. / Size

Includes excavation, backfill, piping, valves, and devices to install a complete compressed air distribution system outside the 5-foot line.

547 Fuel Distribution — Length of Run — Lin. Ft. / Size

Includes excavation, backfill, piping, pipe coatings, valves, and devices to install a complete fuel distribution system outside the 5-foot line.

Figure 22 (Continued)
Cost Model Questionnaire

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548 Exterior Foam System — Length of Run — Lin. Ft. Size

Includes distribution piping, foam storage and proportioning equipment, booster pump, and emergency power.

		Distribution
_01	Bladder Tanks	_11 Above Ground
_02	Standard Tanks	_12 Below Ground
	Amps, Proportioner	_13
_03	100% Back-up Storage	_14
_04	Less than 100%	_15 6% Concentration
	Back-up Storage	_16 3% Concentration
_05	Water Booster Pump	
_06	Emergency Generator	

549 Write-in System

EXTERIOR WATER DISTRIBUTION

551 Water Distribution — Length of Run — Lin. Ft. Size

Includes complete potable water distribution system, i.e., utility service connections, fire hydrants, excavation and backfill, pipe, valves, and fittings. Use when domestic and fire water System 552 are combined in one line.

_01 Firehydrants
_02 Fittings (C.I.)
_03 Tapping sleeves & valves
_04 Valves

552 Fire Protection

Water Distribution — Length of Run — Lin. Ft. Size

Includes complete fire water distribution system, i.e., utility service connection fire hydrants, excavation and backfill, pipe, valves, and fittings.

_01 Firehydrants
_02 Fittings (C.I.)
_03 Tapping sleeves & valves
_04 Valves

553 Salt Water Distribution — Length of Run — Lin. Ft. Size

Includes complete salt water distribution system, i.e., utility service connections, fire hydrants, excavation and backfill, pipe, valves, and fittings.

554 Sanitary Sewers — Length of Run — Lin. Ft. Size

Includes complete sanitary sewer system, i.e., utility service connections, excavation and backfill, sheeting and shoring, dewatering, pipe and fittings, manholes, cleanouts, septic disposal and process, and acid waste system outside the 5-foot line.

Figure 22 (Continued)
Cost Model Questionnaire

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555, 556 Write-in Systems

_01 Manholes
_02 Cleanouts
_03 Dewatering
_04 Sheeting

BULK STORAGE

561 Fuel Storage, Bulk — Capacity — BBL

Includes tanks, foundation, coatings, fire protection, and level controls. Does not include piping or other special foundation work.

_01 Underground
_02 Above ground

562 Fuel Storage, Ready Issue — Capacity — Gal

Includes excavation, backfill, gravel fill, concrete, tie downs, tanks, accessories, piping at tank, coatings, and monitoring wells. (Does not include long runs of supply piping.)

_01 Underground
_02 Above ground

563 Water Storage — Capacity — Gal

Includes ground level and elevated storage tanks, foundation, coatings, lighting, valves, and piping required at tanks. Does not include piling or other special foundation work.

564 - 566 Write-in Systems

_01 Ground level
_02 Elevated

FUELING

571 Vehicle Fueling — Number of Outlets — Each

Includes concrete island, pumps, piping to 5-foot of tank and electrical work to panel.

_01 Single product pump
_02 Dual product pump

572 Aircraft Fueling — Number of Hydrants — Each

Includes hydrants, filter, supports, loading arms, piping, and canopy. System includes truck loading stand for loading of fueling trucks.

573 Marine Fueling — Number of Outlets — Each

Includes hydrants, filters, supports, loading arms, piping, and canopy.

574 - 576 Write-in Systems

Figure 22 (Continued)
Cost Model Questionnaire

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PUMPING

581 Fuel Pumping Station — Pump Capacity GPM / Bldg size

Includes all work to install a complete fuel pumping station to the 5-foot line. Includes excavation and backfill, pit, pump, piping, valves, structure and electrical work at pump, and other miscellaneous work.

_01 Above ground
_02 Below ground
_11 Reinforced concrete
_12 Masonry
_13 Pre engineered
_14 Other

582 Water Pumping Station — Pump Capacity GPM / Bldg size

Includes all work to install a complete water pumping station to the 5-foot line. Includes building, pit excavation and backfill, piping, pumps, electrical, and other miscellaneous work.

_01 Above ground
_02 Below ground
_11 Reinforced concrete
_12 Masonry
_13 Pre engineered
_14 Other

583 Fire Booster Pump — Pump Capacity GPM / Bldg size

Includes all work to install a complete water booster pumping station to the 5-foot line, if within a separate building. If pump is contained within the primary building, include in System 271. Includes pump, piping valves, electrical work at pump, and other miscellaneous work to install a complete fire booster pump.

_01 Above ground
_02 Below ground
_11 Reinforced concrete
_12 Masonry
_13 Pre engineered
_14 Other

584 Sewage Pump Station — Pump Capacity GPM / Bldg size

Includes all work to install a complete sewage pumping station to the 5-foot line. Includes building, wet well, dry well, excavation and backfill, piping, pumps, electrical, and other miscellaneous work.

_01 Above ground
_02 Below ground
_11 Reinforced concrete
_12 Masonry
_13 Pre engineered
_14 Other

585 Sewage Lift Station — Pump Capacity GPM / Bldg Size

Includes all work to install a complete sewage lift station to the 5-foot line. Includes structure, wet well, excavation, backfill, piping, pumps, electrical, and other miscellaneous work.

_01 Above ground
_02 Below ground
_11 Reinforced concrete
_12 Masonry
_13 Pre engineered
_14 Other

Figure 22 (Continued)
Cost Model Questionnaire

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586 Write-in System

TREATMENT FACILITY

591 Water Treatment — Capacity (Million gal/day) — MGD

Includes excavation, backfill, piping, pumps, equipment, controls, and other necessary accessories to install a complete water treatment system.

592 Domestic Sewage Treatment — Capacity (Million gal/day) — MGD

Includes excavation, backfill, piping, pumps, equipment, controls, and other necessary accessories to install a complete domestic sewage treatment system.

593 Industrial Waste Treatment — Capacity (Million gal/day) — MGD

Includes excavation, backfill, piping, pumps, equipment, controls, and other necessary accessories to install a complete industrial waste treatment system.

594 - 596 Write-in Systems

EQUIPMENT BUILDING

611 Mechanical Equipment Building — Gross Bldg. Area — Sq. Ft.

Includes all work to construct a mechanical equipment building as part of a larger building project. Includes building construction, mechanical, and electrical work.

612 Electrical Equipment Building — Gross Bldg. Area — Sq. Ft.

Includes all work to construct an electrical equipment building as part of a larger building project. Includes building construction, mechanical, and electrical work.

613 - 615 Write-in Systems

SUPPORTING STRUCTURES MISCELLANEOUS

621 Security Gate — Opening Width — Lin. Ft.

Includes mechanically operated gate, foundation, supports, operating mechanism controls, and electrical work at gate.

622 Guard House — Gross Bldg. Area — Sq. Ft.

Includes structure, canopy, apron, curb, lighting, and mechanical and electrical work. Use for entrance gate structure as well as other guard house. This system is not intended for large structures that are a primary facility.

623 - 625 Write-in Systems

Figure 22 (Continued)
Cost Model Questionnaire

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POLLUTION ABATEMENT STRUCTURES

641 Oil Water Separators — Capacity — Gal

Includes excavation, backfill, concrete structure, miscellaneous metals, thru wall piping, and other work necessary to install a complete oil water separator.

642 Electro-Static Precipitator — Air Flow — SCFM

Includes all work to install precipitator. Includes breeching (inlet and outlet), structural supports, control system, and ash hopper.

_01 2 Field
_02 3 Field
_03 4 Field
_04 5 Field
_05 6 Field
_11 3 Feet/Second
_12 4 Feet/Second
_13 5 Feet/Second
_14
_15 Single Stage
_16 Two Stage

643 - 645 Write-in Systems

SHOP SUPPORT STRUCTURES

651 Vehicle Grease Rack — Number of Vehicles Racks — Each

Includes all work necessary to install an exterior grease rack. Includes excavation, backfill, concrete, miscellaneous metal, and other required items.

652 Vehicle Wash Platform — Number of Vehicles Bays — Each

Includes all work to install a complete vehicle wash platform. Includes excavation, backfill, paving, piping, accessories, and electrical work to edge of platform. Does not include distribution piping.

653 - 656 Write-in Systems

AIRCRAFT CLEANING FACILITY

661 Aircraft Washing Facility — Paved Area — Sq Yd.

Includes paving, drainage, water supply, washing equipment controls, and other necessary items to install a complete aircraft washing facility. Do not include access paving or long utility supply runs.

662 Aircraft Rinsing Facility — Paved Area — Sq. Yd.

Includes paving, drainage, water supply rinsing equipment, controls, and other necessary items to install a complete aircraft rinsing facility. Do not include access paving or long utility supply runs.

663 - 666 Write-in Systems

SITE PREPARATION & IMPROVEMENTS

Figure 22 (Continued)
Cost Model Questionnaire

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ROADS

711 Concrete Roads — Paved Area — Sq. Yd.

Includes concrete paving, curbs, curbs and gutters, subgrade preparation, fine grading, compaction, sub-base course, base course, edge forms, concrete reinforcement, finishing, expansion and control joints, rails and barriers, wheels stops, and pavement markings.

_01 Thickness (PVMT)
_02 Reinforcing
_03 Curbs & gutter
_04 Thickness (subgrade)

712 Flexible Roads — Paved Area — Sq. Yd.

Includes bituminous concrete paving, tack and seal coats, curbs, curbs and gutters, subgrade preparation, fine grading, compaction, sub-base course, base course, wearing course, finish course, rails and barriers, wheels stops, and pavement markings.

_01 Thickness (PVMT)
_02 Reinforcing
_03 Curbs & gutter
_04 Thickness (subgrade)

713 Overlay Road — Paved Area — Sq.Yd. — Thickness

Includes flexible paving overlay of existing roads. Includes cleaning, sweeping, leveling courses, minor patching, tack coat, and pavement marking.

714 Surface Treatment - Road — Paved Area — Sq. Yd.

Includes bituminous surface treatment over existing pavement or base course. Includes cleaning, sweeping, minor patching, tack or prime coat, and pavement marking.

715 Slurry Seal Road — Paved Area — Sq. Yd.

Includes slurry seal over existing road. Includes cleaning, sweeping, minor patching, and pavement marking.

716, 717 Write-in Systems

PARKING

721 Concrete Parking — Paved Area — Sq. Yd.

Includes concrete paving, curbs, curbs and gutters, fine grading, compaction, sub-base course, base course, edge forms, concrete reinforcement, curbing, finishing, expansion and control joints, rails and barriers, wheel stops, and pavement markings.

_01 Thickness (PVMT)
_02 Reinforcing
_03 Curbs & gutter
_04 Thickness (subgrade)

Figure 22 (Continued)
Cost Model Questionnaire

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722 Flexible Parking — Paved Area — Sq. Yd.

Includes bituminous concrete paving, tack and seal coats, curbs, curbs and gutters, fine grading, compaction, sub-base course, base course, wearing course, finish course, rails and barriers, wheel stops, and pavement markings.

_01 Thickness (PVMT)
_02 Reinforcing
_03 Curbs & gutter
_04 Thickness (subgrade)

723 Overlay Parking — Paved Area — Sq. Yd. — Thickness

Includes flexible paving overlay of existing parking areas. Includes cleaning, sweeping, leveling course, minor patching, tack coat, and pavement marking.

724 Surface Treatment - Parking — Paved Area — Sq. Yd.

Includes bituminous surface treatment over existing pavement or base course. Includes cleaning, sweeping, leveling courses, minor patching, tack or prime coat, and pavement marking.

725 Slurry Seal Parking — Paved Area — Sq. Yd.

Includes slurry seal over existing parking area. Includes cleaning, sweeping, minor patching, and pavement marking.

726, 727 Write-in Systems

WALKS

741 Concrete Walks — Paved Area — Sq. Yd.

Includes concrete paving, base course, edge forms, concrete reinforcement, finishing, edging, and exterior steps.

742 Bituminous Walks — Paved Area — Sq. Yd.

Includes bituminous paving, base course, edging, and exterior steps.

743 Special Walks — Paved Area — Sq. Yd.

Includes brick paving, limestone paving, flagstone paving, other paving materials, base course, and exterior steps.

744 - 746 Write-in Systems

AIRCRAFT PARKING CONCRETE

751 Concrete Runways — Paved Area — Sq. Yd.

Includes concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course, edge forms, reinforcing, finishing, expansion and control joints, rails and barriers, wheel stops, and pavement markings when constructed for aircraft runways.

_01 Thickness (PVMT)
_02 Reinforcing
_03 Curbs & gutter
_04 Thickness (subgrade)

Figure 22 (Continued)
Cost Model Questionnaire

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752 Concrete Taxiways — Paved Area — Sq. Yd.

Includes concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course, edge forms, reinforcing, finishing, expansion and control joints, rails and barriers, wheel stops, and pavement markings when constructed for aircraft taxiing.

_01 Thickness (PVMT)
_02 Reinforcing
_03 Curbs & gutter
_04 Thickness (subgrade)

753 Concrete Aprons — Paved Area — Sq. Yd.

Includes concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course, edge forms, reinforcing, finishing, expansion and control joints, rails and barriers, wheel stops, and pavement markings when constructed for aircraft parking.

_01 Thickness (PVMT)
_02 Reinforcing
_03 Curbs & gutter
_04 Thickness (subgrade)

754 Repairs Concrete Paving — Paved Area — Sq. Ft.

Includes removing and replacing individual slabs, partial slab removal and replacement, spall repairs, full depth repairs, and other miscellaneous patching.

_01 Full depth repair
_02 Spall repair
_03 Corner break
_04 Other

755 Crack and Joint Repair — Length of Repair — Lin. Ft.

Includes cleaning and resealing joints and cracks repairs.

757, 758 Write-in Systems

AIRCRAFT PAVING FLEXIBLE

761 Flexible Runways — Paved Area — Sq. Yd.

Includes bituminous concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course,wearing course, finish course, rails and barriers, wheel stops, and pavement markings when constructed for aircraft runways.

_01 Thickness (PVMT)
_02 Reinforcing
_03 Curbs & gutter
_04 Thickness (subgrade)

Figure 22 (Continued)
Cost Model Questionnaire

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762 Flexible Taxiways — Paved Area — Sq. Yd.

Includes bituminous concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course, wearing course, finish course, rails and barriers, wheel stops, and pavement markings when constructed for aircraft taxiing.

_01 Thickness (PVMT)
_02 Reinforcing
_03 Curbs & gutter
_04 Thickness (subgrade)

763 Flexible Aprons — Paved Area — Sq. Yd.

Includes bituminous concrete paving, curbs, curbs and gutters, subgrade preparation, compaction, base course, sub-base course, wearing course, finish course, rails and barriers, wheel stops, and pavement markings when constructed for aircraft parking.

_01 Thickness (PVMT)
_02 Reinforcing
_03 Curbs & gutter
_04 Thickness (subgrade)

764 Repairs Aircraft Flexible Paving — Paved Area Patched — Sq. Yd.

Includes removing and replacing small area of flexible paving including sub-base course, base course, tack o. prime coat, and surface material.

_01 Full depth repair
_02 surface repair

765 Aircraft Paving Overlay — Paved Area — Sq. Yd.
Thickness

Includes flexible overlays of aircraft paving. Includes cleaning, sweeping, tack coat, milling, minor patching, leveling courses, and pavement marking.

766 Aircraft Paving Slurry Seal — Paved Area — Sq. Yd.

Includes slurry seal applied over existing aircraft paving. Includes cleaning, sweeping, minor patching, tack coat, and pavement markings.

767, 768 Write-in Systems

INCIDENTAL PAVING WORK

771 Pavement Milling — Milled Area — Sq. Yd.
Depth of Milling

Includes bituminous and concrete milling with removal and/or disposal cost. Savings from recycled materials should be reflected in other systems.

772 - 776 Write-in Systems

Figure 22 (Continued)
Cost Model Questionnaire

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EARTHWORK

811 Earthwork — Volume — Cu. Yd.

Includes site grading, site excavation, soil stabilization, soil treatment, and site clearing.

812 Borrow — Volume in Place — Cu. Yd.

Includes all work to provide off site borrow material compacted in place.

813 Topsoil, Seed, Sod — Area Graded — Sq. Yd.

Includes fine grading and leveling, fertilizer and limestone application, spreading and leveling topsoil, seeding, mulching, and sodding.

814 Landscaping — Area Planted — Sq. Yd.

Includes trees, shrubs, ground covers, and planters.

815 Site Irrigation — Number Sprinkler Heads — Each

Includes underground piping, sprinkler heads, and controls to install a site irrigation system.

816 Surcharge — Fill in Surcharge — Cu. Yd.

Includes cost for obtaining, placing, monitoring, and removing to disposal area all required surcharge. If used for site fill does not include cost of compaction.

817 Earth Replacement — Earth Replaced (In place Measurement) — Cu. Yd.

Includes cost of removing and disposal of existing material, obtaining, placing, compacting, and proof rolling new material for improved bearing capacity. (See 846 for contaminated earth removal.)

818 Environmental Protection — Area Protected — Acres

Includes all cost for temporary environmental protection work required due to clearing, grubbing, earthwork, or other related work. (Measure to nearest tenth of acre.)

819 Write-in System

SITE IMPROVEMENTS

821 Site Improvements — Area Developed — Sq. Yd.

Includes retaining walls, terrace and perimeter walls, signs, site furnishings, fountains, pools and watercourse, flagpoles, and other miscellaneous related items.

822 - 825 Write-in Systems

SITE PREPARATION

831 Site Dewatering (Major) Header — Pipe — Lin. Ft.

Includes all costs for site dewatering where the dewatering is a major item and is required for several systems. Includes furnishing, installing, operating, and removal of well pointing system.

Figure 22 (Continued)
Cost Model Questionnaire

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832 - 835 Write-in Systems

DEMOLITION

841 Remove Utilities — Length of Run — Lin. Ft.
Sizes

Includes excavation, utility removal, capping, and backfill as required to remove existing utility lines.

842 Remove Paving and Slabs — Area Removed — Sq. Yd.
Thickness

Includes removal and disposal of material from both bituminous and portland cement concrete paved areas.

843 Remove Structures — Structure Volume — Cu. Ft.

Includes complete cost for removal of structure, foundation, and interior equipment. Removal of hazardous materials is not included. See Systems 844 and 845.

_01 Wood
_02 Masonry
_03 Concrete
_04 Pre-engineered
_05 Other

844 Remove/Dispose of Asbestos (Exterior) — Total Cost — Lump Sum

Includes costs for removal of asbestos from any exterior areas including buildings to be demolished. Included are costs for sealing off areas, asbestos removal, monitoring testing disposal, change areas, disposable suits, respirators, and other related costs. (See System 172 for interior asbestos removal.)

_01 Pipe
_02 Siding
_03 Insulation
_04 Roofing

845 Remove/Dispose of P.C.B. — Total Cost — Lump Sum

Includes costs for removing and disposing of P.C.B. contaminated areas. Included are costs for sealing off area, removal of contaminated materials, monitoring, testing change area, disposal suits, respirators, disposal, and other related cost.

846 Remove/Dispose of Contaminated Earth — Earth Removed — Cu. Yd.

Includes cost of removing and disposing of existing industrial and hazardous materials.

847 Write-in System

Figure 22 (Continued)
Cost Model Questionnaire

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STORM DRAINAGE

851 Storm Drainage Piping — Length of Run — Lin. Ft. / Size

Includes utility service connections, excavation and backfill, sheeting and shoring, dewatering, pipe and fittings, manholes, catch basins, curb inlets, dry wells, ditches and culverts, retention ponds, and headwalls.

_01 Concrete
_02 Corrugated metal
_03 Other

852 Box and Arch Culvert — Length of Culvert — Lin. Ft. / Size

Includes excavation, backfill, forming, concrete, reinforcing steel accessories, steel arch, and bedding materials.

853 Drainage Facing Materials — Surface Area — Sq. Yd. / Type

Includes stone riprap, fabrics, and other materials used as facing materials for outfalls and slope protection from storm water runoff.

854 - 856 Write-in Systems

FENCING

861 Fencing (Perimeter) — Length of Fence — Lin. Ft. / Height

Includes footings, posts, fencing materials, gates, and turnstiles for perimeter fencing. Includes station perimeter and individual facility. Does not include fencing for fields in Systems 872, 874, 875. See 621 for mechanically operated security gate.

_01 Gates
_02 Barb Wire

862 Fencing (Alarm) — Length of Fence — Lin. Ft. / Height

Includes footing post fencing materials gate and turnstiles. When fence is part of a security alarm system.

863 - 865 Write-in Systems

RECREATION EQUIPMENT / FIELDS

871 Playground Equipment — Pieces of Equipment — Each

Includes swings, slides, and other items of recreation-type equipment for childrens' playgrounds.

872 Playing Field Equipment — Pieces of Equipment — Each

Includes costs for playing field equipment not shown elsewhere.

873 Tennis Courts — Number of Courts — Each
Includes paving, fencing, lighting, and nets for a complete tennis court.

Figure 22 (Continued)
Cost Model Questionnaire

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874 Softball/Baseball Fields — Number of Fields — Each

Includes grading, topsoil and seeding drainage, infield, dugouts, fencing, and lighting as required for a complete softball or baseball field.

875 Football/Soccer Fields — Number of Fields — Each

Includes grading, topsoil and seeding, drainage, fencing, goals, benches, and lighting as required for a complete football or soccer field.

876, 877 Write-in Systems

HEAVY CONSTRUCTION

SPECIAL BUILDING FOUNDATIONS

911 Land Piling -- Driven — Length of Piling — Lin. Ft.

Piling driven from land-based equipment used for building foundations or bulkhead tieback systems. Includes mobilization and demobilization, pile cutoff, and piling. Includes pile tests if not a separate bid item.

**** For SYSTEM LAND PILING-DRIVEN ****
**** See Under Foundations(System 111) in Building Systems ****

912 Caissons — Number of Caissons — Each

Includes excavation, concrete, and reinforcing for foundation caissons.

_01 Cased
_02 Uncased
_03 Other
_11 50 - 100 Tons
_12 101 - 150 Tons
_13 151 - 200 Tons
_14 Other

SPECIAL BUILDING FOUNDATIONS

913 Pressure Injected Footings — Number of Footings — Each

Includes drilling, concrete, and reinforcing a complete pressure injected footing system.

_01 Cased
_02 Uncased
_03 Steel Pipe
_04 Other
_11 50 - 100 Tons
_12 101 - 150 Tons
_13 151 - 200 Tons
_14 Other

914 Vibroflotation — Number of Locations — Each

Includes specialized vibratory probe for penetration and extraction, and furnishing and placing sand backfill.

_01 Sand
_02 Stone
_03 Other
_11 50 - 100 Tons
_12 101 - 150 Tons
_13 151 - 200 Tons
_14 Other

Figure 22 (Continued)
Cost Model Questionnaire

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915 Chemical Soil Stabilization — Number of Locations — Each

Includes grouting equipment and specialized chemicals for injection into unstable sub-surface materials.

_01 Cement
_02 Lime
_03 Flyash
_04 Other

916 Dynamic Consolidation — Number of Locations — Each

Includes equipment, consolidation effort, sand, backfill, and testing.

917 Piling -- Cast-in-Place — Length of Piling — Lin. Ft.

Includes drilling, casing concrete, and reinforcing for a complete cast-in-place piling system.

_01 Cased
_02 Uncased
_03 Steel Pipe
_04 Auger Cast
_05 Concrete Filled
_06 Other

_11 0 - 50 Tons
_12 51 - 100 Tons
_13 101 - 150 Tons
_14 Other

RAILROAD

921 Track Work -- Railroad — Length of Track — Lin. Ft.

Includes subgrade preparation, ballast, ties, rail, switches, and other accessories for a complete track system.

922 - 925 Write-in Systems

MARINE STRUCTURES PIERS

931 Piling -- Marine — Length of Piling — Lin. Ft.

Piling driven from floating equipment or land equipment used to support piers or similar waterfront structures. Includes mobilization and demobilization, pile cutoff and piling. Includes pile tests if not a separate bid item.

932 Dolphins — Number of Dolphins — Each

Includes piling, cap and accessories for a complete dolphin.

933 Med Mooring — Total Cost — Lump Sum

Includes chain, anchors, buoys and other accessories for a complete Med mooring.

934 Pier Deck — Area of Deck — Sq. Ft.

Includes mobilization and demobilization, cast-in-place or precast concrete pile caps, trench sections, deck panels, manholes or transformer vaults, embedded conduit, drains, and expansion joints.

935 Fenders — Length of Fender — Lin. Ft.

Includes mobilization and demobilization, wales, chocks, fender piles, rubber fenders, pneumatic fenders, and associated hardware.

Figure 22 (Continued)
Cost Model Questionnaire

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936 Pile Clusters — Number of Clusters — Each

Includes piles and accessories required for a complete pile cluster.

937 Marine Specialties — Total Cost — Lump Sum

Includes mobilization and demobilization, bollards, cleats, brows, camels, oil booms, and other marine specialities.

938 Write-in System

MARINE STRUCTURES SPECIAL

941 Bulkheads/Seawalls — Total Face Area — Sq. Ft.

Includes mobilization and demobilization, sheet piles, concrete cap, relieving platform, and tieback system.

942 Erosion Protection Marine — Area Covered — Sq. Yd.

Includes mobilization and demobilization, filled bags, concrete and asphalt surfaces, ground mats, sand fences, precast structural items (dolos, tetrapads, etc.), and filter cloth.

943 Riprap (Marine) — Area Covered — Sq. Yd.

Includes bed preparation, all layers of stone, and other required work to place riprap for slope protection from wave action.

944 - 946 Write-in Systems

MARINE REPAIRS

951 Underdeck Repair — Total Cost — Lump Sum

Includes mobilization and demobilization, repair of cracks, spalls, beam, and cap repairs.

952 Pile Repairs Marine — Length Repaired — Lin. Ft.

Includes costs for removal and replacement of existing concrete including forms and reinforcing and repair for timber piling. All for marine structures.

953 - 957 Write-in Systems

DREDGING

961 Mobilization/Demobilization Dredging — Total Cost — Lump Sum

Includes outfitting and moving equipment to and from site, dike work, spillways, pipeline laying, and other work not directly associated with actual moving of dredged materials.

962 Hydraulic Dredging — Volume — Cu. Yd.

Includes costs for actual moving of material by hydraulic means from in-place location to disposal area.

963 Bucket Dredging — Volume — Cu. Yd.

Includes pumping of material from scow to disposal area.

Figure 22 (Continued)
Cost Model Questionnaire

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PART 3: CONSTRUCTION COST ESTIMATES

Section 16: CONSTRUCTION COST ESTIMATES

16.1 General

16.1.1 Detailed Cost Estimate Required With Each Submittal. Prepare all estimates in the system format either manually or in the Cost Engineering Systems (CES) format. The estimate detail for each submittal shall be consistent with the level of design required for that submittal. Accurate quantity take-off, inclusion of all elements of construction, and accurate unit prices for the project's geographic location are fundamental to the development of a good cost estimate.

16.1.2 Uses of the Cost Estimate. Properly prepared cost estimates provide a check of plans and specifications for constructability, coordination conflicts, discrepancies, and omissions. Cost estimates are used by the Government to establish and verify budget cost, and to develop historical data for future estimating and verification of contract bid prices.

Final Government estimates are to be classified "For Official Use Only."

16.2 Preliminary (35 Percent) Estimate

16.2.1 Preliminary Estimate Based on Materials Take-off. The estimate for this submittal shall reflect cost based on reasonably accurate take-off of materials and systems consistent with the level of design. For those elements of the project where the status of design does not permit a reasonably accurate take-off of quantities or firm pricing of individual items of work, systems unit prices may be used. Lump sum costs are not acceptable. Use of empirical costs shall be minimized.

16.2.2 Preliminary Estimate Should Monitor the Estimated Construction Cost. The 35 percent estimate may be either in manual (use CES systems) or CES format. The 35 percent estimate, through adjustments, should monitor the estimated construction cost until the prefinal estimate is prepared. It is very important that if indications of cost exceeding allotted funds are confirmed, the Project Manager be contacted for instructions.

16.3 Prefinal Estimate

16.3.1 Prefinal Requirements. Prepare the prefinal estimate from complete drawings and specifications. Full and accurate description of each system shall be provided. Quotations must be obtained for all items of substantial quantity or cost. Documentation must be provided for all line items that represent more than one percent of the contract cost or are of such a nature that a standard price is not available. Documentation may be in the form of written quotations or may be taken by telephone. In either case the documentation must show company name, person contacted, and date of quote.

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The first data for each system shall be a narrative description followed by applicable subsystem numbers (CES standard system and subsystem list). Instructions for inserting the system descriptions when using the CES micro programs are available in the CES Estimates Guide.

16.3.2 Structure of Estimates and Work Requiring Multiple Estimates. Prepare separate estimates for each new non-identical building, structure, or addition costing over $50,000 contract cost. Costs of alteration work to existing buildings will not be included with the building addition costs. When one construction contract contains more than one type of work (e.g., new construction, repair, equipment installation), the estimate shall be structured such that each type of work is identified separately. In addition to an overall or composite summary sheet, each type of work requires a separate summary sheet. Costs from these separate summary sheets must be directly transferable to the composite summary sheet.

16.3.3 CES Estimates. For all projects over $1,000,000, all new buildings or additions over $500,000, a CES estimate is required. The A&E will be required to prepare the estimate using the NAVFACENGCOM CES Version 4.2 which is a part of the Construction Criteria Base (CCB). Hard copies of estimates will be submitted with each submittal. With the 100 percent estimate, a floppy disk containing the complete estimate will also be submitted. The A&E shall be responsible for preparing, loading, printing, and checking the estimate prior to prefinal submission. Contact should be made with the Project Manager prior to prefinal submission to reconfirm programmed construction costs for the contract. Where A&E estimate exceeds programmed construction costs, recommendations for cost reduction or proposed bid items should be provided with the prefinal submission. Minimum computer equipment necessary for running estimates is as follows:

a) Hardware

(1) Microcomputer

(2) 640K main memory

(3) Standard microcomputer printer

(4) Fixed disk (8 megabytes minimum)

b) Minimum Software Requirements

(1) MS-PC/DOS Version 2.0 (or greater)

(2) A copy of COMMAND.COM (from the root directory) placed in the CES directory.

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(3) The file CONFIG.SYS in the root directory with the following lines included:

FILES = 31
BUFFERS = 30

These are the minimum values required to run CES.

(4) System time and date must be accurate. Incorrect time and date may affect processing of estimate reports.

c) Workstation (Representative of NAVFACENGCOM Equipment)

(1) Zenith 248 personal computer

(2) 20 megabyte hard disks with 640K main memory

(3) Laser printer

16.4 Final Estimate. Same as prefinal except for correction and inclusion of prefinal comments. When the final A&E estimate varies by more than plus or minus 10 percent of the low bid, an A&E review and analysis of the estimate will be required at no additional cost to the Government. This analysis shall compare the complete project by specification section tabulating a comparison of the Government estimate and contractor's proposal. Reasons for differences are to be stated. A more detailed format may be obtained from the EFD Code 407.

Examples from a CES estimate are included for illustration in Section 17. The summary sheet and factors sheets are followed by the backup report by systems, summary backup report by specification section, detailed backup report by specification section, and printout of loaded data.

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Section 17: SAMPLE CONSTRUCTION COST ESTIMATE

17.1 Summary Report - System. Figure 23 is an example of a CES summary sheet.

17.2 Input Report. Figure 24 is an example of markup factors and modifier factors.

17.3 Backup Report - System Specification. Figure 25 is a random example of the backup report by systems.

17.4 Summary Report - Specification Section. Figure 26 is an example of the summary backup report by specification section.

17.5 Detailed Backup Report - Specification Section. Figure 27 is a random example of a detailed backup report by specification section.

17.6 Input Listing (Sorted). Figure 28 is an example of a estimate input listing (sorted).

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SUMMARY REPORT
SYSTEM
FINAL 100%

PRINTING DATE : 10/22/91 13
DATA BASE USED : 09/02/91 01
PAGE NUMBER : 1
ESTIMATE NAME : L1900057

ENGINEERING ESTIMATE
N62470-90-B-0057

PROJECT: PUBLIC WORKS FACILITIES
LOCATION: NAVAL HOSPITAL,PORTSMOUTH,VA
ESTIMATORS: LANT NAV FAC ENG COM
PROJECT SIZE: 30330.00 SF
AUTHORIZED CONSTRUCTION FUNDS: 3065000.00

CAT CODE: 219-10
UIC: N00183
P-NO.: P-002
DATE OF ESTIMATE: 10/22/91
BID DATE: 12/03/91

	COST/ SYS BASED ON 30330.00	SYSTEM UNITS	U/M	COST/SYS UNIT	TOTAL MU MATL COST	TOTAL MU LABOR COST	TOTAL MU EQUIP COST	TOTAL CONTRACT COST
BUILDING SYSTEMS								
111 FOUNDATION	3.31	30,330.00	SF	3.31	53,718	39,050	7,583	100,351
112 SLAB ON GRADE	2.79	30,330.00	SF	2.79	46,779	31,441	6,442	84,662
113 STRUCTURAL	1.18	30,330.00	SF	1.18	26,165	9,097	667	35,929
114 SUPPORTED FLOOR	1.46	3,196.00	SF	13.85	27,940	14,978	1,354	44,272
116 ROOF STRUCTURE	4.37	30,330.00	SF	4.37	98,486	25,363	8,556	132,405
141 ROOFING	13.21	30,330.00	SF	13.21	226,424	174,202	98	400,724
142 EXTERIOR WALLS	7.80	12,737.00	SF	18.58	104,729	128,364	3,562	236,654
143 INTERIOR WALLS	3.82	29,510.00	SF	3.93	47,344	66,812	1,800	115,956
144 INTERIOR FINISHES	2.80	30,330.00	SF	2.80	47,278	37,546	0	84,824
145 DOORS	1.17	2,337.00	SF	15.24	29,328	6,288	0	35,617
146 WINDOWS	3.59	440.00	SF	247.59	90,925	18,016	0	108,941
147 SPECIALITIES	0.78	30,330.00	SF	0.78	19,665	3,824	23	23,512
211 PLUMBING - DOMESTIC	4.46	50.00	EA	2,705.78	96,402	38,879	8	135,289
221 HVAC	7.49	26.30	TN	8,635.81	127,999	98,933	190	227,122
223 HEATING AND VENT.	3.31	1,213.00	MBT	82.83	46,458	53,931	84	100,473
224 MECHANICAL VENT.	0.30	24,775.00	CFM	0.37	6,568	2,452	25	9,046
232 MONORAILS	1.03	9.00	TNS	3,465.84	29,694	1,404	95	31,193
238 VERTICAL LADDERS	0.51	40.00	VLF	389.55	9,006	4,893	1,683	15,582
244 COMPRESSED AIR	0.38	1.00	LS	11,630.10	7,419	4,211	0	11,630
251 DUST COLLECTION	0.84	6,400.00	CFM	3.98	19,464	5,889	142	25,495
271 SPRINKLERS	3.08	30,330.00	SF	3.08	32,394	61,027	24	93,445
275 FIRE ALARM	0.26	30,330.00	SF	0.26	4,763	3,248	0	8,010
311 POWER	3.84	300.00	KW	387.89	62,709	53,659	0	116,367
312 LIGHTING	2.34	30,330.00	SF	2.34	40,971	30,120	0	71,092
341 TELEPHONE	0.28	30,330.00	SF	0.28	2,206	6,344	0	8,550
SUBTOTAL	74.42				1,304,834	919,970	32,336	2,257,140
HEAVY CONSTRUCTION								
911 LAND PILING DRIVEN		12,510.00	VLF	11.03	47,437	43,751	46,755	137,943
SUBTOTAL HEAVY CONSTRUCTION					47,437	43,751	46,755	137,943
UTILITIES								
511 ELECTRICAL DISTRIBUTION,		220.00	LF	85.22	11,193	7,468	88	18,749
512 ELECTRICAL DISTRIBUTION,		110.00	LF	42.68	3,474	1,196	26	4,695
513 SUBSTATION/TRANSFORMER		300.00	KVA	44.58	11,850	1,179	344	13,373

Figure 23
Summary Report - System

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SUMMARY REPORT M62470-90-B-0057 PRINTING DATE: 10/22/91 13
SYSTEM DATA BASE DATE: 09/02/91 01
PAGE NUMBER: 2

	COST/ SYS BASED ON 30330.00	SYSTEM UNITS	U/M	COST/SYS UNIT	TOTAL MU MATL COST	TOTAL MU LABOR COST	TOTAL MU EQUIP COST	TOTAL CONTRACT COST
514 AREA LIGHTING		8.00	EA	1,141.64	5,220	3,059	854	9,133
521 FIRE ALARM		990.00	LF	5.00	2,683	2,077	194	4,954
523 COMMUNICATION TELEPHONE		220.00	LF	9.07	1,064	880	51	1,995
542 HEAT DISTRIBUTION UNDERG		100.00	LF	197.93	14,663	4,879	251	19,793
551 WATER DISTRIBUTION		51.00	LF	67.35	2,327	804	304	3,435
554 SANITARY SEWERS		269.00	LF	205.55	31,537	19,109	4,648	55,294
SUBTOTAL UTILITIES					84,011	40,650	6,760	131,420
SITE PREPARATION & IMPROVEMENTS								
712 FLEXIBLE ROADS		4,810.00	SY	17.63	68,262	9,148	7,374	84,784
722 FLEXIBLE PARKING		7,615.00	SY	14.61	83,489	19,540	8,237	111,266
741 CONCRETE WALKS		600.00	SY	30.22	6,998	10,587	546	18,131
811 EARTHWORK		5,940.00	CY	2.03	0	4,849	7,232	12,081
812 BORROW		2,970.00	CY	10.72	9,650	6,062	16,130	31,841
813 TOPSOIL/SEED/SOD		1,200.00	SY	3.38	1,530	2,331	197	4,057
814 LANDSCAPING		350.00	SY	30.44	5,894	4,382	379	10,656
818 ENVIRONMENTAL PROTECTION		3.60	AC	4,644.75	6,300	10,409	11	16,721
842 REMOVE PAVING AND SLABS		12,484.00	SY	1.91	0	7,252	16,566	23,818
SUBTOTAL SITE PREPARATION & IMPROVEMENTS					182,123	74,561	56,671	313,356
TOTAL SUPPORT COSTS(UTIL & SITE PREP)					266,134	115,211	63,431	444,776
BASE BID								
ITEM A					1,618,405	1,078,931	142,522	2,839,859
TOTAL CONTRACT COST.....BASE BID					1,618,405	1,078,931	142,522	2,839,859
TOTAL CONTRACT COST (ROUNDED)								2,840,000
TOTAL ESTIMATE CONTRACT					1,618,405	1,078,931	142,522	2,839,859
TOTAL ESTIMATE CONTRACT (ROUNDED)								2,840,000

Figure 23 (Continued)
Summary Report - System

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```
INPUT REPORT                                          PRINTING DATE   : 10/22/91   13
MARK-UP                                               DATA BASE USED  : 09/02/91   01
FINAL 100%                                            PAGE NUMBER     :          1
                                                      ESTIMATE NAME   : L1900057

                                ENGINEERING ESTIMATE
                                N62470-90-8-0057
          PROJECT: PUBLIC WORKS FACILITIES            CAT CODE: 219-10
          LOCATION: NAVAL HOSPITAL,PORTSMOUTH,VA      UIC: N00183
          ESTIMATORS: LANT NAV FAC ENG COM            P-NO.: P-002
          PROJECT SIZE:   30330.00  SF               DATE OF ESTIMATE: 10/22/91
          AUTHORIZED CONSTRUCTION FUNDS:   3065000.00 BID DATE: 12/03/91

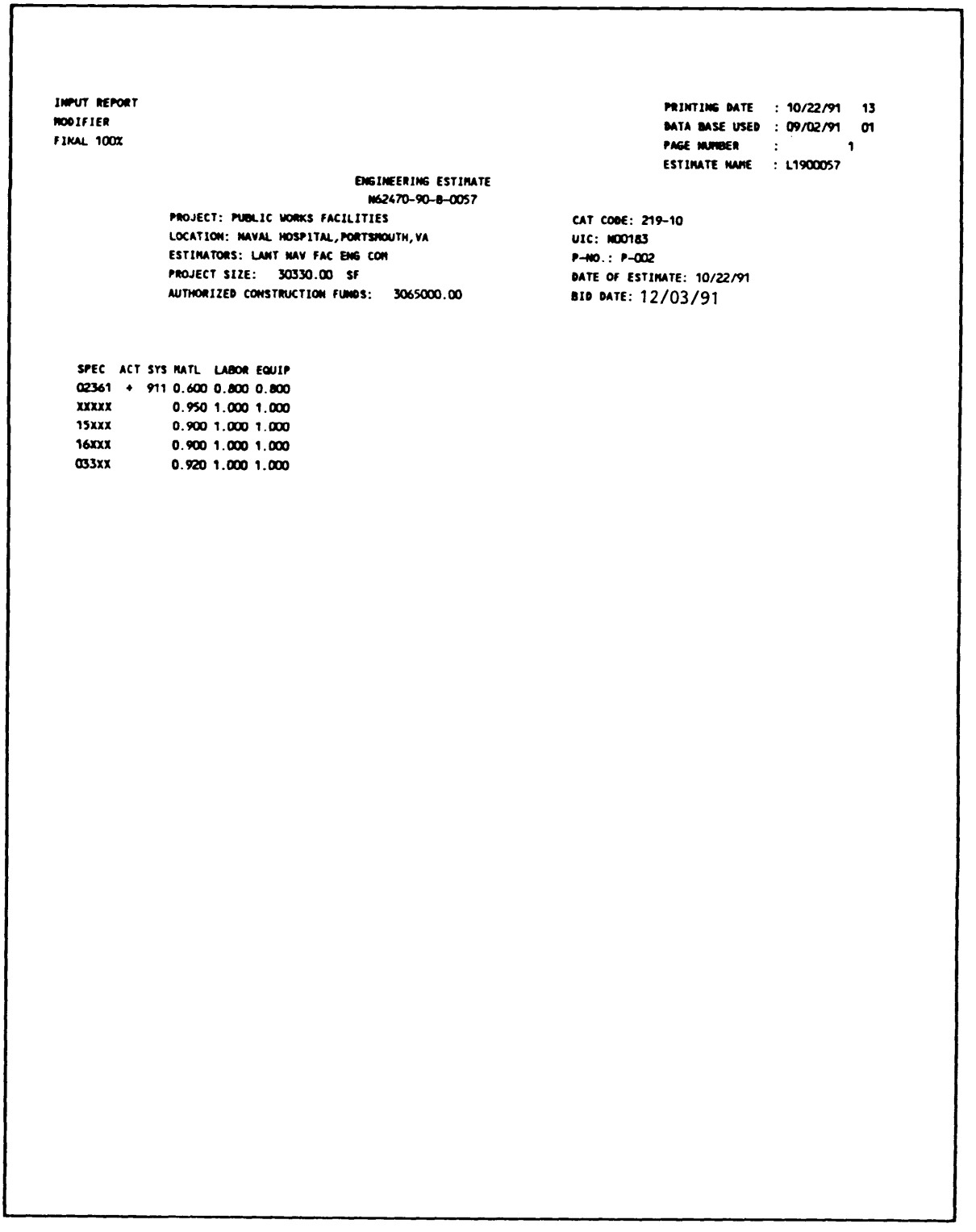
                                                      SPECIFICATION SECTIONS
PRIME MARK-UP                                         MARKED UP FOR PRIME

   DESIGN CONTINGENCIES       0.00%                   XXXXX
   TAX ON MATERIAL            4.5%
   TAX & INSURANCE ON LABOR  20.0%
   TAX ON EQUIPMENT           4.5%
   PRIME OVERHEAD  MAT'L  LABOR  EQUIP
                    12%    12%    12%
   PRIME PROFIT    MAT'L  LABOR  EQUIP
                     8%     8%     8%
   BOND                     1.00%
   MISC. TAXES              0.0%
   CQC                      0.00%
   ESCALATION               0.0%
   PCAS                     0.00%
   CONT                     0.00%
   SIOH                     0.00%
   MATERIAL COMPOSITE MARK-UP  1.277
   LABOR COMPOSITE MARK-UP     1.466
   EQUIPMENT COMPOSITE MARK-UP 1.277

                                                      SPECIFICATION SECTIONS
SUB MARK-UP A                                         MARKED UP FOR SUB
   DESIGN CONTINGENCIES       0.00%                   02XXX 15XXX
   TAX ON MATERIAL            4.5%                    16XXX
   TAX & INSURANCE ON LABOR  20.0%
   TAX ON EQUIPMENT           4.5%
   SUB OVERHEAD    MAT'L  LABOR  EQUIP
                    12%    12%    12%
   SUB PROFIT      MAT'L  LABOR  EQUIP
                     8%     8%     8%
   PRIME OVERHEAD  MAT'L  LABOR  EQUIP
                     4%     4%     4%
   PRIME PROFIT    MAT'L  LABOR  EQUIP
                     3%     3%     3%
   BOND                     1.00%
   MISC. TAXES              0.0%
   CQC                      0.00%
   ESCALATION               0.0%
   PCAS                     0.00%
   CONT                     0.00%
   SIOH                     0.00%
   MATERIAL COMPOSITE MARK-UP  1.368
   LABOR COMPOSITE MARK-UP     1.570
   EQUIPMENT COMPOSITE MARK-UP 1.368
```

Figure 24
Input Report

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INPUT REPORT
MODIFIER
FINAL 100%

PRINTING DATE : 10/22/91 13
DATA BASE USED : 09/02/91 01
PAGE NUMBER : 1
ESTIMATE NAME : L1900057

ENGINEERING ESTIMATE
N62470-90-B-0057

PROJECT: PUBLIC WORKS FACILITIES
LOCATION: NAVAL HOSPITAL,PORTSMOUTH,VA
ESTIMATORS: LANT NAV FAC ENG COM
PROJECT SIZE: 30330.00 SF
AUTHORIZED CONSTRUCTION FUNDS: 3065000.00

CAT CODE: 219-10
UIC: N00183
P-NO.: P-002
DATE OF ESTIMATE: 10/22/91
BID DATE: 12/03/91

SPEC	ACT	SYS	MATL	LABOR	EQUIP
02361	+	911	0.600	0.800	0.800
XXXXX			0.950	1.000	1.000
15XXX			0.900	1.000	1.000
16XXX			0.900	1.000	1.000
033XX			0.920	1.000	1.000

Figure 24 (Continued)
Input Report

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BACKUP REPORT
SYSTEM-SPEC
FINAL 100%

PRINTING DATE : 10/22/91 13
DATA BASE USED : 09/02/91 01
PAGE NUMBER : 1
ESTIMATE NAME : L1900057

ENGINEERING ESTIMATE
N62470-90-B-0057

PROJECT: PUBLIC WORKS FACILITIES
LOCATION: NAVAL HOSPITAL;PORTSMOUTH,VA
ESTIMATORS: LANT NAV FAC ENG COM
PROJECT SIZE: 30330.00 SF
AUTHORIZED CONSTRUCTION FUNDS: 3065000.00

CAT CODE: 219-10
UIC: N00183
P-NO.: P-002
DATE OF ESTIMATE: 10/22/91
BID DATE: 12/03/91

		QUAN U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
111 FOUNDATION						
SYSDC PERIMETER AND INTERIOR WALLS ARE SUPPORTED BY GRADE BEAMS						
SYSDC1 STL COLUMNS ARE SUPPORTDBY PILES AND PILE CAP						
02221 EARTHWORK FOR STRUCTURES & PAVMENTS						
02210 SITE GRADING						
BIGBE	STRUCTURAL EXCAVATION		0.00	2.04	3.13	5.17
	3/4 CY BACKHOE					
	UNCLASSIFIED EARTH	1,494.00 CY	0	3,048	4,676	7,724
BIGFA	FOUNDATION BACKFILL BY					
	HAND W/ COMPACTION BY		0.00	7.69	0.40	8.09
	AIR TAMPER	380.00 CY	0	2,922	152	3,074
	SUBTOTAL-SUBSPEC SECTION 02210		0	5,970	4,828	10,798
	SUBTOTAL-SPEC SECTION 02221		0	5,970	4,828	10,798
	MARKUP		1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02221		0	9,373	6,605	15,978
03300 CAST-IN-PLACE CONCRETE -						
03100 FORMWORK						
EBWRC	FORM GRADE BEAMS WITH		1.05	1.21	0.07	2.33
	PLYWOOD, 3 USES	10,216.00 SF	10,715	12,361	715	23,791
	SUBTOTAL-SUBSPEC SECTION 03100		10,715	12,361	715	23,791
03200 REINFORCING						
EFKF			479.14	437.88*	0.00	917.02
	REBARS GR-60 UP TO #11	14.00 TN	6,708	6,130	0	12,838
	SUBTOTAL-SUBSPEC SECTION 03200		6,708	6,130	0	12,838
03301 CONCRETE PLACEMENT						
EKIMD	PLACE 3000 PSI CONC FOR		54.28	4.41	0.13	58.82
	FTG OR PILE CAP BY CHUTE	170.00 CY	9,228	750	22	9,999
EKWMD	PLACE 3000 PSI CONC FOR					
	WALLS & GRADE BEAMS		54.28	3.53	0.10	57.91
	TO 4' BY CHUTE	284.00 CY	15,416	1,003	28	16,446
	SUBTOTAL-SUBSPEC SECTION 03301		24,643	1,752	51	26,446
	SUBTOTAL-SPEC SECTION 03300		42,066	20,244	766	63,075
	MARKUP		1.277	1.466	1.277	
	TOTAL FOR SPEC SECTION 03300		53,718	29,678	978	84,373
	SUBTOTAL-SYSTEM 111		42,066	26,214	5,594	73,873

Figure 25
Backup Report - System Specification

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SYSTEM-SPEC DATA BASE DATE: 09/02/91 01
PAGE NUMBER: 2

		QUAN U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
TOTAL FOR SYSTEM	111		53,718	39,050	7,583	100,351
COST/SYS UNIT	111					3.31

112 SLAB ON GRADE

SYSDC 5" CONC SLAB REINF WITH WELDED WIREMESH OVER VAPOR BARRIER
SYSDC1 GRAVEL POROUS FILL AND COMPACTED SUB-BASE AND BASE

02250 SOIL TREAT. FOR SUBTER. TERMITE CON

02250 SOIL TREAT. FOR SUBTER. TERMITE CON					
BYAAB SOIL TREATMENT TERMITE					
CONTROL PRETREATMENT		0.02	0.12	0.01	0.15
UNDER SLABS LARGE JOBS	30,330.00 SF	576	3,640	303	4,519
SUBTOTAL-SUBSPEC SECTION 02250		576	3,640	303	4,519
SUBTOTAL-SPEC SECTION 02250		576	3,640	303	4,519
MARKUP		1.368	1.570	1.368	
TOTAL FOR SPEC SECTION 02250		788	5,714	415	6,917

02561 JNTS,REINF, & MOORNG EYES CONC PAVE

02561 JNTS,REINF, & MOORNG EYES CONC PAVE					
B08A JOINTS IN CONC PAVEMENT		0.05	0.52*	0.00*	0.57
(TCJ,LCJ,LCCJ)SAW & SEAL	1,500.00 LF	71	780	0	851
SUBTOTAL-SUBSPEC SECTION 02561		71	780	0	851
SUBTOTAL-SPEC SECTION 02561		71	780	0	851
MARKUP		1.368	1.570	1.368	
TOTAL FOR SPEC SECTION 02561		97	1,225	0	1,322

03300 CAST-IN-PLACE CONCRETE -

02214 BORROW					
EHPF GRAVEL POROUS FILL					
UNDER FLOOR SLABS		14.86	0.74	2.16	17.76
90-100 PASSING 3/4 SIEVE	400.00 CY	5,943	296	864	7,103
SUBTOTAL-SUBSPEC SECTION 02214		5,943	296	864	7,103
03100 FORMWORK					
ECFPHD EDGE FORMS SLAB ON GRADE		1.29	1.09	0.05	2.43
6" TO 12" HIGH 4 USE	1,500.00 LF	1,932	1,635	75	3,642
SUBTOTAL-SUBSPEC SECTION 03100		1,932	1,635	75	3,642
03200 REINFORCING					
EFME		483.00	231.04	0.00	714.04
REBAR #5-#6 SLABS	0.80 TN	386	185	0	571
EGES PLACE WIRE MESH GALV		22.08	9.96	0.00	32.04
SLABS 6X6-W2.9XW2.9	304.00 SQ	6,712	3,028	0	9,740
SUBTOTAL-SUBSPEC SECTION 03200		7,099	3,213	0	10,311
03250 ACCESORIES					
EHHF VAPOR BARRIER, 6 MIL		2.90	3.54	0.00	6.44
POLYETHYLENE	304.00 SQ	881	1,076	0	1,957

Figure 25 (Continued)
Backup Report - System Specification

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BACKUP REPORT N62470-90-B-0057
SYSTEM-SPEC

PRINTING DATE: 10/22/91 13
DATA BASE DATE: 09/02/91 01
PAGE NUMBER: 3

		QUAN	U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
112 SLAB ON GRADE							
EIKH	EXPANSION JT PREMOLDED ASPHALT IMPREG SLAB 1/2 X 6	760.00	LF	0.30 231	0.34 258	0.00 0	0.64 489
EIMN	EXPANSION JT PREMOLDED ASPHALT IMPREG WALL 1/2 X 12	40.00	LF	0.44 18	0.43 17	0.00 0	0.87 35
	SUBTOTAL-SUBSPEC SECTION 03250			1,129	1,352	0	2,481
03301 CONCRETE PLACEMENT							
EKSMN	PLACE CONC 2500 OR 3000# SLAB ON GRADE BY PUMP	350.00	CY	54.28 18,998	4.93 1,726	6.47 2,265	65.68 22,988
EPKS	FINISHING CONC SLAB STEEL TROWEL MACHINE	30,330.00	SF	0.01 279	0.18 5,459	0.05 1,517	0.24 7,255
ERIJ	CURING CONC SLAB SPRAYED MEMBRANE	30,330.00	SF	0.02 558	0.10 3,033	0.00 0	0.12 3,591
	SUBTOTAL-SUBSPEC SECTION 03301			19,835	10,218	3,781	33,834
	SUBTOTAL-SPEC SECTION 03300			35,938	16,713	4,720	57,372
	MARKUP			1.277	1.466	1.277	
	TOTAL FOR SPEC SECTION 03300			45,893	24,502	6,027	76,423
	SUBTOTAL-SYSTEM 112			36,586	21,133	5,023	62,742
	TOTAL FOR SYSTEM 112			46,779	31,441	6,442	84,662
	COST/SYS UNIT 112						2.79
113 STRUCTURAL							
SYSDC	STRUCTURAL MEMBERS ARE STEEL TUBING COLUMN AND WIDE FLANGE STL BEAM						
05120 STRUCTURAL STEEL							
05120 STRUCTURAL STEEL							
GBPE	TS 4"X4"X1/4" COL A36 GALVANIZED	590.00	LB	0.80 471	0.24 142	0.02 12	1.06 624
GBPE	TS 7"X7"X3/16" COL A36 GALVANIZED	2,620.00	LB	0.80 2,091	0.24 629	0.02 52	1.06 2,772
GBPE	W/8X55 STEEL BEAM A36 GALVANIZED	5,720.00	LB	0.80 4,565	0.24 1,373	0.02 114	1.06 6,052
GBPE	W/8X50 STEEL BEAM A36 GALVANIZED	2,700.00	LB	0.80 2,155	0.24 648	0.02 54	1.06 2,857
GBPE	W/0X33 STEEL BEAM A36 GALVANIZED	1,782.00	LB	0.80 1,422	0.24 428	0.02 36	1.06 1,885
GBPE	MISC STL SHAPES A36 (GALVANIZED)	3,496.00	LB	0.80 2,790	0.24 839	0.02 70	1.06 3,699
GBPE	TS 8"X8"X5/8" COL A36 STL GLAVANIZED	1,190.00	LB	0.80 950	0.24 286	0.02 24	1.06 1,259

Figure 25 (Continued)
Backup Report - System Specification

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SYSTEM-SPEC DATA BASE DATE: 09/02/91 01
PAGE NUMBER: 4

		QUAN	U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
113 STRUCTURAL							
GBPE	TS 8"X8"X 1/4" COL A6			0.80	0.24	0.02	1.06
	STL GALVANIZED	260.00	LB	207	62	5	275
GBPE	TS 7"X7"X1/4" COL A36 ST			0.80	0.24	0.02	1.06
	GLAVANIZED	450.00	LB	359	108	9	476
GBPE	W/18X65 STEEL BEAM A36			0.80	0.24	0.02	1.06
	GALVANIZED	1,300.00	LB	1,037	312	26	1,375
GBPE	W/24X84 STEEL BEAM A36			0.80	0.24	0.02	1.06
	GALVANIZED	4,368.00	LB	3,486	1,048	87	4,621
GBPG	MISC STL PLATES A36			0.76	0.24	0.02	1.02
	(GALVANIZED)	1,100.00	LB	836	264	22	1,122
	SUBTOTAL-SUBSPEC SECTION	05120		20,368	6,138	512	27,018
	SUBTOTAL-SPEC SECTION	05120		20,368	6,138	512	27,018
	MARKUP			1.277	1.466	1.277	
	TOTAL FOR SPEC SECTION	05120		26,010	8,999	653	35,662
05500 METAL FABRICATIONS							
05500 METAL FABRICATIONS							
GLBLG	ANCHOR BOLTS 3/4"X 8" LG			4.34	2.39	0.39	7.12
	(GALV)	28.00	EA	122	67	11	199
	SUBTOTAL-SUBSPEC SECTION	05500		122	67	11	199
	SUBTOTAL-SPEC SECTION	05500		122	67	11	199
	MARKUP			1.277	1.466	1.277	
	TOTAL FOR SPEC SECTION	05500		155	98	14	267
	SUBTOTAL-SYSTEM	113		20,489	6,205	522	27,217
	TOTAL FOR SYSTEM	113		26,165	9,097	667	35,929
	COST/SYS UNIT	113					1.18
114 SUPPORTED FLOOR							
SYSDC	SUPPORTED FLOOR CONSIST OF CONC SLAB SUPPORTED BY MASONRY WALL						
03300 CAST-IN-PLACE CONCRETE -							
03100 FORMWORK							
EBFRA	FORM ELEVATED SLAB PLYWD			2.89	1.62	0.06	4.57
	1-USE	3,196.00	SF	9,233	5,178	192	14,602
	SUBTOTAL-SUBSPEC SECTION	03100		9,233	5,178	192	14,602
03200 REINFORCING							
EFME				483.00	231.04	0.00	714.04
	REBAR #5-#6 SLABS	15.60	TN	7,535	3,604	0	11,139
	SUBTOTAL-SUBSPEC SECTION	03200		7,535	3,604	0	11,139
03301 CONCRETE PLACEMENT							
EMFMN	PLACE CONC 5000#			58.42	6.28	8.24	72.94
	ELEVATED SLAB BY PUMP	86.00	CY	5,024	540	709	6,273

Figure 25 (Continued)
Backup Report - System Specification

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BACKUP REPORT M62470-90-B-0057
SYSTEM-SPEC

PRINTING DATE: 10/22/91 13
DATA BASE DATE: 09/02/91 01
PAGE NUMBER: 5

		QUAN	U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
114 SUPPORTED FLOOR							
EPKS	FINISHING CONC SLAB			0.01	0.18	0.05	0.24
	STEEL TROWEL MACHINE	3,196.00	SF	29	575	160	764
ERIJ	CURING CONC SLAB SPRAYED			0.02	0.10	0.00	0.12
	MEMBRANE	3,196.00	SF	59	320	0	378
	SUBTOTAL-SUBSPEC SECTION	03301		5,112	1,435	868	7,416
	SUBTOTAL-SPEC SECTION	03300		21,880	10,217	1,060	33,157
	MARKUP			1.277	1.466	1.277	
	TOTAL FOR SPEC SECTION	03300		27,940	14,978	1,354	44,272
	SUBTOTAL-SYSTEM	114		21,880	10,217	1,060	33,157
	TOTAL FOR SYSTEM	114		27,940	14,978	1,354	44,272
	COST/SYS UNIT	114					13.85
116 ROOF STRUCTURE							
SYSDC	METAL ROOF DECK SUPPORTDBY JOISTS LH-SERIES AND K-SERIES						
02559 PORTLAND CEMENT CONCRETE PAVEMENT							
02559 PORTLAND CEMENT CONCRETE PAVEMENT							
DCBB	STEEL EDGE FORMS UP TO			0.38	0.26	0.00	0.64
	12 INCHES DEEP	346.00	LF	131	90	0	221
	SUBTOTAL-SUBSPEC SECTION	02559		131	90	0	221
	SUBTOTAL-SPEC SECTION	02559		131	90	0	221
	MARKUP			1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION	02559		180	141	0	321
03300 CAST-IN-PLACE CONCRETE -							
03200 REINFORCING							
EGFK	PLACE WIRE MESH SLABS			17.56	10.70	0.00	28.26
	4X4-W2.9XW2.9	65.00	SQ	1,142	696	0	1,837
	SUBTOTAL-SUBSPEC SECTION	03200		1,142	696	0	1,837
03301 CONCRETE PLACEMENT							
EMFMG	PLACE CONC 5000#			58.42	7.80	12.41	78.63
	ELEVATED SLAB BY CRANE	80.00	CY	4,674	624	993	6,290
EPKS	FINISHING CONC SLAB			0.01	0.18	0.05	0.24
	STEEL TROWEL MACHINE	6,120.00	SF	56	1,102	306	1,464
ERVJ	CURING CONC WALLS			0.02	0.10	0.00	0.12
	SPRAYED MEMBRANE	6,120.00	SF	113	612	0	725
	SUBTOTAL-SUBSPEC SECTION	03301		4,843	2,338	1,299	8,479
	SUBTOTAL-SPEC SECTION	03300		5,984	3,033	1,299	10,316
	MARKUP			1.277	1.466	1.277	
	TOTAL FOR SPEC SECTION	03300		7,642	4,447	1,659	13,747

Figure 25 (Continued)
Backup Report - System Specification

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BACKUP REPORT M62470-90-B-0057 PRINTING DATE: 10/22/91 13
SYSTEM-SPEC DATA BASE DATE: 09/02/91 01
PAGE NUMBER: 6

		QUAN	U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
116 ROOF STRUCTURE							
05120 STRUCTURAL STEEL							
05120 STRUCTURAL STEEL							
GB8C	ANGLE CROSS BRIDGING &			0.49	0.19	0.02	0.70
	BRACING A-36	5,410.00	LB	2,673	1,028	108	3,809
GBPE	MISC STL SHAPES A36			0.80	0.24	0.02	1.06
	(GALVANIZED)	4,210.00	LB	3,360	1,010	84	4,454
	SUBTOTAL-SUBSPEC SECTION 05120			6,032	2,038	192	8,263
	SUBTOTAL-SPEC SECTION 05120			6,032	2,038	192	8,263
	MARKUP			1.277	1.466	1.277	
	TOTAL FOR SPEC SECTION 05120			7,703	2,988	246	10,937
05210 STEEL JOISTS							
05210 STEEL JOISTS							
GFU				627.00	140.82	84.10	851.92
	JOIST "K" SERIES	8.40	TN	5,267	1,183	706	7,156
GFW				635.15	144.78	86.76	866.69
	JOIST LH SERIES	48.40	TN	30,741	7,007	4,199	41,948
	SUBTOTAL-SUBSPEC SECTION 05210			36,008	8,190	4,906	49,104
	SUBTOTAL-SPEC SECTION 05210			36,008	8,190	4,906	49,104
	MARKUP			1.277	1.466	1.277	
	TOTAL FOR SPEC SECTION 05210			45,982	12,007	6,264	64,254
05311 STEEL ROOF DECKING							
05311 STEEL ROOF DECKING							
GGPDHH	ROOF DECK MET 16 GA GAVL						
	1-1/2" DP NON-COMPOSITE			1.39	0.13	0.01	1.53
	& NON CELLUAR	6,120.00	SF	8,488	796	61	9,345
GGPFHH	ROOF DECK MET 20 GA GALV						
	1-1/2" DP NON-COMPOSITE			0.85	0.13	0.01	0.99
	& NON CELLUAR	24,210.00	SF	20,470	3,147	242	23,859
	SUBTOTAL-SUBSPEC SECTION 05311			28,958	3,943	303	33,204
	SUBTOTAL-SPEC SECTION 05311			28,958	3,943	303	33,204
	MARKUP			1.277	1.466	1.277	
	TOTAL FOR SPEC SECTION 05311			36,979	5,780	387	43,147
	SUBTOTAL-SYSTEM 116			77,114	17,295	6,700	101,108
	TOTAL FOR SYSTEM 116			98,486	25,363	8,556	132,405
	COST/SYS UNIT 116						4.37
141 ROOFING							
SYSDC	SING PLY RFNG OVER R-10 RIGID INSUL W/GUTTER & DOWNSPOUT						

Figure 25 (Continued)
Backup Report - System Specification

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SUMMARY REPORT:
SPEC SECTION
FINAL 100%

PRINTING DATE : 10/22/91 13
DATA BASE USED : 09/02/91 01
PAGE NUMBER : 1
ESTIMATE NAME : L1900057

ENGINEERING ESTIMATE
N62470-90-B-0057

PROJECT: PUBLIC WORKS FACILITIES
LOCATION: NAVAL HOSPITAL,PORTSMOUTH,VA
ESTIMATORS: LANT NAV FAC ENG COM
PROJECT SIZE: 30330.00 SF
AUTHORIZED CONSTRUCTION FUNDS: 3065000.00

CAT CODE: 219-10
UIC: N00183
P-NO.: P-002
DATE OF ESTIMATE: 10/22/91
BID DATE: 12/03/91

	MATERIAL		LABOR		EQUIPMENT		
	SUB SPEC SECT	SPEC SECT	SUB SPEC SECT	SPEC SECT	SUB SPEC SECT	SPEC SECT	SPEC SECT
02050 DEMOLITION AND REMOVAL							
02052 PAVEMENT & SLABS	251		7,371		16,888		
SUBTOTAL SPEC SECTION 02050		251		7,371		16,888	24,509
02221 EARTHWORK FOR STRUCTURES & PAVMENTS							
02210 SITE GRADING	0		14,222		13,837		
02214 BORROW	9,650		6,062		16,130		
02223 ROADWAY EXCAVATION	0		1,873		3,634		
02270 SLOPE PROTECTION AND EROSION CONTRL	6,300		10,409		11		
SUBTOTAL SPEC SECTION 02221		15,950		32,567		33,612	82,129
02225 EXAV & BACKFLL & COMPACT FOR UTILIT							
02224 EXCAVATION & BACKFILL	0		757		305		
SUBTOTAL SPEC SECTION 02225		0		757		305	1,062
02233 GRADED CRUSHED AGGREGATE BASE COURS							
02233 GRADED CRUSHED AGGREGATE BASE COURS	84,301		4,355		1,596		
SUBTOTAL SPEC SECTION 02233		84,301		4,355		1,596	90,251
02250 SOIL TREAT. FOR SUBTER. TERMITE CON							
02250 SOIL TREAT. FOR SUBTER. TERMITE CON	788		5,714		415		
SUBTOTAL SPEC SECTION 02250		788		5,714		415	6,917
02361 ROUND TIMBER PILES							
02361 ROUND TIMBER PILES	47,437		43,750		46,755		
SUBTOTAL SPEC SECTION 02361		47,437		43,750		46,755	137,943
02559 PORTLAND CEMENT CONCRETE PAVEMENT							
02559 PORTLAND CEMENT CONCRETE PAVEMENT	360		282		0		
SUBTOTAL SPEC SECTION 02559		360		282		0	642
02561 JNTS,REINF, & MOORNG EYES CONC PAVE							
02561 JNTS,REINF, & MOORNG EYES CONC PAVE	97		1,225		0		
SUBTOTAL SPEC SECTION 02561		97		1,225		0	1,322
02573 BITUMINOUS HOT MIX PAVEMENT							
02573 BITUMINOUS HOT MIX PAVEMENT	58,015		8,326		9,924		
SUBTOTAL SPEC SECTION 02573		58,015		8,326		9,924	76,265
02577 PAVEMENT MARKINGS							
02577 PAVEMENT MARKINGS	106		319		0		
SUBTOTAL SPEC SECTION 02577		106		319		0	425
02698 EXT BURIED PREINSULATED WATER PIPNG							
02698 EXT BURIED PREINSULATED WATER PIPNG	1,932		552		50		
SUBTOTAL SPEC SECTION 02698		1,932		552		50	2,534

Figure 26
Summary Report - System Specification

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	MATERIAL		LABOR		EQUIPMENT		
	SUB SPEC SECT	SPEC SECT	SUB SPEC SECT	SPEC SECT	SUB SPEC SECT	SPEC SECT	SPEC SECT
02713 EXTERIOR WATER DISTRIBUTION SYSTEM							
02710 PIPE, WATER DISTRIBUTION	77		38		9		
02711 STRUCTURES-WATER DISTRIBUTION	857		962		360		
02715 EXTERIOR CONDENSATE RETURN SYSTEM	1,538		443		229		
SUBTOTAL SPEC SECTION 02713		2,472		1,443		598	4,514
02714 EXTERIOR STEAM DISTRIBUTION							
02701 PIPE-UNDERGROUND STEAM	7,704		1,337		125		
02702 PIPE-ABOVE GROUND STEAM	380		43		0		
SUBTOTAL SPEC SECTION 02714		8,084		1,379		125	9,589
02722 EXTERIOR SANITARY SEWER SYSTEM							
02723 PIPE-STORM DRAINAGE	31,530		18,932		4,548		
SUBTOTAL SPEC SECTION 02722		31,530		18,932		4,548	55,010
02930 TURF							
02930 TURF	1,530		2,331		197		
SUBTOTAL SPEC SECTION 02930		1,530		2,331		197	4,057
02950 TREES, PLANTS & GROUND COVERS							
02950 TREES, PLANTS & GROUND COVERS	5,894		4,382		379		
SUBTOTAL SPEC SECTION 02950		5,894		4,382		379	10,656
SUBTOTAL SPEC DIVISION 02		258,748		133,685		115,392	507,825
03300 CAST-IN-PLACE CONCRETE -							
02214 BORROW	7,589		434		1,103		
03100 FORMWORK	30,414		30,548		1,388		
03200 REINFORCING	30,169		21,020		0		
03250 ACCESORIES	113,686		159,383		0		
03251 CURB, CURB & GUTTER, & VALLEY GUTTR	9,078		13,697		136		
03252 SIDEWALKS	4,334		6,169		135		
03300 CAST-IN-PLACE CONCRETE -	464		433		15		
03301 CONCRETE PLACEMENT	75,885		28,485		9,595		
04201 CONCRETE FILL FOR MASONRY GROUT	1,495		1,074		284		
04202 REINFORCING FOR MASONRY WALLS	4,318		4,061		0		
SUBTOTAL SPEC SECTION 03300		277,432		265,305		12,657	555,395
SUBTOTAL SPEC DIVISION 03		277,432		265,305		12,657	555,395
04200 UNIT MASONRY							
04200 UNIT MASONRY	121,727		173,532		4,986		
04203 PRECAST CONCRETE FOR MASONRY	1,467		1,589		23		
SUBTOTAL SPEC SECTION 04200		123,195		175,121		5,009	303,324
04400 STONE WORK							
04400 STONE WORK	518		390		0		
SUBTOTAL SPEC SECTION 04400		518		390		0	907
SUBTOTAL SPEC DIVISION 04		123,713		175,510		5,009	304,232
05120 STRUCTURAL STEEL							
05120 STRUCTURAL STEEL							

Figure 26 (Continued)
Summary Report - System Specification

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SPEC SECTION

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	MATERIAL		LABOR		EQUIPMENT		
	SUB SPEC SECT	SPEC SECT	SUB SPEC SECT	SPEC SECT	SUB SPEC SECT	SPEC SECT	SPEC SECT
	34,647		12,590		1,010		
SUBTOTAL SPEC SECTION 05120		34,647		12,590		1,010	48,247
05210 STEEL JOISTS							
05210 STEEL JOISTS	55,962		17,847		6,264		
SUBTOTAL SPEC SECTION 05210		55,962		17,847		6,264	80,074
05311 STEEL ROOF DECKING							
05311 STEEL ROOF DECKING	36,979		5,780		387		
SUBTOTAL SPEC SECTION 05311		36,979		5,780		387	43,147
05400 COLD-FORMED METAL FRAMING							
05400 COLD-FORMED METAL FRAMING	1,556		2,272		68		
SUBTOTAL SPEC SECTION 05400		1,556		2,272		68	3,895
05500 METAL FABRICATIONS							
05500 METAL FABRICATIONS	1,340		404		38		
SUBTOTAL SPEC SECTION 05500		1,340		404		38	1,781
SUBTOTAL SPEC DIVISION 05		130,484		38,893		7,768	177,144
06100 ROUGH CARPENTRY							
06100 ROUGH CARPENTRY	442		657		98		
SUBTOTAL SPEC SECTION 06100		442		657		98	1,197
06200 FINISH CARPENTRY							
06200 FINISH CARPENTRY	214		100		0		
SUBTOTAL SPEC SECTION 06200		214		100		0	313
SUBTOTAL SPEC DIVISION 06		655		756		98	1,510
07220 ROOF INSULATION							
07220 ROOF INSULATION	38,267		8,003		0		
SUBTOTAL SPEC SECTION 07220		38,267		8,003		0	46,270
07230 PERIMETER & UNDER-SLAB INSULATION							
07230 PERIMETER & UNDER-SLAB INSULATION	8,808		3,361		0		
SUBTOTAL SPEC SECTION 07230		8,808		3,361		0	12,169
07530 ELASTROMERIC SHT ROOFING SYS(EPDM)							
07530 ELASTROMERIC SHT ROOFING SYS(EPDM)	67,398		5,809		0		
SUBTOTAL SPEC SECTION 07530		67,398		5,809		0	73,207
07600 FLASHING AND SHEET METAL							
07600 FLASHING AND SHEET METAL	8,074		2,330		0		
SUBTOTAL SPEC SECTION 07600		8,074		2,330		0	10,404
07920 SEALANTS AND CAULKINGS							
07920 SEALANTS AND CAULKINGS	2		14		0		
SUBTOTAL SPEC SECTION 07920		2		14		0	16
SUBTOTAL SPEC DIVISION 07		122,548		19,518		0	142,066
08110 STEEL DOORS AND FRAMES							
08110 STEEL DOORS AND FRAMES							

Figure 26 (Continued)
Summary Report - System Specification

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PAGE NUMBER: 4

	MATERIAL		LABOR		EQUIPMENT		
	SUB SPEC SECT	SPEC SECT	SUB SPEC SECT	SPEC SECT	SUB SPEC SECT	SPEC SECT	SPEC SECT
	16,484		2,986		0		
SUBTOTAL SPEC SECTION 08110		16,484		2,986		0	19,471
08210 WOOD DOORS							
08210 WOOD DOORS	2,420		378		0		
SUBTOTAL SPEC SECTION 08210		2,420		378		0	2,798
08331 OVERHEAD COILING DOORS							
08331 OVERHEAD COILING DOORS	7,279		2,021		0		
SUBTOTAL SPEC SECTION 08331		7,279		2,021		0	9,300
08400 ENTRANCES AND STORE FRONT CONST.							
08400 ENTRANCES AND STORE FRONT CONST.	3,144		902		0		
SUBTOTAL SPEC SECTION 08400		3,144		902		0	4,047
08520 ALUMINUM WINDOWS							
08520 ALUMINUM WINDOWS	90,925		18,016		0		
SUBTOTAL SPEC SECTION 08520		90,925		18,016		0	108,941
SUBTOTAL SPEC DIVISION 08		120,253		24,304		0	144,558
09200 LATHING							
09200 LATHING	236		19		0		
SUBTOTAL SPEC SECTION 09200		236		19		0	255
09250 GYPSUM WALLBOARD							
09250 GYPSUM WALLBOARD	2,941		5,166		0		
SUBTOTAL SPEC SECTION 09250		2,941		5,166		0	8,107
09310 CERAMIC TILE, QUARRY TILE, PAVER TL							
09310 CERAMIC TILE, QUARRY TILE, PAVER TL	8,690		6,943		0		
SUBTOTAL SPEC SECTION 09310		8,690		6,943		0	15,633
09500 ACOUSTICAL TREATMENT							
09500 ACOUSTICAL TREATMENT	9,545		13,151		0		
SUBTOTAL SPEC SECTION 09500		9,545		13,151		0	22,696
09650 RESILIENT FLOORING							
09650 RESILIENT FLOORING	910		1,034		0		
SUBTOTAL SPEC SECTION 09650		910		1,034		0	1,943
09661 VINYL COMPOSITION TILE ON CONCRETE							
09661 VINYL COMPOSITION TILE ON CONCRETE	3,849		930		0		
SUBTOTAL SPEC SECTION 09661		3,849		930		0	4,779
09680 CARPET							
09680 CARPET	14,365		1,908		0		
SUBTOTAL SPEC SECTION 09680		14,365		1,908		0	16,273
09910 PAINTING OF BUILDING (FIELD PNTNG)							
09910 PAINTING OF BUILDING (FIELD PNTNG)	17,278		24,810		0		
SUBTOTAL SPEC SECTION 09910		17,278		24,810		0	42,088
SUBTOTAL SPEC DIVISION 09		57,813		53,961		0	111,774
10162 TOILET PARTITIONS							
10162 TOILET PARTITIONS							

Figure 26 (Continued)
Summary Report - System Specification

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BACKUP REPORT:
SPEC SECTION
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PRINTING DATE : 10/22/91 13
DATA BASE USED : 09/02/91 01
PAGE NUMBER : 1
ESTIMATE NAME : L1900057

ENGINEERING ESTIMATE
N62470-90-B-0057

PROJECT: PUBLIC WORKS FACILITIES
LOCATION: NAVAL HOSPITAL,PORTSMOUTH,VA
ESTIMATORS: LANT NAV FAC ENG COM
PROJECT SIZE: 30330.00 SF
AUTHORIZED CONSTRUCTION FUNDS: 3065000.00

CAT CODE: 219-10
UIC: N00183
P-NO.: P-002
DATE OF ESTIMATE: 10/22/91
BID DATE: 12/03/91

		QUAN	U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
02050	DEMOLITION AND REMOVAL						
02052	PAVEMENT & SLABS						
BCAB	REMOVAL OF BITUMINOUS						
	PAVING UP TO 3 INCHES			0.00	0.37	0.97	1.34
	WITH DISPOSAL	12,484.00	SY	0	4,619	12,109	16,729
BCATA	SAW CUTTING BITUMINOUS			0.22	0.09	0.28	0.59
	PAVING PER INCH OF DEPTH	840.00	LF	184	76	235	494
	SUBTOTAL-SUBSPEC SECTION 02052			184	4,695	12,345	17,223
	SUBTOTAL-SPEC SECTION 02050			184	4,695	12,345	17,223
	MARKUP			1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02050			251	7,371	16,888	24,509
02221	EARTHWORK FOR STRUCTURES & PAVMENTS						
02210	SITE GRADING						
BGAAK	SITE GRADING BALANCE CUT						
	& FILL SANDY CLAY 200FT			0.00	0.52	0.89	1.41
	HAUL 75HP DOZER	5,940.00	CY	0	3,089	5,287	8,375
BIGBE	STRUCTURAL EXCAVATION						
	3/4 CY BACKHOE			0.00	2.04	3.13	5.17
	UNCLASSIFIED EARTH	1,494.00	CY	0	3,048	4,676	7,724
BIGFA	FOUNDATION BACKFILL BY						
	HAND W/ COMPACTION BY			0.00	7.69	0.40	8.09
	AIR TAMPER	380.00	CY	0	2,922	152	3,074
	SUBTOTAL-SUBSPEC SECTION 02210			0	9,059	10,115	19,174
02214	BORROW						
BGCBJ	BORROW FILL SAND COMPACT						
	IN PLACE W/ HAUL TO SITE			2.38	1.30	3.97	7.65
	UP TO 30 MI ROUND TRIP	2,970.00	CY	7,054	3,861	11,791	22,706
	SUBTOTAL-SUBSPEC SECTION 02214			7,054	3,861	11,791	22,706
02223	ROADWAY EXCAVATION						
BIHAD	ROADWAY SUBGRADE PREP						
	SCARIFY, GRADE & COMPACT			0.00	0.09	0.20	0.29
	UNCLASSIFIED EARTH	4,810.00	SY	0	433	962	1,395
BIHAD	PARKING LOT SUBGRADE						
	PREP SCARIFY,GRADE AND			0.00	0.09	0.20	0.29
	COMPACT	7,615.00	SY	0	685	1,523	2,208
BIHBH	ROLL & COMPACT BASE			0.00	0.07	0.16	0.23
	MATERIAL	1,070.00	CY	0	75	171	246

Figure 27
Detailed Backup Report - System Specification

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		QUAN U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
	SUBTOTAL-SUBSPEC SECTION 02223		0	1,193	2,656	3,849
02270	SLOPE PROTECTION AND EROSION CONTRL					
BML	STRAW BALE 3'6"LG X 2'WD					
	X 1'6" SECURED TO GRD		0.95	0.63	0.17	1.75
	W/REBAR (1 BALE=35 LBS)	48.00 LF	46	30	8	84
BMLD	TEMP SEDIMENT FENCE 36"H					
	FABRIC W/6" MESH EMBED		1.90	2.75	0.00	4.65
	4" DP LANTDIV DETAIL SF	2,400.00 LF	4,560	6,600	0	11,160
	SUBTOTAL-SUBSPEC SECTION 02270		4,606	6,630	8	11,244
	SUBTOTAL-SPEC SECTION 02221		11,659	20,743	24,570	56,973
	MARKUP		1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02221		15,950	32,567	33,612	82,129
02225	EXAV & BACKFLL & COMPACT FOR UTILIT					
02224	EXCAVATION & BACKFILL					
BQFAP	TRENCH EXCAV, BACKFILL,		0.00	2.61	1.74	4.35
	& COMPACTION BY UTILITY	10.00 CY	0	26	17	44
BQFAP	TRENCH EXCAV, BACKFILL,		0.00	2.61	1.74	4.35
	& COMPACTION BY UTILITY	42.00 CY	0	110	73	183
BQFKE	TRENCHING & BACKFILL IN					
	SANDY CLAY (NO ROCKS)		0.00	0.63	0.25	0.88
	8" WIDE X 24" DEEP	15.00 LF	0	9	4	13
BQFKE	TRENCHING & BACKFILL IN					
	SANDY CLAY (NO ROCKS)		0.00	0.63	0.25	0.88
	8" WIDE X 24" DEEP	5.00 LF	0	3	1	4
BQFKE	TRENCHING & BACKFILL IN					
	SANDY CLAY (NO ROCKS)		0.00	0.63	0.25	0.88
	8" WIDE X 24" DEEP	385.00 LF	0	243	96	339
BQFKE	TRENCHING & BACKFILL IN					
	SANDY CLAY (NO ROCKS)		0.00	0.63	0.25	0.88
	8" WIDE X 24" DEEP	100.00 LF	0	63	25	88
BQFNB	TRENCH BACKFILL COMPACT					
	VIBRATORY COMPACTION TO		0.00	0.59	0.13	0.72
	95%	3.00 CY	0	2	0	2
BQFNB	TRENCH BACKFILL COMPACT					
	VIBRATORY COMPACTION TO		0.00	0.59	0.13	0.72
	95%	1.00 CY	0	1	0	1
BQFNB	TRENCH BACKFILL COMPACT					
	VIBRATORY COMPACTION TO		0.00	0.59	0.13	0.72
	95%	44.00 CY	0	26	6	32
	SUBTOTAL-SUBSPEC SECTION 02224		0	482	223	705
	SUBTOTAL-SPEC SECTION 02225		0	482	223	705
	MARKUP		1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02225		0	757	305	1,062
02233	GRADED CRUSHED AGGREGATE BASE COURS					
02233	GRADED CRUSHED AGGREGATE BASE COURS					
BWAJ	BASE COURSE TYPE II #21A					
	CRUSHER RUN 8" THICK		4.52	0.20	0.09	4.81
	COMPACTED IN PLACE	7,615.00 SY	34,435	1,523	685	36,643

Figure 27 (Continued)
Detailed Backup Report - System Specification

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		QUAN	U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
BWAL	BASE COURSE TYPE II #21A						
	CRUSHER RUN 10" THICK			5.65	0.26	0.10	6.01
	COMPACTED IN PLACE	4,810.00	SY	27,189	1,251	481	28,920
	SUBTOTAL-SUBSPEC SECTION 02233			61,624	2,774	1,166	65,564
	SUBTOTAL-SPEC SECTION 02233			61,624	2,774	1,166	65,564
	MARKUP			1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02233			84,301	4,355	1,596	90,251
02250 SOIL TREAT. FOR SUBTER. TERMITE CON							
02250 SOIL TREAT. FOR SUBTER. TERMITE CON							
BYAAB	SOIL TREATMENT TERMITE						
	CONTROL PRETREATMENT			0.02	0.12	0.01	0.15
	UNDER SLABS LARGE JOBS	30,330.00	SF	576	3,640	303	4,519
	SUBTOTAL-SUBSPEC SECTION 02250			576	3,640	303	4,519
	SUBTOTAL-SPEC SECTION 02250			576	3,640	303	4,519
	MARKUP			1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02250			788	5,714	415	6,917
02361 ROUND TIMBER PILES							
02361 ROUND TIMBER PILES							
CBABC	TREATED WOOD PILES 40 TO						
	50' LONG 12 INCH BUTT 7"			2.68	0.85	1.54	5.06
	TIP 12 LB CREOSOTE	12,420.00	LF	33,236	10,532	19,077	62,845
CBPAUG	PRE-AUGERED TIMBER PILE						
	TO 20' DEEP ON AVERAGE S			0.00*	2.24*	1.68*	3.92
	OIL, 14" DIAM	5,560.00	VLF	0	12,454	9,341	21,795
CBTEST	45FT LONG TEST PILE TO			720.00*	1,600.00*	2,000.00*	4,320.00
	MAX SOTON TEST LOAD	2.00	EA	1,440	3,200	4,000	8,640
CMODEM	MOBILIZATION & DEMOBILIZ						
	ATION INCLUDE SET UP OF			0.00*	1,680.00*	1,760.00*	3,440.00
	PILING EQUIPMENT	1.00	LS	0	1,680	1,760	3,440
	SUBTOTAL-SUBSPEC SECTION 02361			34,676	27,867	34,178	96,720
	SUBTOTAL-SPEC SECTION 02361			34,676	27,867	34,178	96,720
	MARKUP			1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02361			47,437	43,750	46,755	137,943
02559 PORTLAND CEMENT CONCRETE PAVEMENT							
02559 PORTLAND CEMENT CONCRETE PAVEMENT							
DCBB	STEEL EDGE FORMS UP TO			0.38	0.26	0.00	0.64
	12 INCHES DEEP	346.00	LF	131	90	0	221
DCBB	STEEL EDGE FORMS UP TO			0.38	0.26	0.00	0.64
	12 INCHES DEEP	346.00	LF	131	90	0	221
	SUBTOTAL-SUBSPEC SECTION 02559			263	180	0	443
	SUBTOTAL-SPEC SECTION 02559			263	180	0	443
	MARKUP			1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02559			360	282	0	642

Figure 27 (Continued)
Detailed Backup Report - System Specification

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BACKUP REPORT: M62470-90-B-0057
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	QUAN U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
02561 JNTS,REINF, & MOORNG EYES CONC PAVE					
02561 JNTS,REINF, & MOORNG EYES CONC PAVE					
DDBA JOINTS IN CONC PAVEMENT		0.05	0.52*	0.00*	0.57
(TCJ,LCJ,LCCJ)SAW & SEAL	1,500.00 LF	71	780	0	851
SUBTOTAL-SUBSPEC SECTION	02561	71	780	0	851
SUBTOTAL-SPEC SECTION	02561	71	780	0	851
MARKUP		1.368	1.570	1.368	
TOTAL FOR SPEC SECTION	02561	97	1,225	0	1,322
02573 BITUMINOUS HOT MIX PAVEMENT					
02573 BITUMINOUS HOT MIX PAVEMENT					
DFDE ASPHALT WEARING/SURFACE					
COURSE FOR FLEXIBLE PVMT		2.81	0.35	0.49	3.65
2" THK	7,615.00 SY	21,413	2,665	3,731	27,810
DFDF ASPHALT WEARING/SURFACE					
COURSE FOR FLEXIBLE PVMT		4.21	0.52	0.73	5.46
3" THK	4,810.00 SY	20,243	2,501	3,511	26,255
DFDN		34.20*	6.21	0.54	40.95
FEATHERED ASPHALT EDGES	22.00 TON	752	137	12	901
SUBTOTAL-SUBSPEC SECTION	02573	42,409	5,303	7,255	54,966
SUBTOTAL-SPEC SECTION	02573	42,409	5,303	7,255	54,966
MARKUP		1.368	1.570	1.368	
TOTAL FOR SPEC SECTION	02573	58,015	8,326	9,924	76,265
02577 PAVEMENT MARKINGS					
02577 PAVEMENT MARKINGS					
DFGAA PAVEMENT LINE MARKING		0.02	0.05	0.00	0.07
4" WIDE PAINT	4,064.00 LF	77	203	0	280
SUBTOTAL-SUBSPEC SECTION	02577	77	203	0	280
SUBTOTAL-SPEC SECTION	02577	77	203	0	280
MARKUP		1.368	1.570	1.368	
TOTAL FOR SPEC SECTION	02577	106	319	0	425
02698 EXT BURIED PREINSULATED WATER PIPNG					
02698 EXT BURIED PREINSULATED WATER PIPNG					
DIEBCF 2"PIPE FRP PREINSUL JCKT					
OD 6.14"X70 MIL INS		10.73*	2.67*	0.28*	13.68
INSUL 1.81"THK TRENCH	100.00 LF	1,073	267	28	1,368
DIECNO 2" 90 DEG ELL FRP		33.15*	10.90*	1.13*	45.18
TRENCH	.00 EA	133	44	5	181
DIEGNO 2" FLANGE FRP 125#		51.77*	10.23*	1.06*	63.06
TRENCH	4.00 EA	207	41	4	252
SUBTOTAL-SUBSPEC SECTION	02698	1,412	352	37	1,800
SUBTOTAL-SPEC SECTION	02698	1,412	352	37	1,800
MARKUP		1.368	1.570	1.368	
TOTAL FOR SPEC SECTION	02698	1,932	552	50	2,534

Figure 27 (Continued)
Detailed Backup Report - System Specification

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BACKUP REPORT: M62470-90-B-0057
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		QUAN	U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
02713	EXTERIOR WATER DISTRIBUTION SYSTEM						
02710	PIPE, WATER DISTRIBUTION						
DJFVA	6"X4" REDUCER CI MJ			47.54	17.08	6.41	71.03
	W/ 2 BOLT ACCESSORY KITS	1.00	EA	48	17	6	71
DJGKD	DISINFECTION & TESTING						
	OF WATER DISTRIBUTION			0.09	0.11	0.00	0.20
	SYST BY CHLORINATE WATER	20.00	LF	2	2	0	4
DJGKE	TRACER WIRE 0.10 DIA						
	BARE COPPER OR ALUM WIRE			0.08	0.05	0.00	0.13
	FOR NONMETTALIC PIPE	20.00	LF	2	1	0	3
DJGKE	TRACER WIRE 0.10 DIA						
	BARE COPPER OR ALUM WIRE			0.08	0.05	0.00	0.13
	FOR NONMETTALIC PIPE	60.00	LF	5	3	0	8
DJGKF	TAPE: UTILITY WARNING,						
	IDENTIFICATION, & TRACER			0.06	0.05	0.00	0.11
	3" WIDE FOIL BACKED PLAS	20.00	LF	1	1	0	2
	SUBTOTAL-SUBSPEC SECTION 02710			56	24	6	87
02711	STRUCTURES-WATER DISTRIBUTION						
DJGLI	4' STRIPED GUARD POSTS						
	4"DIA X7'LG SCH 40 STEEL			57.76	60.26	25.87	143.89
	12"DIA X 3'DEEP CONC FTG	5.00	EA	289	301	129	719
DJGLI	4' STRIPED GUARD POSTS						
	4"DIA X7'LG SCH 40 STEEL			57.76	60.26	25.87	143.89
	12"DIA X 3'DEEP CONC FTG	4.00	EA	231	241	103	576
DJGLK	CONCRETE THRUST BLOCK			53.20	35.34	15.17	103.71
	1/2 CY	1.00	EA	53	35	15	104
DJGLK	CONCRETE THRUST BLOCK			53.20	35.34	15.17	103.71
	1/2 CY	1.00	EA	53	35	15	104
	SUBTOTAL-SUBSPEC SECTION 02711			626	613	263	1,502
02715	EXTERIOR CONDENSATE RETURN SYSTEM						
DJFVAA				43.23*	18.55*	7.05*	68.83
	4"X2-1/2" REDUCER	1.00	EA	43	19	7	69
DJFVAB				34.01*	18.55*	7.05*	59.61
	4"X1-1/2" REDUCER	1.00	EA	34	19	7	60
DJGHC	6"GATE VALVE (AWWA)						
	FLANGED WITH ADJ VALVE			274.00	50.07	0.00	324.07
	BOX	1.00	EA	274	50	0	324
DJGLH	VALVE BOX ADJUSTABLE FOR						
	VALVES UP TO 20 INCHES			43.40	18.08	7.76	69.24
	5 FT TRENCH DEPTH	1.00	EA	43	18	8	69
DJGLYF				730.00	80.35	34.50	844.85
	12X6 TAPPING SLEEVE	1.00	EA	730	80	35	845
DJGMYP				0.00	96.42	111.00	207.42
	TAP 6 INCH HOLE IN PIPE	1.00	EA	0	96	111	207
	SUBTOTAL-SUBSPEC SECTION 02715			1,125	282	167	1,574
	SUBTOTAL-SPEC SECTION 02713			1,807	919	437	3,164
	MARKUP			1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02713			2,472	1,443	598	4,514

Figure 27 (Continued)
Detailed Backup Report - System Specification

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SPEC SECTION DATA BASE DATE: 09/02/91 01
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		QUAN	U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
02714	EXTERIOR STEAM DISTRIBUTION						
02701	PIPE-UNDERGROUND STEAM						
DJBACA	2-1/2" SCH 40 STEEL PIPE						
	IN CLASS A CONDUIT SYS			29.45*	9.22*	0.98*	39.65
	3" INSUL FA APPROVED	16.00	LF	471	148	16	634
DKEAAA	2" SCH 40 STEEL PIPE						
	IN CLASS A CONDUIT SYS			39.92	6.93	0.76	47.61
	2-1/2" INSUL FA APPROVED	100.00	LF	3,992	693	76	4,761
DKP	2" "RICWILL" ELBOW, BOLT			356.25*	10.90*	0.00*	367.15
	-UP	1.00	EA	356	11	0	367
DKZ	FACTORY REP TO INSPECT						
	FA APPROVED CLASS A PIPE			270.75	0.00	0.00	270.75
	SYS-LESS THAN 30 DAYS	3.00	DAY	812	0	0	812
	SUBTOTAL-SUBSPEC SECTION 02701			5,632	851	92	6,575
02702	PIPE-ABOVE GROUND STEAM						
DLARAC	GATE VALVE 1/2" FORGED			69.48	6.79	0.00	76.27
	STEEL 800 LB THREADED	4.00	EA	278	27	0	305
	SUBTOTAL-SUBSPEC SECTION 02702			278	27	0	305
	SUBTOTAL-SPEC SECTION 02714			5,910	879	92	6,880
	MARKUP			1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02714			8,084	1,379	125	9,589
02722	EXTERIOR SANITARY SEWER SYSTEM						
02723	PIPE-STORM DRAINAGE						
DPB0B	6"PVC GRAVITY SEWER PIPE						
	12-1/2 FT LAYING LENGTH			1.78	0.76	0.00	2.54
	ASTM 3034 SDR 35 29#/JT	269.00	LF	478	204	0	682
DPBEA	CONNECT NEW SEWER LINE			6.56	269.70	90.96	367.22
	TO EXISTING MANHOLE	2.00	EA	13	539	182	734
DPBED	4" CLEANOUT FOR 6" DIA						
	EXTERIOR SANITARY SEWER			289.28	134.85	42.75	466.88
	LANTDIV DETAIL C0-1	1.00	EA	289	135	43	467
DPPUMP	SEWAGE PUMPING STATION						
	W/200 GPM DUPLEX PUMP			22,135.00*	11,000.00*	3,100.00*	36,235.00
	COMPLETE	1.00	EA	22,135	11,000	3,100	36,235
QLLPA	90 DEG BEND FOR 2-1/2"			0.91*	5.30*	0.00*	6.21
	PVC PIPE	3.00	EA	3	16	0	19
QLLRA	2-1/2" PVC FORCE MAIN			0.64*	0.80*	0.00*	1.44
	LINE	205.00	LF	130	164	0	294
	SUBTOTAL-SUBSPEC SECTION 02723			23,049	12,059	3,325	38,432
	SUBTOTAL-SPEC SECTION 02722			23,049	12,059	3,325	38,432
	MARKUP			1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02722			31,530	18,932	4,548	55,010
02930	TURF						
02930	TURF						
DVAAA	AREA PREPARATION FOR						
	SEEDING (GRADE RAKE &			0.00	0.04	0.02	0.06
	CLEAN)	1,200.00	SY	0	48	24	72

Figure 27 (Continued)
Detailed Backup Report - System Specification

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BACKUP REPORT: M62470-90-B-0057
SPEC SECTION

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		QUAN	U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
DVAAH	SEEDING BY HAND (LEVEL			0.26	0.10	0.00	0.36
	AREA 10# PER 1000 SF)	1,200.00	SY	308	120	0	428
DVAAJ	FERTILIZING			0.02	0.10	0.05	0.17
	(23#/1000 SF)	1,200.00	SY	23	120	60	203
DVAAN	LIMING			0.02	0.10	0.05	0.17
	(70#/1000 SF)	1,200.00	SY	23	120	60	203
DVBE	SPREAD TOPSOIL DELIVERED						
	TO SITE BY HAND			5.46	7.69	0.00	13.15
	(LEVEL SITE)	140.00	CY	765	1,077	0	1,841
	SUBTOTAL-SUBSPEC SECTION	02930		1,118	1,485	144	2,747
	SUBTOTAL-SPEC SECTION	02930		1,118	1,485	144	2,747
	MARKUP			1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION	02930		1,530	2,331	197	4,057
02950	TREES, PLANTS & GROUND COVERS						
02950	TREES, PLANTS & GROUND COVERS						
DVCA	PREPARE PLANT BEDS BY			0.00	3.84	0.00	3.84
	HAND 18" DEEP	350.00	SY	0	1,344	0	1,344
DVCAE				0.26	0.08	0.00	0.34
	3/16X4 STEEL EDGING	250.00	LF	64	20	0	84
DVDB8				1.38	0.35	0.00	1.73
	WOOD CHIP MULCH 2" THICK	350.00	SY	482	123	0	605
DVDBG				2.00	0.38	0.00	2.38
	PEAT MOSS 2" THICK	350.00	SY	698	133	0	831
DVDED	LIRIOPE MUSCARI			1.90	3.84	0.00	5.74
	2" CLUMPS (-30 SF)	53.00	EA	101	204	0	304
DVIAW	ILEX CRENATA HELLERI						
	15"-18"			7.84	2.40	0.00	10.24
	(DWARF JAPANESE HOLLY)	60.00	EA	470	144	0	614
DVIBC	JUNIPERUS HORIZONTALIS						
	PLUMOSA 18"-24"			6.89	2.88	0.00	9.77
	(ANDORRA JUNIPER)	40.00	EA	276	115	0	391
DVIBDC	NANDINA DOMESTICA						
	18"-24"			9.98	2.88	0.00	12.86
	(HEAVENLY BAMBOO)	56.00	EA	559	161	0	720
DVIBDG	ILEX X NELLIE R. STEVENS						
	(NELLIE R. STEVENS			9.98	2.88	0.00	12.86
	HOLLY)	11.00	EA	110	32	0	141
DVIBJ	ILEX CORNUTA ROTUNDA						
	18"-24"			9.98	2.88	0.00	12.86
	(DWARF CHINESE HOLLY)	24.00	EA	239	69	0	309
DVIDU	ILEX CORNUTA BURFORDI						
	30"-36"			12.83	4.05	0.00	16.88
	(BURFORD HOLLY)	6.00	EA	77	24	0	101
DVIFL	JUNIPERUS CHIN TORULOSA						
	3'-4'			15.20	6.59	4.32	26.11
	(HOLLYWOOD JUNIPER)	8.00	EA	122	53	35	209

Figure 27 (Continued)
Detailed Backup Report - System Specification

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BACKUP REPORT: M62470-90-B-0057 PRINTING DATE: 10/22/91 13
SPEC SECTION DATA BASE DATE: 09/02/91 01
PAGE NUMBER: 8

		QUAN	U/M	MUP/ EXT	LUP/ EXT	EUP/ EXT	TOTAL
DVOTD	ACER RUBRUM						
	1-3/4"-2" CAL (11'-13')			61.75	32.97	21.60	116.32
	(RED MAPLE)	4.00	EA	247	132	86	465
DVPRA	LAGERSTROEMIA INDICA						
	8'-10'			66.50	18.32	12.00	96.82
	(CRAPEMYRTLE)	13.00	EA	865	238	156	1,259
	SUBTOTAL-SUBSPEC SECTION 02950			4,309	2,791	277	7,377
	SUBTOTAL-SPEC SECTION 02950			4,309	2,791	277	7,377
	MARKUP			1.368	1.570	1.368	
	TOTAL FOR SPEC SECTION 02950			5,894	4,382	379	10,656
	SUBTOTAL SPEC DIVISION 02			258,748	133,685	115,392	507,825
03300	CAST-IN-PLACE CONCRETE -						
02214	BORROW						
EHPF	GRAVEL POROUS FILL						
	UNDER FLOOR SLABS			14.86	0.74	2.16	17.76
	90-100 PASSING 3/4 SIEVE	400.00	CY	5,943	296	864	7,103
	SUBTOTAL-SUBSPEC SECTION 02214			5,943	296	864	7,103
03100	FORMWORK						
EBFRA	FORM ELEVATED SLAB PLYWD			2.89	1.62	0.06	4.57
	1-USE	3,196.00	SF	9,233	5,178	192	14,602
EBWRC	FORM GRADE BEAMS WITH			1.05	1.21	0.07	2.33
	PLYWOOD, 3 USES	10,216.00	SF	10,715	12,361	715	23,791
ECFPGD	EDGE FORMS SLAB ON GRADE			0.92	0.79	0.05	1.76
	TO 6" HIGH 4 USE	2,106.00	LF	1,938	1,664	105	3,707
ECFPHD	EDGE FORMS SLAB ON GRADE			1.29	1.09	0.05	2.43
	6" TO 12" HIGH 4 USE	1,500.00	LF	1,932	1,635	75	3,642
	SUBTOTAL-SUBSPEC SECTION 03100			23,817	20,838	1,087	45,741
03200	REINFORCING						
EFKF				479.14	437.88*	0.00	917.02
	REBARS GR-60 UP TO #11	14.00	TN	6,708	6,130	0	12,838
EFME				483.00	231.04	0.00	714.04
	REBAR #5-#6 SLABS	0.80	TN	386	185	0	571
EFME				483.00	231.04	0.00	714.04
	REBAR #5-#6 SLABS	15.60	TN	7,535	3,604	0	11,139
EGES	PLACE WIRE MESH GALV			22.08	9.96	0.00	32.04
	SLABS 6X6-W2.9XW2.9	304.00	SQ	6,712	3,028	0	9,740
EGFK	PLACE WIRE MESH SLABS			17.56	10.70	0.00	28.26
	4X4-W2.9XW2.9	65.00	SQ	1,142	696	0	1,837
EGFK	PLACE WIRE MESH SLABS			17.56	10.70	0.00	28.26
	4X4-W2.9XW2.9	65.00	SQ	1,142	696	0	1,837
	SUBTOTAL-SUBSPEC SECTION 03200			23,625	14,338	0	37,963
03250	ACCESORIES						
EHMF	VAPOR BARRIER, 6 MIL			2.90	3.54	0.00	6.44
	POLYETHYLENE	30,330.00	SQ	87,896	107,368	0	195,265

Figure 27 (Continued)
Detailed Backup Report - System Specification

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 COST ENGINEERING SYSTEM          LISTING OF :  ESTIMATE INPUT FOR L1900057          DATE:  10/22/91      PAGE:        1
-------------------------------------------------------------------------------------------------------------------------

** BASIC INFORMATION **
 00 01 02 13
 01 FINAL 100%
 02 N62470-90-B-0057 P-002
 03 PUBLIC WORKS FACILITIES
 04 219-10
 05 NAVAL HOSPITAL,PORTSMOUTH,VA
 06 N00183
 07 12/03/91
 08 LANT NAV FAC ENG COM
 09   3065000.00
 10   30330.00
 11 SF
 12 09/02/91
 13 01
** PRIME CONTRACTOR MARKUP **
 14 A XXXXX  4.5 20.0  4.5 12 12 12  8  8  8 1.00  0.0  0.00   0.0  0.00  0.00  0.00
** SUBCONTRACTOR MARKUP **
 15 A 02XXX  4.5 20.0  4.5 12 12 12  8  8  8  4  4  4  3  3  3  1.00  0.0  0.00  0.0  0.00  0.00  0.00
 15 A 15XXX
 15 A 16XXX
** SPEC SECTION MODIFIERS **
 17 02361 + 9110.600 0.800 0.800
 17 XXXXX      0.950 1.000 1.000
 17 15XXX      0.900 1.000 1.000
 17 16XXX      0.900 1.000 1.000
 17 033XX      0.920 1.000 1.000
** END ITEMS **
 SYS ITEM        QTY  UM       SHPWT R SPEC  SSPEC          MATL       LABR        EQP     DESC
 111 AAAAAA  30330.00 SF         0.00                        0.00       0.00       0.00 FOUNDATION

 111 BIGBE    1494.00 CY         0.00  02221 02210           0.00       2.04       3.13 STRUCTURAL EXCAVATION 3/4 CY BACKHOE
                                                                                        UNCLASSIFIED EARTH
 111 BIGFA     380.00 CY         0.00  02221 02210           0.00       7.69       0.40 FOUNDATION BACKFILL BY HAND W/ COMPACT
                                                                                        ION BY AIR TAMPER
 111 EBWRC   10216.00 SF         4.13  03300 03100           1.14       1.21       0.07 FORM GRADE BEAMS WITH PLYWOOD, 3 USES

 111 EFKF       14.00 TN         0.00  03300 03200         520.80     437.88       0.00 REBARS GR-60 UP TO #11

 111 EKIMD     170.00 CY         0.00  03300 03301          59.00       4.41       0.13 PLACE 3000 PSI CONC FOR FTG OR PILE CAP
                                                                                        BY CHUTE
 111 EKWMD     284.00 CY         0.00  03300 03301          59.00       3.53       0.10 PLACE 3000 PSI CONC FOR WALLS & GRADE B
                                                                                        EAMS TO 4' BY CHUTE
 111 SYSDC       0.00            0.00                        0.00       0.00       0.00 PERIMETER AND INTERIOR WALLS ARE SUPPO
                                                                                        RTED BY GRADE BEAMS
 111 SYSDC1      0.00            0.00                        0.00       0.00       0.00 STL COLUMNS ARE SUPPORTD BY PILES AND PI
                                                                                        LE CAP
 112 AAAAAA  30330.00 SF         0.00                        0.00       0.00       0.00 SLAB ON GRADE

 112 BYAAB   30330.00 SF         0.00  02250 02250           0.02       0.12       0.01 SOIL TREATMENT TERMITE CONTROL PRETREA
                                                                                        TMENT UNDER SLABS LARGE JOBS
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Figure 28
Input Listing (Sorted)

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COST ENGINEERING SYSTEM LISTING OF : ESTIMATE INPUT FOR L1900057 DATE: 10/22/91 PAGE: 2

** END ITEMS **

SYS ITEM	QTY	UM	SHPWT	R	SPEC	SSPEC	MATL	LABR	EQP	DESC
112 BDBA	1500.00	LF	0.00		02561	02561	0.05	0.52	0.00	JOINTS IN CONC PAVEMENT (TCJ,LCJ,LCCJ)SAW & SEAL
112 ECFPHD	1500.00	LF	2.74		03300	03100	1.40	1.09	0.05	EDGE FORMS SLAB ON GRADE 6" TO 12" HIGH 4 USE
112 EFME	0.80	TN	2000.00		03300	03200	525.00	231.04	0.00	REBAR #5-#6 SLABS
112 EGES	304.00	SQ	181.34		03300	03200	24.00	9.96	0.00	PLACE WIRE MESH GALV SLABS 6X6-W2.9X W2.9
112 EMHF	304.00	SQ	3.69		03300	03250	3.15	3.54	0.00	VAPOR BARRIER, 6 MIL POLYETHYLENE
112 EMPF	400.00	CY	0.00		03300	02214	16.15	0.74	2.16	GRAVEL POROUS FILL UNDER FLOOR SLABS 90-100 PASSING 3/4 SIEVE
112 EIKH	760.00	LF	0.53		03300	03250	0.33	0.34	0.00	EXPANSION JT PREMOLDED ASPHALT IMPREG SLAB 1/2 X 6
112 EIMN	40.00	LF	1.07		03300	03250	0.48	0.43	0.00	EXPANSION JT PREMOLDED ASPHALT IMPREG WALL 1/2 X 12
112 EKSMN	350.00	CY	0.00		03300	03301	59.00	4.93	6.47	PLACE CONC 2500 OR 3000# SLAB ON GRADE BY PUMP
112 EPKS	30330.00	SF	0.00		03300	03301	0.01	0.18	0.05	FINISHING CONC SLAB STEEL TROWEL MACHINE
112 ERIJ	30330.00	SF	0.03		03300	03301	0.02	0.10	0.00	CURING CONC SLAB SPRAYED MEMBRANE
112 SYSDC	0.00		0.00				0.00	0.00	0.00	5" CONC SLAB REINF WITH WELDED WIREMESH OVER VAPOR BARRIER
112 SYSDC1	0.00		0.00				0.00	0.00	0.00	GRAVEL POROUS FILL AND COMPACTED SUB-BASE AND BASE
113 AAAAAA	30330.00	SF	0.00				0.00	0.00	0.00	STRUCTURAL
113 GBPE	590.00	LB	1.00		05120	05120	0.84	0.24	0.02	TS 4"X4"X1/4" COL A36 GALVANIZED
113 GBPE	2620.00	LB	1.00		05120	05120	0.84	0.24	0.02	TS 7"X7"X3/16" COL A36 GALVANIZED
113 GBPE	5720.00	LB	1.00		05120	05120	0.84	0.24	0.02	W/8X55 STEEL BEAM A36 GALVANIZED
113 GBPE	2700.00	LB	1.00		05120	05120	0.84	0.24	0.02	W/8X50 STEEL BEAM A36 GALVANIZED
113 GBPE	1782.00	LB	1.00		05120	05120	0.84	0.24	0.02	W/0X33 STEEL BEAM A36 GALVANIZED
113 GBPE	3496.00	LB	1.00		05120	05120	0.84	0.24	0.02	MISC STL SHAPES A36 (GALVANIZED)
113 GBPE	1190.00	LB	1.00		05120	05120	0.84	0.24	0.02	TS 8"X8"X5/8" COL A36 STL GLAVANIZED
113 GBPE	260.00	LB	1.00		05120	05120	0.84	0.24	0.02	TS 8"X8"X 1/4" COL A6 STL GALVANIZED
113 GBPE	450.00	LB	1.00		05120	05120	0.84	0.24	0.02	TS 7"X7"X1/4" COL A36 ST GLAVANIZED
113 GBPE	1300.00	LB	1.00		05120	05120	0.84	0.24	0.02	W/18X65 STEEL BEAM A36 GALVANIZED

Figure 28 (Continued)
Input Listing (Sorted)

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COST ENGINEERING SYSTEM LISTING OF : ESTIMATE INPUT FOR L1900057 DATE: 10/22/91 PAGE: 3

** END ITEMS **

SYS	ITEM	QTY	UM	SHPWT	R	SPEC	SSPEC	MATL	LABR	EQP	DESC
113	GBPE	4368.00	LB	1.00		05120	05120	0.84	0.24	0.02	W/24X84 STEEL BEAM A36 GALVANIZED
113	GBPG	1100.00	LB	1.00		05120	05120	0.80	0.24	0.02	MISC STL PLATES A36 (GALVANIZED)
113	GLBLG	28.00	EA	1.47		05500	05500	4.57	2.39	0.39	ANCHOR BOLTS 3/4"X 8" LG (GALV)
113	SYSDC	0.00		0.00				0.00	0.00	0.00	STRUCTURAL MEMBERS ARE STEEL TUBING COLUMN AND WIDE FLANGE STL BEAM
114	AAAAAA	3196.00	SF	0.00				0.00	0.00	0.00	SUPPORTED FLOOR
114	EBFRA	3196.00	SF	11.37		03300	03100	3.14	1.62	0.06	FORM ELEVATED SLAB PLYWD 1-USE
114	EFME	15.60	TN	2000.00		03300	03200	525.00	231.04	0.00	REBAR #5-#6 SLABS
114	EMFMN	86.00	CY	0.00		03300	03301	63.50	6.28	8.24	PLACE CONC 5000# ELEVATED SLAB BY PUMP
114	EPKS	3196.00	SF	0.00		03300	03301	0.01	0.18	0.05	FINISHING CONC SLAB STEEL TROWEL MACHINE
114	ERIJ	3196.00	SF	0.03		03300	03301	0.02	0.10	0.00	CURING CONC SLAB SPRAYED MEMBRANE
114	SYSDC	0.00		0.00				0.00	0.00	0.00	SUPPORTED FLOOR CONSIST OF CONC SLAB SUPPORTED BY MASONRY WALL
116	AAAAAA	30330.00	SF	0.00				0.00	0.00	0.00	ROOF STRUCTURE
116	DCBB	346.00	LF	0.00		02559	02559	0.40	0.26	0.00	STEEL EDGE FORMS UP TO 12 INCHES DEEP
116	EGFK	65.00	SQ	181.34		03300	03200	19.09	10.70	0.00	PLACE WIRE MESH SLABS 4X4-W2.9XW2.9
116	EMFMG	80.00	CY	0.00		03300	03301	63.50	7.80	12.41	PLACE CONC 5000# ELEVATED SLAB BY CRANE
116	EPKS	6120.00	SF	0.00		03300	03301	0.01	0.18	0.05	FINISHING CONC SLAB STEEL TROWEL MACHINE
116	ERVJ	6120.00	SF	0.03		03300	03301	0.02	0.10	0.00	CURING CONC WALLS SPRAYED MEMBRANE
116	GBBC	5410.00	LB	1.00		05120	05120	0.52	0.19	0.02	ANGLE CROSS BRIDGING & BRACING A-36
116	GBPE	4210.00	LB	1.00		05120	05120	0.84	0.24	0.02	MISC STL SHAPES A36 (GALVANIZED)
116	GFU	8.40	TN	2000.00		05210	05210	660.00	140.82	84.10	JOIST "K" SERIES
116	GFW	48.40	TN	2000.00		05210	05210	668.58	144.78	86.76	JOIST LH SERIES
116	GGPDHH	6120.00	SF	5.00		05311	05311	1.46	0.13	0.01	ROOF DECK MET 16 GA GAVL 1-1/2" DP NON-COMPOSITE & NON CELLUAR
116	GGPFHH	24210.00	SF	5.00		05311	05311	0.89	0.13	0.01	ROOF DECK MET 20 GA GALV 1-1/2" DP NON-COMPOSITE & NON CELLUAR
116	SYSDC	0.00		0.00				0.00	0.00	0.00	METAL ROOF DECK SUPPORTD BY JOISTS LH-SERIES AND K-SERIES

Figure 28 (Continued)
Input Listing (Sorted)

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COST ENGINEERING SYSTEM LISTING OF : ESTIMATE INPUT FOR L1900057 DATE: 10/22/91 PAGE: 4

** END ITEMS **

SYS ITEM	QTY	UM	SHPWT	R SPEC	SSPEC	MATL	LABR	EQP	DESC
141 AAAAAA	30330.00	SF	0.00			0.00	0.00	0.00	ROOFING
141 EHHF	30330.00	SQ	3.69	03300	03250	3.15	3.54	0.00	VAPOR BARRIER, 6 MIL POLYETHYLENE
141 HAQVGZ	700.00	BF	206.00	06100	06100	0.52	0.64	0.11	BLOCKING & NAILERS SYP NO 2 COMMON TREATED
141 IENRF	30330.00	SF	6.80	07220	07220	1.04	0.18	0.00	INSULATION RGD PERLITE ROOF 4.00" TK C -.10 R=10.00 HH-I-529
141 IRCRNV	492.00	LF	0.15	07530	07530	0.10	0.04	0.00	LAP SPLICE CEMENT FOR ELASTOMERIC MEM BRANE ROOFING
141 IRERL	30330.00	SF	0.00	07530	07530	0.90	0.13	0.00	CSPE SINGLE PLY MEMBRANE ROOFING .045 TH (NOT INCL MECH FAST)
141 IRERL2	31000.00	EA	0.00	07530	07530	0.91	0.00	0.00	MECH FASTENER & WASHER FOR SINGLE PLY MEMBRANE 10"LONG(12" OC AT LAPS)
141 ISJWP	711.00	SF	5.00	07600	07600	2.50	0.87	0.00	FLASHING BASE STAINLESS STEEL .018
141 IUEDCH	400.00	LF	5.00	07600	07600	5.43	1.05	0.00	DOWNSPOUT ROUND COPPER 16 OZ 5" DIA
141 IULDCH	492.00	LF	5.00	07600	07600	5.50	1.12	0.00	GUTTER HALF ROUND COPPER 16 OZ 5"
141 SYSDC	0.00		0.00			0.00	0.00	0.00	SING PLY RFNG OVER R-10 RIGID INSUL W/G UTTER & DOWNSPOUT
141 SYSDC1	0.00		0.00			0.00	0.00	0.00	RGD PERLITE INSULATION WOOD NAILER, AN D GUTTER & DOWNSPOUT
142 AAAAAA	12737.00	SF	0.00			0.00	0.00	0.00	EXTERIOR WALLS
142 EFME	2.00	TN	2000.00	03300	04202	525.00	349.80	0.00	REBAR #5-#6 MASONRY WALL HORIZ
142 EFIE	4.00	TN	2000.00	03300	04202	525.00	430.25	0.00	REBAR #5-#6 MASONRY WALL VERT
142 ENEXC	230.00	LF	0.00	03300	04201	0.40	0.23	0.09	CONC FOR BOND BEAM 3000# 8"H X 8"W
142 ENEYC	670.00	LF	0.00	03300	04201	0.58	0.27	0.10	CONC FOR BOND BEAM 3000# 8"H X 10"W
142 ENFXC	84.00	LF	0.00	03300	04201	1.05	0.47	0.18	CONC FOR BOND BEAM 3000# 16"H X 8"W
142 ENFYC	360.00	LF	0.00	03300	04201	1.36	0.94	0.20	CONC FOR BOND BEAM 3000# 16"H X 10"W
142 ENMYC	50.00	SF	106.00	03300	04201	0.89	0.47	0.19	CONC FILL CMU CELLS 10" WIDE 3000#
142 FDBA	12737.00	SF	0.11	04200	04200	0.25	0.47	0.00	MASONRY DAMPPROOFING - ASPHALT PRIMER(1 GAL/SQ) & DAMPPRF COAT(30LBS/SQ)
142 FEKC	12737.00	SF	106.00	04200	04200	2.45	3.18	0.13	BRICK,FACE,STD SIZE 4" VENEER 3/8" JOINT
142 FFAJ	230.00	LF	10607.00	04200	04200	1.66	1.51	0.04	BOND BEAM 8TK 8HGT W/O GROUT OR RE INF
142 FFAL	670.00	LF	10607.00	04200	04200	1.78	1.68	0.04	BOND BEAM 10TK 8HGT W/O GROUT OR RE INF

Figure 28 (Continued)
Input Listing (Sorted)

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PART 4: OTHER TYPES OF ESTIMATES

Section 18: ESTIMATES FOR CHANGE ORDERS

18.1 Estimates for Change Orders. Estimates for change orders to the contract plans and specifications shall be accomplished with adequate backup to negotiate. These estimates must use the "Estimate for Change Order" summary sheet (NAVFAC 4330/43). Estimates and summaries must be prepared for both increases and decreases in contract cost. Documentation must also be provided for "no-cost" changes.

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Section 19: ESTIMATES FOR NEGOTIATED CONSTRUCTION CONTRACTS

19.1 Negotiated Construction Contracts. In addition to requirements established for competitively bid construction contracts, estimates for negotiated construction contracts shall meet the following requirements.

a) Use a computer run CES estimate unless estimated construction cost is less than $50,000.

b) Final estimate must accurately reflect the final plans and specifications. Direct comparison with the construction contractors estimate will be made in quantity takeoffs; quotes for major material; crew size, equipment, and production rates; and area labor rates. Field overhead must be itemized as System 001. Where conflicts exist between A&E estimate and construction contractor estimate, the A&E will provide prompt verification for items in question.

c) Should conflicts between plans and specifications occur, the A&E will promptly clarify intent and prepare a draft "as negotiated" amendment.

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Section 20: ESTIMATES FOR SPECIAL PROJECTS

20.1 Two-Step Procedures. OPNAVINST 11010.20, Facilities Projects Manual provides detailed definitions, policy, cost limitations, preparation, and submittal procedures for special project estimates. Projects will be submitted in two steps. The first step submittal will identify the requirement and appropriate cost estimate. The second step submittal, made when funding is probable, will include the engineering and detailed cost data necessary to define and justify work to be done. The two-step procedure is used only for projects funded from Operation and Maintenance, Navy (O&M,N) or Research, Development, Test and Evaluation, Navy (RDT&E,N) appropriations.

The major claimant determines funding potential from the step one submittal. If the funding is programmed, the activity is required to submit the step-two form, NAVFAC Form 11014/64.

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Section 21: SUPPLEMENTAL INFORMATION

21.1 Blank Forms and References. Refer to Appendix A.

21.1.1 Project Data Sheet. Refer to Appendix A.

21.1.2 DD Form 1391. Refer to Appendix A.

21.1.3 Primary Facility Unit Cost Development. Refer to Appendix A.

21.1.4 Supporting Facilities Cost Development. Refer to Appendix A.

21.1.5 Cost Model Explanation. Refer to Appendix A.

21.1.6 Estimate for Change Order. Refer to Appendix A.

21.1.7 Special Project Request. Refer to Appendix A.

21.2 SI Conversion Factors. Refer to Appendix B.

21.3 Units of Measure and Their Symbols. Refer to Appendix C.

21.4 Acronyms and Abbreviations. Refer to Appendix D.

21.5 Air Conditioning Tonnage Guide. Refer to Appendix E.

21.6 Plumbing Fixture Count. Refer to Appendix F.

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APPENDIX A
BLANK FORMS AND REFERENCES

PROJECT DATA SHEET

COMPONENT NAVY	ESTIMATE YEAR	DATE PREPARED	
INSTALLATION AND LOCATION		PROJECT TITLE	
PROGRAM ELEMENT	CATEGORY CODE	PROJECT NUMBER	PROJECT COST ($000)

ITEM	ESCALATED TO:	ACF: FER:	U/M	QUANTITY	UNIT COST	COST ($000)
PRIMARY FACILITY SUPPORTING FACILITIES SUBTOTAL CONTINGENCY (5%) TOTAL CONTRACT COST SIOH (6.0%) TOTAL REQUEST TOTAL REQUEST ROUNDED EQUIPMENT FROM OTHER APPROPRIATIONS			LS	(NON-ADD)		

DESCRIPTION OF PROPOSED CONSTRUCTION:

REQUIREMENT:

PROGRAMMING DATA:

ACTIVITY UIC:	ALTERNATE HOST:	SUP UNIT:
INVESTMENT PROGRAM:	SIR:	SPEC AREA:
MAJ/SUB CLAIMANT CODE:	INVESTMENT CATEGORY:	MOBILIZATION
ACTIVITY PRIORITY:	CLAIMANT PRIORITY:	INDICATOR:
READINESS RATING:	RESOURCE SPONSOR:	FLEP PRIORITY:

PROJECT DETAIL DATA

CAT CODE	DESCRIPTIONS	SCOPE	U/M	CC	VALIDATION INDICATOR

REQUIREMENTS CERTIFICATION

ACTIVITY _____ DATE _____
MAJOR CLAIMANT _____ DATE _____
NAVFACENGCOMHQ _____ DATE _____

Figure A-1
Project Data Sheet

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APPENDIX A (Continued)

LOCATION AND SITE PLAN

SITING RATIONALE

EFD REVIEW ANALYSIS

REQUIRES FURTHER ACTION ?	NO	YES	COMMENTS
EXPLOSIVE SAFETY			
AIRFIELD SAFETY			
ELECTROMAGNETIC RADIATION			
AICUZ VIOLATION			
CHANGE TO APPROVED MASTER PLAN/CIP			
COASTAL ZONE MANAGEMENT			
NATURAL RESOURCES PLAN			
DREDGING/FILLING PERMITS			
WETLAND/FLOODPLAIN			
HAZARDOUS WASTES ON SITE			
CULTURAL RESOURCES IMPACT			
OTHER SITE PROBLEMS			
UTILITIES SUPPORT			
ROADS, PARKING			
PRELIMINARY ENVIRONMENTAL ASSESSMENT			
PRELIMINARY HAZARDS ANALYSIS			

EFD VALIDATION

SITE APPROVED YES_____ NO _____ DEFERRED _____

NAME_____ DATE _____

Figure A-1 (Continued)
Project Data Sheet

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APPENDIX A (Continued)

1. COMPONENT	FY 19__MILITARY CONSTRUCTION PROJECT DATA	2. DATE
NAVY		

3. INSTALLATION AND LOCATION	4. PROJECT TITLE

5. PROGRAM ELEMENT	6. CATEGORY CODE	7. PROJECT NUMBER	8. PROJECT COST ($000)

9. COST ESTIMATES

ACF: FER: ITEM ESCALATED TO:	U/M	QUANTITY	UNIT COST	COST ($000)

10. DESCRIPTION OF PROPOSED CONSTRUCTION

11. REQUIREMENT:

DD FORM 1 DEC 76 1391

Figure A-2
DD FORM 1391

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APPENDIX A (Continued)

PRIMARY FACILITY UNIT COST DEVELOPMENT

TITLE: _____ P#: _____

LOCATION: _____ DATE: _____

PREPARED BY: _____ PH#: _____

	CAT CODE	SIZE	ACF	DATE mo / yr
PROPOSED FACILITY				
GUIDANCE FACILITY				

HISTORICAL GUIDANCE SOURCE: _____*

HISTORICAL GUIDANCE UNIT COST: _____/SF

HISTORICAL UNIT COST X SIZE FACTOR X AREA COST FACTOR X ESCALATION FACTOR = ADJUSTED UNIT COST

* Show date of source as shown in Figure 21.

ADDITIONAL FUNCTIONAL FEATURES

FUNCTIONAL FEATURES	QTY	UNIT COST	TOTAL COST	ADJUSTED U/C	%CHANGE

FUNCTIONAL FEATURES TOTAL: _____ _____

TOTAL PRIMARY FACILITY UNIT COST: _____ includes built-in equipment

Figure A-3
Primary Facility Unit Cost Development

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APPENDIX A (Continued)

TITLE: DATE:

LOCATION: P# FY:

CESSYS (1)	SYSTEM DESCRIPTION (2)	COMPOSITE QUANTITY	COMPOSITE UNIT COST (3)	TOTAL COST
SPECIAL FOUNDATION FEATURES (91X)				
ELECTRICAL UTILITIES (51X - 52X)				
MECHANICAL UTILITIES (54X - 59X)				
ROADS, PARKING, AND SIDEWALKS (7XX)				
SITE IMPROVEMENTS (8XX)				
DEMOLITION (84X)				

NOTES: Place additional information and excess composite items on back.
(1) Refer to Section 7.2 for CES System Identification
(2) Refer to Table 7 for Supporting Facilities Guidance Cost. Provide backup for cost of additional functional features.
(3) Composite Unit Cost Adjusted by Area Cost Factor, Escalation Factor & Size Factor

Figure A-4
Supporting Facilities Cost Development

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APPENDIX A (Continued)
COST MODEL (FACILITY STUDY)

GENERAL: The cost model questionnaire will establish a measure of the project scope early in the planning stage in a manner which will allow empirical prices to be applied. It is recognized that many design decisions cannot be made at this time; however, if a realistic price is to be developed, the needs of the activity relating to a particular project must be defined. The NAVFACENGCOM method of conceptual estimating requires cost development from guidance cost of comparable construction. Any variation from guidance cost must be supported by a definition and cost of functional features. This support is often necessary to provide adequate project funding.

BUILDING: Under normal conditions, the building cost model will be completed at the design cost certification stage. Therefore, this portion of the cost model is not required for the facilities study.

System 000 Building Construction: This system describes the total function of the building. As such, it is the most important information shown on the cost model.

SUPPORTING FACILITIES: The completed cost model (systems 500 through 999) should state what is planned with emphasis upon scope features that generate significant cost. Refer to additional guidance in the cost model questionnaire.

NOTES:

1. This Cost Model Questionnaire serves two purposes: At the budget stage it serves as a means of describing and quantifying project requirements in such a manner that empirical prices can be used to generate a project cost.

   At the final estimate stage it serves as a means of collecting subsystem numbers, for storage in NAVFACENGCOM historical files.

2. The cost model questionnaire systems list includes numbered systems that are "write-in." These numbers may be used to identify systems in the project that are not identified in the questionnaire.

3. Systems followed by (1) are considered built-in equipment.

4. Exterior Systems and Special Equipment need more scope definition for accurate pricing. Show narrative description as you best understand the scope when detailed lists are not available.

5. Building system quantities should be the area in which the system occurs.

6. A completed supporting facilities cost model questionnaire must be included in the facility study cost submittal.

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APPENDIX A (Continued)

Supporting Facilities (Systems 500 through 999)

7. Show the following criteria suitable for the application of historical cost under each applicable system heading:

   a. Size (assume if unknown) and length of utility runs. Include storm drainage requirement. Indicate valves, fire hydrants, manholes, catch basins, and other significant extras. Depth of cut (when over 6 feet) and dewatering requirements must be shown.

   b. Area and assumed section for paved areas. Identify each type and section variation.

   c. Description of work required for grading, cut, and borrow. Describe assumed nature of materials to be moved. If rock excavation is required, state assumptions on removal requirements (e.g., rip, blast, etc.).

   d. Indicate approximate quantities of clearing and grubbing in acres, and whether heavy, medium, or light.

   e. Describe any functional features required outside the building 5-foot line such as sewage lift stations, wash racks and water tanks, and unsuitable soil conditions.

   f. If hazardous materials are present, state type, extent and method of disposal.

   g. See System 911 for Land Piling - Driven.

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EXTERIOR ELECTRICAL

511 Electrical Distribution, Primary	Length of Run (LF) _____
512 Electrical Distribution, Secondary	Length of Run (LF) _____
513 Substation/Transformer	Capacity (KVA) _____
514 Area Lighting	Number of Fixtures (EA) _____
515 Airfield Lighting	Length of Run (LF) _____
516 Lightning Protection	Points of Protection (EA) _____
517, 518 Write-in Systems	

EXTERIOR COMMUNICATION

521 Fire Alarm	Length of Run (LF) _____
522 Security Alarm	Length of Run (LF) _____
523 Communication, Telephone	Length of Run (LF) _____
524 Exterior EMCS (Energy Monitoring Control System)	Length of Run (LF) _____
525, 526 Write-in Systems	

EXTERIOR MECHANICAL

541 Heat Distribution, Overhead	Length of Run (LF) _____
542 Heat Distribution, Underground	Length of Run (LF) _____
543 Chilled Water Distribution	Length of Run (LF) _____
544 Condensate Collection (only)	Length of Run (LF) _____
545 Gas Distribution	Length of Run (LF) _____
546 Compressed Air Distribution	Length of Run (LF) _____
547 Fuel Distribution	Length of Run (LF) _____
548 Exterior Foam	Length of Run (LF) _____
549 Write-in System	

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EXTERIOR WATER DISTRIBUTION

551 Water Distribution	Length of Run (LF) _____
552 Fire Protection Water Distribution	Length of Run (LF) _____
553 Salt Water Distribution	Length of Run (LF) _____
554 Sanitary Sewers	Length of Run (LF) _____
555, 556 Write-in Systems	

BULK STORAGE

561 Fuel Storage, Bulk	Capacity (BBL) _____
562 Fuel Storage, Ready Issue	Capacity (GAL) _____
563 Water Storage	Capacity (GAL) _____
564 - 566 Write-in Systems	

FUELING

571 Vehicle Fueling	Number of Outlets (EA) _____
572 Aircraft Fueling	Number of Hydrants (EA) _____
573 Marine Fueling	Number of Outlets (EA) _____
574 - 576 Write-in Systems	

PUMPING

581 Fuel Pumping Station	Pump Capacity (GPM) _____
582 Water Pumping Station	Pump Capacity (GPM) _____
583 Fire Booster Pump	Pump Capacity (GPM) _____
584 Sewage Pump Station	Pump Capacity (GPM) _____
585 Sewage Lift Station	Pump Capacity (GPM) _____
586 Write-in System	

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TREATMENT FACILITY

591 Water Treatment — Capacity (MGD) _____

592 Domestic Sewage Treatment — Capacity (MGD) _____

593 Industrial Waste Treatment — Capacity (MGD) _____

594 - 596 Write-in Systems

EQUIPMENT BUILDING

611 Mechanical Equipment Building — Gross M.E. Bldg Area (SF)_____

Building required to house mechanical equipment when not connected to the main building and not included in gross area shown under the primary facility.

612 Electrical Equipment Building — Gross E.E. Bldg Area (SF)_____

613 - 615 Write-in Systems

SUPPORTING STRUCTURES MISCELLANEOUS

621 Security Gate — Opening Width (LF) _____

622 Guard House — Gross Building Area (SF) _____

623 - 625 Write-in Systems

POLLUTION ABATEMENT STRUCTURES

641 Oil Water Separators — Capacity (GAL) _____

642 Electrostatic Precipitator — Airflow (SCFM) _____

643 - 645 Write-in Systems

SHOP SUPPORT STRUCTURES

651 Vehicle Grease Racks — Number of Vehicle Racks (EA) ___

652 Vehicle Wash Platforms — Number of Vehicle Bays (EA) ____

653 - 656 Write-in Systems

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AIRCRAFT CLEANING FACILITY

661 Aircraft Washing Facility — Paved Area (SY) _____

662 Aircraft Rinsing Facility — Paved Area (SY) _____

663 - 666 Write-in Systems

PAVING

711 Concrete Roads — Paved Area (SY) _____

712 Flexible Roads — Paved Area (SY) _____

713 Overlay Roads — Paved Area (SY) _____

714 Surface Treatment Roads — Paved Area (SY) _____

715 Slurry Seal Road — Paved Area (SY) _____

716, 717 Write-in Systems

PARKING

721 Concrete Parking — Paved Area (SY) _____

722 Flexible Parking — Paved Area (SY) _____

723 Overlay - Parking — Paved Area (SY) _____

724 Surface Treatment - Parking — Paved Area (SY) _____

725 Slurry Seal Parking — Paved Area (SY) _____

726, 727 Write-in Systems

WALKS

741 Concrete Walks — Paved Area (SY) _____

742 Bituminous Walks — Paved Area (SY) _____

743 Special Walks — Paved Area (SY) _____

744 - 746 Write-in Systems

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AIRCRAFT PARKING CONCRETE

751 Concrete Runways	Paved Area (SY) _____
752 Concrete Taxiways	Paved Area (SY) _____
753 Concrete Aprons	Paved Area (SY) _____
754 Repairs Concrete Paving	Paved Area (SF) _____
755 Crack and Joint Repair	Length of Repair (LF) _____
757, 758 Write-in Systems	

AIRCRAFT PAVING FLEXIBLE

761 Flexible Runways	Paved Area (SY) _____
762 Flexible Taxiways	Paved Area (SY) _____
763 Flexible Aprons	Paved Area (SY) _____
764 Repair Aircraft Flexible Paving	Paved Area Patched (SY)_____
765 Aircraft Paving Overlays	Paved Area (SY) _____
766 Aircraft Paving Slurry Seal	Paved Area (SY) _____
767, 768 Write-in Systems	

INCIDENTAL PAVING WORK

771 Pavement Milling	Milled Area (SY) _____
772 - 776 Write-in Systems	

EARTHWORK

811 Earthwork	Volume (CY) _____
812 Borrow	Volume in Place (CY) _____
813 Topsoil, Seed, Sod	Area Graded (SY) _____
814 Landscaping	Area Planted (SY) _____
815 Site Irrigation	Number of Sprinkler Heads (EA)___

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EARTHWORK (Continued)

816 Surcharge	Fill in Surcharge (CY) _____
817 Earth Replacement	In-Place Measurement (CY) _____
818 Environmental Protection	Area Protected (ACRES) _____
819 Write-in System	

SITE IMPROVEMENTS

821 Site Improvements	Area Developed (SY) _____
822 - 825 Write-in Systems	

SITE PREPARATION MISCELLANEOUS

831 Site Dewatering (Major)	Header Pipe (LF) _____
832 - 835 Write-in Systems	

DEMOLITION

841 Remove Utilities	Length of Run (LF) _____
842 Remove Paving and Slabs	Area Removed (SY) _____
843 Remove Structures *** See attached for bldgs to be demolished ***	Structure Volume (CF) _____
844 Remove/Dispose of Asbestos (Exterior)	Total Cost (LS) _____
845 Remove/Dispose of P.C.B.	Total Cost (LS) _____
846 Remove/Dispose of Contaminated Earth	Volume of Earth Removed (CY) ___
847 Write-in System	

STORM DRAINAGE

851 Storm Drainage Piping	Length of Run (LF) _____
852 Box and Arch Culvert	Length of Culvert (LF) _____
853 Drainage Facing Materials	Surface Area (SY) _____
854 - 856 Write-in Systems	

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FENCING

861 Fencing (Perimeter)	Length of Fence (LF) _____
862 Fencing (Alarm)	Length of Fence (LF) _____
863 - 865 Write-in Systems	

RECREATION EQUIPMENT/FIELDS

871 Playground Equipment	Pieces of Equipment (EA) _____
872 Playing Field Equipment	Pieces of Equipment (EA) _____
873 Tennis Courts	Number of Courts (EA) _____
874 Softball/Baseball Fields	Number of Fields (EA) _____
875 Football/Soccer Fields	Number of Fields (EA) _____
876, 877 Write-in Systems	

SPECIAL BUILDING FOUNDATIONS

911 Piling - Land	Total Length (LF) _____
912 Caissons	Number of Caissons (EA) _____
913 Pressure Injected Footing	Number of Footings (EA) _____
914 Vibroflotation	Number of Locations (EA) _____
915 Chemical Soil Stabilization	Number of Locations (EA) _____
916 Dynamic Consolidation	Number of Locations (EA) _____
917 Piling Cast-In-Place Concrete	Length of Piling (LF) _____

RAILROAD

921 Track Work - Railroad	Length of Track (LF) _____
922 - 925 Write-in Systems	

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MARINE STRUCTURES PIERS

931 Piling - Marine	Total Length (LF) _____
932 Dolphins	Number of Dolphins (EA) _____
933 Med Mooring	Total Cost (LS) _____
934 Pier Deck	Area of Deck (SF) _____
935 Fenders	Length of Fender (LF) _____
936 Pile Clusters	Number of Clusters (EA) _____
937 Marine Specialties	Total Cost (LS) _____
938 Write-in Systems	

MARINE STRUCTURES SPECIAL

941 Bulkheads/Seawalls	Total Face Area (SF) _____
942 Erosion Protection - Marine	Area Covered (SY) _____
943 Riprap (Marine)	Area Covered (SY) _____
944 - 946 Write-in Systems	

MARINE REPAIRS

951 Underdeck Repairs	Total Cost (LS) _____
952 Pile Repairs - Marine	Length Repaired (LF) _____
953 - 957 Write-in Systems	

DREDGING

961 Mobilization/Demobilization Dredging	Total Cost (LS)_____
963 Hydraulic Dredging	Volume (CY) _____
964 Bucket Dredging	Volume (CY) _____
965 - 966 Write-in Systems	

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TOWERS/ANTENNAS

971 Towers	Length (LF) _____
972 Antennas	Total Cost (LS) _____
973 - 976 Write-in Systems	

TUNNEL/PIPE JACKING

981 Tunneling	Volume Removed (CY) _____
982 Pipe Jacking	Length Jacked (LF) _____
983 - 986 Write-in Systems	

HEAVY CONSTRUCTION MISCELLANEOUS

991 Bridges	Deck Area (SF) _____
992 Overpasses	Deck Area (SF) _____
993 - 996 Write-in Systems	

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APPENDIX A (Continued)

PROPOSAL/ESTIMATE FOR CONTRACT MODIFICATION **DATE:**

NAVFAC 4330/43 (8/88)

CONTRACT TITLE: **CONTRACT NO:**

ROICC OFFICE:

DESCRIPTION:

		PRIME CONTRACTOR'S WORK			Revisions/Comments
1. Direct Materials					
2. Sales Tax on Materials	% of line 1	%			
3. Direct Labor					
4. Insurance, Taxes, and Fringe Benefits	% of line 3	%			
5. Rental Equipment					
6. Sales Tax on Rental Equipment	% of line 5	%			
7. Equipment Ownership and Operating Expenses					
8. SUBTOTAL (add lines 1 - 7)					
9. Field Overhead	10.00% of line 8	%			
10. SUBTOTAL (Add Lines 8 & 9)					

Prime Remarks:

		SUB-CONTRACTOR'S WORK			
11. Direct Materials					
12. Sales Tax on Materials	% of line 11	%			
13. Direct Labor					
14. Insurance, Taxes, and Fringe Benefits	% of line 13	%			
15. Rental Equipment					
16. Sales Tax on Rental Equipment	% of line 15	%			
17. Equipment Ownership and Operating Expenses					
18. SUBTOTAL (add lines 11 - 17)					
19. Field Overhead	10.00% of line 18	%			
20. SUBTOTAL (add lines 18 & 19)					
21. Home Office Overhead	3.00% of line 20	%			
22. Profit	% of line 20	%			
23. SUBTOTAL (Add Lines 20 - 22)					

Sub's Remarks:

		SUMMARY			
24. Prime Contractor's Work (from line 10)					
25. Sub-contractor's Work (from line 23)					
26. SUBTOTAL (add lines 24 & 25)					
27. Prime Overhead on sub-contractor	5.00% of line 25	%			
28. Prime's Home Office Overhead	3.00% of line 24	%			
29. Prime's Profit	% of line 26	%			
30. SUBTOTAL (add lines 26 - 29)					
31. Prime Contractor's Bond Premium	% of line 30	%			
32. TOTAL COST (Add Lines 30 & 31)					

Estimated time extension and justification

Prime Contractor name:

Sub-contractor name:

Signature & Title of preparer

Date

Figure A-5
Estimate for Change Order

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APPENDIX A (Continued)

INSTRUCTIONS FOR PREPARING PROPOSAL/ESTIMATE FOR CONTRACT MODIFICATION

All Contract Modification Proposals shall be addressed to the Resident Officer in Charge of Construction. Proposals must clearly state the conditions and scope of the modification and shall be accompanied by a breakdown of cost, as indicated. Lump sum costs will not be accepted in either the prime or sub-contractor's breakdown of direct cost. The total cost for labor, material, and equipment rental (or ownership) for each item shall be transferred to the corresponding item on the front of this form. At the contractor's option, the overhead rates printed on the front of this form may be used for proposals under $ 500,000 in lieu of detailed itemized estimates of overhead costs. The proposal should also include a request for an extension of time, in calendar days, only if overall completion of contract is impacted by the proposed modification. The contractor shall not proceed with any of the work included in the modification prior to receipt of an executed modification of contract (SF30).

BREAKDOWN OF DIRECT COSTS Contract No._____ Date:_____

ITEMS OF WORK FOR Prime Contractor	QTY	UNIT	MATERIAL		LABOR		R	EQUIPMENT		
			Unit Cost	Total Cost	Unit Cost	Total Cost	O	Days	Rate	Total
DIRECT Prime Contractor's TOTALS							R	Total (Rental)		
							O	Total (Owned)		

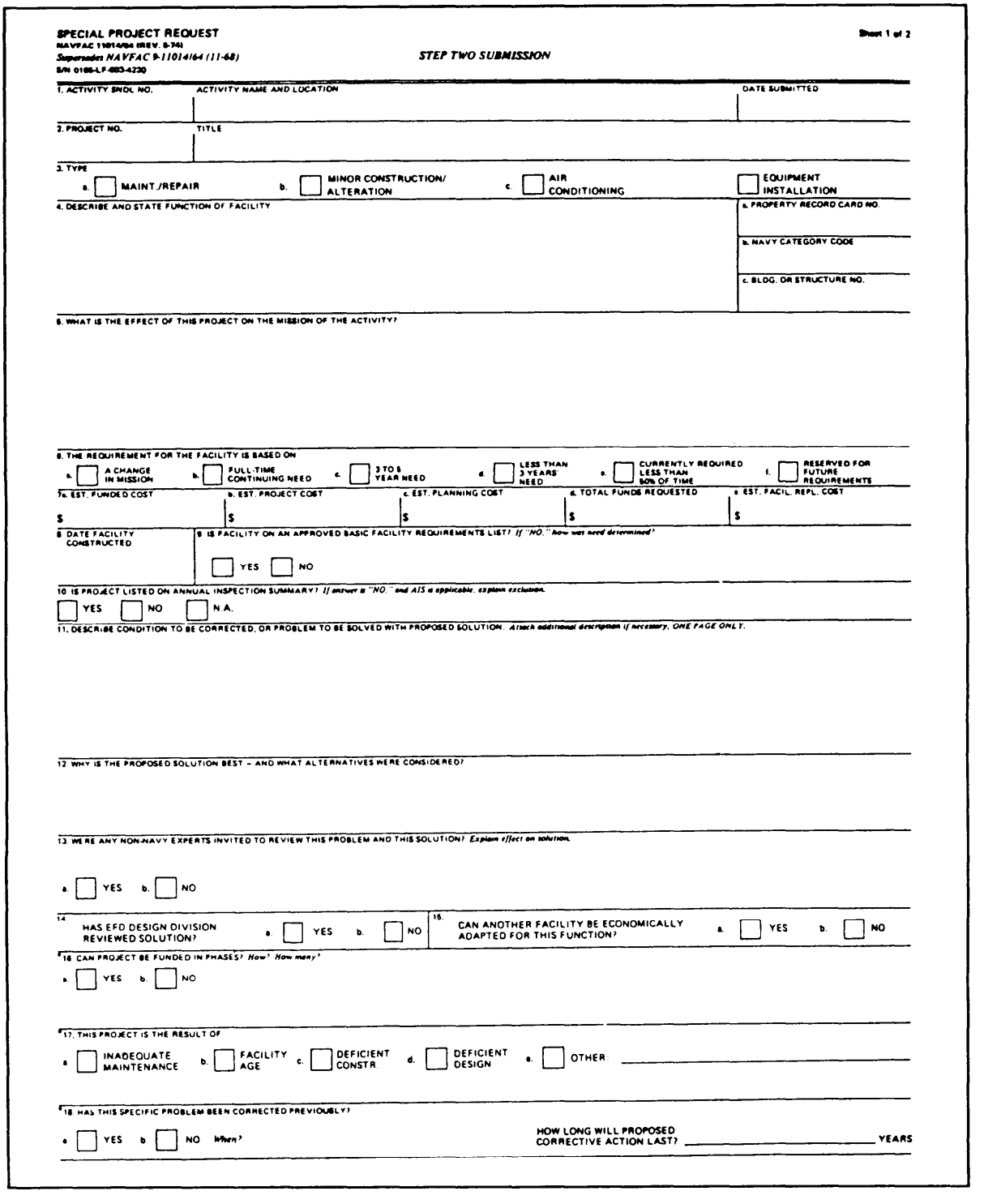
ITEMS OF WORK FOR Sub-contractor	QTY	UNIT	MATERIAL		LABOR		R	EQUIPMENT		
			Unit Cost	Total Cost	Unit Cost	Total Cost	O	Days	Rate	Total
DIRECT Sub-contractor's TOTALS							R	Total (Rental)		
							O	Total (Owned)		

NAVFAC 4330/43 (Back)

Figure A-5 (Continued)
Estimate for Change Order

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APPENDIX A (Continued)



SPECIAL PROJECT REQUEST
NAVFAC 11014/64 (REV. 8-74)
Supersedes NAVFAC 9-11014/64 (11-68)
S/N 0105-LF-003-4230

Sheet 1 of 2

STEP TWO SUBMISSION

1. ACTIVITY SNOL NO. | ACTIVITY NAME AND LOCATION | DATE SUBMITTED

2. PROJECT NO. | TITLE

3. TYPE
a. ☐ MAINT./REPAIR b. ☐ MINOR CONSTRUCTION/ ALTERATION c. ☐ AIR CONDITIONING ☐ EQUIPMENT INSTALLATION

4. DESCRIBE AND STATE FUNCTION OF FACILITY

a. PROPERTY RECORD CARD NO.
b. NAVY CATEGORY CODE
c. BLDG. OR STRUCTURE NO.

5. WHAT IS THE EFFECT OF THIS PROJECT ON THE MISSION OF THE ACTIVITY?

6. THE REQUIREMENT FOR THE FACILITY IS BASED ON
a. ☐ A CHANGE IN MISSION b. ☐ FULL-TIME CONTINUING NEED c. ☐ 3 TO 8 YEAR NEED d. ☐ LESS THAN 3 YEARS NEED e. ☐ CURRENTLY REQUIRED LESS THAN 50% OF TIME f. ☐ RESERVED FOR FUTURE REQUIREMENTS

7a. EST. FUNDED COST | b. EST. PROJECT COST | c. EST. PLANNING COST | d. TOTAL FUNDS REQUESTED | e. EST. FACIL. REPL. COST
$ | $ | $ | $ | $

8. DATE FACILITY CONSTRUCTED

9. IS FACILITY ON AN APPROVED BASIC FACILITY REQUIREMENTS LIST? *If "NO," how was need determined?*
☐ YES ☐ NO

10. IS PROJECT LISTED ON ANNUAL INSPECTION SUMMARY? *If answer is "NO" and AIS is applicable, explain exclusion.*
☐ YES ☐ NO ☐ N.A.

11. DESCRIBE CONDITION TO BE CORRECTED, OR PROBLEM TO BE SOLVED WITH PROPOSED SOLUTION. *Attach additional description if necessary, ONE PAGE ONLY.*

12. WHY IS THE PROPOSED SOLUTION BEST – AND WHAT ALTERNATIVES WERE CONSIDERED?

13. WERE ANY NON-NAVY EXPERTS INVITED TO REVIEW THIS PROBLEM AND THIS SOLUTION? *Explain effect on solution.*
a. ☐ YES b. ☐ NO

14. HAS EFD DESIGN DIVISION REVIEWED SOLUTION? a. ☐ YES b. ☐ NO

15. CAN ANOTHER FACILITY BE ECONOMICALLY ADAPTED FOR THIS FUNCTION? a. ☐ YES b. ☐ NO

16. CAN PROJECT BE FUNDED IN PHASES? *How? How many?*
a. ☐ YES b. ☐ NO

17. THIS PROJECT IS THE RESULT OF
a. ☐ INADEQUATE MAINTENANCE b. ☐ FACILITY AGE c. ☐ DEFICIENT CONSTR. d. ☐ DEFICIENT DESIGN e. ☐ OTHER: _____

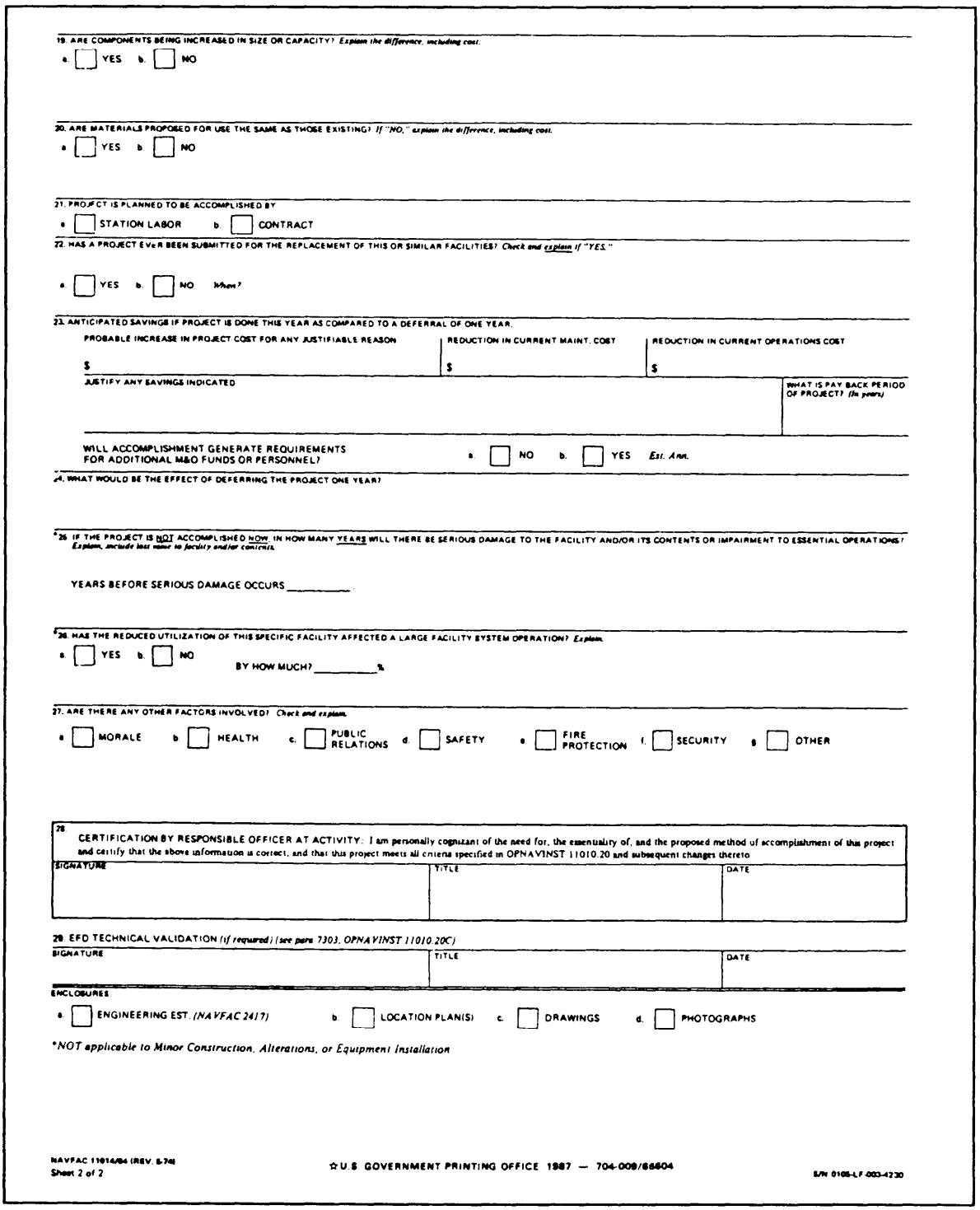
18. HAS THIS SPECIFIC PROBLEM BEEN CORRECTED PREVIOUSLY?
a. ☐ YES b. ☐ NO *When?*

HOW LONG WILL PROPOSED CORRECTIVE ACTION LAST? _____ YEARS

Figure A-6
Special Project Request

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APPENDIX A (Continued)



19. ARE COMPONENTS BEING INCREASED IN SIZE OR CAPACITY? *Explain the difference, including cost.*

a. ☐ YES b. ☐ NO

20. ARE MATERIALS PROPOSED FOR USE THE SAME AS THOSE EXISTING? *If "NO," explain the difference, including cost.*

a. ☐ YES b. ☐ NO

21. PROJECT IS PLANNED TO BE ACCOMPLISHED BY

a. ☐ STATION LABOR b. ☐ CONTRACT

22. HAS A PROJECT EVER BEEN SUBMITTED FOR THE REPLACEMENT OF THIS OR SIMILAR FACILITIES? *Check and explain if "YES."*

a. ☐ YES b. ☐ NO *When?*

23. ANTICIPATED SAVINGS IF PROJECT IS DONE THIS YEAR AS COMPARED TO A DEFERRAL OF ONE YEAR.

PROBABLE INCREASE IN PROJECT COST FOR ANY JUSTIFIABLE REASON	REDUCTION IN CURRENT MAINT. COST	REDUCTION IN CURRENT OPERATIONS COST
$	$	$

JUSTIFY ANY SAVINGS INDICATED

WHAT IS PAY BACK PERIOD OF PROJECT? *(In years)*

WILL ACCOMPLISHMENT GENERATE REQUIREMENTS FOR ADDITIONAL M&O FUNDS OR PERSONNEL? a. ☐ NO b. ☐ YES *Est. Amt.*

24. WHAT WOULD BE THE EFFECT OF DEFERRING THE PROJECT ONE YEAR?

*25. IF THE PROJECT IS NOT ACCOMPLISHED NOW, IN HOW MANY YEARS WILL THERE BE SERIOUS DAMAGE TO THE FACILITY AND/OR ITS CONTENTS OR IMPAIRMENT TO ESSENTIAL OPERATIONS? *Explain, include loss value to facility and/or contents.*

YEARS BEFORE SERIOUS DAMAGE OCCURS _____.

*26. HAS THE REDUCED UTILIZATION OF THIS SPECIFIC FACILITY AFFECTED A LARGE FACILITY SYSTEM OPERATION? *Explain.*

a. ☐ YES b. ☐ NO BY HOW MUCH? _____%

27. ARE THERE ANY OTHER FACTORS INVOLVED? *Check and explain.*

a. ☐ MORALE b. ☐ HEALTH c. ☐ PUBLIC RELATIONS d. ☐ SAFETY e. ☐ FIRE PROTECTION f. ☐ SECURITY g. ☐ OTHER

28. CERTIFICATION BY RESPONSIBLE OFFICER AT ACTIVITY: I am personally cognizant of the need for, the essentiality of, and the proposed method of accomplishment of this project and certify that the above information is correct, and that this project meets all criteria specified in OPNAVINST 11010.20 and subsequent changes thereto

SIGNATURE	TITLE	DATE

29. EFD TECHNICAL VALIDATION *(if required) (see para 7303, OPNAVINST 11010.20C)*

SIGNATURE	TITLE	DATE

ENCLOSURES:

a. ☐ ENGINEERING EST. *(NAVFAC 2417)* b. ☐ LOCATION PLAN(S) c. ☐ DRAWINGS d. ☐ PHOTOGRAPHS

**NOT applicable to Minor Construction, Alterations, or Equipment Installation*

NAVFAC 11014/84 (REV. 5-74)
Sheet 2 of 2

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S/N 0105-LF-003-4230

Figure A-6 (Continued)
Special Project Request

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APPENDIX B

SI CONVERSION FACTORS

Quantity	Multiply	By	to Obtain	
Length	inch	25.400	millimeter	mm
	foot	0.304800	meter	m
	yard	0.914400	meter	m
	mile (U.S. Statute)	1.609347	kilometer	km
	millimeter	0.039370079	inch	in
	meter	3.280840	foot	ft
	meter	1.093613	yard	yd
	kilometer	0.621370	mile	mi
Area	square inch	645.16	square millimeter	mm^2
	square foot	0.092903	square meter	m^2
	square yard	0.836127	square meter	m^2
	square mile (U.S. Statute)	2.589998	square kilometer	km^2
	acre	4046.873	square meter	m^2
	acre	0.404687	hectare	
	square millimeter	0.001550003	square inch	in^2
	square meter	10.763910	square foot	ft^2
	square meter	1.195990	square yard	yd^2
	square kilometer	0.386101	square mile	mi^2
	square meter	0.000247104	acre	
	hectare	2.471044	acre	
Volume	cubic inch	16387.06	cubic millimeter	mm^3
	cubic foot	0.02831685	cubic meter	m^3
	cubic yard	0.764555	cubic meter	m^3
	gallon (U.S.liquid)	3.785412	liter	l
	quart (U.S.liquid)	0.946353	liter	l
	cubic millimeter	0.000061023759	cubic inch	in^3
	cubic meter	35.314662	cubic foot	ft^3
	cubic meter	1.307951	cubic yard	yd^3
	liter	0.264172	gallon (U.S.liquid)	gal
	liter	1.056688	quart (U.S.liquid)	qt
Mass	ounce (avoirdupois)	28.34952	gram	g
	pound (avoirdupois)	0.453592	kilogram	kg
	short ton	907.185	kilogram	kg
	gram	0.035273966	ounce (avoirdupois)	oz av
	kilogram	2.204622	pound (avoirdupois)	lb av
	kilogram	0.001102311	short ton	

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APPENDIX C

UNITS OF MEASURE AND THEIR SYMBOLS

Symbol	Meaning	Notes
AC	Acres	
BD	Hospital Beds, Normal Capacity	
BL	Barrels (42 gals each)	
BX	Boxes (fire alarm etc.)	
CF	Cubic Feet	
CM	CF/Minute	1
CP	Candlepower 1	
CY	Cubic Yards Tide	1
DS	Depth of Water Over Sill @ High Tide	1
DW	Depth of Water @ Low Tide	
EA	Each	
FA	Family Units (housing)	
FB	Feet of Berthing (ships/boats/small craft)	
FP	Firing Point (firing ranges)	1
GA	Gallons	
GM	Gallons Per Minute	
HO	Holes (golf course)	1
KG	Thousands of Gallons Per Day, Capacity	
KV	Kilovolt-Amperes, Capacity (KVA)	
KW	Kilowatts	1
LA	Lanes (bowling)	1
LC	Light Care (hospital spaces)	1
LF	Linear Feet	
MB	Millions of BTU Per Hr (capacity)	
ME	Meters (length)	1
MG	Millions of Gallons	1
MI	Statute Miles	
NF	Net Square Feet (housing)	1, 2
NS	Net square Feet (storage)	1, 3
OL	Outlets (number of)	
OU	Operating Units	
PH	Pounds Per Hour	1
PH	Persons, Designed Capacity	

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APPENDIX C (Continued)

Symbol	Meaning	Notes
SE	Seats (number of)	
SF	Square Feet	
SH	Stacking Height	1, 3
SI	Sites	1
SP	Service Points	1
SY	Square Yards	
TC	Total Cubic Feet	1, 3
TH	Tons Per Hour (capacity)	1
TN	Tons (capacity)	
VE	Vehicles (number of)	

1 Not listed in DODINST 4165.3 of 24 October 1978, as amended
2 As defined by NAVFACENGCOM for housing purposes
3 As defined by NAVSUPSYSCOM for supply facilities

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APPENDIX D

ACRONYMS AND ABBREVIATIONS

ABSLA	Approved Basic Stock Level of Ammunition
AC	Area Cost Factor
ADCA	Anticipated Date of Contract Award
A&E	Architect and Engineer
AGI	Activity General Information
AH	Alternate Host
AICUZ	Air Installations Compatible Use Zones
AIS	Annual Inspection Summary
ASD	Assistant Secretary of Defense
BAM	Baseline Assessment Memorandum
BASE REP	Shore Base Readiness Report
BEAP	Base Exterior Architecture Plan
BEQ	Bachelor Enlisted Quarters
BESEP	Base Electronics Systems Engineering Plan
BFR	Basic Facility Requirements
BLS	Base Loading System
BOQ	Bachelor Officer Quarters
CAUSE	Computer Assisted Utility Systems Evaluation
CCD	Category Code Directory (data base)
CCN	Category Code Number
CCTV	Closed Circuit Television
CES	Computerized Estimating System
CES-H	Computerized Estimating System - Historical
CIP	Capital Improvements Plan
CL3-D	Class 3 (Relocatable); planned for disposition (planning action)
CL3-I	Class 3 (Relocatable); used on interim basis (planning action)
CL3-U	Class 3 (Relocatable); temporary use (planning action)
CMC	Commandant, Marine Corps
CNET	Chief of Naval Education and Training
CNO	Chief of Naval Operations
CO	Commanding Officer
COMNAVFACENGCOM	Commander, Naval Facilities Engineering Command
COMNISCOM	Naval Investigative Service Command
CONSTR	New Construction (planning action)
CONUS	Continental United States
CONVFR	Convert From (planning action)
CONVTO	Convert To (planning action)
CPS	(Final) Consolidated Personnel Summary
CPV	Current Plant Value
CQM	Contractor Quality Management
CRD	Certified, Ready for Design
CY	Cubic Yards (unit of measure)

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APPENDIX D (Continued)

D	Planning Action Designator
DASN	Deputy Assistant Secretary of the Navy
DCNO	Deputy Chief of Naval Operations
DDESB	Department of Defense Explosives Safety Board
DEMOL	Demolition (planning action)
DF	Discount Factor
DISPOS	Disposable Asset (planning action)
DPS	Decision Package Sets
DRA	Defense Relocation Account (funds)
DOD	Department of Defense
EA	Economic Analysis
EA	Each (unit of measure)
EAF	Environmental Adjustment Factor
ECIP	Energy Conservation Investment Program
EE	Engineering Evaluation
EFD	Engineering Field Division
EMCS	Energy Monitoring and Control Systems
EOD	Explosives Ordnance Disposal
FACSO	Facilities Systems Office (Port Hueneme, CA)
FEC	Foreign Exchange Cost
FEP	Facility Energy Plans
FER	Foreign Exchange Rate
FHS	Family Housing Plans
FPD	Facility Planning Document
FRP	Facilities Requirements Plan
FUR	Facilities Utilization Report
GEMS	Graphics Engineering and Mapping System
GSA	General Services Administration
H/T	Host/Tenant
HCA	Hazard Control Assessment
HVAC	Heating, Ventilation, and Air Conditioning
IBOP	International Balance of Payments
ID	Planning Action Identifiers
IDS	Intrusion Detection System
IP	Investment Program
JFAI	Joint Formal Acceptance Inspection
LS	Lump Sum (unit of measure)
MAGIC	Master Activity General Information and Control
MARCORPS	Marine Corps
MBTU	Million British Thermal Units

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APPENDIX D (Continued)

MCNR	Military Construction Naval Reserve
MCON	Military Construction, Navy
MCP/MIS	Military Construction Programming Management Information System
MILCON	Military Construction
MILPERS	Naval Military Personnel Command
MODIFY	Modification (planning action)
NACIP	Navy Assessments and Control of Installation Pollutants
NADEP	Naval Aviation Depot (formerly Naval Air Rework Facility (NARF))
NAF	Nonappropriated Funded; Nonappropriated Fund
NATO	North Atlantic Treaty Organization
NAVAIRSYSCOM	Naval Air Systems Command
NAVCOMPT	Comptroller of the Navy
NAVELEXCEN	Naval Electronic Systems Engineering Center
NAVELEXSECCEN	Naval Electronic Systems Security Engineering Center
NAVFAC	Naval Facilities Engineering Command (for directives and publications)
NAVFACENGCOM	Naval Facilities Engineering Command
NAVFACENGCOMHQ	Naval Facilities Engineering Command, Headquarters
NAVMEDCOM	Naval Medical Command
NAVOSH	Navy Occupational Safety and Health
NAVRESSO	Navy Resale and Services Support Office
NAVSEASYSCOM	Naval Sea Systems Command
NAVSUPSYSCOM	Naval Supply Systems Command
NCR	National Capital Region
NFADB	Navy Facility Assets Data Base
NFH	Navy Family Housing
NMCRC	Navy Marine Corps Reserve Centers
NPV	Net Present Value
NS	Net Square Feet (unit of measure)
O&M,N	Operations & Maintenance, Navy (funds)
OCONUS	Outside of Continental United States
OCR	Occupational Safety and Health Control Report
OMB	Office of Management and Budget
OMSI	Operation and Maintenance Support Information
OPN	Other Procurement Navy
OPNAV	Office of the Chief of Naval Operations
OSD	Office of the Secretary of Defense
OUTG-C	Outgrant, continue (planning action)
OUTG-R	Outgrant, retrieve (planning action
PART FRP	Partial Facilities Requirements Plan
PCAS	Post Construction Award Services
PCR	Pollution Control Report
PCS	Permanent Change of Station
PDS	Project Data Sheet

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APPENDIX D (Continued)

PEA	Preliminary Environmental Assessment
PED	Project Engineering Documentation
PHA	Preliminary Hazard Analysis
PID	Project Identification Document
POM	Program Objectives Memorandum
PPBS	Planning, Programming, and Budgeting System
PVA	Present Value Analysis (economic analysis)
PWC	Public Works Center
QAP	Quality Assurance Plan
RAC	Risk Assessment Code
RCRA	Resource Conservation and Recovery Act
RDT&E	Research, Development, Test & Evaluation
REASFR	Reassignment From (planning action)
REASTO	Reassignment To (planning action)
RENOV	Renovation (planning action)
REPLCE	Replace (planning action)
RETAIN	Retain For Contingency (planning action)
RETR	Retrieve (planning action identifier)
RJE	Remote Job Entry
RL	Requirements List
SA	Special Area
SCF	Supplemental Category Flag
SECDEF	Secretary of Defense
SECNAV	Secretary of the Navy
SF	Square Feet, Gross (unit of measure)
SFMR	Supply Facility Management Report
SFPB	Shore Facilities Programming Board
SFPS	Shore Facilities Planning System
SH	Stacking Height (unit of measure)
SIMA	Shore Intermediate Maintenance Activity
SIOH	Supervision, Inspection, and Overhead
SIR	Savings to Investment Ratio (economic analysis)
SNDL	Standard Navy Distribution List
SPP	Sponsor Program Proposal
SY	Square Yards (unit of measure)
SYDP	Six Year Defense Program
TCF	Total Cubic Feet (unit of measure)
TDY	Temporary Duty
TOA	Total Obligation Authority
TPAA	Total Proposed Adequate Assets
UIC	Unit Identification Code
UM	Unit of Measure

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APPENDIX D (Continued)

UPH	Unaccompanied Personnel Housing (now called bachelor housing)
UPS	Uninterruptible Power Source
USA	Utility System Assessments
USE	Continued Use (planning action)
UTS	Utility Technical Studies
VAC	Vacant (planning action identifier)
VI	Validation Indicator

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APPENDIX E

AIR CONDITIONING TONNAGE GUIDE

Category Code	Title	Square Feet of Floor Area/Ton of A/C
171-15	Reserve Centers	450 - 600
171-20	Applied Instruction Building	NO STANDARD
171-25	Auditorium	0.04 - 0.06 tons/seat
171-35	Flight Simulator	100 - 300
610-10	Administration Building	350 - 400
721-11	BEQ	1000 - 1200
722-10	Dining Facility	175 - 450
724-11	BOQ	1000 - 1200
730-83	Chapel	300 - 500; or 0.02 - 0.03 tons/seat
740-02	Location Exchange	NO STANDARD
740-20	Temporary Lodging	550 - 750
740-23	Commissary	800 - 900
740-25	Family Service Center	375 - 400
740-40	Bowling Alley	0.8 - 1.4 tons/alley
740-63	Enlisted Men's & Officers Clubs	600 - 650
740-74	Child Care Center	350 - 450
	Classrooms	400 - 500
	Computer Room	50 - 150
	Dispensaries	450 - 550
	Hospital Patient Rooms	450 - 550
	Married Personnel Quarters	900 - 1275
	Recreation Rooms	375 - 450
	Shops (Precision Equipment)	450 - 500

Estimating Electrical Loads for Air Conditioning Systems

It is frequently necessary to estimate the power requirements for air conditioning systems when preparing a MILCON project and in other instances. For years, a rough rule of thumb was to use one kilowatt per ton of cooling for any type system. Use of this rule neglects variation in cooling system efficiencies and the power consumption of auxiliaries such as fans and pumps.

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APPENDIX F

PLUMBING FIXTURE COUNT

Floor Drains (All Sizes)	1/3 Fixture
Roof Drains (All Sizes)	1 Fixture
Floor Sink (All Sizes)	1/3 Fixture
Water Closet	1 Fixture
Lavatories	1 Fixture
Kitchen Sink	1 Fixture
Service Sink	1 Fixture
Bidets	1 Fixture
Electric Water Cooler	1 Fixture
Drinking Fountain	1/2 Fixture
Hand Wash Fountain (180)	3 Fixture
Hand Wash Fountain (360)	4 Fixture
Bathtub	1 Fixture
Water Heater	0 Fixture
Urinal	1 Fixture
Single Shower Fixture	1/2 Fixture
Prefab. Shower (incl drain & fix)	1 Fixture
Multi-head Shower (SS)	1 Fixture
Emergency Shower & Eye Wash	1 Fixture
Eye Wash	1/2 Fixture
Emergency Shower	1/2 Fixture
Dishwasher Connection	1/2 Fixture
Washing Machine Connection	1/2 Fixture
Garbage Disposal	1/4 Fixture
Rough-in Piping (HW, CW, Waste)	1/2 Fixture
Mop Sink	1 Fixture

The above table is to be used to determine number of fixtures for Standard System Descriptions, Systems 211 and 213.

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REFERENCES

NOTE: THE FOLLOWING REFERENCED DOCUMENTS FORM A PART OF THIS HANDBOOK TO THE EXTENT SPECIFIED HEREIN. USERS OF THIS HANDBOOK SHOULD REFER TO THE LATEST REVISIONS OF CITED DOCUMENTS UNLESS OTHERWISE DIRECTED.

FEDERAL/MILITARY SPECIFICATIONS, STANDARDS, BULLETINS, AND HANDBOOKS:

HANDBOOKS

MIL-HDBK-1190	Facility Planning and Design Guide
MIL-HDBK-1195	Radio Frequency Shielded Enclosures

Unless otherwise indicated, copies are available from the Naval Publications and Forms Center, Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

DESIGN MANUALS AND NAVFAC P-PUBLICATIONS:

DM-13.02	Commercial Intrusion Detection Systems (IDS)
P-72	Department of the Navy Facility Category Codes

(Unless otherwise indicated, copies are available from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120; private organizations may purchase design manuals and NAVFAC P-publications from the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

OTHER GOVERNMENT DOCUMENTS AND PUBLICATIONS:

FAR	Federal Acquisition Regulation

(Available from the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

NAVFACINST 11010.44	Shore Facilities Planning Manual

(Available from Commander, Naval Facilities Engineering Command, 200 Stovall Street, Alexandria, VA 22332-2300.)

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OPNAVINST 11010.16	Command Responsibility for Shore Activity Land and Facilities
OPNAVINST 11010.20	Facilities Projects Manual

(Available from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

CUSTODIAN
NAVY YD

PREPARING ACTIVITY
NAVY YD

PROJECT NO.
FACR-1075

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-HDBK 1010A	2. DOCUMENT DATE (YYMMDD) 920801

3. DOCUMENT TITLE
Cost Engineering: Policy and Procedures

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Naval Facilities Engineering Command	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON 564-9895
c. ADDRESS (Include Zip Code) Atlantic Division, Code 04A4 Norfolk, VA 23511-6287	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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